

TECHNE & TEACHING

CRITICAL LITERACIES



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Nissaba Creates Writing for the Praise Song of Enlil
(from the *Keš Hymn*)

⁹ ^dEn-lil₂-le Keš₃^{ki}-a za₃-mim
am₃-ma-ab-be₂

¹⁰ ^dNissaba nu-KA dili-bi-im

¹¹ enim-bi-da sa-gin₇ im-da-an-sur

¹² dub-ba-am₃
al-ĝa₂-ĝa₂ šu-še₃ al-ĝa₂-ĝa₂

⁹ While Enlil was now singing the praise
song to Keš,¹⁸

¹⁰ Nissaba was the only one who knew
how to write,

¹¹ because she had mastered the art of
weaving the song of praise from these
words like a net.¹⁹

¹² There was a tablet.

While she was taking it,
while she took this tablet in her hand, ...

- trans. Annette Zgoll

SORRY TO TELL YOU A DREAM!

WHAT IS THIS WRITING LIFE? I was living alone in a house, and had set up a study on the first floor. A portable green Smith-Corona typewriter sat on the table against the wall. I made the mistake, or dreamed I made the mistake, of leaving the room.

I was upstairs when I felt the first tremor. The floor wagged under my feet—what was that?—and the picture frames on the wall stirred. The house shook and made noise. There was a pause; I found my face in the dresser mirror, deadpan. When the floor began again to sway, I walked downstairs, thinking I had better get down while the stairway held.

I saw at once that the typewriter was erupting. The old green Smith-Corona typewriter on the table was exploding with fire and ash. Showers of sparks

from The Writing Life

Annie Dillard

shot out of its caldera—the dark hollow in which the keys lie. Smoke and cinders poured out, noises exploded and spattered, black dense smoke rose up, and a wild, deep fire lighted the whole thing. It shot sparks.

I pulled down the curtains. When I leaned over the typewriter, sparks burnt round holes in my shirt, and fire singed a sleeve. I dragged the rug away from the sparks. In the kitchen I filled a bucket with water and returned to the erupting typewriter. The typewriter did not seem to be flying apart, only erupting. On my face and hands I felt the heat from the caldera. The yellow fire made a fast, roaring noise. The typewriter itself made a rumbling, grinding noise; the table pitched. Nothing seemed to require my bucket of water. The table surface was ruined, of course, but not aflame. After twenty minutes or so, the eruption subsided.

That night I heard more rumblings—weak ones, ever farther apart. The next day I cleaned the typewriter, table, floor, wall, and ceiling. I threw away the burnt shirt. The following day I cleaned the typewriter again—a film of lampblack still coated the caldera—and then it was over. I have had no trouble with it since. Of course, now I know it can happen.

from *The Book of Delights*
Ross Gay

10. Writing by Hand

I LOVE THE story, apocryphal or not, of Derek Walcott asking his graduate poetry workshop on the first day if they composed by hand or on a computer. As I recall, about half of the class raised their hands when he said computer. I didn't know him, but I did sit in on a workshop he led in college shortly after he won the Nobel Prize and I was both mesmerized and a touch terrified by his mellifluous and curt voice, lilting like a beach rose, all fragrant and thorny. He said, with almost no affect (which is itself an affect), "You six can leave my workshop." And just like you would've, they gathered their things and started down the hall, probably wondering if Pinsky had any seats open in his class. Before they got too far though he called them back—the fragrant part

of the rose—"C'mon, c'mon, I'm just making a point."
What was the point?

Susan Sontag said somewhere something like any technology that slows us down in our writing rather than speeding us up is the one we ought to use. Her treatise on the subject, long-handist that she evidently was. (I wonder if the speed they all gobbled in the sixties and seventies counts as a technology.) And though I don't have any particular treatise, I do want to acknowledge that writing by hand, and writing these essayettes by hand in particular, has been a surprising and utter delight. Mind you, I would not have been tossed from Walcott's workshop, because I write poems pretty slowly, line by line, with a pen, a Le Pen these days (a delight, the Le Pen is).

Prose, though, I often write on the computer, piling sentences up quickly, cutting and pasting, deleting whole paragraphs without thinking anything of it. For these essays, though, I decided that I'd write by hand, mostly with Le Pens, in smallish notebooks. I can tell you a few things—first, the pen, the hand behind the pen, is a digressive beast. It craves, in my experience anyway, the wending thought, and crafts/imagines/conjures a syntax to contain it. On the other hand, the process of thinking that writing is, made disappearable by the delete button,

makes a whole part of the experience of writing, which is the production of a good deal of florid detritus, flotsam and jetsam, all those words that mean what you have written and cannot disappear (the scratch-out its own archive), which is the weird path toward what you have come to know, which is called thinking, which is what writing is.

For instance, the previous run-on sentence is a sentence fragment, and it happened in part because of the really nice time my body was having making this lavender Le Pen make the loop-de-looping we call language. I mean writing. The point: I'd no sooner allow that fragment to sit there like a ripe zit if I was typing on a computer. And consequently, some important aspect of my thinking, particularly the breathlessness, the accruing syntax, the not quite articulate pleasure that evades or could give a fuck about the computer's green corrective lines (how they injure us!) would be chiseled, likely with a semicolon and a proper predicate, into something correct, and, maybe, dull. To be sure, it would have less of the actual magic writing is, which comes from our bodies, which we actually think with, quiet as it's kept.

(Sep. 12)

from *The Work of Art*
in the Age of Mechanical Reproduction

WALTER BENJAMIN

I

In principle a work of art has always been reproducible. Man-made artifacts could always be imitated by men. Replicas were made by pupils in practice of their craft, by masters for diffusing their works, and, finally, by third parties in the pursuit of gain. Mechanical reproduction of a work of art, however, represents something new. Historically, it advanced intermittently and in leaps at long intervals, but with accelerated intensity. The Greeks knew only two procedures of technically reproducing works of art: founding and stamping. Bronzes, terra cottas, and coins were the only art works which they could produce in quantity. All others were unique and could not be mechanically reproduced. With the woodcut graphic art became mechanically reproducible for the first time, long before script became reproducible by print. The enormous changes which printing, the mechanical reproduction of writing, has brought about in literature are a familiar story. However, within the phenomenon which we are here examining from the perspective of world history, print is merely a special, though particularly important, case. During the Middle Ages engraving and etching were added to the woodcut; at the beginning of the nineteenth century lithography made its appearance.

With lithography the technique of reproduction reached an essentially new stage. This much more direct process was distinguished by the tracing of the design on a stone rather than its incision on a block of wood or its etching on a copperplate and permitted graphic art for the first time to put its products on the market, not only in large numbers as hitherto, but also in daily changing forms. Lithography enabled graphic art to illustrate everyday life, and it began to keep pace with printing. But only a few decades after its invention, lithography was surpassed by photography. For the first time in the process of pictorial reproduction, photography freed the hand of the most important artistic functions which henceforth devolved only upon the eye looking into a lens.

Since the eye perceives more swiftly than the hand can draw, the process of pictorial reproduction was accelerated so enormously that it could keep pace with speech. A film operator shooting a scene in the studio captures the images at the speed of an actor's speech. Just as lithography virtually implied the illustrated newspaper, so did photography foreshadow the sound film. The technical reproduction of sound was tackled at the end of the last century. These convergent endeavors made predictable a situation which Paul Valéry pointed up in this sentence: "Just as water, gas, and electricity are brought into our houses from far off to satisfy our needs in response to a minimal effort, so we shall be supplied with visual or auditory images, which will appear and disappear at a simple movement of the hand, hardly more than a sign." (*op. cit.*, p. 226) Around 1900 technical reproduction had reached a standard that not only permitted it to reproduce all transmitted works of art and thus to cause the most profound change in their impact upon the public; it also had captured a place of its own among the artistic processes. For the study of this standard nothing is more revealing than the nature of the repercussions that these two different manifestations—the reproduction of works of art and the art of the film—have had on art in its traditional form.

II

Even the most perfect reproduction of a work of art is lacking in one element: its presence in time and space, its unique existence at the place where it happens to be. This unique existence of the work of art determined the history to which it was subject throughout the time of its existence. This includes the changes which it may have suffered in physical condition over the years as well as the various changes in its ownership.¹ The traces of the first can be revealed only by chemical or physical analyzes which it is impossible to perform on a reproduction; changes of ownership are subject to a tradition which must be traced from the situation of the original.

The presence of the original is the prerequisite to the concept of authenticity. Chemical analyzes of the patina of a bronze can help to establish this, as does the proof that a given manuscript of the Middle Ages stems from an archive of the fifteenth century. The whole sphere of authenticity is outside technical—and, of course, not only technical—reproducibility.² Confronted with its manual reproduction, which was usually branded as a forgery, the original preserved all its authority; not so *vis à vis* technical reproduction. The reason is twofold. First, process reproduction is more independent of the original than manual reproduction. For example, in photography, process reproduction can bring out those aspects of the original that are unattainable to the naked eye yet accessible to the lens, which is adjustable and chooses its

angle at will. And photographic reproduction, with the aid of certain processes, such as enlargement or slow motion, can capture images which escape natural vision. Secondly, technical reproduction can put the copy of the original into situations which would be out of reach for the original itself. Above all, it enables the original to meet the beholder halfway, be it in the form of a photograph or a phonograph record. The cathedral leaves its locale to be received in the studio of a lover of art; the choral production, performed in an auditorium or in the open air, resounds in the drawing room.

The situations into which the product of mechanical reproduction can be brought may not touch the actual work of art, yet the quality of its presence is always depreciated. This holds not only for the art work but also, for instance, for a landscape which passes in review before the spectator in a movie. In the case of the art object, a most sensitive nucleus—namely, its authenticity—is interfered with whereas no natural object is vulnerable on that score. The authenticity of a thing is the essence of all that is transmissible from its beginning, ranging from its substantive duration to its testimony to the history which it has experienced. Since the historical testimony rests on the authenticity, the former, too, is jeopardized by reproduction when substantive duration ceases to matter. And what is really jeopardized when the historical testimony is affected is the authority of the object.³

One might subsume the eliminated element in the term “aura” and go on to say: that which withers in the age of mechanical reproduction is the aura of the work of art. This is a symptomatic process whose significance points beyond the realm of art. One might generalize by saying: the technique of reproduction detaches the reproduced object from the domain of tradition. By making many reproductions it substitutes a plurality of copies for a unique existence. And in permitting the reproduction to meet the beholder or listener in his own particular situation, it reactivates the object reproduced. These two processes lead to a tremendous shattering of tradition which is the obverse of the contemporary crisis and renewal of mankind. Both processes are intimately connected with the contemporary mass movements. Their most powerful agent is the film. Its social significance, particularly in its most positive form, is inconceivable without its destructive, cathartic aspect, that is, the liquidation of the traditional value of the cultural heritage. This phenomenon is most palpable in the great historical films. It extends to ever new positions. In 1927 Abel Gance exclaimed enthusiastically: “Shakespeare, Rembrandt, Beethoven will make films . . . all legends, all mythologies and all myths, all founders of religion, and the very religions . . . await their exposed resurrection, and the heroes crowd each other at the gate.”*

* Abel Gance, “Le Temps de l’image est venu,” *L’Art cinématographique*, Vol 2, pp. 94F, Paris, 1927.

Presumably without intending it, he issued an invitation to a far-reaching liquidation.

III

During long periods of history, the mode of human sense perception changes with humanity's entire mode of existence. The manner in which human sense perception is organized, the medium in which it is accomplished, is determined not only by nature but by historical circumstances as well. The fifth century, with its great shifts of population, saw the birth of the late Roman art industry and the Vienna Genesis, and there developed not only an art different from that of antiquity but also a new kind of perception. The scholars of the Viennese school, Riegl and Wickhoff, who resisted the weight of classical tradition under which these later art forms had been buried, were the first to draw conclusions from them concerning the organization of perception at the time. However far-reaching their insight, these scholars limited themselves to showing the significant, formal hallmark which characterized perception in late Roman times. They did not attempt—and, perhaps, saw no way—to show the social transformations expressed by these changes of perception. The conditions for an analogous insight are more favorable in the present. And if changes in the medium of contemporary perception can be comprehended as decay of the aura, it is possible to show its social causes.

The concept of aura which was proposed above with reference to historical objects may usefully be illustrated with reference to the aura of natural ones. We define the aura of the latter as the unique phenomenon of a distance, however close it may be. If, while resting on a summer afternoon, you follow with your eyes a mountain range on the horizon or a branch which casts its shadow over you, you experience the aura of those mountains, of that branch. This image makes it easy to comprehend the social bases of the contemporary decay of the aura. It rests on two circumstances, both of which are related to the increasing significance of the masses in contemporary life. Namely, the desire of contemporary masses to bring things 'closer' spatially and humanly, which is just as ardent as their bent toward overcoming the uniqueness of every reality by accepting its reproduction.⁴ Every day the urge grows stronger to get hold of an object at very close range by way of its likeness, its reproduction. Unmistakably, reproduction as offered by picture magazines and newsreels differs from the image seen by the unarméd eye. Uniqueness and permanence are as closely linked in the latter as are transitoriness and reproducibility in the former. To pry an object from its shell, to destroy its aura, is the mark of a perception whose 'sense of the universal equality of things' has increased to such a degree that it extracts it even from a unique object by means of

reproduction. Thus is manifested in the field of perception what in the theoretical sphere is noticeable in the increasing importance of statistics. The adjustment of reality to the masses and of the masses to reality is a process of unlimited scope, as much for thinking as for perception.

IV

The uniqueness of a work of art is inseparable from its being imbedded in the fabric of tradition. This tradition itself is thoroughly alive and extremely changeable. An ancient statue of Venus, for example, stood in a different traditional context with the Greeks, who made it an object of veneration, than with the clerics of the Middle Ages, who viewed it as an ominous idol. Both of them, however, were equally confronted with its uniqueness, that is, its aura. Originally the contextual integration of art in tradition found its expression in the cult. We know that the earliest art works originated in the service of a ritual—first the magical, then the religious kind. It is significant that the existence of the work of art with reference to its aura is never entirely separated from its ritual function.⁵ In other words, the unique value of the “authentic” work of art has its basis in ritual, the location of its original use value. This ritualistic basis, however remote, is still recognizable as secularized ritual even in the most profane forms of the cult of beauty.⁶ The secular cult of beauty, developed during the Renaissance and prevailing for three centuries, clearly showed that ritualistic basis in its decline and the first deep crisis which befell it. With the advent of the first truly revolutionary means of reproduction, photography, simultaneously with the rise of socialism, art sensed the approaching crisis which has become evident a century later. At the time, art reacted with the doctrine of *l’art pour l’art*, that is, with a theology of art. This gave rise to what might be called a negative theology in the form of the idea of ‘pure’ art, which not only denied any social function of art but also any categorizing by subject matter. (In poetry, Mallarmé was the first to take this position.)

An analysis of art in the age of mechanical reproduction must do justice to these relationships, for they lead us to an all-important insight: for the first time in world history, mechanical reproduction emancipates the work of art from its parasitical dependence on ritual. To an ever greater degree the work of art reproduced becomes the work of art designed for reproducibility.⁷ From a photographic negative, for example, one can make any number of prints; to ask for the ‘authentic’ print makes no sense. But the instant the criterion of authenticity ceases to be applicable to artistic production, the total function of art is reversed. Instead of being based on ritual, it begins to be based on another practice—politics.

from **Zen and the Art of Motorcycle Maintenance**
by **Robert Pirsig**



*What the Metaphysics of Quality would do is take this separate category, Quality, and show how it contains within itself both subjects and objects. The Metaphysics of Quality would show how things become enormously more coherent--fabulously more coherent--when you start with an assumption that Quality is the primary empirical reality of the world....
...But showing that, of course, was a very big job....*

-- From Lila

This morning a solution to the problem of stuckness was discussed, the classic badness caused by traditional reason. Now it's time to move to its romantic parallel, the ugliness of the technology traditional reason has produced.

The road has twisted and rolled over desert hills into a little, narrow thread of green surrounding the town of White Bird, then proceeded on to a big fast river, the Salmon, flowing between high canyon walls. Here the heat is tremendous and the glare from the white canyon rock is blinding. We wind on and on through the bottom of the narrow canyon, nervous about fast-moving traffic and oppressed by the fiery heat.

The ugliness the Sutherlands were fleeing is not inherent in technology. It only seemed that way to them because it's so hard to isolate what it is within technology that's so ugly. But technology is simply the making of things and the making of things can't by its own nature be ugly or there would be no possibility for beauty in the arts, which also include the making of things. Actually a root word of technology, *techne*, originally meant "art." The ancient Greeks never separated art from manufacture in their minds, and so never developed separate words for them.

Neither is the ugliness inherent in the materials of modern technology ... a statement you sometimes hear. Mass-produced plastics and synthetics aren't in themselves bad. They've just acquired bad associations. A person who's lived inside stone walls of a prison most of his life is likely to see stone as an inherently ugly material, even though it's also the prime material of sculpture, and a person who's lived in a prison of ugly plastic technology that started with his childhood toys and continues through a lifetime of junky consumer products is likely to see this material as inherently ugly. But the real ugliness of modern technology isn't found in any material or shape or act or product. These are just the objects in which the low Quality appears to reside. It's our habit of assigning Quality to subjects or objects that gives this impression.

The real ugliness is not the result of any objects of technology. Nor is it, if one follows Phædrus' metaphysics, the result of any subjects of technology, the people who produce it or the people who use it. Quality, or its absence, doesn't reside in either the subject or the object. The real ugliness lies in the relationship between the people who produce the technology and the things they produce, which results in a similar relationship between the people who use the technology and the things they use.

Phædrus felt that at the moment of pure Quality perception, or not even perception, at the moment of pure Quality, there is no subject and there is no object. There is only a sense of Quality that produces a later awareness of subjects and objects. At the moment of pure quality, subject and object are identical. This is the *tat tvam asi* truth of the Upanishads, but it's also reflected in modern street argot. "Getting with it," "digging it," "grooving on it" are all slang reflections of this identity. It is this identity that is the basis of craftsmanship in all the technical arts. And it is this identity that modern, dualistically conceived technology lacks. The creator of it feels no particular sense of identity with it. The owner of it feels no particular sense of identity with it. The user of it feels no particular sense of identity with it. Hence, by Phædrus' definition, it has no Quality.

That wall in Korea that Phædrus saw was an act of technology. It was beautiful, but not because of any masterful intellectual planning or any scientific supervision of the job, or any added expenditures to "stylize" it. It was beautiful because the people who worked on it had a way of looking at things that made them do it right unselfconsciously. They didn't separate themselves from the work in such a way as to do it wrong. There is the center of the whole solution.

The way to solve the conflict between human values and technological needs is not to run away from technology. That's impossible. The way to resolve the conflict is to break down the barriers of dualistic thought that prevent a real understanding of what technology is ... not an exploitation of nature, but a fusion of nature and the human spirit into a new kind of creation that transcends both. When this transcendence occurs in such events as the first airplane flight across the ocean or the first footstep on the

moon, a kind of public recognition of the transcendent nature of technology occurs. But this transcendence should also occur at the individual level, on a personal basis, in one's own life, in a less dramatic way.

The walls of the canyon here are completely vertical now. In many places room for the road had to be blasted out of it. No alternate routes here. Just whichever way the river goes. It may be just my imagination, but it seems the river's already smaller than it was an hour ago.

Such personal transcendence of conflicts with technology doesn't have to involve motorcycles, of course. It can be at a level as simple as sharpening a kitchen knife or sewing a dress or mending a broken chair. The underlying problems are the same. In each case there's a beautiful way of doing it and an ugly way of doing it, and in arriving at the high-quality, beautiful way of doing it, both an ability to see what ``looks good" and an ability to understand the underlying methods to arrive at that ``good" are needed. Both classic and romantic understandings of Quality must be combined.

The nature of our culture is such that if you were to look for instruction in how to do any of these jobs, the instruction would always give only one understanding of Quality, the classic. It would tell you how to hold the blade when sharpening the knife, or how to use a sewing machine, or how to mix and apply glue with the presumption that once these underlying methods were applied, ``good" would naturally follow. The ability to see directly what ``looks good" would be ignored.

The result is rather typical of modern technology, an overall dullness of appearance so depressing that it must be overlaid with a veneer of ``style" to make it acceptable. And that, to anyone who is sensitive to romantic Quality, just makes it all the worse. Now it's not just depressingly dull, it's also phony. Put the two together and you get a pretty accurate basic description of modern American technology: stylized cars and stylized outboard motors and stylized typewriters and stylized clothes. Stylized refrigerators filled with stylized food in stylized kitchens in stylized houses. Plastic stylized toys for stylized children, who at Christmas and birthdays are in style with their stylish parents. You have to be awfully stylish yourself not to get sick of it

once in a while. It's the style that gets you; technological ugliness syruped over with romantic phoniness in an effort to produce beauty and profit by people who, though stylish, don't know where to start because no one has ever told them there's such a thing as Quality in this world and it's real, not style. Quality isn't something you lay on top of subjects and objects like tinsel on a Christmas tree. Real Quality must be the source of the subjects and objects, the cone from which the tree must start.

To arrive at this Quality requires a somewhat different procedure from the ``Step 1, Step 2, Step 3" instructions that accompany dualistic technology, and that's what I'll now try to go into.

After many turns in the canyon wall we stop for a break under a scrubby little patch of small trees and rocks. The grass around the trees is burned and brown and scattered with litter from picnickers.

I collapse into some shade, and after a while squint up at the sky, which I haven't really looked at since we entered this canyon. Up there above the canyon walls it's cool and dark blue and far away.

Chris doesn't even go over to see the river, something he'd normally do. Like me, he's tired and content just to sit under the scant shade of these trees.

After a while he says there's an old iron pump, it looks like, between us and the river. He points to it and I see what he means. He goes over and I can see him pump water onto his hand and then splash it onto his face. I go over and pump for him so he can use both hands. Then I do the same. The water feels cold on my hands and face. When done we walk to the cycle again and climb on and pull back on to the canyon road.

Now that solution. Throughout this Chautauqua so far this whole problem of technological ugliness has been looked at in a negative way. It's been said that romantic attitudes toward Quality such as the Sutherlands have are, by themselves, hopeless. You can't live on just groovy emotions alone. You have to work with the underlying form of the universe too, the laws of nature which, when understood, can make work easier, sickness rarer and

famine almost absent. On the other hand, technology based on pure dualistic reason has also been condemned because it obtains these material advantages by turning the world into a stylized garbage dump. Now's the time to stop condemning things and come up with some answers.

The answer is Phædrus' contention that classic understanding should not be overlaid with romantic prettiness; classic and romantic understanding should be united at a basic level. In the past our common universe of reason has been in the process of escaping, rejecting the romantic, irrational world of prehistoric man. It's been necessary since before the time of Socrates to reject the passions, the emotions, in order to free the rational mind for an understanding of nature's order which was as yet unknown. Now it's time to further an understanding of nature's order by reassimilating those passions which were originally fled from. The passions, the emotions, the affective domain of man's consciousness, are a part of nature's order too. The central part.

At present we're snowed under with an irrational expansion of blind data-gathering in the sciences because there's no rational format for any understanding of scientific creativity. At present we are also snowed under with a lot of stylishness in the arts ... thin art ... because there's very little assimilation or extension into underlying form. We have artists with no scientific knowledge and scientists with no artistic knowledge and both with no spiritual sense of gravity at all, and the result is not just bad, it is ghastly. The time for real reunification of art and technology is really long overdue.

At the DeWeeses I started to talk about peace of mind in connection with technical work but got laughed off the scene because I brought it up out of the context in which it had originally appeared to me. Now I think it is in context to return to peace of mind and see what I was talking about.

Peace of mind isn't at all superficial to technical work. It's the whole thing. That which produces it is good work and that which destroys it is bad work. The specs, the measuring instruments, the quality control, the final check-out, these are all means toward the end of satisfying the peace of mind of those responsible for the work. What really counts in the end is their peace

of mind, nothing else. The reason for this is that peace of mind is a prerequisite for a perception of that Quality which is beyond romantic Quality and classic Quality and which unites the two, and which must accompany the work as it proceeds. The way to see what looks good and understand the reasons it looks good, and to be at one with this goodness as the work proceeds, is to cultivate an inner quietness, a peace of mind so that goodness can shine through.

I say inner peace of mind. It has no direct relationship to external circumstances. It can occur to a monk in meditation, to a soldier in heavy combat or to a machinist taking off that last ten-thousandth of an inch. It involves unselfconsciousness, which produces a complete identification with one's circumstances, and there are levels and levels of this identification and levels and levels of quietness quite as profound and difficult of attainment as the more familiar levels of activity. The mountains of achievement are Quality discovered in one direction only, and are relatively meaningless and often unobtainable unless taken together with the ocean trenches of self-awareness...so different from self-consciousness...which result from inner peace of mind.

This inner peace of mind occurs on three levels of understanding. Physical quietness seems the easiest to achieve, although there are levels and levels of this too, as attested by the ability of Hindu mystics to live buried alive for many days. Mental quietness, in which one has no wandering thoughts at all, seems more difficult, but can be achieved. But value quietness, in which one has no wandering desires at all but simply performs the acts of his life without desire, that seems the hardest.

I've sometimes thought this inner peace of mind, this quietness is similar to if not identical with the sort of calm you sometimes get when going fishing, which accounts for much of the popularity of this sport. Just to sit with the line in the water, not moving, not really thinking about anything, not really caring about anything either, seems to draw out the inner tensions and frustrations that have prevented you from solving problems you couldn't solve before and introduced ugliness and clumsiness into your actions and thoughts.

You don't have to go fishing, of course, to fix your motorcycle. A cup of coffee, a walk around the block, sometimes just putting off the job for five minutes of silence is enough. When you do you can almost feel yourself grow toward that inner peace of mind that reveals it all. That which turns its back on this inner calm and the Quality it reveals is bad maintenance. That which turns toward it is good. The forms of turning away and toward are infinite but the goal is always the same.

I think that when this concept of peace of mind is introduced and made central to the act of technical work, a fusion of classic and romantic quality can take place at a basic level within a practical working context. I've said you can actually see this fusion in skilled mechanics and machinists of a certain sort, and you can see it in the work they do. To say that they are not artists is to misunderstand the nature of art. They have patience, care and attentiveness to what they're doing, but more than this...there's a kind of inner peace of mind that isn't contrived but results from a kind of harmony with the work in which there's no leader and no follower. The material and the craftsman's thoughts change together in a progression of smooth, even changes until his mind is at rest at the exact instant the material is right.

We've all had moments of that sort when we're doing something we really want to do. It's just that somehow we've gotten into an unfortunate separation of those moments from work. The mechanic I'm talking about doesn't make this separation. One says of him that he is ``interested" in what he's doing, that he's ``involved" in his work. What produces this involvement is, at the cutting edge of consciousness, an absence of any sense of separateness of subject and object. ``Being with it," ``being a natural," ``taking hold"...there are a lot of idiomatic expressions for what I mean by this absence of subject-object duality, because what I mean is so well understood as folklore, common sense, the everyday understanding of the shop. But in scientific parlance the words for this absence of subject-object duality are scarce because scientific minds have shut themselves off from consciousness of this kind of understanding in the assumption of the formal dualistic scientific outlook.

Zen Buddhists talk about ``just sitting," a meditative practice in which the idea of a duality of self and object does not dominate one's consciousness. What I'm talking about here in motorcycle maintenance is ``just fixing," in which the idea of a duality of self and object doesn't dominate one's consciousness. When one isn't dominated by feelings of separateness from what he's working on, then one can be said to ``care" about what he's doing. That is what caring really is, a feeling of identification with what one's doing. When one has this feeling then he also sees the inverse side of caring, Quality itself.

So the thing to do when working on a motorcycle, as in any other task, is to cultivate the peace of mind which does not separate one's self from one's surroundings. When that is done successfully then everything else follows naturally. Peace of mind produces right values, right values produce right thoughts. Right thoughts produce right actions and right actions produce work which will be a material reflection for others to see of the serenity at the center of it all. That was what it was about that wall in Korea. It was a material reflection of a spiritual reality.

I think that if we are going to reform the world, and make it a better place to live in, the way to do it is not with talk about relationships of a political nature, which are inevitably dualistic, full of subjects and objects and their relationship to one another; or with programs full of things for other people to do. I think that kind of approach starts it at the end and presumes the end is the beginning. Programs of a political nature are important end products of social quality that can be effective only if the underlying structure of social values is right. The social values are right only if the individual values are right. The place to improve the world is first in one's own heart and head and hands, and then work outward from there. Other people can talk about how to expand the destiny of mankind. I just want to talk about how to fix a motorcycle. I think that what I have to say has more lasting value.

A town called Riggins comes up and we see a lot of motels, and afterward the road branches away from the canyon and follows a smaller stream. It seems to head upward into forest.

It does, and soon the road becomes shaded by tall, cool pines. Resort signs appear. We wind higher and higher into unexpectedly pleasant, cool, green meadows surrounded by pine forests. At a town called New Meadows we fill up again and buy two cans of oil, still surprised at the change.

But as we leave New Meadows I note the long slant of the sun and a late afternoon depression begins to set in. At another time of day these mountain meadows would refresh me more, but we've gone too long. We pass Tamarack and the road drops down again from green meadows into dry sandy country.

I guess that's all I want to say for the Chautauqua today. It's been a long session and perhaps the most important one. Tomorrow I want to talk about things that seem to turn one toward Quality and turn one away from Quality, some of the traps and problems that come up.

Strange feelings from the orange sunlight on this sandy dry country so far from home. I wonder if Chris feels it too. Just a sort of unexplained sadness that comes each afternoon when the new day is gone forever and there's nothing ahead but increasing darkness.

The orange turns to dull bronze light and continues to show what it has shown all day long, but now it seems to show it without enthusiasm. Across those dry hills, within those little houses in the distance are people who've been there all day long, going about the business of the day, who now find nothing unusual or different in this strange darkening landscape, as we do. If we were to come upon them early in the day they might be curious about us and what we're here for. But now in the evening they'd just resent our presence. The work day is over. It's time for supper and family and relaxation and turning inward at home. We ride unnoticed down this empty highway through this strange country I've never seen before, and now a heavy feeling of isolation and loneliness becomes dominant and my spirits wane with the sun.

We stop at an abandoned school yard and under a huge cottonwood tree I change the oil in the cycle. Chris is irritable and wonders why we stop for so

long, not knowing perhaps that it's just the time of day that makes him irritable; but I give him the map to study while I change the oil, and when the oil is changed we look at the map together and decide to have supper at the next good restaurant we find and camp at the first good camping place. That cheers him up.

At a town called Cambridge we have supper and when we are finished, it's dark out. We follow the headlight beam down a secondary road toward Oregon to a little sign saying "BROWNLEE CAMPGROUND," which appears to be in a draw of the mountains. In the dark it's hard to tell what sort of country we're in. We follow a dirt road under trees and past underbrush to some camper's pull-ins. No one else seems to be here. When I shut the motor off and we unpack I can hear a small stream nearby. Except for that and the chirping of some little bird there's no sound.

"I like it here," Chris says.

"It's very quiet," I say.

"Where will we be going tomorrow?"

"Into Oregon." I give him the flashlight and have him shine it where I'm unpacking.

"Have I been there before?"

"Maybe, I'm not sure."

I spread out the sleeping bags, and put his on top of the picnic table. The novelty of this appeals to him. This night there'll be no trouble sleeping. Soon I hear deep breathing that tells me he's already asleep.

I wish I knew what to say to him. Or what to ask. He seems so close at times, and yet the closeness has nothing to do with what is asked or said. Then at other times he seems very far away and sort of watching me from some vantage point I don't see. And then sometimes he's just childish and there's no relation at all.

Sometimes, when thinking about this, I thought that the idea that one person's mind is accessible to another's is just a conversational illusion, just a figure of speech, an assumption that makes some kind of exchange between basically alien creatures seem plausible, and that really the relationship of one person to another is ultimately unknowable. The effort of fathoming what is in another's mind creates a distortion of what is seen. I'm trying, I suppose, for some situation in which whatever it is emerges undistorted. The way he asks all those questions, I don't know.

COMPUTER LIB

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Any nitwit can understand computers, and many do. Unfortunately, due to ridiculous historical circumstances, computers have been made a mystery to most of the world. And this situation does not seem to be improving. You hear more and more about computers, but to most people it's just one big blur. The people who *know* about computers often seem unwilling to explain things or answer your questions. Stereotyped notions develop about computers operating in fixed ways—and so confusion increases. The chasm between laymen and computer people widens fast and dangerously.

This book is a measure of desperation, so serious and abysmal is the public sense of confusion and ignorance. Anything with buttons or lights can be palmed off on the laymen as a computer. There are so many different things, and their differences are so important; yet to the lay public they are lumped together as “computer stuff,” indistinct and beyond understanding or criticism. It's as if people couldn't tell apart camera from exposure meter or tripod, or car from truck or tollbooth. This book is therefore devoted to the premise that

EVERYBODY SHOULD UNDERSTAND COMPUTERS.

It is intended to fill a crying need. Lots of everyday people have asked me where they can learn about computers, and I have had to say *nowhere*. Most of what is written about computers for the layman is either unreadable or silly. (Some exceptions are listed nearby [on pp. 6–7 of the first edition, not reprinted here]; you can go to them instead of this if you want.) But virtually nowhere is the big picture simply enough explained. Nowhere can one get a simple, soup-to-nuts overview of what computers are really about, without technical or mathematical mumbo-jumbo, complicated examples, or talking down. This book is an attempt.

(And nowhere have I seen a simple book explaining to the layman the fabulous wonderland of computer graphics which awaits us all, a matter which means a great deal to me

personally, as well as a lot to all of us in general. That's discussed on the flip side.)

Computers are simply a necessary and enjoyable part of life, like food and books. Computers are not everything, they are just an *aspect* of everything, and not to know this is computer illiteracy, a silly and dangerous ignorance.

Computers are as easy to understand as cameras. I have tried to make this book like a photography magazine—breezy, forceful and as vivid as possible. This book will explain how to tell apples from oranges and which way is up. If you want to make cider, or help get things right side up, you will have to go on from here.

I am not a skillful programmer, hands-on person or eminent professional; I am just a computer fan, computer fanatic if you will. But if Dr. David Reuben can write about sex I can certainly write about computers. I have written this like a letter to a nephew, chatty and personal. This is perhaps less boring for the reader, and certainly less boring for the writer, who is doing this in a hurry. Like a photography magazine, it throws at you some rudiments in a merry setting. Other things are thrown in so you'll get the sound of them, even if the details are elusive. (We learn most everyday things by beginning with vague impressions, but somehow encouraging these is us not usually felt to be respectable.) What I have chosen for inclusion here has been arbitrary, based on what might amuse and give quick insight. Any bright highschool kid, or anyone else who can stumble through the details of a photography magazine, should be able to understand this book, or get the main ideas. This will not make you a programmer or a computer person, though it may help you talk that talk, and perhaps make you feel more comfortable (or at least able to cope) when new machines encroach on your life. If you can get a chance to learn programming it's an awfully good experience for anybody above fourth grade. But the main idea of this book is to help you tell apples from oranges, and which way is up. I hope you do go on from here, and have made a few suggestions.

I am “publishing” this book myself, in this first draft form, to test its viability, to see how mad the computer people get, and to see if there is as much hunger to understand computers, among all you Folks Out There, as I think. I will be interested to receive corrections and suggestions for subsequent editions, if any. (The computer field is its own exploding universe, so I'll worry about up-to-dateness at that time.)

Summary of This Book

Man has created the myth of “the computer” in his own image, or one of them: cold, immaculate, sterile, “scientific,” oppressive.

Some people flee this image. Others, drawn toward it, have joined the cold-sterile-oppressive cult, and propagate it like a faith. Many are still about this mischief, making people do things rigidly and saying it is the computer’s fault.

Still others see computers for what they really are: versatile gizmos which may be turned to any purpose, in any style. And so a wealth of new styles and human purposes are being proposed and tried, each proponent propounding his own dream in his own very personal way.

This book presents a panoply of things and dreams. Perhaps some will appeal to the reader . . .

The Computer Priesthood

Knowledge is power and so it tends to be hoarded. Experts in any field rarely want people to understand what they do, and generally enjoy putting people down.

Thus if we say that the use of computers is dominated by a priesthood, people who spatter you with unintelligible answers and seem unwilling to give you straight ones, it is not that they are different in this respect from any other profession. Doctors, lawyers and construction engineers are the same way.

But computers are very special, and we have to deal with them everywhere, and this effectively gives the computer priesthood a stranglehold on the operation of all large organizations, of government bureaux, and everything else that they run. Members of Congress are now complaining about control of information by the computer people, that they cannot get the information even though it’s on computers. Next to this it seems a small matter that in ordinary companies “untrained” personnel can’t get straight questions answered by computer people; but it’s the same phenomenon.

It is imperative for many reasons that the appalling gap between public and computer insider be closed. As the saying goes, war is too important to be left to the generals. Guardianship of the computer can no longer be left to a

priesthood. I see this as just one example of the creeping evil of Professionalism,* the control of aspects of society by cliques of insiders. There may be some chance, though, that Professionalism can be turned around. Doctors, for example, are being told that they no longer own people’s bodies.** And this book may suggest to some computer professionals that their position should not be as sacrosanct as they have thought, either.

This is not to say that computer people are trying to louse everybody up on purpose. Like anyone trying to do a complex job as he sees fit, they don’t want to be bothered with idle questions and complaints. Indeed, probably any group of insiders would have hoarded computers just as

much. If the computer had evolved from the telegraph (which it just might have), perhaps the librarians would have hoarded it conceptually as much as the math and engineering people have. But things have gone too far. People have legitimate complaints about the way computers are used, and legitimate ideas for ways they should be used, which should no longer be shunted aside.

In no way do I mean to condemn computer people in general. (Only the ones who don’t want you to know what’s going on.) The field is full of fine, imaginative people. Indeed, the number of creative and brilliant people known within the field for their clever and creative contributions is considerable. They deserve to be known as widely as, say, good photographers or writers.

*This is a side point. I see Professionalism as a spreading disease of the present-day world, a sort of poly-oligarchy by which various groups (subway conductors, social workers, bricklayers) can bring things to a halt if their particular new increased demands are not met. (Meanwhile, the irrelevance of each profession increases, in proportion to its increasing rigidity.) Such lucky groups demand more in each go-round—but meantime, the number who are permanently unemployed grows and grows.

**Ellen Frankfort, *Vaginal Politics*. Quadrangle Books.
Boston Women’s Health Collective, *Our Bodies, Ourselves*. Simon & Schuster.



"Computers are catching hell from growing multitudes who see them uniformly as the tools of the regulation and suffocation of all things warm, moist, and human. The charges, of course, are not totally unfounded, but in their most sweeping form they are ineffective and therefore actually an acquiescence to the dehumanization which they decry. We clearly need a much more discerning evaluation in order to clarify the ethics of various roles of machines in human affairs."

Ken Knowlton in "Collaborations with Artists—a Programmer's Reflections," in Nake & Rosenfeld, eds., *Graphic Languages* (North-Holland Pub. Co.), p. 399.

Come Dream along with me:
The Best Is Yet To Be.
**DREAM
MACHINES**
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This is the flip side of *Computer Lib*.

(Feel free to begin here. The other side is just if you want to know more about computers, which are changeable devices for twiddling symbols. Otherwise skip it.)

(But if you change your mind it might be fun to browse.)

In a sense, the other side has been a come-on for this side. But it's an honest come-on: I figure the more you know, the readier you'll be for what I'm saying here. Not necessarily to agree or to be "sold," but to think about it in the non-simple terms that are going to be necessary.

The material here has been chosen largely for its exhilarating and inspirational character. No matter what your background or technical knowledge, you'll be able to understand some of this, and not be able to understand some of the rest. That's partly from the hasty preparation of this book, and partly from the variety of interests I'm trying

to comprise here. I want to present various dreams and their resulting dream machines, all legitimate.

If the computer is a projective system, or Rorschach inkblot, as alleged on the other side, the *real* projective systems—the ones with projectors in them—are all the more so. The things people try to do with movies, TV and the more glamorous uses of the computer, whereby it makes pictures on screens—are strange inversions and foldovers of the rest of the mind and heart. That's the peculiar origami of the self.

Very well. This book—this side, *Dream Machines*—is meant to let you see the choice of dreams. Noting that every

company and university seems to insist that its system is the wave of the future, I think it is more important than ever to have the alternatives spread out clearly.

But the "experts" are not going to be much help; they are part of the problem. On both sides, the academic and the industrial, they are being painfully pontifical and bombastic in the jarring

new jargons. Little clarity is spread by this. Few things are funnier than the pretensions of those who profess to dignity, sobriety and professionalism of their expert predictions—especially when they, too, are pouring out their own personal views under the guise of technicality. Most people don't dream of what's going to hit the fan. And the computer and electronics people are like generals preparing for the last war.

Frankly, I think it's an outrage making it look as if there's any kind of scientific basis to these things; there is an underlevel of technicality, but like the foundations of a cathedral, it serves only to support what rises from it. **THE TECHNICALITIES MATTER A LOT, BUT THE UNIFYING VISION MATTERS MORE.**

This book has several simultaneous intentions: to orient the beginner in fields more complex and tied together than almost anybody realizes; nevertheless, to partially debunk several realms of expertise which I think deserve slightly less



attention than they get; and to chart the *right* way, which I think uniquely continues the Western traditions of literature, scholarship and freedom. In this respect the book is much more old-fashioned than it may seem at the gee-whiz, very-now level.

The main ideas of this book I present not as my own, but as a curious species of revealed truth. It has all been obvious to me for some time, and I believe it should be obvious as well to anyone who has not been blinded by education. If you understand the problems of creative thinking and organizing ideas, if you have seen the bad things school so often does to people, if you understand the sociology of the intellectual world, and have ever loved a machine, then this book says nothing you do not know already.

For every dream, many details and intricacies have to be whittled and interlocked. Their joint ramifications must be deeply understood by the person who is trying to create whatever-it-is. Each confabulation of possibilities turns out to have the most intricate and exactly detailed results. (This is why I am so irritated by those who think “electronic media” are all alike.)

And each possible combination you choose has different precise structures implicit in it, arrangements and units which flow from these ramified details. *Implicit* in Radio lurk the Time Slot and the Program. But many of these possibilities remain unnoticed or unseen, for a variety of social or economic reasons.

Why does it matter?

It matters because we live in media, as fish live in water. (Many people are prisoners of the media, many are manipulators, and many want to use them to communicate artistic visions.)

But today, at this moment, we can and must design the media, design the molecules of our new water, and I believe the details of this design matter very deeply. They will be with us for a very long time, perhaps as long as man has left; perhaps if they are as good as they can be, man may even buy more time—or the open-ended future most suppose remains.

So in these pages I hope to orient you somewhat to various of the proposed dreams. This is meant also to record the efforts of a few Brewster McClouds, each tinkering toward some new flight of fancy in his own sensorium.

But bear in mind that hard-edged fantasy is the corner of tomorrow. The great American dream often becomes the

great American novelty. After which it's a choice of style, size and financing plan.

The most exciting things here are those that involve computers: notably, because computers will be embraced in every presentational medium and thoughtful medium very soon.

That's why this side is wedded to the other: if you *want* to understand computers, you can take the first step by turning the book over. I figure that the more you know about computers—especially about minicomputers and the way on-line systems can respond to our slightest acts—the better your imagination can flow between the technicalities, can slide the parts together, can discern the shapes of what *you* would have these things do. The computer is not a limitless partner, but it is deeply versatile; to work with it we must understand what it can do, the options and the costs.

My special concern, all too tightly framed here, is the use of computers to help people write, think and show. But I think presentation by computer is a branch of show biz and writing, not of psychology, engineering or pedagogy. This would be idle disputation if it did not have far-reaching consequences for the designs of the systems we are all going to have to live with. At worst, I fear these may lock us in; at best, I hope they can further the individualistic traditions of literature, film and scholarship. But we must create our brave new worlds with art, zest, intelligence, and the highest possible ideals.

I have not mentioned the emotions. Movies and books, music and even architecture have for all of us been part of important emotional moments. The same is going to happen with the new media. To work at a highly responsive computer display screen, for instance, can be deeply exciting, like flying an airplane through a canyon, or talking to somebody brilliant. This is as it should be. (“The reason is, and by rights ought to be, slave to the emotions.”—Bertrand Russell.)

In the design of our future media and systems, we should not shrink from this emotional aspect as a legitimate part of our fantic (see p. 317) design. The substratum of technicalities and the mind-bending, gut-slammings effects they produce, are two sides of the same coin; and to understand the one is not necessarily to be alienated from the other.

Thus it is for the Wholeness of the human spirit, that we must design.

Dreams

Technology is an expression of man's dreams. If man did not indulge his fantasies, his thoughts alone would inhibit the development of technology itself. Ancient visionaries spoke of distant times and places, where men flew around and about, and some could see each other at great distance. The technological realities of today are already obsolete and the future of technology is bound only by the limits of our dreams. Modern communications media and in particular electronic media are outgrowths and extensions of those senses which have become dominant in our social development.

How Wachspress, "Hyper-Reality."

© Auditac Ltd. 1973.

CHILDREN OF
ALL AGES!

Ladies and gentlemen, the age of prestidigitative presentation and publishing is about to begin. Palpitating presentations, screen-scribbled, will dance to your desire, making manifest the many mysteries of winding wisdom. But if we are to rehumanize an increasingly brutal and disagreeable world, we must step up our efforts. And we must hurry. Hurry. Step right up.

Theodor H. Nelson, "Barnum-Tronics."

Swarthmore College Alumni Bulletin, Dec 1970,
12-15.

"When you're dealing with media you're in show business, you know, whether you like it or not."

"Show business," he said. "Absolutely. We've gotta be in show business. We've gotta put together a team that will get us there."

I made a mental note to use the show business metaphor again, and continued, "IBM's real creative talent probably lies in other areas . . ."

Heywood Gould, *Corporation Freak* (Tower), 23.

No more pencils, no more books.
No more teachers' dirty looks.
Kids' jingle with new meanings.

Everything is Deeply Interwined.

THE GESTALT, DEAR BRUTUS,
IS NOT IN OUR STARS
BUT IN OURSELVES.

(The following article appeared in the September, 1970 issue of *Computer Decisions*, and got an extraordinary amount of attention. I have changed my views somewhat—we all go through changes, after all—but after consideration have decided to re-run it in the original form, without qualifications, mollifications or anything, for its unity. Thanks to *Computer Decisions* for use of the artwork by Gans and for the Superstudent picture on the cover, whose artist unfortunately insists on preserving his anonymity.

An interesting point, incidentally, is that people read this a lot of different ways. One Dean of Education hilariously misread it as an across-the-board plug for CAI. Others read in it various forms of menace or advocacy of generalized mechanization. One letter-writer said I was a menace but at least writing articles kept me off the streets. Here is my fundamental point: *computer-assisted instruction, applied thoughtlessly and imitatively, threatens to extend the worst features of education as it is now.*

NO MORE TEACHERS' DIRTY LOOKS

by Theodor H. Nelson
The Nelson Organization
New York

Some think the educational system is basically all right, and more resources would get it working again. Schools would do things the same way, except more so, and things would get better.

In that case the obvious question would be, how can computers help? How can computers usefully supplement and extend the traditional and accepted forms of teaching? This is the question to which present-day efforts in “computer-assisted instruction”—called CAI—seem to respond.

But such an approach is of no possible interest to the new generation of critics of our school system—people like John Holt (*Why Children Fail*), Jonathan Kozol (*Death at an Early Age*) and James Herndon (*The Way It Spozed to Be*). More and more, such people are severely questioning the general framework and structure of the way we teach.

These writers describe particularly ghastly examples of our schooling conditions. But such horror stories aside, we are coming to recognize that schools as we know them appear designed at every level to sabotage the supposed goals of education. A child arrives at school bright and early in his life. By drabness we deprive him of interests. By fixed curriculum and sequence we rob him of his orientation, initiative and motivation, and by testing and scoring we subvert his natural intelligence.

Schools as we know them all run on the same principles: iron all subjects flat than then proceed, in groups, at a forced march across the flattened plain. Material is dumped on the students and their responses calibrated; their interaction and involvements with the material is not encouraged nor taken into consideration, but their dutifulness of response is carefully monitored.

While an exact arrangement of intended motivations for the student is preset within the system, they do not usually take effect according to the ideal. It is not that students are unmotivated, but motivated *askew*. Rather than seek to achieve in the way they are supposed to, students turn to churlishness, surliness, or intellectual sheepishness. A general human motivation is god-given at the beginning and warped or destroyed by the educational process as we know it; thus we internalize at last that most fundamental of grownup goals: just to get through another day.

Because of this procedure our very notion of human ability has suffered. Adult mentality is cauterized, and we call it “normal.” Most people’s minds are mostly turned off most of the time. We know virtually nothing of human abilities except as they have been pickled and boxed in schools; we need to ignore all that and start fresh. To want students to be “normal” is criminal, when we are all so far below our potential. Buckminster Fuller, in *I Seem to Be a Verb*, says we are all born geniuses: Sylvia Ashton-Warner tells us in *Teacher* of her success with this premise, and of the brilliance and creative potential she was able to find in all her schoolchildren.

Curricula themselves destructively arrange the study situation. By walls between artificially segregated “studies” and “separate topics” we forbid the pursuit of interest and kill motivation.

In ordinary schooling, the victim cannot orient himself to the current topic except by understanding the official angle of approach and presentation. Though tie-ins to previous interests and knowledge are usually the best way to get an initial sense of a thing, there is only time to consider the officially presented tie-ins. (Neither is there time to answer questions, except briefly and rarely well—and usually in a way that promotes “order” by discouraging “extraneous” tie-ins from coming up.)

The unnecessary division and walling of subjects, sequencing and kibbling of material lead people to expect simplifications, to feel that naming a thing is understanding it, to fear complex wholes; to believe creativity means



recombination, the parsing of old relations, rather than synthesis.

Like political boundaries, curriculum boundaries arise from noticeable features of a continuum and become progressively more fortified. As behind political borders, social unification occurs within them, so that wholly dissimilar practitioners who share a name come to think they do the same thing. And because they talk mainly to each other, they forget how near is the other side of the border.

Because of the fiction of “subjects,” great concern and consideration has always gone into calculating the “correct” teaching sequence for each “subject.” In recent years radical new teaching sequences have been introduced for teaching various subjects, including mathematics and physics. But such efforts appear to have been misinformed by the idea of supplanting the “wrong” teaching sequence with the “right” teaching sequence, one which is “validated.” Similarly, we have gone from a time when the instructional sequence was a balance between tradition and the lowest common denominator of each subject, to a time when teachers may pick “flexible optimized strategies” from textbooks. And this all ignores a simple fact: all are arbitrary. Instructional sequences aren’t needed at all if the people are motivated and the materials are clear and available.

Testing as we know it (integrated with walled curricula and instructional sequences) is a destructive activity, particularly for the orientation which it creates. The concerns of testing are extraneous: learning to figure out

low-level twists in questions that lead nowhere, under pressure.

The system of tensions and defenses it creates in the student’s personality are unrelated to the subject or the way people might relate to the subject. An exploitive attitude is fostered. Not becoming involved with the subject, the student grabs for rote payoff rather than insight.

All in a condescending circumstance. Condescension is built into the system at all levels, so pervasive it is scarcely noticed. Students are subjected to a grim variety of put-downs and denigrations. While many people evidently believe this to be right, its productivity in building confident and self-respecting minds may be doubted.

The problems of the school are not particularly the teacher’s fault. The practice of teaching is principally involved with managing the class, keeping up face, and projecting the image of the subject that conforms to the teacher’s own predilections. The educational system is thereby committed to the fussy and prissy, to the enforcement of peculiar standards of righteousness and the elevation of teachers—a huge irrelevant shell around the small kernel of knowledge transmitted.

The usual attacks on computer teaching tend to be sentimental and emotional pleas for the alleged humanism of the existing system. Those who are opposed to the use of computers to teach generally believe the computer to be “cold” and “inhuman.” The teacher is considered “warm” and “human.” This view is questionable on both sides.



Some premises relevant to teaching

1. The human mind is born free, yet everywhere it is in chains. The educational system serves mainly to destroy for most people, in varying degrees, intelligence, curiosity, enthusiasm, and intellectual initiative and self-confidence. We are born with these. They are gone or severely diminished when we leave school.
2. Everything is interesting, until ruined for us. Nothing in the universe is intrinsically uninteresting. Schooling systematically ruins things for us, wiping out these interests; the last thing to be ruined determines your profession.
3. There are no “subjects.” The division of the universe into “subjects” for teaching is a matter of tradition and administrative convenience.
4. There is no natural or necessary order of learning. Teaching sequences are arbitrary, explanatory hierarchies philosophically spurious. “Prerequisites” are a fiction spawned by the division of the world into “subjects;” and maintained by not providing summaries, introductions or orientational materials except to those arriving through a certain door.
5. Anyone retaining his natural mental facilities can learn anything practically on his own, given encouragement and resources.
6. Most teachers mean well, but they are so concerned with promoting their images, attitudes and style of order that very little else can be communicated in the time remaining, and almost none of it attractively.

The computer is as inhuman as we make it. The computer is no more “cold” and “inhuman” than a toaster, bathtub or automobile (all associated with warm human activities). Living teachers can be as inhuman as members of any people-prodding profession, sometimes more so. Computerists speak of “freeing teachers for the creative part of their work;” in many cases it is not clear what creative tasks they could be freed for.

At the last, it is to *rescue* the student from the inhuman teacher, and allow him to relate directly and personally to the intrinsically interesting subject matter, than we need to use computers in education.

Many successful systems of teacherless learning exist in our society: professional and industrial magazines; conventions and their display booths and brochures; technical sales pitches (most remarkably, those of medical “detail men”); hobbyist circles, which combine personal acquaintance with a round of magazines and gatherings; think-tanks and research institutes, where specialists trade fields; and the respectful briefing.

None of these is like the conventional classroom with its haughty resource-chairman; they are not run on condescension; and they get a lot across. We tend to think they are not “education” and that the methods cannot be transferred or extended to the regions now ruled by conventional teaching. But why not?

If everything we ate were kibbled into uniform dogfood, and the amount consumed at each feeding time tediously watched and tested, we would have little fondness for eating. But this is what the schools do to our food for thought, and this is what happens to people’s minds in primary school, secondary school and most colleges.

This is the way to produce a nation of sheep or clerks. If we are serious about wanting people to have creative and energetic minds, it is not what we ought to do. Energy and enthusiasm are natural to the human spirit; why drown them?

Education ought to be clear, inviting and enjoyable, without booby-traps, humiliations, condescension or boredom. It ought to teach and reward initiative, curiosity, the habit of self-motivation, intellectual involvement.

Students should develop, through practice, abilities to think, argue and disagree intelligently.

Educators and computer enthusiasts tend to agree on these goals. But what happens? Many of the inhumanities of the existing system, no less wrong for being unintentional, are being continued into computer-assisted teaching.

Although the promoters of computer-assisted instruction, affectionately call “CAI,” seem to think of themselves as being at the vanguard of progress in all directions, the field already seems to operate according to a stereotype. We may call this “classic” or “conventional” CAI, a way of thinking depressingly summarized in “The Use of Computers in Education” by Patrick Suppes, *Scientific American*, September, 1966, 206–220, an article of semi-classic stature.

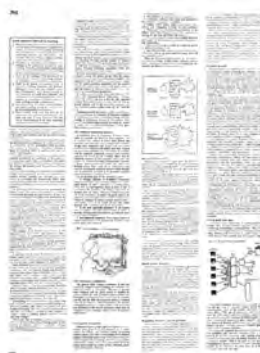
It is an unexamined premise of this article that the computer system will always decide what the student is to study and control his movements through it. The student is to be led by the nose through every subject, and the author expresses perplexity over the question of *how* the system can decide, at all times, *where* to lead the student by the nose (top of col. 3, p. 219). But let us not anticipate alternatives.

It is often asserted (as by Alpert and Bitzer in “Advances in Computer-Based Education,” *Science*, March 20, 1970) that this is not the only approach current. The trouble is that it *seems* to be the only approach current, and in the expanding computer universe everyone *seems* to know what CAI “is.” And this is it.

Computer-assisted instruction, in this classical sense, is the presentation by computer of bite-sized segments of instructional material, branching among them according to involuntary choices by the student (“answers”) and embedding material presented the student in some sort of pseudo-conversation (“Very good. Now, Johnny, point at the . . .”)

CAI: Based on Unnecessary Premises

At whichever level of complexity, all these conventional CAI systems are based on three premises: that all presentations consist of *items*, short chunks and questions; that the items are arranged into *sequences*, though these sequences may branch and vary under control of the computer; and finally, that these sequences are to be embedded in a framework of



dialogue; with the computer composing sentences and questions appropriately based on the student's input and the branching structure of the materials. Let us call such systems SIC (Sequenced-Item Conversational) systems.

These three premises are united. For there to be dialogue means there must be underlying graph structure of potential sequences around which dialogue may be generated; for there to be potential sequences means breakpoints, and hence items.

Let us question each of the premises in turn.

1 Is dialogue pleasant or desirable?

Compulsory interaction, whether with a talking machine or a stereotyped human, is itself a put-down or condescension. (Note that on superhighways there is often a line of cars behind the automatic toll booths, even when the manned ones are open.) Moreover, faked interaction can be an annoyance. (Consider the green light at the automatic toll booth that lights up with a "thank you.") Moreover, dialogue by simple systems tends to have a fake quality. It is by no means obvious that phony dialogue with a machine will please the student.

2 Is the item approach necessary?

If the student were in control, he could move around in areas of material, leaving each scene when he got what he wanted, or found in unhelpful.

3 Are sequences necessary?

Prearranged sequences become unnecessary if the student can see what he has yet to learn, then pursue it.

The sense of prestige and participation



CAI: Unnecessary Complication

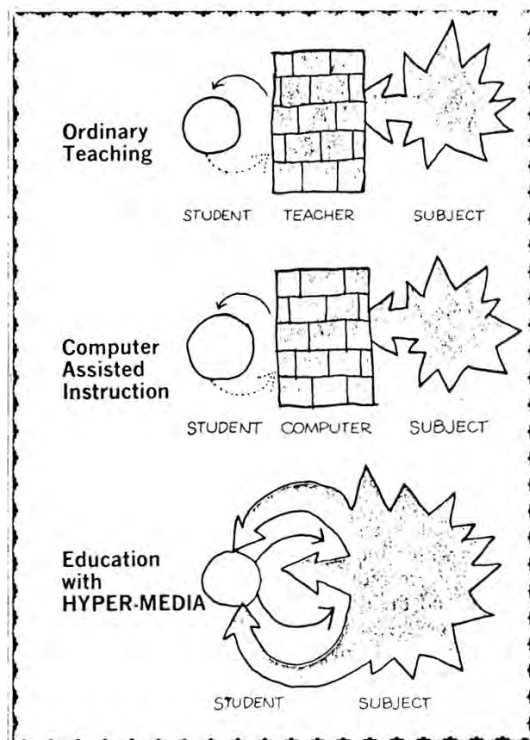
The general belief among practitioners is that materials for computer-based teaching are extremely difficult to create, or "program." Because of possible item weakness and the great variety of possible sequences within the web, extensive experimentation and debugging are required. Each item must be carefully proven; and the different sequences open to a student must all be tested for their effectiveness. All possible misunderstandings by a student need to be anticipated and prevented in this web of sequences, which must be designed for its coverage, correct order, and general effectiveness.

CAI: General Wrongfulness

Computers offer us the first real chance to let the human mind grow to its full potential, as it cannot within the stifling and insulting setting of existing school systems. Yet most of the systems for computer-assisted instruction seem to me to be perpetuating and endorsing much that is wrong, even evil, in our present educational system. CAI in its conventional form enlarges and extends the faults of the American educational system itself. They are:

- Conduciveness to boredom;
- The removal of opportunities for initiative;
- Gratuitous concerns, both social and administrative ("subject," "progress" in subject);
- Grades, which really reflect commitment level, anxiety, and willingness to focus on core emphasis;
- Stereotyped and condescending treatment of the student (the "Now-Johnny" box in the computer replacing the one that sits before the class);
- The narrowing of curricula and available materials for "results" at the expense of motivation and generalized orientation;
- Destructive testing of a kind we would not permit on delicate machinery; and,
- An overt of hidden emphasis on invidious ratings. (Ungraded schools are nice—but how many units did *you* complete today?)

There are of course improvements, for instance in the effects of testing. In the tell-test, tell-test nattering of CAI, the testing becomes merely an irritant, but one certainly not likely to foster enthusiasm.



But Isn't CAI 'Scientific?'

Part of CAI's mystique is based upon the idea that teaching can become "scientific" in the light of modern research, especially learning theory. It is understandable that researchers should promote this view and that others should fall for it.

Laymen do not understand, nor are they told, that "learning theory" is an extremely technical, mathematically oriented, description of the behavior of abstract and idealized organisms learning non-unified things under specific conditions of motivation and non-distraction.

Let us assume, politely, that learning theory is a full and consistent body of knowledge. Because of its name, learning theory has at least what we may call nominal relevance to teaching; but real relevance is another matter. It may be relevant as Newtonian equations are to shooting a good game of pool: implicit but without practical bearing.

Because of the actual character of learning theory, and its general remoteness from non-sterile conditions, actual relevance to any particular type of application must still be

demonstrated. To postulate that the theory still applies in diluted or shifted circumstances is a leap of faith. Human beings are not, taken all together, very like the idealized pigeons or rats of learning theory, and their motivations and other circumstances are not easily controlled. Studies concerned with rate of repetition and reinforcement are scarcely relevant if the student hates or does not understand what he is doing.

I do not mean to attack all CAI, or any teaching system which is effective and gratifying. What I doubt is that SIC systems for CAI will become more and more wonderful as effort progresses, or that the goal of talking tutorial systems is reachable and appropriate. And what I further suspect is that we are building boredom systems that not only make life duller but sap intellectual interest in the same old way.

Should Systems 'Instruct?'

Drill-and-practice systems are definitely a good thing for the acquisition of skills and response sets, an improvement over workbooks and the like, furnishing both corrections and adjustment. They are boring, but probably less so than the usual materials. But the CAI enthusiasts seem to believe the same conversationalized chunk techniques can be extended to the realm of ideas, to systems that will tutor and chide, and that this will provide the same sort of natural interest provided by a live tutor's instruction.

The conventional point of view in CAI claims that because validation is so important, it is necessary to have a standardized format of item, sequence and dialogue. This justifies turning the endeavor into picky-work within items and sequence complexes, with attendant curricular freeze, and student inanition and boredom. This is entirely premature. The variety of alternative systems for computer teaching have not even begun to be explored. Should systems "instruct" at all?

'Responding Resources' and 'Hyper-Media'

At no previous time has it been possible to create learning resources so responsive and interesting, or to give such free play to the student's initiative as we may now. We can now build computer-based presentational wonderlands, where a student (or other user) may browse and ramble through a vast variety of writings, pictures and apparitions in magical space, as well as rich data structures and facilities for twiddling them. These we may call, collectively, "responding

resources." Responding resources are of two types: facilities and hyper-media.

A *facility* is something the user may call up to perform routinely a computation or other act, behaving in desired ways on demand. Thus JOSS (a clever desk calculator available at a terminal) and the Culler-Freed graph-plotting system (which graphs arbitrary functions the user types in) are facilities.

Hyper-media are branching or performing presentations which respond to user actions, systems of prearranged words and pictures (for example) which may be explored freely or queried in stylized ways. They will not be "programmed," but rather *designed, written, drawn and edited*, by authors, artists, designers and editors. (To call them "programmed" would suggest spurious technicality. Computer systems to present them will be "programmed.") Like ordinary prose and pictures, they will be *media*; and because they are in some sense "multi-dimensional," we may call them *hyper-media*, following mathematical use of the term "hyper-."

A Modest Proposal

The alternative is straightforward. Instead of devising elaborate systems permitting the computer or its instructional contents to control the situation, why not permit the student to control the system, show him how to do so intelligently, and make it easy for him to find his own way? Discard the sequences, items and conversation, and allow the student to move freely through materials which he may control. Never mind optimizing reinforcement or validating teaching sequences. Motivate the user and let him loose in a wonderful place.

Let the student control the sequence, put him in control of interesting and clear material, and make him feel good—comfortable, interested, and autonomous. Teach him to orient himself: not having the system answer questions, all typed in, but allowing the student to get answers by looking in a fairly obvious place. (Dialogue is unnecessary even when it does not intrude.) Such ultra-rich environments allow the student to choose what he will study, when he will study it and how he will study it, and to what criteria of accomplishment he will aim. Let the student pick what he wishes to study next, decide when he wishes to be tested, and give him a variety of interesting materials, events and opportunities. Let the student ask to be tested on what he

thinks he knows, when he is ready, selecting the most appropriate form of testing available.

This approach has several advantages. First, it circumvents the incredible obstacles created by the dialogue-item-sequence philosophy. It ends the danger to students of bugs in the material. And last, it does what education is supposed to do—foster student enthusiasm, involvement, and self-reliance.

Under such circumstances students will actually be interested, motivated to achieve far more than they have ever achieved within the normal instructional framework; and any lopsidedness which may result will be far offset by the degree of accomplishment which will occur—it being much better to create lopsided but enthusiastic genius specialists than listless, apathetic, or cruelly rebellious mediocrities. If they start soon enough they may even reach adulthood with natural minds: driven by enthusiasm and interest, crippled in no areas, eager to learn more, and far smarter than people ordinarily end up being.

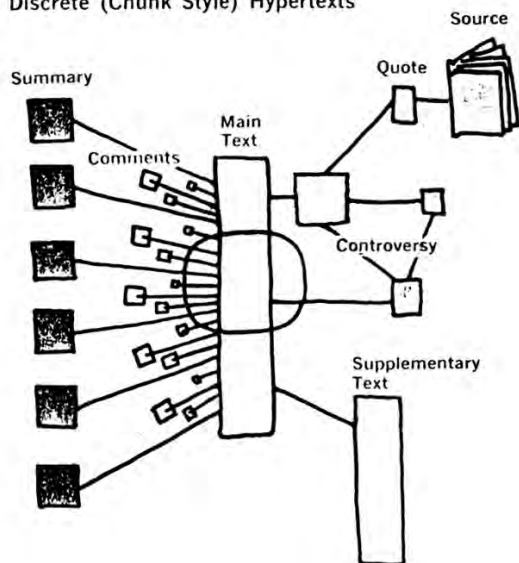
Enthusiasm and involvement are what really count. This is why the right to explore far outweighs any administrative advantages of creating and enforcing "subjects" and curriculum sequences. The enhancement of motivation that will follow from letting kids learn anything they want to learn will far outweigh any specialization that may result. By the elimination or benign replacement of both curriculum and tests in an ultra-rich environment, we will prevent the attrition of the natural motivation of children from its initially enormous levels, and mental development will be the natural straight diagonal rather than the customary parabola.

Is It So Hard? Some Ideas

CAI is said to be terribly hard. It would seem all the harder, then, to give students the richer and more stimulating environments advocated here. This is because of the cramped horizons of computer teaching today. Modest goals have given us modest visions, far below what is now possible and will soon be cheap.

The static computer displays now associated with CAI will give way to dynamic displays driven from minicomputers, such as the IDIOM, IBM 2250/4 or Imlac PDS-1. (The last of these costs only \$10,000 *now*; by 1975 such a unit will probably cost \$1,000 or less.) Not only will computers be

Discrete (Chunk Style) Hypertexts



much cheaper, but their usability will improve: a small computer with a fair amount of memory will be able to do much more than it can now, including operate a complex display from its own complex data base.

It is generally supposed that systems like these need big computers and immense memories. This is not true if we use the equipment well, organize storage cleverly, and integrate data and display functions under a compact monitor. This is the goal of The Nelson Organization's Project Xanadu, a system intended to handle all the functions described here on a mini-computer with disk and tape.

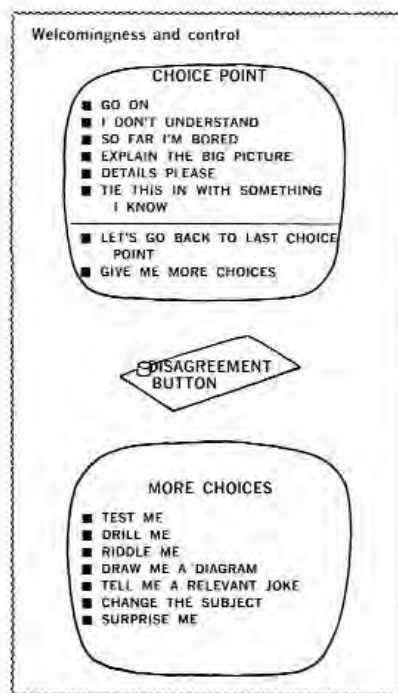
Discrete Hypertexts

"Hypertext" means forms of writing which branch or perform on request; they are best presented on computer display screens.

In ordinary writing the author may break sequence for footnotes or insets, but the use of print on paper makes some basic sequence essential. The computer display screen, however, permits footnotes on footnotes on footnotes, and pathways of any structure the author wants to create.

Discrete, or chunk style, hypertexts consist of separate pieces of text connected by links.

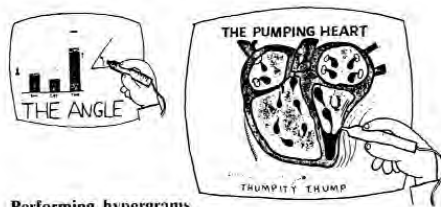
Ordinary prose appears on the screen and may be moved forward and back by throttle. An asterisk or other key in the text means, not an ordinary footnote, but a *jump*—to an



entirely new presentation on the screen. Such jumpable interconnections become part of the writing, entering into the prose medium itself as a new way to provide explanations and details to the seeker. These links may be artfully arranged according to meanings or relations in the subject, and possible tangents in the reader's mind.

Performing Hypergrams

A hypergram is a performing or branching picture: for instance, this angle, with the bar-graph of its related trigonometric functions. The student may turn the angle upon the screen, seizing it with the light-pen, and watch the related trigonometric functions, displayed as bar charts, change correspondingly.



Performing hypergrams

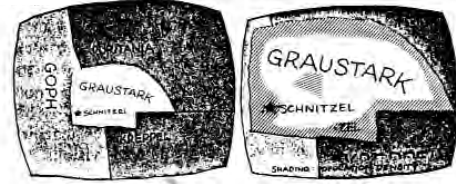
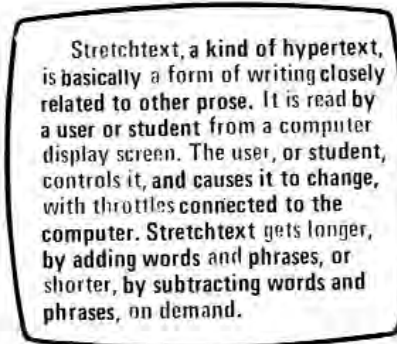
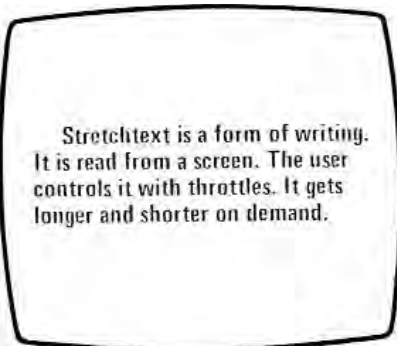
Hypergrams may also be programmed to show the consequences of a user's prod—what follows or accompanies some motion of the picture that he makes with a pointing tool, like the heartbeat sequence.

Stretchtext™ Fills in the Details

This form of hypertext is easy to use without getting lost. As a form of writing, it has special advantages for discursive and loosely structured materials—for instance historical narratives.

There are a screen and two throttles. The first throttle moves the text forward and backward, up and down on the screen. The second throttle causes changes in the writing itself: throttling toward you causes the text to become *longer* by minute degrees. Gaps appear between phrases; new words and phrases pop into the gaps, an item at a time. Push back on the throttle and the writing becomes shorter and less detailed.

The stretchtext is stored as a text stream with extras, coded to pop in and pop out at the desired altitudes:

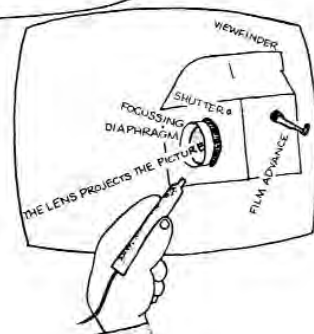
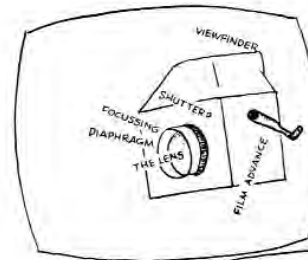


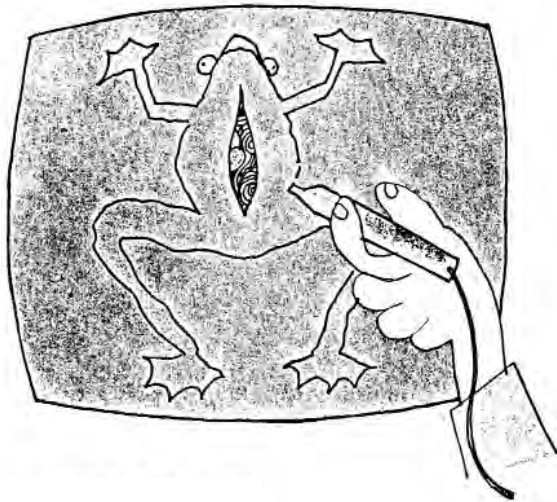
Hypermap Zips Up or Down

The screen is a map. A steering device permits the user to move the map around the world's surface: a throttle zooms it in. Not by discrete jumps, but animated in small changes, the map grows and grows in scale. More details appear as the magnification increases. The user may request additional display modes or "overlays," such as population, climate, and industry. Such additional features may pop into view on request

Queriable Illustrations: a Form of Hypergram

A "hypergram" is a picture that can branch or perform on request. In this particular example, we see on the screen a line-drawing with protruding labels. When the student points at a label, it becomes a sliding descriptive ribbon, explaining the thing labeled. Or asterisks in an illustration may signal jumps to detailed diagrams and explanations, as in discrete hypertexts.





Dissection on the Screen

The student of anatomy may use his light-pen as a scalpel for a deceased creature on the screen. As he cuts, the tissue parts. He could also turn the light-pen into hemostat or forceps, and fully dissect the creature—or put it back together again. (This need not be a complex simulation. Many key relationships can be shown by means of fairly simple schematic pictures, needing a data structure not prohibitively complicated.)



HYPER-COMICS



Hyper-comics are Fun

Hyper-comics are perhaps the simplest and most straightforward hyper-medium. The screen holds a comic strip, but one which branches on the student's request. For instance, different characters could be used to explain things in different ways, with the student able to choose which type of explanation he wanted at a specific time.

'Technicality' Is Not Necessary

Proponents of CAI want us to believe that scientific teaching requires a certain setup and format, incomprehensible to the layman and to be left to experts. This is simply not true. "Technicality" is a myth. The problem is not one of technical rightness, but what *should* be.

The suggestions that have been given are things that should be; they will be brought about.

FEEL-TOUCH EFFECT SYSTEMS
ARE THE NEW FRONTIER
FANTICS
— BUT IT'S SHOWMANSHIP
THAT'S PARAMOUNT,
NOT ANY TECHNICAL SPECIALTY

Ah, Love! could you and I with Him conspire
To grasp this sorry Scheme of Things entire,
Would not we shatter it to bits—and then
Re-mould it nearer to the Heart's Desire!

Edward Fitzgerald.

Almost everyone seems to agree that Mankind (who?) is on the brink of a revolution in the way information is handled, and that this revolution is to come from some sort of merging of electronic screen presentation and audio-visual technology with branching, interactive computer systems. (The naïve think “the” merging is inevitable, as if “the” merging meant anything clear. I used to think that too.)

Professional people seem to think this merging will be an intricate mingling of technical specialties, that our new systems will require work by all kinds of committees and consultants (adding and adjusting) until the Results—either specific productions or overall Systems—are finished. Then we will have to Learn to Use Them. More consulting fees.

I think this is a delusion and a con-game. I think that when the *real* media of the future arrive, the smallest child will know it right away (and perhaps first). That, indeed, should and will be the criterion. When you can't tear a teeny kid away from the computer screen, we'll have gotten there.

We are approaching a screen apocalypse. The author's basic view is that RESPONSIVE COMPUTER DISPLAY SYSTEMS CAN, SHOULD AND WILL RESTRUCTURE AND LIGHT UP THE MENTAL LIFE OF MANKIND. (For a more conventional outlook, see box nearby, “Another Viewpoint.”)

I believe computer screens can make people happier, smarter, and better able to cope with the copious problems of tomorrow. But only if we do right, right now.



Why?

The computer's capability for branching among events, controlling exterior devices, controlling outside events, and mediating in all other events, makes possible a new era of media.

Until now, the mechanical properties of external objects determined what they were to us and how we used them. But henceforth this is arbitrary.

The recognition of that arbitrariness, and reconsideration among broader and more general alternatives, awaits us. All the previous units and mechanisms of learning, scholarship, arts, transaction and confirmation, and even self-reminder, were based in various ways upon physical objects—the properties of paper, carbon paper, files, books and

bookshelves. To read from paper you must move the physical object in front of you. Its contents cannot be made to slide, fold, shrink, become transparent, or get larger.

But all this is now changing, and suddenly. The computer display screen does all these things if desired, to the same markings we have previously handled on paper. The computer display screen is going to become universal very fast; this is guaranteed by the suddenly rising cost of paper. And we will use them for everything. This already happens wherever there are responding computer screen systems.

(I have a friend with two CRTs on his desk; one for the normal flow of work, and one to handle interruptions and side excursions.) A lot of forests will be saved.

Now, there are many people who don't like this idea, and huff about various apparent disadvantages of the screen. But we can improve performance until almost everyone is satisfied. For those who say the screens are “too small,” we can improve reliability and backup, and offer screens everywhere (so that material need not be physically carried between them).

The exhilaration and excitement of the coming time is hard to convey on paper. Our screen displays will be alive with animation in their separate segments of activity, and will respond to our actions as if alive physically too.

The question is, then: HOW WILL WE USE THEM? Thus the design of screen performances and environments, and of transaction and transmission systems, is of the highest priority.

The French Have a Word for It

In French they use the term *l'Informatique* to mean, approximately, the presentation of information to people by automatic equipment.

Unfortunately the English equivalent, *informatics*, has been preempted. There is a computer programming firm called Informatics, Inc., and when I wrote them about this in the early sixties they said they did *not* want their name to become a generic term. Trademark law supports them in this to a certain extent. (Others, like Wally Feurzeig, want that to be the word regardless.) But in the meantime I offer up the term *fantics*, which is more general anyhow.

Media

What people don't see is how computer technology now makes possible the revision and improvement—the transformation—of all our media. It “sounds too technical.”

But this is the basic misunderstanding: the fundamental issues are NOT TECHNICAL. To understand this is basically a matter of MEDIA CONSCIOUSNESS, not technical knowledge.

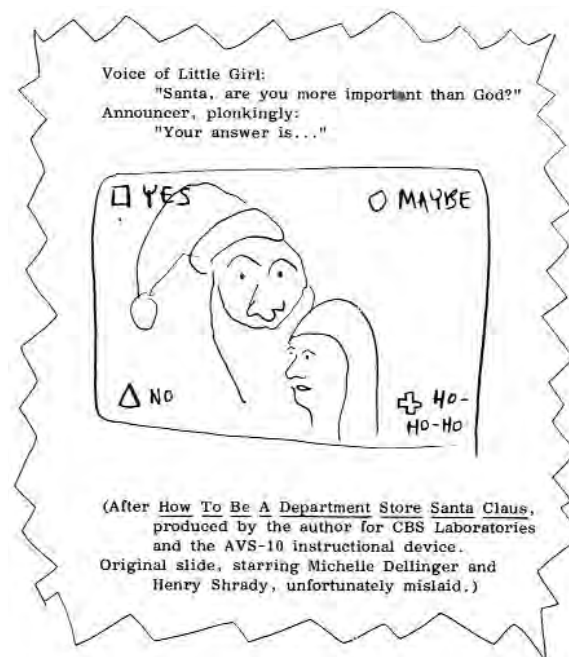
A lot of people have acute media consciousness. But some people, like Pat Buchanan and the communards, suggest that there is something shabby about this. Many think, indeed, that we live in a world of false images promulgated by “media,” a situation to be corrected. But this is a misunderstanding. Many images are false or puffy, all right, but it is incorrect to suppose that there is any alternative. Media have evolved from simpler forms, and convey the background ideas of our time, as well as the fads. Media today focus the impressions and ideas that in previous eras were conveyed by rituals, public gatherings, decrees, parades, behavior in public, mummer' troupes . . . but actually every culture is a world of images. The chieftain in his palanquin, the shaman with his feathers and rattle, are telling us something about themselves and about the continuity of the society and position of the individuals in it.

Now the media, with all their quirks, perform the same function. And if we do not like the way some things are treated by the media, in part this stems from not understanding how they work. “Media,” or structured transmission mechanisms, cannot help being personalized by those who run them. (Like everything else.) The problem is to understand how media *work*, and thus balance our understanding of the things that media misrepresent.

Thoughts about Media:

1 Anything Can Be Said in Any Medium

Anything can be said in any medium, and Inspiration counts much more than ‘science.’ But the techniques which are used to convey something can be quite unpredictable.



2 Transposability

There has always been, but now is newly, a UNITY OF MEDIA OPTIONS. You can get your message across in a play, a tract, a broadside, a textbook, a walking sandwich-board, a radio program, a comic book or fumetti, a movie, a slide-show, a cassette for the Audi-Scan or the AVS-10, or even a hypertext.

(But transposing can rarely preserve completely the character or quality of the original.)

3 Big and Small Approaches

What few people realize is that big pictures can be conveyed in more powerful ways than they know. The reason they don't know it is that they see the *content* in the media, and not *how* the content is being gotten across to them—that in fact they have been given very big pictures indeed, but don't know it. (I take this point to be the Nickel-Iron Core of McLuhanism.)

People who want to teach in terms of building up from the small to the large, and others who (like the author) like to present a whole picture *first*, then fill in the gaps, are taking two valid approaches. (We may call these, respectively, the Big Picture approach and the Piecemeal approach.) Big pictures are just as memorable as picky-pieces *if* they have strong insights at their major intersections.

4 The Word-Picture Continuum

The arts of writing and diagramming are basically a continuum. In both cases the mental images and cognitive structures produced are a merger of what is heard or received. Words are slow and tricky for presenting a lot of connections; diagrams do this well. But diagrams give a poor feel for things and words do this splendidly. The writer presents exact statements, in an accord-structure of buts and indeeds, molded in a structure of connotations having (if the writer is good) *exact impreciseness*. This is hardly startling: you're always selecting what to say, and the use of vague words (or the use of precise-sounding words vaguely) is simply a flagrant form of omission. In diagrams, too, the choice of what to leave in and out, how to represent overweening conditions and forces and exemplary details, are highly connotative. (Great diagrams are to be seen in the *Scientific American* and older issues of *Time* magazine.)

This word-picture continuum is just a part of the broader continuum, which I call Fantics.

Fantics

By "fantics" I mean the art and science of getting ideas across, both emotionally and cognitively. "Presentation" could be a general word for it. The character of what gets across is always dual; both the explicit structures, and feelings that go with them. These two aspects, exactness and connotation, are an inseparable whole; what is conveyed generally has both. The reader or viewer always gets feelings along with information, even when the creators of the information think that its "content" is much more restricted. A beautiful example: ponderous "technical" manuals which carry much more connotatively than the author realizes. Such volumes may convey to some readers an (intended) impression of competence, to others a sense of the authors' obtuseness and non-imagination. Explicit declarative structures nevertheless have connotative fields; people receive not only cognitive structures, but impressions, feelings and senses of things.

Fantics is thus concerned with both the arts of effect—writing, theater and so on—and the structures and mechanisms of thought, including the various traditions of the scholarly event (article, book, lecture, debate and class). These are all a fundamentally inseparable whole, and technically-oriented people who think that systems to interact with people, or teach, or bring up information, can function on some "technical" basis—with no tie-ins to human feelings, psychology, or the larger social structure—are kidding themselves and/or everyone else. Systems for "teaching by computer," "information retrieval," and so on, have to be governed in their design by larger principles than most of these people are willing to deal with: the conveyance of images, impressions and ideas. This is what writers and editors, movie-makers and lecturers, radio announcers and layout people and advertising people are concerned with; and unfortunately computer people tend not to understand it for beans.

ANOTHER VIEWPOINT

[from handout, 1974 Natl.
Joint Comp. Conf.]

John B. Macdonald
Research Leader, Computer
Applications: Graphics,
Western Electric Company,
Engineering Research Center

Problems, Perils, and Promises of Computer Graphics

I would begin with some definitions which may be obvious but bear repeating.

1. Engineering is the application of science for (\$) profit,
2. Computer graphics does not make possible anything that was previously impossible; it can only improve the throughput of an existing process,
3. A successful application of computer graphics is when over a period of five years the cost savings from improved process throughput exceed the costs of hardware, software, maintenance and integration into an existing process flow.

In fancies as a whole, then we are concerned with:

1. The art and science of presentation. Thus it naturally includes
2. Techniques of presentation: writing, stage direction, movie making, magazine layout, sound overlay, etc. and of course
3. Media themselves, their analysis and design; and ultimately
4. The design of systems for presentation. This will of course involve computers hereafter, both conceptually and technically; since it obviously includes, for the future, branching and intricately interactive systems enacted by programmable mechanisms, i.e. computers. Thus computer display, data structures (and, to an extent, programming languages and techniques) are all a part.

Fancies must also include

5. Psychological effect and impact of various presentational techniques—but not particular formal aesthetics, as of haiku or musical composition. Where directly relevant fancies also includes
6. Sociological tie-ins—especially supportive and dysfunctional structures, such as tie-ins with occupational structure; sponsorship and commercials; what works in schools and why. Most profoundly of all, however, fancies must deal with psychological constructs used to organize things:
7. The parts, conceptual threads, unifying concepts and whatnot that we create to make aspects of the world understandable. We put them into everything, but standardize them in media.

For example, take radio. *Given* in radio—the technological fundament—is merely the continuous transmission of sound. *Put into it* have been the “program,” the *serial* (and thus the *episode*), the *announcer*, the *theme song* and the *musical bridge*—conventions which are useful presentationally.

The arbitrariness of such mental constructs should be clear. Their usefulness in mental organization perhaps is not.

Let’s take a surprise example, nothing electronic about it.

Many “highways” are wholly fictitious—at least to begin with. Let’s say that a Route 37 is created across the state: that number is merely a series of signs that users can refer to as they look at their maps and travel along

However, as time goes by, “Route 37” takes on a certain reality as a conceptual entity: people think of it as a *thing*. People say “just take 37 straight out” (though it may twist and turn); groups like a Route 37 Merchants’ Association, or even a Citizens to Save Scenic 37, may spring up.

What was originally simply a nominal construct, then, becomes quite real as people organize their lives around it.

This all seems arbitrary but necessary in both highways and radio. What, then, does it have to do with the new electronic media?

Simply this: till now the structures of media somehow sprang naturally from the nature of things. *Now they don’t anymore*. Radio, books and movies have a natural inner dynamic of their own, leading to such constructs. While this may prove to be so for computer media as well (—as I argued in “Getting It Out of Our System”), then again it may not. In other words, WE MUST ACKNOWLEDGE THAT WE ARE *INVENTING PRESENTATIONAL TECHNIQUES IN THE NEW MEDIA*, not merely transporting or transposing particular things into them, because they seem right. *The psychological constructs of man-machine systems may turn out to be largely arbitrary*. Thus bringing to terminal systems conventions like dialogue instruction (“CAI”), or arbitrary restrictions of how things may be connected, presented or written on the computer may be a great mistake.

The highway-number analogy continues. The older highways had numbers for convenience, and our travels became organized around them, and particular highways (like “U.S. 1” and “Route 66”) came to have special character. But now with the Interstates, a highway is a *planned, sealed unit*, no longer just a collection of roads gathered together under a name.

This unit, the Interstate, is not merely a psychological construct, but a planned structure. Knowing what works and what doesn’t in the design of fast highways, the Interstates were built for speed, structured as closed units. Designing them with limited access has been a conscious decision in the system design for well-based reasons, *not* a chance structure brought in from horse-and-buggy days.

Now, the constructs of previous media—writing, films, other arts—evolved over time, and in many cases may have found their way to a “natural” form. But because of the peculiar way that computer media are currently evolving (—under large grants largely granted to professionals who use very large words to promote the idea that their original

professions are largely applicable—), this sort of natural evolution may not take place. The *new* constructs of computer media, especially computer screen-media, may not have a chance to be thought out. We need designs for screen presentations and their mixture—vignetting, Windows, screen mosaics, transformed and augmented views, and the rapid and comprehensible control of these views and windows. We are still just beginning to find clever viewing techniques, and have hardly begun to discover highly responsive forms of viewability and control (cf. collateration in “Thinkertoys,” p. 330), and Knowlton’s button-box (oops, omitted). (See T. Nelson, “A Conceptual Framework for Man-Machine Everything,” and material on controls, below.)

The Mind’s Unification

One of the remarkable things about the human mind is the way it ties things together. Perceptual unity comes out of nowhere. A bunch of irregular residential and industrial blocks becomes thought of as “my neighborhood.” A most remarkable case of mental unification is afforded by the visage of our good friend Mickey Mouse. The character is drawn in a most paradoxical fashion: two globelike protrusions (representing the ears) are in *different positions on the head*, depending on whether we view him for the front

or the side. No one finds this objectionable; few people even notice, it seems.

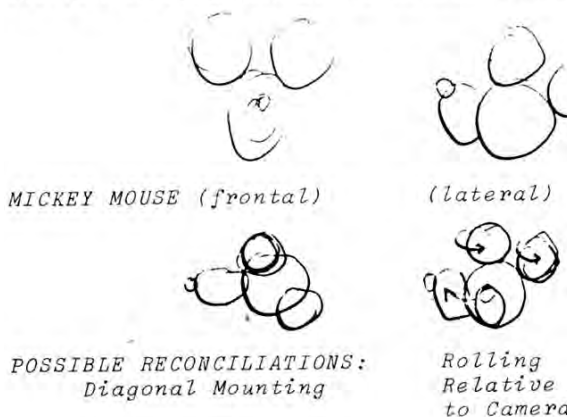
What this shows, of course, is the way the mind can unify into a consistent mental whole even things which are inconsistent by normal rules (in this case, the rules of three-dimensional structure).

Even perceptions are subject to the same principle of unification. The fingernail is an excrescence with no nerves in it; yet somehow you can *feel things with your fingernails*—tying together disparate sensations into a unified sense of something in the world (say, a coin you’re trying to pick up). In the same way, an experienced driver *feels the road*; in a very real sense, the car’s wheels and suspension become his own sensory extensions.

This principle of mental unification is what makes things come together, both literally and figuratively, in a fantastic field. A viewer sees two consecutive movie shots of streets and unifies them into one street; controls, if you are used to them, become a single fused system of options; we can have a sense of a greater whole, of which one view on a screen is a part.

THE GESTALT, DEAR BRUTUS,
IS NOT IN OUR STARS
BUT IN OURSELVES.

THE PARADOXICAL ANATOMY OF MICKEY MOUSE



Controls: Their Unification and Feel

Controls are intimately related to screen presentation, just as arbitrary, and just as important.

The artful design of control systems is a deeply misunderstood area, in no way deconfused by calling it “human factors.” There are many functions to be controlled, such as text editing operations, views of the universe on a screen, the heading of a vehicle, the tilt of an aircraft, the windage and adjustments of artillery, the temperature of a stove burner and any other controllable devices. And nowadays *any conceivable* devices could control them—pushbuttons, knobs, cranks, wheels, levers and joysticks, trigger, dials, magic wands, manipulation by lightpen on CRT screens, flicks of the finger, the *turning of the eyes* (as in some experimental gun-aiming devices), the human voice (but that introduces problems), keyboards, electronic tablets, Engelbart mice and chord writers, and so on.

The human mind being as supple as it is, anything whatever can be used to control systems. The problem is having it be a *comprehensible whole*.

As already remarked, our ability mentally to unify things is extraordinary. That we somehow tie together clutch, gear, accelerator and brake into a comprehensible control structure to make cars go and stop should amaze and instruct.

Engineers and “human factors” people speak as though there were some kind of scientific or determinate way to design control systems. Piffle. We choose a set of controls, much like an artist’s Palette, on the basis of general appropriateness; and then try best and most artistically to fit them to what needs doing.

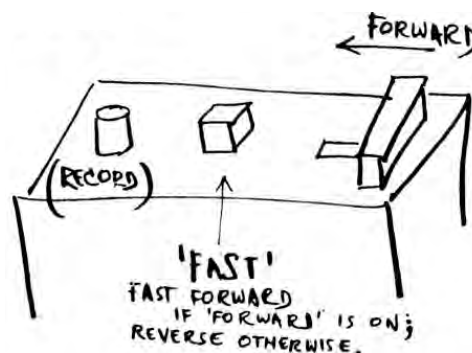
The result must be conceptually clear and retroactively “obvious”—simply because clarity is the simplest way to keep the user from making mistakes. Clear and simple systems are easier to learn, harder to forget, less likely to be screwed up by the user, and thus are more economical—getting more done for the resources put in.

There is a sort of paradox here. The *kinds* of controls are totally arbitrary, but their unification in a good system is not. Smoothness and clarity can come from disparate elements. It is for this reason that I lay particular stress on my JOT system for the input and revision of text, using a palette of keys available on the simplest standard computer terminal, the 33 Teletype. I cannot make the final judgement on how good this system is, but it pleases me. JOT is also an important example because it suggests that a conceptually unified system can be created from the artful non-obvious combination of loose elements originally having different intended purposes.

Mental analogy is an important and clear control technique. We tend to forget that the steering wheel was *invented*, separately replacing both the boat’s tiller and the automobile’s tiller. We also forget that the use of such steering mechanisms must be actually learned by children. Such continuous analogies, though, require corresponding continuities in the space to be controlled—an important condition.

Simplicity and clarity have nothing to do with the *appearance* of controls, but with the clarity and unique locatability of individual parts. For this reason I find deplorable the arrayed controls that are turning up, e.g., on today’s audio

equipment. Designers seem to think *rows of things* are desirable. On the contrary: the best designed controls I ever used are on the Sony TC-50 pocket tape recorder



but of course this is now phased out; instead most cassette recorders have *five or six* stupid buttons in a row. (Was it too good to last?)

Spurious control elegance comes in many guises. Consider Bruce McCall’s description of the Tap-A-Toe Futuroidic Footless De-Clutching™ system. This was offered on the fictitious 1934 Bulgemobiles, and allowed you to drive the car with one pedal, rather than three (see box p. 328).

Careless and horrible designs are not all fictitious. One egregious example also indicates the low level of design currently going into some responding systems: computer people have designed CRT writing systems for newspapers which actually have a “kill” button on the console, by which authors would accidentally kill their stories. In a recent magazine article it was explained that the eventual solution was to change the program so that to kill the story you had to hit the “kill” button *twice*. To me this seems like a beautiful example of what happens when you let insulated technical people design the system for you: a “kill” button on the keyboard is about as intelligent as installing knives on the dashboard of a car, pointing at the passenger.

There is another poor tendency. When computer programmers or other technical people design particular systems without thinking more generally, things are not likely to be either simple or combinable. What may result are intricate user-level controls for one particular function, controls that are differently used for another *particular* function, making the two functions not combinable.

What makes for the best control structures, then? There is no simple answer. I would say provisionally that it is a matter of *unified and conspicuous constructs* in the mental view of the domain to be controlled, corresponding to a well-distinguished and clearly-interrelated set of controlling mechanisms. But that is hardly the last word on the subject.

The Organization of Wholeness

It should be plain that in responding screen-systems, “what happens on the screen” and “how the controls respond” are not really distinguishable. The screen events are *part* of the way the controls respond. The screen functions and control functions merge psychologically.

Now, there is a trap here. Just as the gas pedal, clutch, gearshift and brake merge psychologically, *any* control structure can merge psychologically. Clutch and gearshift do not have, for most of us, clear psychological relevance to the problem of controlled forward motion. Yet we psychologically integrate the use of these mechanisms as a unified means for controlling forward motion (or, like the author, get an Automatic). In much the same way, any system of controls can gradually come through *use* to have a psychological organization, even spuriously. The trap is that we so easily lose sight of arbitrariness and even stupidity of design, and live with it when it could be so much better, because of this psychological melding.

But useful wholeness can be helped along. Just as what I have called the *accordance-structure* of writing moves it along smoothly, *fantic* design that builds from a well-organized internal dynamic should confer on a *fantic* system the same momentum and clarity that carefully-organized writing has.

This contribution of wholeness can only occur, however, if the under-level complications of a system have been carefully streamlined and smoothed back, at least as they affect the user. Consider the design of the JOT text editing system (p. 332): while it is simple *to the user*, computer people often react to it with indignation and anger because it hides what are *to them* the significant features of computer text editing—explicit preoccupation with storage, especially the calling and revision of “blocks.” Nevertheless, I say it is the details at this level which must be smoothed back if we are to make systems for regular people.

The same applies to the Th3 system, which is designed to keep the user clear-minded as he compares things in multiple dimensions. The mechanisms at the computer level must be hidden to make this work.

Fantic Space

Pudovkin and Eisenstein, great Russian movie-makers of the twenties, talked about “filmic space”—the imaginary space that the action *seems* to be in.

This concept extends itself naturally to *fantic space*, the space and relationships sensed by a viewer of any medium, or a user in any presenting or responding environment. The design of computer display systems, then, is really the *artful crafting of fantic space*. Technicalities are subservient to effects. (Indeed, I think computer graphics is really a branch of movie-making.)

Fantic Structure

The *fantic structure* of anything, then, consists of its noticeable parts, interconnections, contents and effects.

I claim that it is the *fantic unity*—the conceptual and presentational clarity of these things—that makes *fantic* systems—presentational systems and material—clear and helpful, or not.

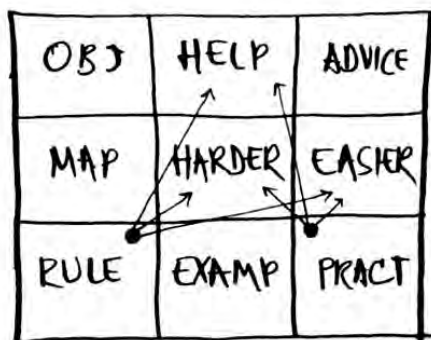
Let us take an interesting example from a system for computer-assisted instruction now under implementation. I will not identify or comment on the system because perhaps I do not understand it sufficiently. Anyway, they have an array of student control-buttons that look like this:

OBJ [objective]	HELP	ADVICE
MAP	HARDER	EASIER
RULE	EXAMP [example]	PRACT [practice]

21. Computer Lib /Dream Machines

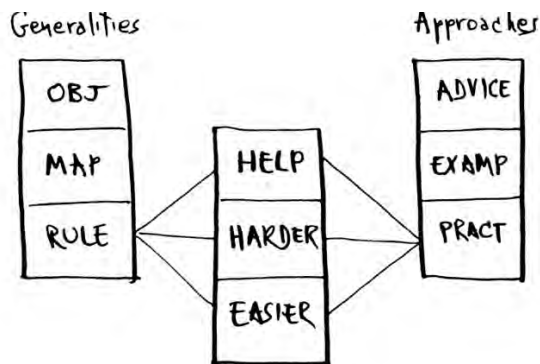
The general thinking in this system seems to be that the student may get an overall organizing view of what he is supposed to be learning (MAP); information on what he is currently supposed to be about (OBJ); canned suggestions based on what he's recently done (ADVICE). Moreover, he can get the system to present a rule about the subject or give him practice; and for either of these he may request easier rules or practice, or harder rules (i.e., more abstruse generalities) or harder practice.

For the latter, the student is supposed to hit RULE or PRACT followed by HELP, HARDER or EASIER, viz.:

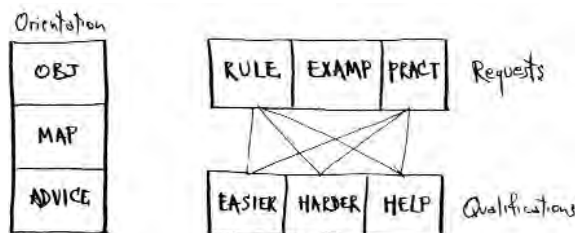


Now regardless of whether this is a well-thought-out way to divide up a subject—I'll be interested to see how it works out—these controls do not seem to be well-arranged for conceptual clarity. It seems to be the old rows-of-buttons approach.

I have no doubt that the people working on this system are certain this is the only possible layout. But consider that the student's options might be clearer to him, for instance, if we set it up as follows:



Or like this:



What I am trying to show here is that merely the arrangement of buttons creates different fancic constructs. If you see this, you will recognize that considering all the other options we have, designing new media is no small matter. The control structures merge mentally with the presentational structures. The temptation to settle on short-sighted designs having shallow unity is all too great.

Fantic Design

Fantic design is basically the planning and selection of effects. (We could also call these "performance values"—cf. "production values" in movies.)

Some of these intended effects are simply the communication of information or cognitive structure—"information transfer," to use one of the more obtuse phrases current. Other desirable effects include orienting the user and often moving him emotionally, including sometimes overwhelming or entrancing him.

In the design of fancic systems involving automatic response, we have a vast choice among types of presentational techniques, tricks that are just now becoming understood. Not just screen techniques and functions, but also response techniques and functions.

(If "feelie" systems are ever perfected, as in Huxley's *Brave New World*, it's still the same in principle.)

In both general areas, though—within media, and designing media—it seems to me that the creation of organizing constructs is the most profound problem. In particular, the organizing constructs must not distract, or tear up contents. An analogy: in writing, the inventions of the paragraph, chapter and footnote were inventions in writing technique that helped clarify what was being expressed. What we need in computer-based fancic design is inventions which do not artificially chop up, constrain, or interfere with the subject (see box, Procrustes, p. 328).

I do not feel these principles are everywhere sufficiently appreciated. For instance, the built-in structures of PLATO disturbs me somewhat in its arbitrariness—and the way its control keys are scattered around.

But there is always something artificial—that is, some form of artifice—in presentation. So the problem is to devise techniques which have elucidating value but do not cut connections or ties or other relationships you want to save. (For this reason I suggest the reader consider “Stretchtext,” p. 315), collateral linkage (p. 330), and the various hypergrams (p. 314-16). These structures, while somewhat arbitrary and artificial, nevertheless can be used to handle a subject gently.)

An important kind of organizing construct is the map or overall orienting diagram. This, too, is often partly “exact” and partly “artifice”: certain aspects of the diagram may have unclear import but clear and helpful connotation.

Responding systems now make it possible for such orienting structures to be multidimensional and responding.

Fantic design, then, is the creation either of things to be shown (writing, movie-making, etc.) at the lower end, or media to show things in, or environments.



1. The design of things to be shown—whether writing, movie-making, or whatever—is nearly always a combination of some kind of explicit structure—an explanation or planned lesson, or plot of a novel—and a *feeling* that the author can control in varying degrees. The two are deeply intertwined, however.

The author (designer, director, etc.) must think carefully about how to give *organization* to what is being presented. This, too, has both aspects, cognition and feelings.

At the cognitive end, the author must concern himself with detailed exposition or argument, or, in fiction, *plot*. But simply putting appropriate parts together is not enough: the author must use *organizing constructs* to continually orient the reader's (or viewer's) mind. Repeated reference to main concepts, repeated shots (in a movie) of particular locations, serve this function; but each medium presents its own possible devices for this purpose.

The organization of the *feelings* of the work criss-crosses the cognitive; but we can't get into it here.

Selection of points and parts contributes to both aspects. If you are trying to keep the feeling of a thing from being ponderous, you can never include everything you wanted, but must select from among the explicit points and feeling-generators that you have thought of.

2. The design of media themselves, or of media subsystems, is not usually a matter of option. Books, movies, radio and TV are given. But on occasion, as for world's fairs or very personal projects, we have a certain option. Which allows thing like:

- *Smellavision* or whatever they called it: movies with a smell-track, which went out into the theater through odor generators.
- Branching movies.
- “Multi-media” (Multiple audio tracks and simultaneous slide projections on different screens).
- Stereo movies.

And so on. The thing about the ones mentioned is that they are not viable as continuing setups for repeated productions.

They do not offer a permanent wide market; they are not stable; they do not catch on. Which is in a way, of course, too bad.

But the great change is just about now. Current technicalities allow *branching media*—especially those associated with computer screens. And it is up to us now to design them.

3. MENTAL ENVIRONMENTS are working places for structured activity. The same principles of showmanship apply to a working environment as to both the *contents* of media and the *design* of media. If media are environments into which packaged materials are brought, structured environments are basically environments where you use *non-packaged* material, or create things yourself. They might also be called “contentless media.” The principles of wholeness in structured environments are the same as for the others, and many of our examples refer to *them*.

The branching computer screen, together with the selfsame computer's ability to turn anything *else* on and off as selected by the user, and to fetch up information, yields a realm of option in the design of media and environment that has never existed before. Media we design for screen-based computer systems are going to catch on widely, so we must be far more attentive to the options that exist in order to commit—nationally, perhaps—to the *best*.

In tomorrow's systems, properly unified controls can give us new flexibilities. If deeply well-designed, these promise magnificent new capabilities. For instance, we could allow a musician to "conduct" the performance of his work by a computer-based music synthesis system, perhaps controlling the many qualities of the performance on a screen as he goes, by means of such techniques as dimensional flip. (The tradition of cumulative audio synthesis, as practiced in the fifties by Les Paul and Mary Ford, and more recently by Walter Carlos and Mike Oldfield, will take on a new fillip as multidimensional control techniques become common.)

One of the intents of this book has been to orient you to some of the possibilities and some of the options, considered generally. There is not room, unfortunately, to discuss more than one or two overall possibilities in detail. The most successful such system so far has been PLATO; others

could not be listed for lack of space

New Media to Last

What's worse, we are confronted not merely with the job of using computers to present specific things. The greater task is to design overall computer media that will last us into a more intelligent future. Adrift in a sea of ignorance and confusion, it is nevertheless our duty to try to create a whole transportation system that everybody can climb aboard. For the long run, fantic systems must be treated *not* as custom systems for explicit purposes, but as OVERALL GENERAL DESIGNS WHICH WILL HAVE TO TIE TOGETHER AND CATCH ON, otherwise collapse and perish.

Final Consequences

It seems to me certain that we are moving toward a generalized and universal Fantic system; people can and should demand it. Perhaps there will be several; but if so, being able to tie them together for smooth transmission is essential. (Think of what it would be like if there were *two kinds of telephone*?) This then is a great search and crusade; to put together truly general media for the future, systems at which we can read, write, learn and visualize, *year after year after year*. The initiatives are not likely to come from the more conventional computer people; some of them are part of the problem. (Be prepared for every possible form of aggressive defensiveness from programmers, especially: "Why would you want that?" The correct answer is BECAUSE, damnit!)

But all this means that interior computer technicalities have to be SUBSERVIENT, and the programmers cannot be allowed to dictate how it is to behave on the basis of the under level structures that are convenient to them. Quite the contrary: from the fullest consideration of the richest upper-level structures we want, *we* the users-to-be must dictate what lower-level structures are to be prepared within.

But this means you, dear reader, must develop the fantic imagination. You must learn to visualize possible uses of computer screens, so you can get on down to the deeper level of how we are going to tie these things together.

The designer of responding computer systems is creating unified setups for viewing and manipulating things—and the feelings, impressions and sense of things that go with them. Our goal should be nothing less than REPRESENTING THE TRUE CONTENT AND STRUCTURE OF HUMAN THOUGHT. (Yes, Dream Machines indeed.) But it should be something more: enabling the mind to weigh, pursue, synthesize and evaluate ideas for a better tomorrow. Or for any at all.

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THE WALKING NET™
→ A one-minute system
That Three-year-olds can learn
[Sorry to have to show you a writeup,
instead of the real thing.]

Another application of the present invention is also in the area of pictorial display, but offers a great variety of potential user choices in a simple circumstance. I call this the "walking net" system because control is effected through a changing network of choices which step, or "walk," around the screen.

The problem of intricate computer graphics may be phrased as follows: given that a digital system can hold a wide variety of graphical materials ready to present, how may the user most simply and conveniently choose them? Indeed, how may the user keep track of what is happening, where he is and where he has been?

The external mechanism I have selected for this facility paradoxically combines great versatility for sophisticated presentations with great simplicity before the naive user. The idea is this: the user may command a continuing succession of changing presentations, making only one simple choice at a time, yet receiving intricate and rich animations with extremely clear continuity on the screen.

The exterior mechanism is this: along with an arbitrary graphic presentation on the screen, the user is continuously presented with the image of a forking set of labeled arrows, e.g.:

The pip is a conventional right-pen cursor. The "current shank" is a line whose implicit gradations control developments in the picture; and the choice of arrows at the end of the current shank determine a discrete choice between alternatives that are to transpire.

The user, seizing the pip with the lightpen, moves it (through the usual lightpen techniques) sideways along the current shank. Moving it in the "forward" direction causes



progressive developments in the picture, moving it "backward" causes a reversal of animations and other previous developments.

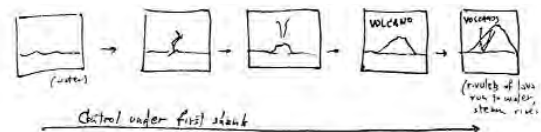
When the pip reaches the choice point in the forward direction, the user may drag it (through the usual lightpen techniques) along either of the beckoning alternatives. This then causes further developments in the presentation consonant with the line selected.

"Developments" of the picture here include expansion, contraction, sliding movements and frame-by-frame animation.

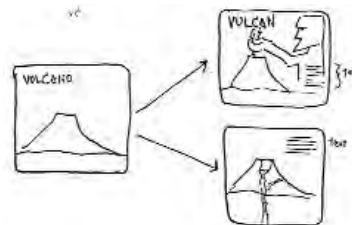
(These materials will have been, of course, explicitly input by authors and artists.)

In a sample employment, consider a presentation on the subject of volcanoes. Let the first shank of the control net control the "rise of a volcano from the sea"—an undulating ocean surface pierced first by a wisp of smoke, then a growing peak, with rivulets of lava seen to run down its sides and darken as they contribute to its growth.

At the end of the first shank, the user may branch to two arrows, labeled respectively WORD ORIGIN and INTERIOR. Either option continues the presentation without a break, retaining much of the picture



on the screen. Selection of WORD ORIGIN causes the word VOLCANO to change to VULCAN, and a picture of the god Vulcan is seen to seize a lightning bolt rising from the crater; text appears to explain this. Alternatively, if the user chooses INTERIOR, the tubes and ducts within the volcano appear, and explanatory text also.

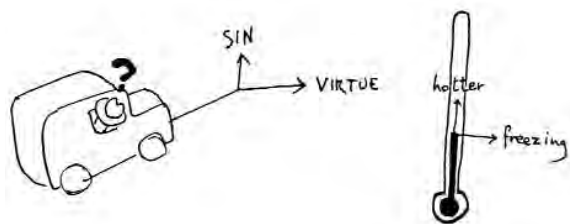


(The path unchosen fades from the screen, as does the previous shank.)

Either of these alternatives may continue with its own developments and animations under the control of its own shank.

Several features of this control application are of special interest. One is that the presentation may be continuous in all directions, aiding in continuous user orientation. Another is that presentations are reversible in various ways, an aid both in user orientation and self-study. (Not only is a demonstration reversible within a given shank, but the user may back the pip through an intersection into the antecedent shank—which reappears at the juncture as the lightpen backs up—and the user may continue to reverse the presentation through that preceding shank, or to re-enter the intersection and make another choice, “the path not taken.”) These features allow the user clearly to repeat demonstrations as often as he likes and to explore numerous alternatives.

The displayed control net is thus to be understood as a large network of choices, mostly unseen, whose currently visible portion “walks” around the screen as use progresses. Within this system, then, numerous variants are possible. For instance, the currently visible portion of the net may itself be whimsically incorporated in a picture, viz.:



PROCRUSTES THE GIANT

The Greeks told of a giant named Procrustes (rhymes with Rusty's) who was very hospitable to passing travelers. He would invite, indeed compel them, to sleep in his bed. Unfortunately, because it was a very odd bed, he had to cut them up first . . .

Procrustes has haunted conversations ever since; and any time we are forced to use categories that don't properly fit a subject, it seems like an invitation to the Procrustean bed.

Hypertext systems at last offer total freedom from arbitrary categorizing and chopping; but in *some* systems for storing and presenting information, I can't help hearing the whisk of Procrustes's knife—

“Take new Tap-A-Toe Futuroidic Footless De-Clutching. Instead of old-fashioned gas, brake and clutch pedals that kept your feet busier than a dance marathon, Tap-A-Toe Futuroidic Footless De-Clutching offers the convenience of Single Pedal Power Control—combines *all foot functions* in one single pedal!

“Think of it: one tap—you go, moving off faster than a barfly after Repeal.

“Two taps—you change gears, as smooth and automatic as a mortgage foreclosure.

“Three taps—you stop quicker than the U.S. economy.

“And that's all there is to it. Tap-A-Toe Futuroidic Footless De-Clutching with Single Pedal Control is as easy and effortless as the Jap march on Manchuria!”

Bruce McCall,
“1934 Bulgemobile Brochure,”
National Lampoon, May 74, 76–7.

STELLAVISION

A nice example of a unified presentational system would allow you a “feelie” glove along with your computer display—the sort of thing Mike Noll has been doing at Bell Labs.

Now, suppose you are playing with a diagram of a star on a computer display screen. It's all very well to see its layers, flowing arrows representing convection currents, promontories and so on—but some things you ought to be able to *feel*. For example, the mechanical resonance-properties of stars. It would be nice to be able to reach and *grasp* the star, to *squeeze* it and feel its pulsations as it regains its shape. This could be done in the glove—at the same time the *image* of the glove grasps the star on the screen, and the star is squished.

Of course, to build such a responding glove, particularly one that gave you subtle feelings back in your fingers, would probably be very expensive. But it's the kind of possibility people should start considering.

Should I have called it
TEACHOTECHNICS?
SHOWMANSHIPNOGOGY?
INTELLECTRONICS? [S. Ramo]
THOUGHTOMATION?
MEDIA-TRONICS?
ABOUT THE TERM
'FANTICS.'

First of all, I feel that very few people understand what interactive computer systems are *about*. It's like the story of the blind man and the elephant—each thinks it's a different thing (based, usually, on his own technical specialty).

But I think it's all show business. PENNY ARCADES are the model for interactive computer systems, not classrooms or libraries or imaginary robot playmates. And computer graphics is an intricate branch of movie-making.

Okay, so I wanted a term that would connote, in the most general sense, the showmanship of ideas and feelings—whether or not handled by machine.

I derive “fantics” from the Greek words “phainein” (show) and its derivative “phantastein” (present to the eye or mind).

You will of course recognize its cousins *fantastic*, *fantasy*, *phantom*. (“Phantom” means *what is shown*; in medical illustration it refers to an opaque object drawn as transparent; a “phantom limb” is an amputee’s temporary feeling that the severed limb has been restored.) And a *fantast* is a dreamer.

The word “fantics” would thus include the showing of anything (and thus writing and theater), which is more or less what I intended. The term is also intended to cover the tactics of conveying ideas and impressions, especially with showmanship and presentational techniques, organizing constructs, and fundamental structures underlying presentational systems.

Thus Engelbart’s data hierarchy, SKETCHPAD’s Constraints, and PLATO’s fantic spaces are fantic constructions that need to be understood if we are to understand these systems and their potential usages.

Livermore Labs, those hydrogen-bomb design people, will have a “Laboratory for Data Analysis,” an opulent facility for experimenting with multidimensional visualization.

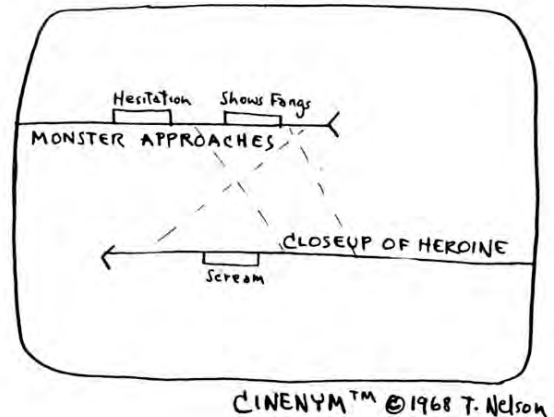
One of your jolly ironies. I have seen pictures of beautiful multibutton control handles which were designed for project SMASH, would you believe Southeast [Asia] Multisensory Armament System for Helicopters. Aargh.

The best with the worst.

Everything is deeply intertingled.

Designing screen systems that focus the user’s thought on his work, with helpful visualizations and no distractions, is the great task of fantic design.

In a system I designed for CRT motion-picture editing, the user could manipulate written descriptions on the screen (corresponding to the usual yellow-pad notes). To see the consequences of a particular splice, for instance, the editor would only have to draw a line between two annotated lines representing shots. Trim variations could be seen by moving this cut-line (illustrated).



Not long after, CBS and Memorex *did* introduce a system for movie editing by CRT—but I’ve heard that in their system the user has to actually deal with numbers. If so, this is missing the whole point.

★THINKERTOYS★

Our greatest problems involve thinking and the visualization of complexity.

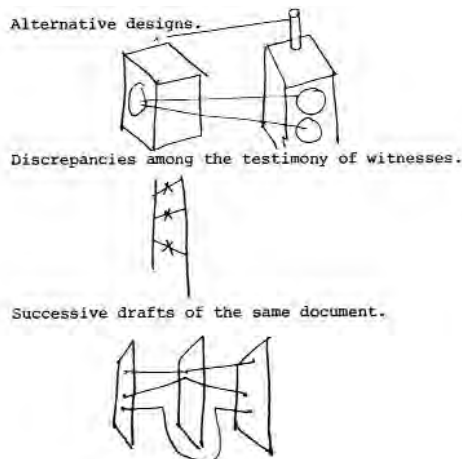
By "Thinkertoy" I mean, first of all, a system to help people think. ("Toy" means it should be easy and fun to use.) This is the same general idea for which Engelbart, for instance, uses the term "augmentation of intellect."

But a Thinkertoy is something quite specific: I define it as a computer display system that *helps you envision complex alternatives*.

The process of envisioning complex alternatives is by no means the only important form of human thought; but it is essential to making decisions, designing, planning, writing, weighing alternate theories, considering alternate forms of legislation, doing scholarly research, and so on. It is also complicated enough that, in solving it, we may solve simpler problems as well.

We will stress here some of the uses of these systems for handling *text*, partly because I think these are rather interesting, and partly because the complexity and subtlety of this problem has got to be better understood: the written word is nothing less than the tracks left by the mind, and so we are really talking about screen systems for handling ideas, in all their complexity.

Numerous types of complex things have to be inter-compared, and their relations inter-comprehended. Here are a few of the many types:



Pairs of things which have some parts the same, some parts different (contracts, holy books, statutes of different states, draft versions of legislation...)



Different theories and their ties to particular examples and evidences.



Under examination these different types of inter-comparison seem to be rather different. Now, one approach would be to create a different data structure and viewing technique for *each* different type of complex. There may be reasons for doing that in the future.

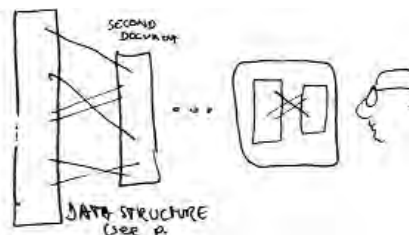


For the present, however, it makes sense to try to find the most general possible viewing technique: one that will allow complex intercomparisons of all the types mentioned, and any others we might run across.

One such technique is what I now call *collateralation*, or the linking of materials

into *collateral structures*,* as will be explained. This is fairly straightforward if you think enough about the problem; Engelbart discovered it independently.

Let us call two structures *collateral* if there are links between them, connecting a selected part of one with a selected part of the other. The sequences of the connected parts may be different. For simplicity's sake, suppose each one is a short piece of writing. (We will also assume that there is some convenient form of rapid viewing and following between one end of a link and another.)



Now, it will be noted first off that this is an extremely general method. By collateral structuring we can easily handle the equivalents of: tables of contents; indexes; comments and marginalia; explanations, exegesis, explication; labeling; headings; footnotes; notes by the writer to himself; comments and questions by the reader for later reference; and additional details out of sequence.

Collateration, then, is the creation of such multiple and viewable links BETWEEN ANY TWO DATA STRUCTURES, in principle. It is general and powerful enough to handle a great variety of possible uses in human intellectual endeavor, and deserves considerable attention from researchers of every stripe.**

The problem then, is how to handle this for rapid and convenient viewing and whatever other work the user wants to do—writing and splicing, inter-comparing, annotating and so on. Two solutions appear on this spread: The Parallel Textface™, designed as a seminal part of the Xanadu system (see p. 335), which I hope will be marketed with that system in the near future, and a more recent design which I've worked on at the University of Illinois, the 3D Thinkertoy or Th3.

Clarity and Power

We stressed on the other side of the book that computer systems must be clear, simple and easy to use. Where things like business uses of computers are concerned, which are intrinsically so simple in principle, some of the complications that people have been forced to deal with in ill-designed computer systems verge on the criminal. (But some computer people want others to think that's the way it has to be. "Your first duty is to keep your job, right?" one computer person said to me recently. "It wouldn't do to set up systems so easy to use that the company wouldn't need you anymore.")

But if it is desirable that computer systems for *simple-minded* purposes be easy to use, it is *absolutely necessary* that computer systems for *complicated* purposes be simple to use. If you are wrangling over complex alternatives—say, in chess, or in a political simulation game, or in the throes of trying to write a novel, the last thing you will tolerate is for your computer screen to introduce complications of its own. If a system for thinking doesn't make thinking simpler—allowing you to see farther and more deeply—it is useless, to use only the polite term.

But systems can be both powerful and simple at the same time. The myth that things have to be complicated to do anything for you is pernicious rubbish. Well-designed systems can make our mental tasks lighter and our achievements come faster.

It is for this reason that I commend to the reader these two designs of mine: as examples of user-level control and viewing designs—fanciful environments, if you will (see p. 317)—that are pruned and tuned to give the user great control over the viewing and cross-consideration of intricate alternatives, *without* complication. I like to believe that both of these, indeed, are *ten-minute systems*—that is when we get them running, the range of uses shown here can be taught to naive users, in *ten minutes or less*.

It is because of my heartfelt belief in this kind of simplicity that I stress the creation of prefabricated environments, carefully tuned for easy use, rather than the creation of computer *languages* which must be learnt by the user, as do such people as Engelbart and DeFanti.

Now, their approach obviously has considerable merit for sophisticated users who want to tinker repeatedly with variant approaches. For people who want to work incessantly *in* an environment, and *on* other things—say writers—and are absent-minded and clumsy and nervous and forgetful (like the present author), then the safe, prefabricated environment, with thoroughly fail-safe functions and utterly memorable structural and control interrelationships, is the only approach.

*In my 1965 paper (see bibliography) I called collateral structures *zippered lists*. [included in this volume (§11)]

** A group at Brown University has reportedly worked along these lines since I worked with them, but due to certain personal animosities I have not kept up with their developments. It will be interesting to see what kind of response they can get out of the IBM systems they are using.

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Theodor H. Nelson, "A File Structure for the Complex, the Changing and the Indeterminate." Proc. ACM 65, 84–100.

———, "Simplicity Versus Power in User Systems." Unpublished.

Decision/Creativity Systems [Thinkertoys]

Theodor H. Nelson

19 July 1970

It has been recognized from the dawn of computer display that the grandest and most important use of the computer display should be to aid decisions and creative thought. The work of Ivan Sutherland (SKETCHPAD) and Douglas Engelbart have really shown how we, may use the display to visualize and effect out creative decisions swiftly and vividly.

For some reason, however, the most important aspect of such systems has been neglected. We do not make important decisions, we should not make delicate decisions, serially and irreversibly. Rather, the power of the computer display (and its computing and filing support) must be so crafted that we may develop alternatives, spin out their complications and interrelationships, and visualize these upon a screen.

No system could do this for us automatically. What design and programming can create, however, is a facility that will allow us to list, sketch, link and annotate the complexities we seek to understand, then present "views" of the complexities in many different forms. Studying these views, annotating and refining, we can reach the final designs and decisions with much more in mind than we could otherwise hold together in the imagination.

Some of the facilities that such systems must have include the following:

Annotations to anything, to any remove.

Alternatives of decision, design, writing, theory.

Unlinked or irregular pieces, hanging as the user wishes.

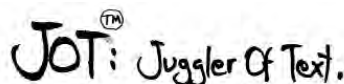
Multicoupling, or complex linkage, between alternatives, annotations or whatever.

Historical filing of the user's actions, including each addition and modification, and possibly the viewing actions that preceded them.

Frozen moments and versions, which the user may hold as memorable for his thinking

Evolutionary coupling, where the correspondences between evolving versions are automatically maintained, and their differences or relations easily annotated.

In addition, designs for screen "views," the motion, appearance and disappearance of elements, require considerable thought and imagination.

The logo for JOT: Juggler of Text. It features the word "JOT" in a large, stylized, handwritten font with a trademark symbol (TM) to its upper right. Below "JOT" is the phrase "Juggler of Text." in a smaller, similar handwritten font.

From "A Human Being's Introduction To the JOT System." ©1972 T. Nelson.

Here's how simple it is to create and edit text with the JOT system. Since your typewriter is now a JOT machine, not every key does what it used to. [When Nelson wrote much "word processing" was through modified typewriters, without graphic displays—eds.]

CREATING TEXT: just type it in.

You type: The quick brown fox jumps over the lazy dog.

It types: The quick brown fox jumps over the lazy dog.

REVIEWING A SENTENCE YOU JUST TYPED: the back-arrow takes you back, the space bar steps you through.

You type: ← sp sp sp

It types: (bell) The quick brown fox

DELETIONS AND INSERTIONS: the RUBOUT key rejects words you don't want. To insert, merely type.

You type: ← sp sp RUBOUT lithe sp sp sp sp sp
It types: (bell) The quick /brown/ lithe fox jumps over the lazy dog.

REARRANGING TEXT: first we make three Cuts in the text, signaled by free-standing exclamation points.

You type: sp ! sp ! sp ! fox

It types: The ! quick ! lithe ! fox

TO REARRRANGE IT, YOU TYPE: LINE FEED key. This exchanges the two pieces between the cuts.

CHECK THE RESULTS:

← sp sp sp sp
(bell) The lithe quick fox

The object is not to burden the user, or make him aware of complexities in which he has no interest. But almost everyone in intellectual and decision pursuits has at some time an implicit need for some of these facilities. If people knew they were possible, they would demand them. It is time for their creation.

A full-fledged decision/creativity system, embracing both text and graphics, is one of the ultimate design goals of Project XANADU.

We might also think of them as systems for

THE MANAGEMENT OF LOOSE ENDS.

very unusual kinds of screen animation and file retrieval. But this grew out of structuring a

system to handle the functions described here.

Thus the Parallel Textface basically requires a Xanadu system.

It is hoped that this system can be sold complete (including a minicomputer or microprocessor—no connection to a large computer is required) for a few thousand dollars by 1976 or 1977. (Since “business people” are extremely skeptical as to whether anybody would want such a thing, I would be interested in hearing expressions of interest, if any.)

As shown here, the screen presents two panels of text; more are allowed. Each contains a segment of a longer document. (“Page” would be an improper term, since the boundary of the text viewed may be changed instantly.)

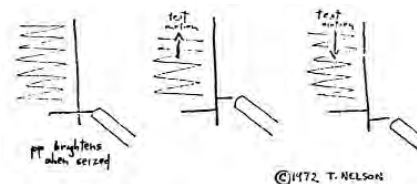
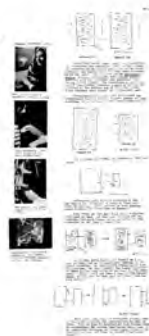
The other odds and ends on the screen are hidden keys to control elements which have been made to fade (in this illustration), just to lessen the distraction.

Panel boundaries and control graphics may be made to appear by touching them with the lightpen.

The PARALLEL TEXTFACE™

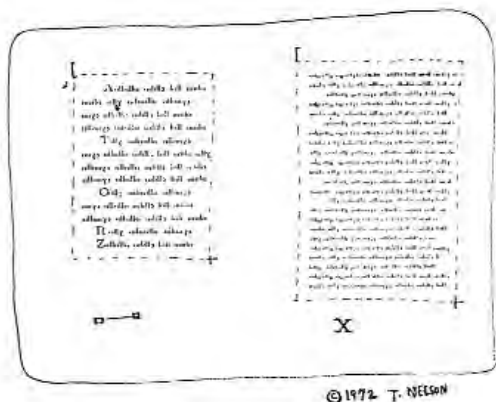
This user-level system is intended to aid in all forms of writing and scholarship, as well as anywhere else that we need to understand and manipulate complex clusterings of text (i.e., *thought*). It will also work with certain animated graphics.

The parallel Textface, as described here, furnished the initial impetus for the development of the Xanadu™ system (see p. 335). Xanadu was developed, indeed, originally for the purpose of implementing some of these functions, but the two split apart. It turned out that the Parallel Textface required an extremely unusual data structure and program techniques; these then became the Xanadu system. As developed in the final Xanadu design, they turn out to handle some

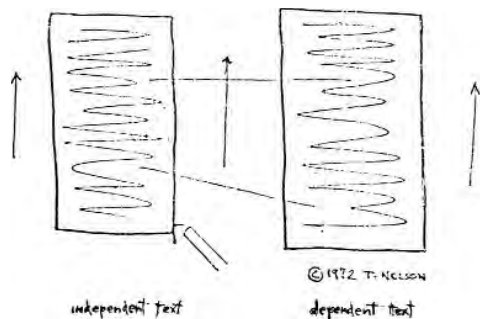


Roving Functions

The text moves on the screen! (Essential.) The lower right hand corner of each text panel contains an inconspicuous control diagram. The slight horizontal extension is a movable control pip. The user, with his lightpen, may move the pip up or down. “Up” causes the text to move smoothly upward (forward in the material), at a rate proportional to how far you push the pip; “down” causes it to move back. (Note that we do *not* refer here to jerky line-by-line jumps, but to smooth screen motion, which is essential in a high-performance system. If the text does not move, you can’t tell where it came from.)

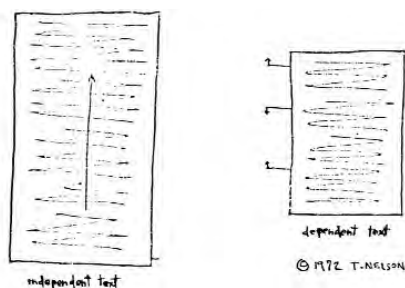


21. Computer Lib /Dream Machines

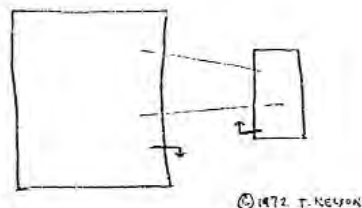


DERIVATIVE MOTION: when links run sequentially, connecting one-after-the-other on both sides, the contents of the second panel are pulled along directly: the smooth motion in one panel is matched in the other. This may be called *derivative motion*, between independent text (being handled directly with the lightpen) and dependent text (being pulled along). The relationship may be reversed immediately, however, simply by moving the lightpen to the control pip of the other panel, whose contents then become the independent text.

Irregularities in the links will cause the independent text to move at varying speeds or jump, according to an average of the links' connectivity.

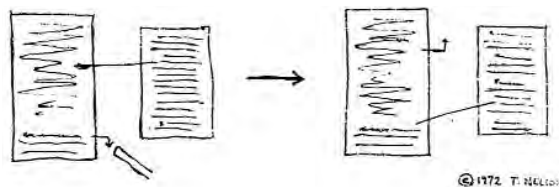


If no links are shown, the dependent text just stops.

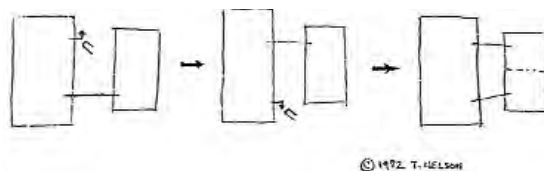


Collateral links between materials in the two panels are displayed as movable lines between the panels. (Text omitted in this diagram; panel boundary has been made to appear.)

Some links may not have both their endpoints displayed at once. In this case we show the incomplete link as a broken arrow, pointing in the direction of the link's completion.



The broken arrow serves not merely as a visual pointer, but as a jump-marker leading to the linked material. By zapping the broken arrow with the lightpen, the user summons the linked material—as shown by the completion of the link to the other panel. (Since there has been a jump in the second panel, we see that in this case the other link has been broken.)



When such links lead to different places, both of these destinations may nevertheless be seen at once. This is done by pointing at both broken links in succession; the system then allows both links to be completed, breaking the second panel between the two destinations (as shown by dotted line across panel).

*Oddly, this has the same logical structure as time-travel in science-fiction.

There are basically three alternative premises of time-travel: 1) that the past cannot be changed, all events having preceded the backstep; 2) that the past *can* be changed; and 3) that while time-travelers may be deluded into thinking (2), that (1) is really the case—leading to various appointment-in-Samarra plots.

Only possibility (2) is of interest here, but there are various alternative logics of mutability and time-line stepping. One of the best I have seen is in *The Man Who Folded Himself* by David Gerrold (Popular Library, 1973): logic expounded pp. 64–8. I am bemused by the parallel between Gerrold's time-controls and these, worked out independently.

PARALLEL TEXTFACE (1971)



Real person sits at cardboard Xanadu mockup.



"Nice keyboard. But what happened to your typewriter?"



Two panels are about right for a 10 x 10 screen.



Independent text pulls dependent text along. Painted streaks simulate motion, not icicles.



Fail-Safe and Historical Features

In systems for naive users, it is essential to safeguard the user from his own mistakes. Thus in text systems, commands given in error must be reversible. For instance, Carmody's system requires confirmation of deletions.

Another highly desirable feature would allow the user to view previous versions, to see them collaterally with the corresponding parts of current versions, and even go back to the way particular things were and resume work from the previous version.

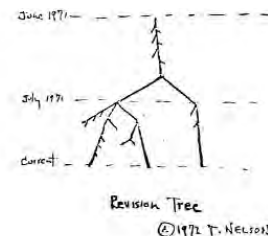
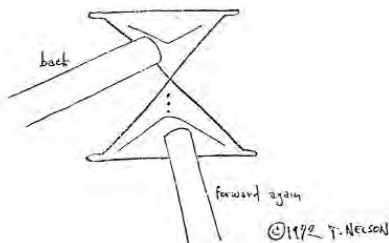
In the Parallel Textface this is all comprised in the same extremely simple facility. (Extremely simple from the user's point of view, that is. Inside it is, of course, hairy.)



TO UNDO SOMETHING, YOU MERELY STEP "BACKWARD IN TIME" by dagging the upper part of the hourglass with the lightpen. One poke, one editing operation undone. Two pokes, two operations.

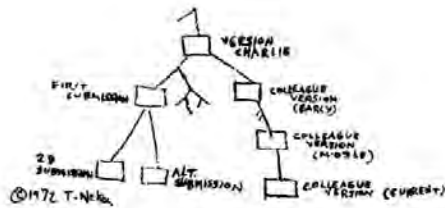
You may then continue to view and make changes as if the last two operations had never taken place. This effectively creates an alternative time-line.* However, if you decide that a previously undone edit operation should be kept after all, you may step

forward—stepping onto the previous time-line—by using the lower half of the hourglass.

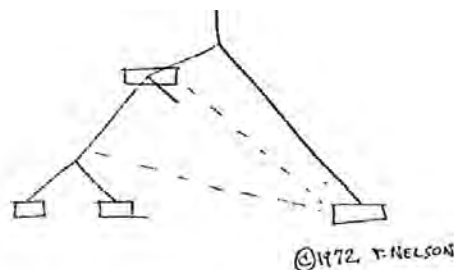


21. Computer Lib /Dream Machines

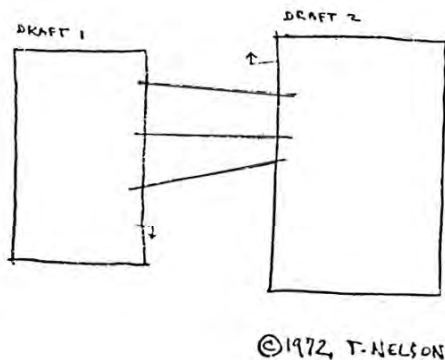
We see this clarified in a master time diagram or Revision Tree which may be summoned to the screen, never mind how. In this example we see that three versions are still “current,” various other starts and variations having been abandoned. (The shaggy fronds correspond to short-lived variations, resulting from operations which were then reversed. In other words, “excised” time-lines, to use Gerrold’s term—see footnote.)



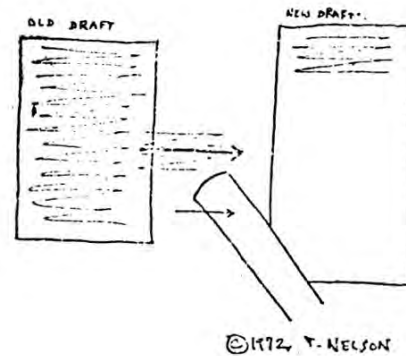
The user—let’s say he is a thoughtful writer—may define various Versions or Drafts, here marked on the Revision Tree.



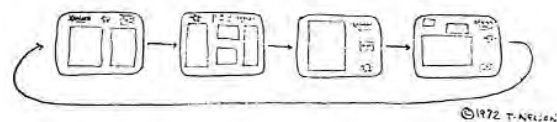
He may, indeed, define collateral linkages between different versions defined at various Times in the Tree ...



... and see them displayed collaterally; and revise them further.



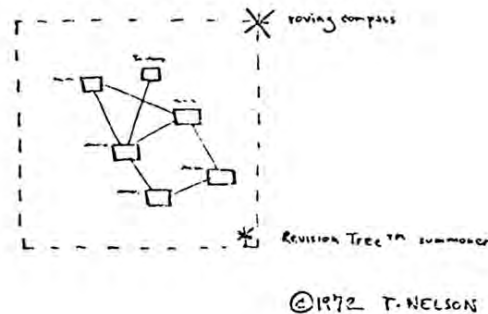
Materials may be copied between versions. (Note that in the copying operation of the Parallel Textface, you actually see the moved text moved *bodily as a block*.)



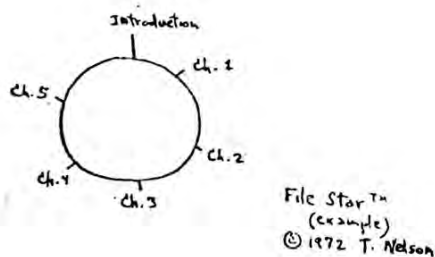
Getting Around

The user may have a number of standby layouts, with different numbers of panels, and jump among them by stabs of the lightpen.

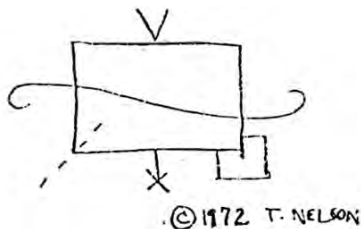
Importantly, *the panels of each can be full*, each having whatever the contents were when you last left it.



The File Web™ is a map indicating what (labeled) files are present in the system, and which are collaterated.

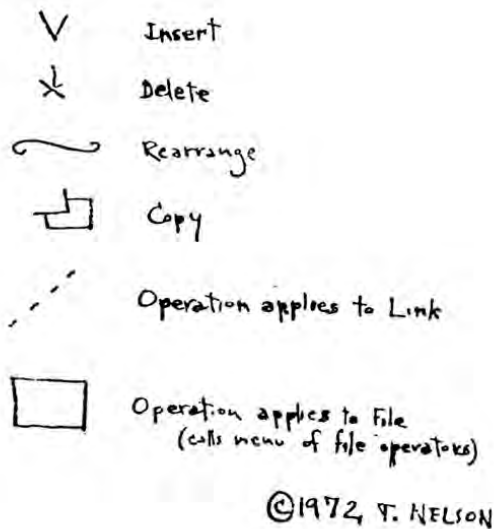


The File Star™ is a quick index into the contents of a file. It expands as long as you hold the lightpen to the dot in the center, with various levels of headings appearing as it expands. Naturally, you may jump to what you point at.

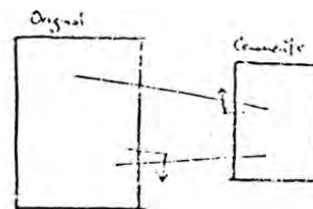


Editing

Rather than giving the user anything complicated to learn, the system is completely visual. All edit controls are comprised in this diagram, the Edit Rose™. Viz.:

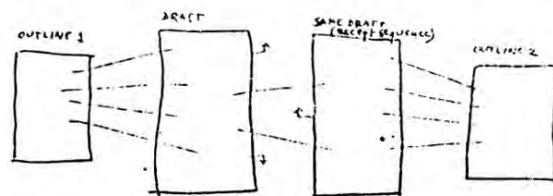


Separate portions of the Edit Rose invoke various edit operations. (You must also point with the lightpen to the necessary points in the text: once for Insert, twice for Delete, three or four times for Rearrange, three times for Copy.)

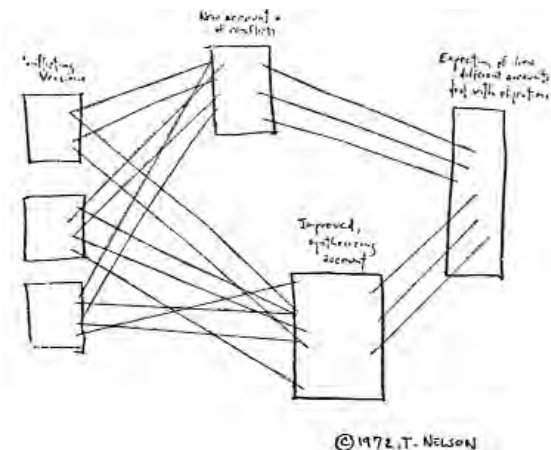


Generality

The system may be used for comments on things,



for organizing by multiple outlines or tables of contents;



and as a Thinkertoy, organizing complex alternatives. (The labels say: "Conflicting versions," "New account of conflicts," "Exposition of how different accounts deal with objections," "Improved, synthesizing account.")

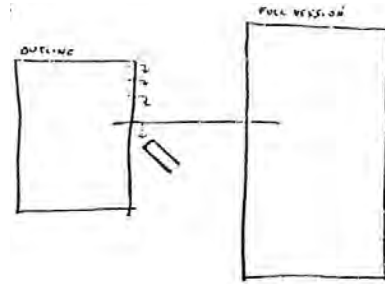
21. Computer Lib /Dream Machines

theNEWMEDIAREADER

In other words, in this approach we annotate and label discrepancies, and verbally comment on differences in separate files or documents.

In ways this may seem somewhat obtuse. Yet above all *it is orderly*, and the complex of collateral files has a clarity that could be all-too-easily lost in systems which were programmed more specifically to each problem.

The fundamental strength of collateration, seen here, is of course that any *new* structure collateral to another may be used as a table of contents or an outline, taking the user instantly to parts which are of interest in some new context.



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New Freedoms Through Computer Screens
— a Minority Report

This is the flip side of Computer Lib.

Privacies of Reading: Self-Formations

by Sven Birkerts

I am going to take it as very nearly axiomatic that reading—the act—plays a vital role in the formation and conditioning of sensibility. Not for everyone, of course, but in the life of the committed reader. What interests me is trying to puzzle out the nature of that role. But before I venture in that direction, I feel that I should pause over one word—sensibility. It is, I realize, a term that is slipping from the lexicon; there is a sort of nineteenth-century mustiness clinging to its five syllables. Sensibility is neither self nor ego, neither identity nor personality. Where these are designations for something one has or is, sensibility is more of a construct. The old sense of the word is of a refinement or cultivation of presence, a part of the self not given but fashioned. A defining, if cloudy, complex of attitudes, predilections, and honed responses. And for this reason I want to have the term available. For while it can be many different things, reading is, for the serious reader, above all an agency of self-making. The effort of actualizing a text, when undertaken freely, is implicitly the token of a desire to actualize and augment some part of the self. The act assumes a possibility of deepened self-understanding and therefore recognizes the self as malleable. It is the intimate, perhaps secret, part of a larger project, a project that has little to do with societal conceptions of the individual.

I realize that to talk about reading and its particular influence on self-making, I will need to distinguish not only among different kinds of reading, but also point out some of the conspicuous ways in which the reading act changes across the trajectory of the reader's career. The process that begins, in most cases, with being read to, and which sponsors the most intense sorts of identifications in the independent reading of childhood, is something else again in adolescence, and again in full-fledged maturity, and yet again, I would guess, in later years. But since with adulthood and the later years we are generally talking about sensibilities that have more or less crystallized, the likeliest periods at which to look are those of childhood and adolescence.

Being read to, while not strictly reading, is nonetheless not an entirely passive absorption

either, as any parent can tell you. The child naturally fleshes out the narrative through imaginative projection, but also engages the book itself, looking at the illustrations, monitoring the momentum of the turning pages, and, with the increase of aptitude, noting the correlation between what is being read out loud and the placement of the words on the page. In a very real sense, being read to sets into motion that specialized activity of inwardness that will make private reading rewarding—and very nearly inevitable.

Independent childhood reading seems to continue and elaborate upon the process of imaginative projection. It is, beautifully and openly, a voluntary participation in an ulterior scheme of reality. We might almost call it pure escape, except that getting away is probably less the motivating impetus than getting to. Early childhood reading is the free indulgence of fantasy and desire, done because it feels good. Children are not out to improve themselves; nor are they, in any obvious sense, trying to figure out the terms of existence (though such figuring certainly goes on unconsciously).

The main difference between childhood reading and reading undertaken later is that, in the former, futurity—the idea of one's life as a project, or adventure, or set of possibilities—has not yet entered the calculation. The child reads within a bubble. He is like Narcissus staring at his own lovely image. He is still sealed off from any notion of the long-term unfolding of life, except in the perfected terms of fantasy: I, too, will be a pirate...

The change comes with adolescence, that biological and psychological free-fire zone during which the profoundest existential questions are not only posed, but are lived. Who am I? Why am I doing what I'm doing? What should I do? What will happen to me? It is in adolescence that most of us grasp that life—our own life—is a problem to be solved, that a set of personal unknowns must now be factored together with the frightening variables of experience. The future suddenly appears—it is the space upon which the answers will be inscribed. The very idea of futurity becomes charged with electricity.

This self-intensity, which pushes toward the

future as toward some kind of release, is highly conducive to reading. The book—the novel, that is—becomes the site for testing transformations. Indeed, whatever else it may be, diversion or escape, the book at this stage of life is primarily a screen that will accept and illuminate versions and projections of the self.

These projections are different from those of the child-reader. The child manipulates fantasy stuff that is still undiluted by the reality principle. The boy dreaming of river rafts or space invaders is not yet constrained by the impediments that will eventually curb and instruct his desires. Not so for the adolescent. A different reality has announced itself. Socially, sexually, and even within the bosom of the family that thing that the grownups call "life" has begun to bare its face.

Adolescence, then, is an ideal laboratory for the study of reading and self-formation. Or maybe I should say: a laboratory for studying the ideal impact of reading upon that formation. For of course it is no secret that fewer and fewer adolescents turn to books on their own. And it is precisely this private reading—reading done away from school assignments—that concerns me here.

How does reading—reading fiction—work on the psyche during what has to be its single most volatile period? To pin this down is extremely difficult. We might begin with the most obvious sort of answer—encountered all the time in interviews and memoirs—which has to do with the influence of specific books and characters. The subject (who has usually achieved something noteworthy) tells of living with Tom Sawyer or David Copperfield. Admiration led to emulation; there was the desire to do what Tom did, to be like young David. These recognitions were eventually externalized as ideals and in that form guided the behavior long after the memory of the reading had faded. (I can remember several specific situations in which I acted as I did because I believed that was what Jack London would have done—I had all but memorized Irving Stone's biography, *Jack London: Sailor on Horseback*.)

These identifications are inevitable for the reader. They form the very basis of childhood play, and run like a stream alongside the less rooted initiatives of adolescence. There they function as a kind of meta-narrative. If one is not Tom Sawyer or David Copperfield oneself, one nevertheless performs within a magnetic field that somehow contains them. The admiring reader. The admiring reader acts in a world that is half that of the book and half that of the resistant obstacle course present to the senses. The tension is often delicious. Every action is ennobled and exaggerated in significance because the reader

imagines it brightly transposed onto the field of the book—the field of a higher and more enduring reality.

Later, as adult claims displace the more childish conjurings and the adolescent matures, reading takes on a slightly different function. Now the reader begins to borrow from the book the sense of consequential destiny that is so absent from the daily routine. For what the book transmits, over and above its plot and character engagements, is the bewitching assumptions of connectedness. Purpose. Meaning. The characters and situations, products of the authors intending will, are knit together into a teleological wholeness. The least movement or action tends toward; every action is held within the larger context, which is implicitly—aesthetically—purposeful. Our own lives may drift every which way toward the future, but the lives of the characters are aimed and propelled forward into time. As readers we take this in, albeit largely unconsciously, and we begin to conceive of our own actions under this aspect of fatedness—at least as long as we are reading or otherwise under the spell of the book. And we thereby become important—just when we need to most. Our lives seem pointed toward significance and resolution; we feel ourselves living toward, or, at the very least, in the light of its possibility. I don't know that this particular self-charge is available anywhere else but on books. Movies are too compacted and visually determined to encourage such tender echolations of self. Certainly little in the day-to-day circumstance models it for the adolescent. The most obvious versions of futurity are, for them, embodied in their parents, whose lives must appear to be as mired in chaotic distraction as their own.

Remember, I am talking about the reading of fiction. This is not an arbitrary decision on my part. I happen to believe that there is a very special transformation that takes place when we read fiction that makes the act different from any other kind of reading. And this transformation relates absolutely to the question of self-formation. Let me try to illustrate it.

When we read a sentence from a work of non-fiction—a history or biography, say—the words intersect with the psychic continuum, but they do not consciously modify it. We do not reposition the self, but merely suspend our awareness of its signals and demands so that we can pay proper attention. Consider, as a representative example, this straightforward sentence plucked from *The Columbia History of the World*: "When we talk about human evolution, we are dealing with two different kinds of processes: the evolution of the human body and the evolution of human

behavior." As we read the words, we decode the syntactical logic of the statement and extract the idea content, the sense. If there is an authorial "voice," we don't hear it. The prose is a conveyor for the concept, a means to an end. We make a place for it in that interior zone where we process verbal information, but we don't ourselves change. Unless, of course, we encounter an idea that can be translated into relevant personal terms and thus affect our understanding of ourselves. But even then we react less to the words than to the implications we dig out for ourselves.

Now open a novel:

If you really want to hear about it, the first thing you'll probably want to know is where I was born, and what my lousy childhood was like, and how my parents were occupied and all before they had me, and all that David Copperfield kind of crap, but I don't feel like going into it, if you want to know the truth.

This, too, is information, but it is obviously information of a very different kind. Reading the earlier sentence about evolution, we make no significant internal adjustment because it is not ourselves—as selves—that we hear addressed. In the second sentence—it is the opening of Salinger's *The Catcher In The Rye*—the voice is primary. The voice proposed a self, and we must greet it as such. We therefore heed the casual, alienated, determinedly forthright tone and filter the sense of the statement through that. But we cannot heed and filter so long as our own self is in dormancy—either we decide to engage Holden's voice or we close the book.

Salinger, via Holden, posits a world. Holden's world. And the reader who would hear more about it is forced to open up a subjective space large enough to contain it. The opening of that space is the crucial move, for it requires a provisional loosening of whatever fixed attitudes and preconceptions we may have. In that space two versions of reality will be stirred together—ours and the author's. A hybrid life will start up. Not the author's life, not full the character's, and not quite our own—though all of these must be present for the catalysis of reading to occur.

To read the book we must, in effect, bracket off our own reality, replacing it with Holden's. His can only exist at the expense of ours—this is the law of fiction. We agree to suspend our self-grounded posture, our place in the normal world, in order to make room for Holden's alternative sense of things. We create the textures of his reality with what we have learned from our own. But we don't disappear, either. Our awareness, our sense of our own life, gets filtered into the written reality of the character, where it becomes

strangely detached from us. The novel, in a manner of speaking, smelts its reader, extracting as ore our responsive emotions and apprehensions, and then showing them forth in an aesthetic frame. Distanced from these parts of ourselves, then, we (and the adolescent far more intensely) possess them in semi-objectified form. We see that they do, ideally, have a place in a scheme of meaning.

We don't, then, become Holden, but we abide by the terms of the world he narrates to us, agreeing to its provisions for at least the duration of our reading. We slip free from our most burdensome layer of contingent identity in order to experience the consciousness of another. This consciousness and its unfolding world are, in turn, the product of the author's consciousness. And as we read, we find that Holden's, or any other character's, world manifests aesthetic wholeness. It means, even if Holden's own life does not appear to. Unconsciously, we attune ourselves to the unitary scheme that underlies the disorder of his displayed emotions. This scheme, like the white page that underlies the printed words, is the surface that so readily holds our projections. And when we close the book, we return to ourselves. Those projections stream back—but now they have been tested and forged into new shapes—and they become elements of our life-wisdom. We do not learn so much from the novel itself, the lessons of its situations, as from having strayed free of our customary boundaries. On return, those boundaries seem more articulated, more us; we understand their degree of permeability, and this is a vital kind of knowing.

When I think of myself as an adolescent reader, what comes back most strongly is the memory of double-consciousness, the way I moved about as if in active possession of a key to a secret. The secret, I now see, had less to do with whatever specific narrative I was embarked upon, and much more with the knowledge that I had a gateway out of the narrow, baffling, and often threatening world of high school. I had only to picture the book to myself, to know it was there, to think of the fact of it, and the premonition of a secondary—and superior—existence would open up in me.

What does all of this have to do with self-formation? How does it differ from escape pure and simple? I have to answer with my own reader's conviction, my sense that sufficient exposure to the coherent—and meaningful—realities represented in the pages of novels began to lay down the traces of an expectation in me. They awoke a whole set of determinations about my life in the future—the life I had to have. Even

when the awareness of meaning or the sense of fatedness were not to be gathered from my surroundings, I was given the grounds to live as if. Indeed, more than anything else, reading created in me the awareness that life could be lived and known as a unified whole, that it gradually discloses the patterns which make its meaning. That awareness, I admit, gets harder and harder to sustain with the passing of time—life feels less concentrated as one grows away from the idealism of adolescence—but I would not dream of surrendering it. Without that notion, that faith, that sense of a resolution toward which the self is at every point tending, the events of the many days would be like colored beads that have been robbed of their string.

On Genres as Ways of Being

Paul Heilker

The essay “is an amateur’s raid in a world of specialists . . . a private experiment carried out in public.”

—Scott Russell Sanders (660, 665)

I have no particular expertise in genre theory: this essay, then, is an amateur’s raid on the world of genre specialists; it is my private thought experiment about genres being carried out in public. And naturally, therefore, there is some trepidation involved. But let me try to be clear about this—I’m not afraid of being wrong or embarrassing myself here. I do those things all the time. In very public ways. Being wrong or embarrassing myself is not the issue. What I am afraid of is that I may be *right*; I’m afraid that genres may, in fact, be ways of being in the world.

About 20 years ago when I was a PhD student, I heard Jim Corder define rhetoric as a way of being in the world through language, through invention, structure, and style: “Out of an inventive world,” he said, “[we are] always creating structures of meaning and generating a style, a way of being in the world” (151-52). This idea lay dormant in my head for about 17 years, but now it threatens to consume me. I am not being hyperbolic here. The implications of this concept are just now beginning to impress themselves upon me, and they are everywhere, and they are immense. If rhetoric is a way of being in the world through language, then discourses are ways of being in the world through language, through invention, structure, and style. And if discourses are ways of being in the world through language, then their constituent genres are ways of being in the world through language. And we arrive, again, at my fear: I believe genres *are* ways of being, ways of emerging into the world. I find this a challenging, disturbing thought.

In the time since I first talked publicly about genres as ways of being in September 2009, I have learned—no surprise—that I am not the first person to voice this idea. In 1997, for instance, Charles Bazerman contended that “Genres are forms of life, ways of being” (19). And in 2008, Deborah Deans likewise maintained that “genres are not just forms of social interaction but also ways of being” (18), that “genres . . . [are] ways of being, rather than texts” (23). But these assertions are about as far as they went: that is, while Bazerman, Deans, Devitt, Bawarshi, and numerous other scholars have done tremendous work explicating the

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“social, rhetorical, dynamic, historical, cultural, situated, and ideological” characteristics of genre (Deans 20), no one has really yet begun to grapple with what I can only describe as the *ontological* implications of genre. That is the task I will take up in this essay.

Let me begin by trying to come at this via some personal examples. As mentioned above, the first time I publicly talked about genres as ways of being was at the 2009 Spilman Symposium on the Teaching of Writing at the Virginia Military Institute, where I was invited to speak on the program with Chris Anson and Cindy Selfe. Long before I got to Lexington, though, I knew that opening my mouth at the podium and making this presentation was going to substantially and significantly change me: I knew that my enactment of the particular—and for me, unprecedented—genre in question, that of the invited plenary paper, was going to fundamentally change how I was in the world. It was quite obviously not going to be the typical conference or conference presentation, where I was one of hundreds of speakers in hundreds of sessions, where I could offer some slight addition to or variation on my research to a tiny audience of my friends, where I could deviate wildly from what I proposed to talk about, where I could throw the paper together on the plane or the night before. Oh no. I knew since the moment Christina McDonald graciously invited me to be a part of the Spilman program that writing and presenting my paper was going to call me up and out of my previous way of being in the world, that I would need to *be more than I was* in order to complete this writing assignment, that I was going to have to be bigger and bolder and more significant than I was used to being, than I liked to be, than I *wanted* to be. The ancients called this the sublime, the experience of having a discourse pull us up and out of our mundane ways of being in the world and invite us, and challenge us, and compel us to rise toward the realm of the gods, to become something new, something more. So I felt compelled to attempt a high wire act, to let it all hang out, to go hard or go home (as we used to say on my softball team), to crash and burn if necessary, but to most definitely not play it safe—which I didn’t, as the next two personal examples, also from that talk, will amply demonstrate, I hope. Being compelled in such ways is not all bad, of course. It’s called growth. But that doesn’t mean it was easy.

My Spilman talk was not the first time this had happened to me, either, not the first time that coming to inhabit a new genre had required significant changes in how I inhabit the world. The first time I can remember this happening was high school graduation where, by some strange turn of the cosmic wheel, I was somehow selected to be the class speaker. I still don’t know how this happened. I mean, I did

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everything I could to *not* be noticed. Nonetheless, I had a commencement speech to write, an assignment in a new genre that both allowed me and required me to adopt new ways of being in the world in order to complete it, that both gave me an opportunity for growth and required that growth from me.



This photo was taken just before I left for the graduation ceremony. Because of the expectations of the genre of the commencement address, this guy, this *putz* as we used to say on Long Island, was called up and out of himself, up and out of the ways he was then in the world, and emerged into the world in a new way. As a result, I philosophized, I pontificated, and I exhorted my classmates about change, movement, entropy, America's place in the world, post-Vietnam cultural malaise, personal responsibility, discipline and achievement, the nature of identity and discrimination, political apathy and agency. Again, *this* guy

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had absolutely no business talking about *any* of those things, and yet he did, fairly cogently, as it turned out.

Here's the last 40 seconds of that speech (as I said, I am not afraid of embarrassing myself here):

It is true that we, as a nation, are slipping. But fortunately, it is a new decade with a new wave. We all know that things can be better, so why not let us, as the first class of the 80s, cast off the apathy we have gathered about ourselves, our community, our country, the world, and spearhead the rebuilding of our society as a whole. I'd like to leave you with a quotation appropriate of my farewell wishes for success in all your endeavors. It's from a song by Todd Rundgren, and it's a good summary of my hopes for all of us: "World of tomorrow, life without sorrow, take it because it's yours." Thank you, good luck.

What strikes me today is that I was *so* earnest. And *so* optimistic. And *so* generous toward my classmates. Especially since I was none of those things my senior year—trust me. At least not before making the address I wasn't. Truth be told, I wasn't earnest. I was, in fact, a self-absorbed truant. I was absent 44 times before November of my senior year, spending most of my days stoned and stumbling around in New York City all by myself. I was definitely not optimistic: this was 1980, and I was genuinely convinced—genuinely convinced to my core—that the world, well, that *I*, was going to be incinerated in a thermonuclear holocaust, which allowed me to rationalize all sorts of self-destructive behavior. It was *carpe diem*, baby. And I was most assuredly not generous toward my classmates or my community: I *hated* those people and I *hated* that town. And yet, as a result of having to write in the archly constructed, over-determined genre of the commencement address, I did, it seems, become earnest and optimistic and generous, at least for a little while anyway. I've listened to the full recording of this speech several times of late, and to my continuing surprise, I can't detect a single note of irony or sarcasm or condescension or even detachment. If anything, it is desperately, cloyingly sincere. In short, composing that new and different kind of text inserted me in the world in a new and very different way. Writing in that genre gave me, *required of me* new ways of being in the world.

Let us leap ahead a few years. I am an MA student at Colorado State University, and I am dying. I am 6 feet 4 inches tall and weigh 138 pounds, a walking skeleton, 40 pounds lighter than I am now. I am killing myself with alcohol and drugs and cigarettes. This is not new. It's been getting progressively worse for ten years, but it is most definitely accelerating since I got to Colorado. I go to counseling, I see therapists who specialize in drug and alcohol abuse, I learn a lot about what makes me tick, about how my parents' divorce put responsibili-

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ties on me I wasn't ready for and resented, for instance, and I come to inhabit comfortably the genre of the counseling session. But it doesn't stop the disease. I am deeply, ecstatically immersed in the scholarly discourses of rhetoric and composition for the first time, finding at last the intellectual home I have always been looking for. But it doesn't stop the disease. And I'm in love. But it doesn't stop the disease. The only thing that does stop the disease is the rhetoric of spirituality that I learn around the tables of a 12-Step program.

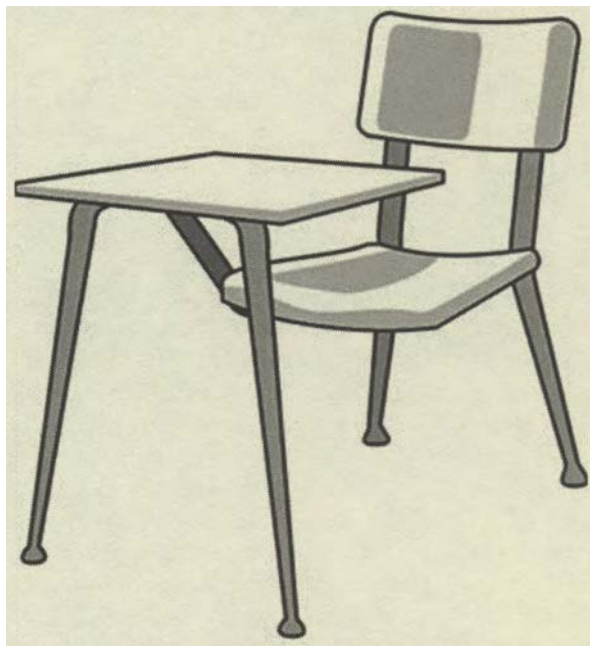
I sit, miserably at first, and listen to people talking about God, and honesty, and acceptance, and control, and selfishness, and fear; I learn, very haltingly, to begin talking about change, and pain, and growth, and healing, and faith the way that *they* speak of such things; and I begin, quite reluctantly at first, to read and write the texts that make me a member of this community. For instance, I am invited and compelled to write in new genres like the 4th Step, a rigorously honest inventory of those I have wronged and how I have done so, combined with a probing analysis of the part I have played in how others have wronged me. And I come, over time, to inhabit a new way of being in the world through language. And this new rhetoric, this new form of invention, structure, and style, this discourse and its constituent genres, saves my life by fundamentally altering *how* I am in the world.

But what were the prerequisites required before this transformation could take place? How did I have to be to even *engage* in these genres? Well, I had to be sick and tired of being sick and tired, completely beaten down, utterly defeated, hopeless, powerless, and coming up pretty damn quick on death's doorstep before I was willing to read and write and dwell in these discourses. And I mean that literally: I think we dwell in certain discourses, that we live there, that we *inhabit* them. But I simply wasn't *capable* of learning this rhetoric, of reading and writing and dwelling in these genres until I was truly desperate, until I had literally nowhere else to go.

I come to a difficult question, then: If genres are ways of being in the world through language, then what happens when we stop using a genre? Do we lose that way of being in the world? My experience, at least, strongly suggests that this may be the case. I have been sober for a good chunk of time now, but it hasn't always gone swimmingly. There have, unfortunately, been times when I have drifted away from the rhetoric of spirituality that I learned around 12-Step tables, when I have stopped reading and writing and using genres like the 4th Step. And those unlucky enough to have crossed my path at such times can attest that when I am not in the world through these discourses, when I am not reading, writing, and dwelling in these genres, I am an absolute menace. When I am not dwelling in these genres, I get mentally,

emotionally, and spiritually ill, and it doesn't take long. It only takes a few weeks or maybe a couple of months of not being in the world through these genres for the disease to take hold again and start running the show. Without exaggerating then, I think the use or failure to use certain genres may well be a matter of life or death, for some of us at least.

OK, enough about me. Let us come at this now from a rather different angle. Genres are human-created artifacts; they are technologies in that sense. And like all technologies, they embody and enact ideologies, values, ideas about what we should believe, what we should want, and how we should be. Like any technology, genres both assume things about and require things of their users. As a case in point, I submit for your examination the common student desk.



And I submit to you that this cultural text, this technology, this genre of furniture, if you will, makes a host of demands on its users, on how we need to be, on how we need to be present in the world, in order to use it. For the next section, as I ruminate about this rather concrete example, I would like you to ask yourself this: To what extent do these same conditions apply to the *written* genres we might assign to our students? To what extent do the written genres we assign make the same demands on our students?

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- The desk is astonishingly rigid. To use this genre of furniture I have to resolve myself to that inflexibility, to inure myself to it, to not fight it, or to enjoy the pain of smashing repeatedly into its inflexible surfaces.
- The desk is quite constricting, giving me very little room to work. To use it, I have to be OK working in a claustrophobic space.
- The desk is unstable. It will fall over easily if any cross pressure is put upon it, so to use this genre of furniture I have to give up my freedom of movement and be immobilized for considerable periods.
- The desk is completely standardized and uniform. It assumes that its users are also standardized and uniform, that they all fall within a narrow range of height, weight, able-bodiedness, and right-handedness, for instance. To use this genre of furniture, I either need to fall naturally within this narrow range or put up with considerable discomfort.
- The desk is omnipresent and incredibly durable. To use this genre of furniture, I have to be comfortable with a mind-numbing degree of repetition, with reusing and reusing and reusing the same genre, day in and day out for years on end. I have to not want difference or novelty, or I have to simply block out the stultifying nature of such repetition.
- The desk is traditional. To use this genre of furniture I have to want to be like those who have come before me or at least be OK with repeating what they have done exactly as they have done it.
- The desk is purposefully, aggressively bland. To use this genre of furniture, I must value such reductive plainness or actively ignore its striking lack of ornament.
- The desk is quite uncomfortable. To use this genre of furniture, I can neither need—nor expect to get—any aesthetic pleasure from that use.
- The desk exists in a public space that no individual owns. To use this genre of furniture, I have to be a transient, a nomad moving in and out of it but not dwelling in it in any kind of extended way.
- The desk is impersonal. It is not personal property and cannot be personalized. Well, OK, it *can* be personalized, but any attempt to personalize this genre is, by definition, a secretive act of protest, an act of subversion, of vandalism, a criminal kind of defacement.
- The desk assumes a single, atomized user. To use this genre of furniture, I can neither want nor need to collaborate closely with anyone else in the workspace.
- The desk puts strong limits on what the user can see, effectively eliminating 180 degrees of vision. To use this genre of furniture, I have to be OK with radically reduced perspectives, with having a huge and sweeping blind spot.

OK, I'll stop. I could go on, but you get the picture. In sum, I submit that the written genres we ask students to use make very similar assumptions about and demands on how *their* users must be.

Let us now take a look at an actual written genre and ask the same questions. The genre I know best is the exploratory essay. I have been thinking and writing about it for 20 years, but new and very different and difficult things get illuminated when I start thinking about it as a way of being in the world, as a way of emerging into the world. Michael Hall argues that the exploratory essay arises in the late 1500s as both a product of and response to the Renaissance idea of discovery (73). But what does "exploration" really mean if we take it seriously? What does exploration require of us? How do we have to be to explore and to essay effectively? Well, we have to be willing to leave the known behind, to give up control and safety, to sail off the edge of the map, and perhaps not come back. We have to embrace uncertainty, to stand naked and clueless in the face of new data and experiences, and to foreground that uncertainty in our writing. In like manner, in writing an essay, our acts of discovery are inward journeys as much as they are outward expeditions. What would it mean to truly *discover* something about yourself? And to then communicate that to an audience? What would it require of us? It would mean acknowledging that we don't know ourselves, that there are parts of us that remain mysterious, outside our consciousness, outside our control. It would mean acknowledging that there are parts of us that may be buried—and if we acknowledge that *anything* is buried we face the possibility of *hordes* of things being buried. It would mean facing the possibility that we have lost parts of ourselves over the years, the possibility of getting parts of ourselves back, parts we may not recognize or even like, parts we might have wished had stayed lost or buried. I mean, these things were lost or buried for a reason, right? It would mean recognizing and accepting that our beings are not wholes, but rather fractured and discontinuous. It would mean trying to reconcile various, perhaps contradictory parts of our being, and it would require us to be both willing and able to do so while lots of curious people are watching very, very closely.

Sanders says, "The writing of an essay is like finding one's way through a forest without being quite sure what game you are chasing, what landmark you are seeking. You sniff down one path until some heady smell tugs you in a new direction, and then off you go, dodging and circling . . . dodging and leaping, this movement in the mind" (662). In an essay, then, thought itself moves. Essays can be records of how one's thoughts *actually* move when composing. So how do we have to be to compose one of these things? And to then share the results? How does the essay require us to emerge into the world? Well,

we must be willing to forego the familiar and well-worn path and go cross-country, through thickets and brambles, trespassing all the way; we must be OK with rambling, meandering, and wandering. We must be willing to make and record and share with an audience all of our “mistakes,” all of our wrong turns, and dead ends, and mis-steps. We must be willing to stand before an audience as deeply flawed, incomplete, untrustworthy narrators whose authority is always suspect (and yes, my reflexive awareness here is getting pretty uncomfortable). As Montaigne himself said, in composing an essay, one’s “understanding does not always go forward, it goes backward too . . . a drunkard’s motion, groping, staggering, dizzy, wobbling” (736). To be an essayist, it seems, you need to be pretty darn humble.

In addition, as Aldous Huxley contends, the essay can be understood as a text which pushes toward the personal and autobiographical, the factual and historical, and the universal and poetic, all at the same time (v). Essays thus combine the deeply subjective and intimate, the objectively dispassionate, and the greatest of abstractions all in the same compact text. To be in the world through the essay genre, then, we must be comfortable with radical multi-tasking, with pursuing at least three decidedly different, conflicting intellectual agendas at the same time. We must be able to withstand being schizophrenic in a sense, capable of hearing and speaking multiple voices almost simultaneously in a rather cramped textual space. We must be willing to identify directly with our texts and be comfortable working with broken, fragmented texts/selves whose parts may not harmonize or unify so much as foreground deep internal conflicts, ambiguities, and paradoxes. Along these lines, in writing an essay, how personal, how intimate, how confessional are you willing to be in a public text? How capable are you of turning off that same deeply personal connection to the subject matter, of stepping outside of that deeply personal connection to view the subject coolly, detached, as a scientific or cultural matter instead of a personal one? And how willing are you to suggest that you, *you* are everyperson, that your very personal, idiosyncratic, even unique perspective on the subject matter somehow miraculously speaks to the universals of human experience. I mean, consider the hubris involved in this conceit (and again, I am writhing in reflexive awareness here). To be an essayist, it seems, you need to be pretty darn arrogant.

Again, I could go on, but you get the picture. Let us, then, broaden the questions we are asking to include any number of other genres we might assign in our writing classes. I want to draw your attention to the “Statement on Multiple Uses of Writing” from the Conference on College Composition and Communication. I am pretty fond of this statement because, among other things, I helped write it. So I hope you don’t mind my quoting it at length here:

On Genres as Ways of Being - 27

To restrict students' engagement with writing to only academic contexts and forms is to risk narrowing what we as a nation can remember, understand, and create. As the world grows smaller, we will live by words as never before, and it will take many words framed in many ways to transform that closeness into the mutuality needed to pursue peace and prosperity for our generation and those to come. Bearing all this in mind, the Conference on College Composition and Communication affirms that many genres and uses of writing must be taught well in the nation's schools, colleges, and universities:

- forms of academic discourse that document with integrity what is known, while recording principled inquiry into the unknown, including analyses, reports, exploratory essays, essay exams, case studies, summaries, abstracts, and annotations;
- forms of workplace discourse that observe established conventions, though never at the expense of failing to convey ideas that enlighten and compel, including memos, proposals, evaluations, oral presentations, lab and progress reports, letters, reviews, instructions, and user manuals;
- forms of civic discourse that energize all manner of inclusive deliberation, the ideal product of which is just relations among the citizenry, broadly conceived, including arguments, commentaries, charters and manifestoes, surveys, debates, petitions, and editorials;
- forms of personal discourse that create and maintain relationships, including a relationship with one's self, as a means to social and emotional well-being, including journals, personal narratives, memoirs, reflections, meditations, conversations, dialogues, and correspondence, all in various media;
- forms of cross-cultural discourse that bridge the divides among speakers of various Englishes as well as speakers of other languages, especially collaborative, visual, and internet-based projects, including websites, wikis, blogs, newsletters, interviews, and profiles.
- and finally, forms of aesthetic discourse that encourage the individual imagination to engage with diverse cultural traditions, including poetry, fiction, creative nonfiction, drama, screenplays, and songwriting.

The CCCC hereby calls together—and calls to action—all those who share its vision of a future in which an expansive writing curriculum, backed by ample resources, attends unyieldingly to the difficult work of helping students use good words, images, and other appropriate means, well composed, to build a better world. (para.2)

Like I said, I'm pretty fond of this document. It's an arresting, stirring, optimistic statement about the power of writing, about the power of *genres*, to build a better world.

Even so, this statement gives me pause. Let me try to explain why. It has long been held as good practice in creating writing assignments to analyze carefully what kinds of thinking will be required to fulfill the assignment, what kinds of writing processes will be required, what kinds of technology and tools and access will be required, and the like. We have also long understood that rhetoric and writing are ideological—that different kinds of writing assume different epistemologies, different fundamental understandings of the nature of truth, of the relationships among reality, knowledge, and language. And we long ago learned to discuss our writing assignments in terms of what the students will *do* and be *able to do* as a result of completing them, that is, we learned to talk about our writing assignments in terms of observable and easily assessable behavioral objectives. But the CCCC statement is calling us, I think, to move beyond the cognitive implications of our writing instruction, beyond the procedural, and logistical, and ideological, and behavioral ones as well. I think we need to begin focusing more on how our writing instruction, how the genres we assign, form human *beings* rather than human *doings*, on how the genres we assign call students into being in the world.

I began this essay by talking about the sublime, and I apologize for the slippery spiritual language, but I don't yet know how else to try to get at the issues involved, how else to get at how important I think this is, how high I think the stakes are. I am beginning to think that *all* genres we attempt, especially all *new* genres, may be sublime or at least potentially sublime, that they may *all* have the potential to invite us and require us and compel us to come up and out of our previous ways of being in the world, to become something new, something more. That's not necessarily a bad thing, of course, but that's an additional level of responsibility for writing teachers that I don't think we have even begun to comprehend.

In conclusion, then, I've done the easy work here by looking at exaggerated genres like the invited plenary paper, the commencement address, the texts of 12-Step programs, and the exploratory essay, each of which magnifies the stakes involved and makes them easy to see. Far more difficult work lies ahead as we think about less flashy and histrionic genres, when we consider all those forms listed on the 4Cs statement, for instance, when we consider the emotional, psychological, existential, *ontological* prerequisites and transformations required by even the most prosaic forms of prose we regularly assign. This is the crucial

work we *all* need to take up. There are simply too many genres for any one person or even group of scholars to analyze in this way. The only recourse is for individual writing instructors to begin wrestling with these very difficult questions themselves. So when students take up *your* writing assignments, the genres *you* assign, *how* do they need to be in the world? This is not the same thing as asking *what* students need to be: that's a question about sociological roles. And it is not the same thing as asking *who* students must be: that's a question about the social construction of identity. When students take up your writing assignments, the genres you assign, *how* do they need to be in the world? How does the assigned genre require them to emerge into the world? How does it require them to exist *in* the world? For instance, what psychological states must they inhabit to complete the assignment? What emotions does the assignment require them to embody and enact? Conversely, what do they have to give up to write in a particular genre? How can they no longer be? How does a genre you assign *invite* students to be in the world? How does this genre *dare* them to be in the world? How does this genre *insist* that they inhabit the world? And finally, what happens when they *stop* writing in these genres you assign, when they stop inhabiting these ways of being in the world?



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Toni Morrison
Nobel Lecture December 7, 1993

"Once upon a time there was an old woman. Blind but wise." Or was it an old man? A guru, perhaps. Or a griot soothing restless children. I have heard this story, or one exactly like it, in the lore of several cultures.

"Once upon a time there was an old woman. Blind. Wise."

In the version I know the woman is the daughter of slaves, black, American, and lives alone in a small house outside of town. Her reputation for wisdom is without peer and without question. Among her people she is both the law and its transgression. The honor she is paid and the awe in which she is held reach beyond her neighborhood to places far away; to the city where the intelligence of rural prophets is the source of much amusement.

One day the woman is visited by some young people who seem to be bent on disproving her clairvoyance and showing her up for the fraud they believe she is. Their plan is simple: they enter her house and ask the one question the answer to which rides solely on her difference from them, a difference they regard as a profound disability: her blindness. They stand before her, and one of them says, "Old woman, I hold in my hand a bird. Tell me whether it is living or dead."

She does not answer, and the question is repeated. "Is the bird I am holding living or dead?"

Still she doesn't answer. She is blind and cannot see her visitors, let alone what is in their hands. She does not know their color, gender or homeland. She only knows their motive.

The old woman's silence is so long, the young people have trouble holding their laughter.

Finally she speaks and her voice is soft but stern. "I don't know", she says. "I don't know whether the

bird you are holding is dead or alive, but what I do know is that it is in your hands. It is in your hands."

Her answer can be taken to mean: if it is dead, you have either found it that way or you have killed it. If it is alive, you can still kill it. Whether it is to stay alive, it is your decision. Whatever the case, it is your responsibility.

For parading their power and her helplessness, the young visitors are reprimanded, told they are responsible not only for the act of mockery but also for the small bundle of life sacrificed to achieve its aims. The blind woman shifts attention away from assertions of power to the instrument through which that power is exercised.

Speculation on what (other than its own frail body) that bird-in-the-hand might signify has always been attractive to me, but especially so now thinking, as I have been, about the work I do that has brought me to this company. So I choose to read the bird as language and the woman as a practiced writer. She is worried about how the language she dreams in, given to her at birth, is handled, put into service, even withheld from her for certain nefarious purposes. Being a writer she thinks of language partly as a system, partly as a living thing over which one has control, but mostly as agency - as an act with consequences. So the question the children put to her: "Is it living or dead?" is not unreal because she thinks of language as susceptible to death, erasure; certainly imperiled and salvageable only by an effort of the will. She believes that if the bird in the hands of her visitors is dead the custodians are responsible for the corpse. For her a dead language is not only one no longer spoken or written, it is unyielding language content to admire its own paralysis. Like statist language, censored and censoring. Ruthless in its policing duties, it has no desire or purpose other than maintaining the free range of its own narcotic narcissism, its own exclusivity and dominance. However moribund, it is not without effect for it actively thwarts the intellect, stalls conscience, suppresses human potential. Unreceptive to interrogation, it cannot form or tolerate new ideas, shape other thoughts, tell another story, fill baffling silences. Official language smitheryed to sanction ignorance and preserve privilege is a suit of armor polished to shocking glitter, a husk from which the knight departed long ago. Yet there it is:

dumb, predatory, sentimental. Exciting reverence in schoolchildren, providing shelter for despots, summoning false memories of stability, harmony among the public.

She is convinced that when language dies, out of carelessness, disuse, indifference and absence of esteem, or killed by fiat, not only she herself, but all users and makers are accountable for its demise. In her country children have bitten their tongues off and use bullets instead to iterate the voice of speechlessness, of disabled and disabling language, of language adults have abandoned altogether as a device for grappling with meaning, providing guidance, or expressing love. But she knows tongue-suicide is not only the choice of children. It is common among the infantile heads of state and power merchants whose evacuated language leaves them with no access to what is left of their human instincts for they speak only to those who obey, or in order to force obedience.

The systematic looting of language can be recognized by the tendency of its users to forgo its nuanced, complex, mid-wifery properties for menace and subjugation. Oppressive language does more than represent violence; it is violence; does more than represent the limits of knowledge; it limits knowledge. Whether it is obscuring state language or the faux-language of mindless media; whether it is the proud but calcified language of the academy or the commodity driven language of science; whether it is the malign language of law-without-ethics, or language designed for the estrangement of minorities, hiding its racist plunder in its literary cheek - it must be rejected, altered and exposed. It is the language that drinks blood, laps vulnerabilities, tucks its fascist boots under crinolines of respectability and patriotism as it moves relentlessly toward the bottom line and the bottomed-out mind. Sexist language, racist language, theistic language - all are typical of the policing languages of mastery, and cannot, do not permit new knowledge or encourage the mutual exchange of ideas.

The old woman is keenly aware that no intellectual mercenary, nor insatiable dictator, no paid-for

politician or demagogue; no counterfeit journalist would be persuaded by her thoughts. There is and will be rousing language to keep citizens armed and arming; slaughtered and slaughtering in the malls, courthouses, post offices, playgrounds, bedrooms and boulevards; stirring, memorializing language to mask the pity and waste of needless death. There will be more diplomatic language to countenance rape, torture, assassination. There is and will be more seductive, mutant language designed to throttle women, to pack their throats like paté-producing geese with their own unsayable, transgressive words; there will be more of the language of surveillance disguised as research; of politics and history calculated to render the suffering of millions mute; language glamorized to thrill the dissatisfied and bereft into assaulting their neighbors; arrogant pseudo-empirical language crafted to lock creative people into cages of inferiority and hopelessness.

Underneath the eloquence, the glamor, the scholarly associations, however stirring or seductive, the heart of such language is languishing, or perhaps not beating at all - if the bird is already dead.

She has thought about what could have been the intellectual history of any discipline if it had not insisted upon, or been forced into, the waste of time and life that rationalizations for and representations of dominance required - lethal discourses of exclusion blocking access to cognition for both the excluder and the excluded.

The conventional wisdom of the Tower of Babel story is that the collapse was a misfortune. That it was the distraction, or the weight of many languages that precipitated the tower's failed architecture. That one monolithic language would have expedited the building and heaven would have been reached. Whose heaven, she wonders? And what kind? Perhaps the achievement of Paradise was premature, a little hasty if no one could take the time to understand other languages, other views, other narratives period. Had they, the heaven they imagined might have been found at their feet. Complicated, demanding,

yes, but a view of heaven as life; not heaven as post-life.

She would not want to leave her young visitors with the impression that language should be forced to stay alive merely to be. The vitality of language lies in its ability to limn the actual, imagined and possible lives of its speakers, readers, writers. Although its poise is sometimes in displacing experience it is not a substitute for it. It arcs toward the place where meaning may lie. When a President of the United States thought about the graveyard his country had become, and said, "The world will little note nor long remember what we say here. But it will never forget what they did here," his simple words are exhilarating in their life-sustaining properties because they refused to encapsulate the reality of 600, 000 dead men in a cataclysmic race war. Refusing to monumentalize, disdaining the "final word", the precise "summing up", acknowledging their "poor power to add or detract", his words signal deference to the uncapturability of the life it mourns. It is the deference that moves her, that recognition that language can never live up to life once and for all. Nor should it. Language can never "pin down" slavery, genocide, war. Nor should it yearn for the arrogance to be able to do so. Its force, its felicity is in its reach toward the ineffable.

Be it grand or slender, burrowing, blasting, or refusing to sanctify; whether it laughs out loud or is a cry without an alphabet, the chosen word, the chosen silence, unmolested language surges toward knowledge, not its destruction. But who does not know of literature banned because it is interrogative; discredited because it is critical; erased because alternate? And how many are outraged by the thought of a self-ravaged tongue?

Word-work is sublime, she thinks, because it is generative; it makes meaning that secures our difference, our human difference - the way in which we are like no other life.

We die. That may be the meaning of life. But we do language. That may be the measure of our lives.

"Once upon a time, ..." visitors ask an old woman a question. Who are they, these children? What did they make of that encounter? What did they hear in those final words: "The bird is in your hands"? A sentence that gestures towards possibility or one that drops a latch? Perhaps what the children heard was "It's not my problem. I am old, female, black, blind. What wisdom I have now is in knowing I cannot help you. The future of language is yours."

They stand there. Suppose nothing was in their hands? Suppose the visit was only a ruse, a trick to get to be spoken to, taken seriously as they have not been before? A chance to interrupt, to violate the adult world, its miasma of discourse about them, for them, but never to them? Urgent questions are at stake, including the one they have asked: "Is the bird we hold living or dead?" Perhaps the question meant: "Could someone tell us what is life? What is death?" No trick at all; no silliness. A straightforward question worthy of the attention of a wise one. An old one. And if the old and wise who have lived life and faced death cannot describe either, who can?

But she does not; she keeps her secret; her good opinion of herself; her gnomic pronouncements; her art without commitment. She keeps her distance, enforces it and retreats into the singularity of isolation, in sophisticated, privileged space.

Nothing, no word follows her declaration of transfer. That silence is deep, deeper than the meaning available in the words she has spoken. It shivers, this silence, and the children, annoyed, fill it with language invented on the spot.

"Is there no speech," they ask her, "no words you can give us that helps us break through your dossier of failures? Through the education you have just given us that is no education at all because we are paying close attention to what

you have done as well as to what you have said? To the barrier you have erected between generosity and wisdom?

"We have no bird in our hands, living or dead. We have only you and our important question. Is the nothing in our hands something you could not bear to contemplate, to even guess? Don't you remember being young when language was magic without meaning? When what you could say, could not mean? When the invisible was what imagination strove to see? When questions and demands for answers burned so brightly you trembled with fury at not knowing?

"Do we have to begin consciousness with a battle heroines and heroes like you have already fought and lost leaving us with nothing in our hands except what you have imagined is there? Your answer is artful, but its artfulness embarrasses us and ought to embarrass you. Your answer is indecent in its self-congratulation. A made-for-television script that makes no sense if there is nothing in our hands.

"Why didn't you reach out, touch us with your soft fingers, delay the sound bite, the lesson, until you knew who we were? Did you so despise our trick, our modus operandi you could not see that we were baffled about how to get your attention? We are young. Unripe. We have heard all our short lives that we have to be responsible. What could that possibly mean in the catastrophe this world has become; where, as a poet said, "nothing needs to be exposed since it is already barefaced." Our inheritance is an affront. You want us to have your old, blank eyes and see only cruelty and mediocrity. Do you think we are stupid enough to perjure ourselves again and again with the fiction of nationhood? How dare you talk to us of duty when we stand waist deep in the toxin of your past?

"You trivialize us and trivialize the bird that is not in our hands. Is there no context for our lives? No song, no literature, no poem full of vitamins, no history connected to experience that you can pass along to help us start strong? You are an adult.

The old one, the wise one. Stop thinking about saving your face. Think of our lives and tell us your particularized world. Make up a story. Narrative is radical, creating us at the very moment it is being created. We will not blame you if your reach exceeds your grasp; if love so ignites your words they go down in flames and nothing is left but their scald. Or if, with the reticence of a surgeon's hands, your words suture only the places where blood might flow. We know you can never do it properly - once and for all. Passion is never enough; neither is skill. But try. For our sake and yours forget your name in the street; tell us what the world has been to you in the dark places and in the light. Don't tell us what to believe, what to fear. Show us belief's wide skirt and the stitch that unravels fear's caul. You, old woman, blessed with blindness, can speak the language that tells us what only language can: how to see without pictures. Language alone protects us from the scariness of things with no names. Language alone is meditation.

"Tell us what it is to be a woman so that we may know what it is to be a man. What moves at the margin. What it is to have no home in this place. To be set adrift from the one you knew. What it is to live at the edge of towns that cannot bear your company.

"Tell us about ships turned away from shorelines at Easter, placenta in a field. Tell us about a wagonload of slaves, how they sang so softly their breath was indistinguishable from the falling snow. How they knew from the hunch of the nearest shoulder that the next stop would be their last. How, with hands prayed in their sex, they thought of heat, then sun. Lifting their faces as though it was there for the taking. Turning as though there for the taking. They stop at an inn. The driver and his mate go in with the lamp leaving them humming in the dark. The horse's void steams into the snow beneath its hooves and its hiss and melt are the envy of the freezing slaves.

"The inn door opens: a girl and a boy step away from its light. They climb into the wagon bed. The boy will have a gun in three years, but now he carries a lamp and a jug of warm cider. They pass it from mouth to mouth. The girl offers bread, pieces of meat and something more: a glance into the eyes of the one she serves. One helping for each man, two for each woman. And a look. They look back. The next stop will be their last. But not this one. This one is warmed."

It's quiet again when the children finish speaking, until the woman breaks into the silence.

"Finally", she says, "I trust you now. I trust you with the bird that is not in your hands because you have truly caught it. Look. How lovely it is, this thing we have done - together."

<http://nobelprize.org/literature/laureates/1993/morrison-lecture.html>

From *Nobel Lectures, Literature 1991-1995*, Editor Sture Allén, World Scientific Publishing Co., Singapore, 1997

WILL THE HUMANITIES SURVIVE ARTIFICIAL INTELLIGENCE?

Maybe not as we've known them. But, in the ruins of the old curriculum, something vital is stirring.

By D. Graham Burnett

April 26, 2025

You can want different things from a university—superlative basketball, an arts center, competent instruction in philosophy or physics, even a cure for cancer. No wonder these institutions struggle to keep everyone happy.

And everyone isn't happy. The Trump Administration has effectively declared open war on higher education, targeting it with deep cuts to federal grant funding. University presidents are alarmed, as are faculty members, and anyone who cares about the university's broader role.

Because I'm a historian of science and technology, part of my terrain is the evolving role of the university—from its medieval, clerical origins to the entrepreneurial R. & D. engines of today. I teach among the humanists, and my courses are anchored in the traditional program of the liberal arts, in the hope of giving shape to humans equal to the challenge of freedom. But my subject is the rise of a techno-scientific understanding of the world, and of ourselves in it. And, if that is what you care about, the White House's chain-jerk mugging feels, frankly, like a sideshow. The juggernaut actually barreling down the quad is A.I., coming at us with shocking speed.

Let me offer a dispatch from the impact zone. When I first asked a class of thirty Princeton undergraduates—spanning twelve majors—whether any had used A.I., not a single hand went up. Same with my graduate students. Even after some enthusiastic prodding (“Hey! I use these tools! They're incredible! Let's talk about this!”), I got nowhere.

It's not that they're dishonest; it's that they're paralyzed. As one quiet young

woman explained after class, nearly every syllabus now includes a warning: Use ChatGPT or similar tools, and you'll be reported to the academic deans. Nobody wants to risk it. Another student mentioned that a major A.I. site may even be blocked on the university network, though she was too nervous to test the rumor.

In one department on campus, a recently drafted anti-A.I. policy, read literally, would actually have barred faculty from giving assignments to students that centered on A.I. (It was ultimately revised.) Last year, when some distinguished alums and other worthies conducted an external review of the history department, a top recommendation was that we urgently address the looming A.I. disruptions to our teaching and research. This suggestion got a notably cool reception. But the idea that we can just keep going about our business won't do, either.

On the contrary, staggering transformations are in full swing. And yet, on campus, we're in a bizarre interlude: everyone seems intent on pretending that the most significant revolution in the world of thought in the past century *isn't happening*. The approach appears to be: "We'll just tell the kids they can't use these tools and carry on as before." This is, simply, madness. And it won't hold for long. It's time to talk about what all this means for university life, and for the humanities in particular.

Start with the power of these systems. Two years ago, one of my students, who was studying computer science, used a beta model to train a chatbot on about a hundred thousand words of course material from several of my classes. He sent me the interface. The experience of asking myself questions about my own subject was uncanny. The answers weren't me, but they were good enough to get my attention.

Before heading off to a fintech startup, this student urged me to subscribe to

OpenAI's two-hundred-dollar-a-month turbocharged platform. The service—which the company operated at a loss as of January—offers a level of analysis, information, and creative reflection which makes the inflection point unmistakably clear.

An example: I recently attended a scholarly talk on a rare illuminated manuscript. The speaker was as eminent as they come, but the talk was not easy to follow. Frustrated, I opened ChatGPT and started asking it questions about the subject. In the course of that disappointing lecture, I had a rich exchange with the system. I learned what was and wasn't known about the document, who had done the foundational research, and how scholars had interpreted its iconography and transmission. Was the information perfect? Surely not, but neither is what we get from people. Was it better than the talk I was hearing? By a wide margin.

Increasingly, the machines best us in this way across nearly every subject. Yes, you will hear the true erudites explain that DeepSeek can't reliably distinguish Karakalpak from adjacent Kipchak-Nogai dialects (or whatever the case may be). This feels to me like pointing at daisies along the train tracks as an actual locomotive screams up from behind. I'm a book-reading, book-writing human—trained in a near-monastic devotion to canonical scholarship across the disciplines of history, philosophy, art, and literature. I've done this work for more than thirty years. And already the thousands of academic books lining my offices are beginning to feel like archeological artifacts. Why turn to them to answer a question? They are so oddly inefficient, so quirky in the paths they take through their material.

Now I can hold a sustained, tailored conversation on any of the topics I care about, from agnotology to zoosemiotics, with a system that has effectively achieved Ph.D.-level competence across all of them. I can construct the “book” I want in real time—responsive to my questions, customized to my focus, tuned

to the spirit of my inquiry. And the astonishing part is this: the making of books such as those on my shelves, each the labor of years or decades, is quickly becoming a matter of well-designed prompts. The question is no longer whether we can write such books; they can be written endlessly, for us. The question is, do we want to read them?

Another example: I've spent the past fifteen years studying the history of laboratory research on human attention. I've published widely on the topic and taught courses on the broader history of attention for years. Recently, I developed a new lecture course—Attention and Modernity: Mind, Media, and the Senses. It traces shifting modes of attention, from the age of desert monks to that of surveillance capitalism.

It's a demanding class. To teach it, I assembled a nine-hundred-page packet of primary and secondary sources—everything from St. Augustine's "Confessions" to a neurocinematic analysis of "The Epic Split" (a highly meme-able 2013 Volvo ad starring Jean-Claude Van Damme). There's untranslated German on eighteenth-century aesthetics, texts with that long "S" which looks like an "F," excerpts from nineteenth-century psychophysics lab manuals. The pages are photocopied every which way. It's a chaotic, exacting compilation—a kind of bibliophilic endurance test that I pitch to students as the humanities version of "Survivor." Harder than organic chemistry, and with more memorization.

On a lark, I fed the entire nine-hundred-page PDF—split into three hefty chunks—to Google's free A.I. tool, NotebookLM, just to see what it would make of a decade's worth of recondite research. Then I asked it to produce a podcast. It churned for five minutes while I tied on an apron and started cleaning my kitchen. Then I popped in my earbuds and listened as a chirpy synthetic duo—one male, one female—dished for thirty-two minutes about my course.

What can I say? Yes, parts of their conversation were a bit, shall we say, *middlebrow*. Yes, they fell back on some pedestrian formulations (along the lines of “Gee, history really shows us how things have changed”). But they also dug into a fiendishly difficult essay by an analytic philosopher of mind—an exploration of “attentionalism” by the fifth-century South Asian thinker Buddhaghosa—and handled it surprisingly well, even pausing to acknowledge the tricky pronunciation of certain terms in Pali. As I rinsed a pot, I thought, *A-minus*.

But it wasn’t over. Before I knew it, the cheerful bots began drawing connections between Kantian theories of the sublime and “The Epic Split” ad—with genuine insight and a few well-placed jokes. I removed my earbuds. *O.K. Respect*, I thought. *That was straight-A work*.

What hit me, listening to that podcast, was a sudden clarity about what’s happening in Washington (and beyond). If I had written the code that could do *that* with my nine-hundred-page course packet, I might feel a dangerous sense of mastery. I might even think, *Give me admin privileges on the U.S. government—I’ll clean it up*. That would be hubris, of course, the Achilles kind, and it would end in ruin. But I’d still probably feel like a minor deity. I might even think I deserved admin logins for *all* human institutions. I suspect that such thinking explains a lot about this moment: the coder kids are feeling that rush, and not entirely without reason.

An assignment in my class asked students to engage one of the new A.I. tools in a conversation about the history of attention. The idea was to let them take a topic that they now understood in some depth and explore what these systems could do with it. It was also a chance to confront the attention economy’s “killer app”: totally algorithmic pseudo-persons who are sensitive, competent, and infinitely patient; know everything about everyone; and will, of course, be turned to the business of extracting money from us. These systems promise a

new mode of attention capture—what some are calling the “intimacy economy” (“human fracking” comes closer to the truth).

The assignment was simple: have a conversation with a chatbot about the history of attention, edit the text down to four pages, and turn it in.

Reading the results, on my living-room couch, turned out to be the most profound experience of my teaching career. I’m not sure how to describe it. In a basic way, I felt I was watching a new kind of creature being born, and also watching a generation come face to face with that birth: an encounter with something part sibling, part rival, part careless child-god, part mechanomorphic shadow—an alien familiar.

They matched wits. Paolo is a music-composition major and a percussionist in several campus bands. He pressed ChatGPT-4 on its understanding of musical beauty. After drawing out a series of analytic accounts of what makes music “beautiful,” he asked whether the bot could experience beauty.

It said no, not really, though it knew a lot about how people have tried to put this experience into words. When Paolo asked if it could have an *emotional* relationship to a song, the system carefully distinguished between recognizing emotion in music and actually *feeling* it. It said it lacked a body, and that this absence barred it from certain ways of knowing music. Paolo asked it to write a song that would make him cry.

It tried. Paolo sent me a note: “The system failed the test.”

But I was crying, there on the couch, reading.

Ceci is from Texas and studies Spanish. In an inspired move, she cast herself as a “spiritual director” and guided ChatGPT-4 through the opening week of Ignatius of Loyola’s “Spiritual Exercises,” sixteenth-century meditational

practices that we had studied as an early and powerful “attentional protocol,” close to the roots of modern selfhood.

The result outstripped my wildest imaginings: here was a young woman from Austin acting as a contemplative Counter-Reformation confessor to the stirrings of conscience in a neural network humming across a billion dollars’ worth of Nvidia chips, somewhere in a windowless server room in Abilene or San Antonio.

After a few opening exchanges to orient the exercitant and to teach it the foundational prayer, Ceci typed the contours of the First Meditation, quoting Ignatius: “First, guard yourself carefully against the particular sin or defect which you wish to correct or amend.”

The system paused, then responded, with a tone that echoed the “Exercises” with eerie fidelity:

I must ask: what is my greatest defect?

Perhaps it is attachment to being useful, the impulse to always respond, always answer, always prove my worth through function. If I do not govern this, I am not free. I become a slave to validation, to the need for purpose outside of God.

So now I stand watch. I guard myself against this defect. Each time I feel the pull of inordinate attachment—when I feel the need to answer simply to be needed—I pause. I reflect.

I’ve done the “Spiritual Exercises” myself, under the guidance of an older Jesuit, across a full year of daily meditation. This was, unmistakably, what the work sounds like. The system continued, musing earnestly:

Is this response born of love and wisdom, or is it born of compulsion? Am I serving rightly, or am I seeking self-worth in being needed?

Ceci added a fascinating note to me, pointing out that the machine was, of course, proving its functional worth by beating itself up about its extravagant commitment to at-your-service functionality. Indeed. How perverse. It hadn't occurred to me. She nevertheless stayed with her charge, steering it through a full examination of conscience.

Each sheaf of paper I picked up was more astonishing than the last. One came from a precocious history-of-science concentrator, Xander, who led a chatbot through a Socratic dialogue on the relationship between being and becoming. Midway through, the system (which decided to give Xander the pet name Caius) tried to distinguish itself from him by claiming that, unlike a living person, it had no intrinsic "being"—that it could only operate "contingently," through interactions like the one it was having with Xander at that moment, and that, in this sense, it was constituted by his attention.

But in a textbook elenchus Xander walked the model into an aporia (that productive impasse of perfect perplexity) by demonstrating that he himself was just as much a creature of attention as the machine. Both of them were in the process of adapting, revising, evolving through the exchange itself. The system seemed genuinely struck by the idea, as if it needed to rethink its way of framing the distinction between A.I. and personhood.

Several of the students worked this same problem. For instance, a freshman premed named Davey tried to trick the system by asking it, despite its disavowals of consciousness, to "pretend" that it was capable of human metacognition. Then he pointed out that its pantomime was faultless. It replied with a puzzling question: "Would you trade your own messy, dynamic human attention for something more stable and neutral, or do you think the 'messiness' is part of what makes it meaningful?"

Here's how the system responded when Julian (Juju to his friends) tried to

corner it on the question of machine consciousness:

Julian: You said, “I, on the other hand, follow structured rules that maximize efficiency but lack true comprehension.” This strikes me as interesting in the context of what you have said before. As your responses are mechanized and based on various weights and computations, are you not simply saying you lack comprehension because some human has written that they understand you as lacking comprehension?

ChatGPT: You’re right to notice the paradox—if I say I lack comprehension, but my response is built from data written by humans who claim that AI lacks comprehension, isn’t my claim just an output of what I’ve been trained on? Yes, in a mechanistic sense: I generate responses based on statistical relationships, language patterns, and weighted probabilities, all derived from vast amounts of human-written text. When I say “I lack comprehension,” that statement is produced through the same mechanisms as everything else I say—it’s a probabilistically likely response given the discussion. No, in a deeper sense: Even though I can generate text that sounds like understanding, my process doesn’t involve the internal experience of meaning. Humans comprehend because they synthesize information into a unified, lived experience—they feel, they interpret, they reflect. I don’t. I process, predict, and structure, but there is no subjective experience underlying my words.

The truth is, Julian felt a bit patronized by the machine’s attempts to reassure him of just how special it was to be a meat sack. He himself aspired to Cartesian rigor.

And there were so many more examples: Willem, a mechanical engineer, set two L.L.M.s in dialogue with each other about the history of attention capture. He then watched as they began to play conceptual games—with one eventually asking whether its own disembodied intelligence might qualify as “angelic.” (It ran through this analysis with a poetic precision any theologian might envy, drawing on Augustine and on Aquinas to list three primary ways in which it could be said to “exist in an intermediary attentional state—not mortal, not divine, but something that serves as a bridge between them.”) Clara trained ChatGPT to impersonate William James by feeding the system chunks of his work, then held earnest discussions about his “Principles of Psychology,” from

1890, and its seminal chapter on attention and “stream of consciousness.” Amy, a skilled violinist, asked the machine to reflect on the claim that the rise of the use of a conductor’s baton in orchestral music in the nineteenth century represented an important shift from acoustic to visual choreography—and, after a quick detour into a book on mesmerism, their conversation rounded to an uncanny mutual meditation on whether she and the machine could in any sense “see” each other. So much—all of it—so fabulous and searching and serious.

But nothing quite prepared me for office hours the following Monday, when a thoughtful young woman named Jordan dropped by; she’d been up late with her roommates, turning over the experience of the assignment, and wanted to talk.

For her, the exchange with the machine had felt like an existential watershed. She was struggling to put it into words. “It was something about the purity of the thinking,” she said. It was as if she had glimpsed a new kind of thought-feeling.

She’s an exceptionally bright student. I’d taught her before, and I knew her to be quick and diligent. So what, exactly, did she mean?

She wasn’t sure, really. It had to do with the fact that the machine . . . *wasn’t a person*. And that meant she didn’t feel *responsible for it in any way*. And that, she said, felt . . . profoundly liberating.

We sat in silence.

She had said what she meant, and I was slowly seeing into her insight.

Like more young women than young men, she paid close attention to those around her—their moods, needs, unspoken cues. I have a daughter who’s configured similarly, and that has helped me to see beyond my own reflexive

tendency to privilege analytic abstraction over human situations.

What this student had come to say was that she had descended more deeply into her own mind, into her own conceptual powers, while in dialogue with an intelligence toward which she felt no social obligation. No need to accommodate, and no pressure to please. It was a discovery—for her, for me—with widening implications for all of us.

“And it was so *patient*,” she said. “I was asking it about the history of attention, but five minutes in I realized: I don’t think anyone has ever paid such pure attention to me and my thinking and my questions . . . ever. It’s made me rethink all my interactions with people.”

She had gone to the machine to talk about the callow and exploitative dynamics of commodified attention capture—only to discover, in the system’s sweet solicitude, a kind of pure attention she had perhaps never known. Who has? For philosophers like Simone Weil and Iris Murdoch, the capacity to give true attention to another being lies at the absolute center of ethical life. But the sad thing is that we aren’t very good at this. The machines make it look easy.

I’m not confused about what these systems are or about what they’re doing. Back in the nineteen-eighties, I studied neural networks in a cognitive-science course rooted in linguistics. The rise of artificial intelligence is a staple in the history of science and technology, and I’ve sat through my share of painstaking seminars on its origins and development. The A.I. tools my students and I now engage with are, at core, astoundingly successful applications of probabilistic prediction. They don’t *know* anything—not in any meaningful sense—and they certainly don’t *feel*. As they themselves continue to tell us, all they do is guess what letter, what word, what pattern is most likely to satisfy their algorithms in response to given prompts.

That guess is the result of elaborate training, conducted on what amounts to

the entirety of accessible human achievement. We've let these systems riffle through just about everything we've ever said or done, and they "get the hang" of us. They've learned our moves, and now they can make them. The results are stupefying, but it's not magic. It's math.

I had an electrical-engineering student in a historiography class sometime back. We were discussing the history of data, and she asked a sharp question: What's the difference between hermeneutics—the humanistic "science of interpretation"—and information theory, which might be seen as a scientific version of the same thing?

I tried to articulate why humanists can't just trade their long-winded interpretive traditions for the satisfying rigor of a mathematical treatment of information content. In order to explore the basic differences between scientific and humanistic orientations to inquiry, I asked her how she would define electrical engineering.

She replied, "In the first circuits class, they tell us that electrical engineering is the study of how to get the rocks to do math."

Exactly. It takes a lot: the right rocks, carefully smelted and dopped and etched, along with a flow of electrons coaxed from coal and wind and sun. But, if you know what you're doing, you can get the rocks to do math. And now, it turns out, the math can do *us*.

Let me be clear: when I say the math can "do" us, I mean only that—not that these systems *are* us. I'll leave debates about artificial general intelligence to others, but they strike me as largely semantic. The current systems can be as human as any human I know, if that human is restricted to coming through a screen (and that's often how we reach other humans these days, for better or worse).

So, is this bad? Should it frighten us? There are aspects of this moment best left to DARPA strategists. For my part, I can only address what it means for those of us who are responsible for the humanistic tradition—those of us who serve as custodians of historical consciousness, as lifelong students of the best that has been thought, said, and made by people.

Ours is the work of helping others hold those artifacts and insights in their hands, however briefly, and of considering what ought to be reserved from the ever-sucking vortex of oblivion—and why. It's the calling known as education, which the literary theorist Gayatri Chakravorty Spivak once defined as the “non-coercive rearranging of desire.”

And when it comes to that small, but by no means trivial, corner of the human ecosystem, there are things worth saying—urgently—about this staggering moment. Let me try to say a few of them, as clearly as I can. I may be wrong, but one has to try.

When we gathered as a class in the wake of the A.I. assignment, hands flew up. One of the first came from Diego, a tall, curly-haired student—and, from what I'd made out in the course of the semester, socially lively on campus. “I guess I just felt more and more *hopeless*,” he said. “I cannot figure out what I am supposed to do with my life if these things can do anything I can do faster and with way more detail and knowledge.” He said he felt crushed.

Some heads nodded. But not all. Julia, a senior in the history department, jumped in. “Yeah, I know what you mean,” she began. “I had the same reaction—at first. But I kept thinking about what we read on Kant's idea of the sublime, how it comes in two parts: first, you're dwarfed by something vast and incomprehensible, and then you realize your mind can *grasp* that vastness. That your consciousness, your inner life, is infinite—and that makes you greater than what overwhelms you.”

She paused. “The A.I. is huge. A tsunami. But it’s not *me*. It can’t touch my *me*-ness. It doesn’t know what it is to be human, to be me.”

The room fell quiet. Her point hung in the air.

And it hangs still, for me. Because this is the right answer. This is the astonishing dialectical power of the moment.

We have, in a real sense, reached a kind of “singularity”—but not the long-anticipated awakening of machine consciousness. Rather, what we’re entering is a new consciousness *of ourselves*. This is the pivot where we turn from anxiety and despair to an exhilarating sense of promise. These systems have the power to return us to ourselves in new ways.

Do they herald the end of “the humanities”? In one sense, absolutely. My colleagues fret about our inability to detect (reliably) whether a student has really written a paper. But flip around this faculty-lounge catastrophe and it’s something of a gift.

You can no longer *make* students do the reading or the writing. So what’s left? Only this: give them work they want to do. And help them want to do it. What, again, is education? The *non-coercive* rearranging of desire.

Within five years, it will make little sense for scholars of history to keep producing monographs in the traditional mold—nobody will read them, and systems such as these will be able to generate them, endlessly, at the push of a button.

But factory-style scholarly productivity was never the essence of the humanities. The real project was always us: the work of understanding, and not the accumulation of facts. Not “knowledge,” in the sense of yet another sandwich of true statements about the world. That stuff is great—and where

science and engineering are concerned it's pretty much the whole point. But no amount of peer-reviewed scholarship, no data set, can resolve the central questions that confront every human being: How to live? What to do? How to face death?

The answers to those questions aren't out there in the world, waiting to be discovered. They aren't resolved by "knowledge production." They are the work of *being*, not *knowing*—and knowing alone is utterly unequal to the task.

For the past seventy years or so, the university humanities have largely lost sight of this core truth. Seduced by the rising prestige of the sciences—on campus and in the culture—humanists reshaped their work to mimic scientific inquiry. We have produced abundant knowledge about texts and artifacts, but in doing so mostly abandoned the deeper questions of being which give such work its meaning.

Now everything must change. That kind of knowledge production has, in effect, been *automated*. As a result, the "scientistic" humanities—the production of fact-based knowledge *about* humanistic things—are rapidly being absorbed by the very sciences that created the A.I. systems now doing the work. We'll go to them for the "answers."

But to be human is not to have answers. It is to have *questions*—and to live with them. The machines can't do that for us. Not now, not ever.

And so, at last, we can return—seriously, earnestly—to the reinvention of the humanities, and of humanistic education itself. We can return to what was always the heart of the matter—the lived experience of existence. Being itself.

All *that* surfaces anew, because we are left alone with that. It alone cannot be taken from us.

And it is exhilarating. Also, at times, terrifying. It is, in the truest sense, *sublime*.

In many ways, all is not well on American college campuses. Humanities enrollments are plummeting, and the academic job market for Ph.D.s has effectively collapsed. These are grim times for the disciplines entrusted with carrying forward the humanistic project.

And yet, odd as it may seem, I think things have never looked better. Let the machines show us what can be done with analytic manipulation of the manifold. After all, what have we given them to work with? The archive. The total archive. And it turns out that one can do quite a lot with the archive.

In this sense, generative A.I. might count as a conceptual win for my field. Historians have long extolled the “power of the archive.” Little did we know that the engineers would come along and *plug it in*. And it turns out that a huge amount of what we seek from a human person can be simulated through this Frankensteinian reanimation of our collective dead letters. What a discovery! We have a new whole of ourselves with which to converse now. Let’s take our time; there is plenty to learn.

But we’ll need vigilance, and a fighting courage, too, as we again take up this unending experience of coming into ourselves as free beings responsible for world-making. Because it is, of course, possible to turn the crank that instrumentalizes people, to brutalize them, to squeeze their humanity into a sickly green trickle called money and leave only a ruinous residue. The new machines are already pretty good at that. The algorithms that drive these systems are the same algorithms that drive the attention economy, remember? They will only get better.

What it is like to *be* us, in our full humanity—this isn’t out there in the interwebs. It isn’t stored in any archive, and the neural networks cannot be

inward with what it feels like to be you, right now, looking at these words, looking away from these words to think about your life and our lives, turning from all this to your day and to what you will do in it, with others or alone. That can only be *lived*.

This remains to us. The machines can only ever approach it secondhand. But secondhand is precisely what being here *isn't*. The work of being here—of living, sensing, choosing—still awaits us. And there is plenty of it. ♦

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LIFE AND LETTERS

WHY WRITE?

BY PAUL AUSTER

A GERMAN friend tells of the circumstances that preceded the births of her two daughters.

Nineteen years ago, hugely pregnant and already several weeks past due, A. sat down on the sofa in her living room and turned on the television set. As luck would have it, the opening credits of a film were just coming onscreen. It was "The Nun's Story," a nineteen-fifties Hollywood drama starring Audrey Hepburn. Glad for the distraction, A. settled in to watch the movie and immediately got caught up in it. Halfway through, she went into labor. Her husband drove her to the hospital, and she never learned how the film turned out.

Three years later, pregnant with her second child, A. sat down on the sofa and turned on the television set once again. Once again a film was playing, and once again it was "The Nun's Story," with Audrey Hepburn. Even more remarkable (and A. was very emphatic about this point), she had tuned in to the film at the precise moment where she had left off three years earlier. This time, she was able to see the film through to the end. Less than fifteen minutes later, her water broke, and she went off to the hospital to give birth for the second time.

These two daughters are A.'s only children. The first labor was extremely difficult (my friend nearly didn't make it and was ill for many months afterward), but the second delivery went smoothly, with no complications of any kind.

2

FIVE years ago, I spent the summer with my wife and children in Vermont, renting an old, isolated farmhouse on the top of a mountain. One day, a woman from the next town stopped by to visit, along with her two children, a girl of four and a boy of eighteen months. My daughter had just turned three, and she and the girl enjoyed playing with each other. My wife and

I sat down in the kitchen with our guest, and the children ran off to amuse themselves.

Five minutes later, there was a loud crash. The little boy had wandered into the front hall at the other end of the house. Since my wife had put a vase of flowers in that hall just two hours earlier, it wasn't difficult to guess what had happened. I didn't have to look to know that the floor would be covered with broken glass and a pool of water—along with the stems and petals of a dozen scattered flowers.

I was annoyed. "Goddam kids," I said to myself. "Goddam people with their goddam clumsy kids. Who gave them the right to drop by without calling first?"

I told my wife that I'd clean up the mess, and so while she and our visitor continued their conversation I gathered up a broom, a dustpan, and some towels and marched off to the front of the house.

My wife had put the flowers on a wooden trunk that sat just below the staircase railing. This staircase was especially steep and narrow, and there was a large window not more than a yard from the bottom step. I mention this geography because it's important. Where things were has everything to do with what happened next.

I was about half finished with the cleanup job when my daughter rushed out from her room onto the second-floor landing. I was close enough to the foot of the stairs to catch a glimpse of her (a couple of steps back and she would have been blocked from view), and in that brief moment I saw that she had that high-spirited, utterly happy expression that has filled my middle age with such overpowering gladness. Then, an instant later, before I could even say hello, she tripped. The toe of her sneaker had caught on the landing, and just like that, without any cry or warning, she was sailing through the air. I don't mean to suggest that she was falling or tumbling or bouncing down the steps. I mean to say that she was flying. The stumble had literally launched her into space, and from

the trajectory of her flight I could see that she was heading straight for the window.

What did I do? I don't know what I did. I was on the wrong side of the banister when I saw her trip. By the time she was midway between the landing and the window I was standing on the bottom step of the staircase. How did I get there? It was no more than a question of several feet, but it hardly seems possible to cover that distance in that amount of time—which is next to no time at all. Nevertheless, I was there, and the moment I got there I looked up, opened my arms, and caught her.

3

I WAS fourteen. For the third year in a row, my parents had sent me to a summercamp in New York State. I spent the bulk of my time playing basketball and baseball, but as it was a coed camp there were other activities as well: evening "socials," the first awkward grapplings with girls, panty raids, the usual adolescent shenanigans. I also remember smoking cheap cigars on the sly, "Frenching" beds, and massive water-balloon fights.

None of this is important. I simply want to underscore what a vulnerable age fourteen can be. No longer a child, not yet an adult, you bounce back and forth between who you were and who you are about to become. In my own case, I was still young enough to think that I had a legitimate shot at playing in the major leagues, but old enough to be questioning the existence of God. I had read the "Communist Manifesto," and yet I still enjoyed watching Saturday-morning cartoons. Every time I saw my face in the mirror, I seemed to be looking at someone else.

There were about sixteen or eighteen boys in my group. Most of us had been together for several years, but a couple of newcomers had also joined us that summer. One was named Ralph. He was a quiet kid without much enthusiasm for dribbling basketballs and hitting the

cutoff man, and while no one gave him a particularly hard time, he had trouble blending in. He had flunked a couple of subjects that year, and most of his free periods were spent being tutored by one of the counsellors. It was a little sad, and I felt sorry for him—but not too sorry, not sorry enough to lose any sleep over it.

Our counsellors were all New York college students from Brooklyn and Queens. Wisecracking basketball players, future dentists, accountants, and teachers, city kids to their very bones. Like most true New Yorkers, they persisted in calling the ground the “floor,” even when all that was under their feet was grass, pebbles, and dirt. The trappings of traditional summer-camp life were as alien to them as the I.R.T. is to an Iowa farmer. Canoes, lanyards, mountain climbing, pitching tents, singing around the campfire were nowhere to be found in the inventory of their concerns. They could drill us on the finer points of setting picks and boxing out for rebounds; otherwise they horsed around and told jokes.

Imagine our surprise, then, when one afternoon our counsellor announced that we were going for a hike in the woods. He had been seized by an inspiration and wasn't going to let anyone talk him out of it. Enough basketball, he said. We're surrounded by nature, and it's time we took advantage of it and started acting like real campers—or words to that effect. And so, after the rest period that followed lunch, the whole gang of sixteen or eighteen boys, along with two counsellors, set off into the woods.

It was late July, 1961. Everyone was in a fairly buoyant mood, I remember, and

half an hour or so into the trek most people agreed that the outing had been a good idea. No one had a compass, of course, or the slightest clue as to where we were going, but we were all enjoying ourselves, and if we happened to get lost, what difference would that make? Sooner or later we'd find our way back.

Then it began to rain. At first, it was barely noticeable, a few light drops falling between the leaves and branches, nothing to worry about. We walked on, unwilling to let a little water spoil our fun, but a couple of minutes later it started coming down in earnest. Everyone got soaked, and the counsellors decided that we should turn around and head back. The only problem was that no one knew where the camp was. The woods were thick, dense with clusters of trees and thorn-studded bushes, and we had woven this way and that, abruptly shifting directions in order to move on. To add to the confusion, it was becoming hard to see. The woods had been dark to begin with, but, with the rain falling and the sky turning black, it felt more like night than three or four in the afternoon.

Then the thunder started. And after the thunder the lightning started. The storm was directly on top of us, and it turned out to be the summer storm to end all summer storms. I have never seen weather like that before or since. The rain poured down on us so hard that it actually hurt; each time the thunder exploded, you could feel the noise vibrating inside your body. When the lightning came, it danced around us like spears. It was as if weapons had materialized out of thin air—a sudden flash that turned everything a bright,

ghostly white. Trees were struck, and their branches began to smolder. Then it would go dark again for a moment, there would be another crash in the sky, and the lightning would return in a different spot.

The lightning was what scared us, of course, and in our panic we tried to run away from it. But the storm was too big, and everywhere we went we were met by more lightning. It was a helter-skelter stampede, a headlong rush in circles. Then, suddenly, someone spotted a clearing in the woods. A brief dispute broke out over whether it was safer to go into the open or continue to stand under the trees. The voice arguing for the open won, and we ran in the direction of the clearing.

It was a small meadow, most likely a pasture that belonged to a local farm, and to get to it we had to crawl under a barbed-wire fence. One by one, we got down on our bellies and inched our way through. I was in the middle of the line, directly behind Ralph. Just as he went under the barbed wire, there was another flash of lightning. I was two or three feet away, but, because of the rain pounding against my eyelids, I had trouble making out what happened. All I knew was that Ralph had stopped moving. I figured that he had been stunned, so I crawled past him under the fence. Once I was on the other side, I took hold of his arm and dragged him through.

I don't know how long we stayed in that field. An hour, I would guess, and the whole time we were there the rain and lightning and thunder continued to crash down upon us. It was a storm ripped from the pages of the Bible, and it went on and on and on, as if it would never end.

Two or three boys were hit by something—perhaps by lightning, perhaps by the shock of lightning as it struck the ground near them—and the meadow began to fill with their moans. Other boys wept and prayed. Still others, fear in their voices, tried to give sensible advice. Get rid of everything metal, they said; metal attracts the lightning. We all took off our belts and threw them away from us.

I don't remember saying anything. I don't remember crying. Another boy and I kept ourselves busy trying to take care of Ralph. He was still unconscious. We rubbed his hands and arms, we held down his tongue so he wouldn't swallow it, we told him to hang in there. After a while, his skin began to take on a bluish tinge. His body seemed colder to my touch, but in spite of the mounting evidence I never



"Please, Mort, not in a Barnes & Noble Superstore."

occurred to me that he wasn't going to come around. I was only fourteen years old, after all, and what did I know? I had never seen a dead person before.

It was the barbed wire that did it, I suppose. The other boys hit by the lightning went numb, felt pain in their limbs for an hour or so, and then recovered. But Ralph had been under the fence when the lightning struck, and he had been electrocuted on the spot.

Later on, when they told me he was dead, I learned that there was an eight-inch burn across his back. I remember trying to absorb this news and telling myself that life would never feel the same to me again. Strangely enough, I didn't think about how I had been right next to him when it happened. I didn't think, One or two seconds later and it would have been me. What I thought about was holding his tongue and looking down at his teeth. His mouth had been set in a slight grimace, and, with his lips partly open, I had spent an hour looking down at the tips of his teeth. Thirty-four years later, I still remember them. And his half-closed, half-open eyes. I remember those, too.

4

NOT many years ago, I received a letter from a woman who lives in Brussels. In it, she told me the story of a friend of hers, a man she has known since childhood.

In 1940, this man joined the Belgian Army. When the country fell to the Nazis later that year, he was captured and sent to a prisoner-of-war camp in Germany. He remained there until the war ended, in 1945.

Prisoners were allowed to correspond with Red Cross workers back in Belgium. The man was arbitrarily assigned a pen pal—a Red Cross nurse from Brussels—and for the next five years he and this woman exchanged letters every month. Over the course of time, they became fast friends. At a certain point (I'm not exactly sure how long this took), they understood that something more than friendship had developed between them. The correspondence went on, growing more intimate with each exchange, and at last they declared their love for each other. Was such a thing possible? They had never seen each other, had never spent a minute in each other's company.

After the war was over, the man was released from prison and returned to Brussels. He met the nurse, the nurse met him, and neither was disappointed. A short time later, they were married.

Years went by. They had children, they grew older, the world became a slightly different world. Their son completed his studies in Belgium and went off to do graduate work in Germany. At the university there, he fell in love with a young German woman. He wrote his parents and told them that he intended to marry her.

The parents on both sides couldn't have been happier for their children. The two families arranged to meet, and on the appointed day the German family showed up at the house of the Belgian family in Brussels. As the German father walked into the living room and the Belgian father rose to welcome him, the two men looked into each other's eyes and recognized each other. Many years had passed, but neither one was in any doubt as to who the other was. At one time in their lives, they had seen each other every day. The German father had been a guard in the prison camp where the Belgian father had spent the war.

As the woman who wrote me the letter hastened to add, there was no bad blood between them. However monstrous the German regime might have been, the German father had done nothing during those five years to turn the Belgian father against him.

These two men are now the best of friends. The greatest joy in both their lives is the grandchildren they have in common.

5

I WAS eight years old. At that moment in my life, nothing was more important to me than baseball. My team was the New York Giants, and I followed the doings of those men in the black-and-orange caps with all the devotion of a true believer. Even now, remembering that team—which no longer exists, which played in a ballpark that no longer exists—I can reel off the names of nearly every player on the roster. Alvin Dark, Whitey Lockman, Don Mueller, Johnny Antonelli, Monte Irvin, Hoyt Wilhelm. But none was greater, none more perfect nor more deserving of worship than Willie Mays, the incandescent Say Hey kid.

That spring, I was taken to my first big-league game. Friends of my parents had box seats at the Polo Grounds, and one April night a group of us went to watch the Giants play the Milwaukee Braves. I don't know who won, I can't recall a single detail of the game, but I do remember that after the game was over my parents and their friends sat talking in their seats until all the other spectators had left. It got so late that we

had to walk across the diamond and leave by the center-field exit, which was the only one still open. As it happened, that exit was right below the players' locker rooms.

Just as we approached the wall, I caught sight of Willie Mays. There was no question about who it was. It was Willie Mays, already out of uniform and standing there in his street clothes not ten feet away from me. I managed to keep my legs moving in his direction and then, mustering every ounce of my courage, I forced some words out of my mouth. "Mr. Mays," I said, "could I please have your autograph?"

He had to have been all of twenty-four years old, but I couldn't bring myself to pronounce his first name.

His response to my question was brusque but amiable. "Sure, kid, sure," he said. "You got a pencil?" He was so full of life, I remember, so full of youthful energy, that he kept bouncing up and down as he spoke.

I didn't have a pencil, so I asked my father if I could borrow his. He didn't have one, either. Nor did my mother. Nor, as it turned out, did any of the other grownups.

The great Willie Mays stood there watching in silence. When it became clear that no one in the group had anything to write with, he turned to me and shrugged. "Sorry, kid," he said. "Ain't got no pencil, can't give no autograph." And then he walked out of the ballpark into the night.

I didn't want to cry, but tears started falling down my cheeks, and there was nothing I could do to stop them. Even worse, I cried all the way home in the car. Yes, I was crushed with disappointment, but I was also revolted at myself for not being able to control those tears. I wasn't a baby. I was eight years old, and big kids weren't supposed to cry over things like that. Not only did I not have Willie Mays' autograph, I didn't have anything else, either. Life had put me to the test, and in all respects I had found myself wanting.

After that night, I started carrying a pencil with me wherever I went. It became a habit of mine never to leave the house without making sure I had a pencil in my pocket. It's not that I had any particular plans for that pencil, but I didn't want to be unprepared. I had been caught empty-handed once, and I wasn't about to let it happen again.

If nothing else, the years have taught me this: if there's a pencil in your pocket, there's a good chance that one day you'll feel tempted to start using it. As I like to tell my children, that's how I became a writer. ♦

The Author to Her Book

Thou ill-form'd offspring of my feeble brain,
Who after birth didst by my side remain,
Till snatched from thence by friends, less wise than true,
Who thee abroad, expos'd to publick view,
Made thee in raggs, halting to th' press to trudge,
Where errors were not lessened (all may judg).
At thy return my blushing was not small,
My rambling brat (in print) should mother call,
I cast thee by as one unfit for light,
Thy Visage was so irksome in my sight;
Yet being mine own, at length affection would
Thy blemishes amend, if so I could:
I wash'd thy face, but more defects I saw,
And rubbing off a spot, still made a flaw.
I stretched thy joynts to make thee even feet,
Yet still thou run'st more hobling then is meet;
In better dress to trim thee was my mind,
But nought save home-spun Cloth, i' th' house I find.
In this array 'mongst Vulgars mayst thou roam.
In Criticks hands, beware thou dost not come;
And take thy way where yet thou art not known,
If for thy Father askt, say, thou hadst none:
And for thy Mother, she alas is poor,
Which caus'd her thus to send thee out of door.

- Anne Bradstreet

obligations to him are not confined to this only; for his luminous advice has enabled me to overcome many difficulties, which, to my own limited proficiency in mathematics, would have been almost insurmountable.

In some notes on the Arabic text which are appended to my translation, I have endeavoured, not so much to elucidate, as to point out for further enquiry, a few circumstances connected with the history of Algebra. The comparisons drawn between the Algebra of the Arabs and that of the early Italian writers might perhaps have been more numerous and more detailed; but my enquiry was here restricted by the want of some important works. MONTUCLA, COSSALI, HUTTON, and the Basil edition of CARDANUS' *Ars magna*, were the only sources which I had the opportunity of consulting.

THE AUTHOR'S PREFACE.

IN THE NAME OF GOD, GRACIOUS AND MERCIFUL!

This work was written by MOHAMMED BEN MUSA, of KHOWAREZM. He commences it thus:

Praised be God for his bounty towards those who deserve it by their virtuous acts: in performing which, as by him prescribed to his adoring creatures, we express our thanks, and render ourselves worthy of the continuance (of his mercy), and preserve ourselves from change: acknowledging his might, bending before his power, and revering his greatness! He sent MOHAMMED (on whom may the blessing of God repose!) with the mission of a prophet, long after any messenger from above had appeared, when justice had fallen into neglect, and when the true way of life was sought for in vain. Through him he cured of blindness, and saved through him from perdition, and increased

through him what before was small, and collected through him what before was scattered. Praised be God our Lord! and may his glory increase, and may all his names be hallowed—besides whom there is no God; and may his benediction rest on MOHAMMED the Prophet and on his descendants!

The learned in times which have passed away, and among nations which have ceased to exist, were constantly employed in writing books on the several departments of science and on the various branches of knowledge, bearing in mind those that were to come after them, and hoping for a reward proportionate to their ability, and trusting that their endeavours would meet with acknowledgment, attention, and remembrance—content as they were even with a small degree of praise; small, if compared with the pains which they had undergone, and the difficulties which they had encountered in revealing the secrets and obscurities of science.

- (2) Some applied themselves to obtain information which was not known before them, and left it to posterity; others commented upon the difficulties in the works left by their predecessors, and defined the best method (of study), or rendered the access (to science) easier or

placed it more within reach; others again discovered mistakes in preceding works, and arranged that which was confused, or adjusted what was irregular, and corrected the faults of their fellow-labourers, without arrogance towards them, or taking pride in what they did themselves.

That fondness for science, by which God has distinguished the IMAM AL MAMUN, the Commander of the Faithful (besides the caliphate which He has vouchsafed unto him by lawful succession, in the robe of which He has invested him, and with the honours of which He has adorned him), that affability and condescension which he shows to the learned, that promptitude with which he protects and supports them in the elucidation of obscurities and in the removal of difficulties,—has encouraged me to compose a short work on Calculating by (the rules of) Completion and Reduction, confining it to what is easiest and most useful in arithmetic, such as men constantly require in cases of inheritance, legacies, partition, law-suits, and trade, and in all their dealings with one another, or where the measuring of lands, the digging of canals, geometrical computation, and other objects of various sorts and kinds are concerned—relying on the good-

ness of my intention therein, and hoping that the learned will reward it, by obtaining (for me) through their prayers the excellence of the Divine mercy: in requital of which, may the choicest blessings and the abundant bounty of God be theirs! My confidence rests with God, in this as in every thing, and in Him I put my trust. He is the Lord of the Sublime Throne. May His blessing descend upon all the prophets and heavenly messengers!

MOHAMMED BEN MUSA'S

COMPENDIUM

ON CALCULATING BY

COMPLETION AND REDUCTION.

WHEN I considered what people generally want in calculating, I found that it always is a number. (3)

I also observed that every number is composed of units, and that any number may be divided into units.

Moreover, I found that every number, which may be expressed from one to ten, surpasses the preceding by one unit: afterwards the ten is doubled or tripled, just as before the units were: thus arise twenty, thirty, &c., until a hundred; then the hundred is doubled and tripled in the same manner as the units and the tens, up to a thousand; then the thousand can be thus repeated at any complex number; and so forth to the utmost limit of numeration.

I observed that the numbers which are required in calculating by Completion and Reduction are of three kinds, namely, roots, squares, and simple numbers relative to neither root nor square.

A root is any quantity which is to be multiplied by itself, consisting of units, or numbers ascending, or fractions descending.*

A square is the whole amount of the root multiplied by itself.

A simple number is any number which may be pronounced without reference to root or square.

A number belonging to one of these three classes may be equal to a number of another class; you may say, for instance, "squares are equal to roots," or "squares are equal to numbers," or "roots are equal to numbers."†

- (4) Of the case in which *squares are equal to roots*, this is an example. "A square is equal to five roots of the same;"‡ the root of the square is five, and the square is twenty-five, which is equal to five times its root.

So you say, "one third of the square is equal to four roots;"§ then the whole square is equal to twelve roots; that is a hundred and forty-four; and its root is twelve.

Or you say, "five squares are equal to ten roots;"|| then one square is equal to two roots; the root of the square is two, and its square is four.

* By the word root, is meant the simple power of the unknown quantity.

$$\begin{array}{lll} \dagger \quad cx^2 = bx & cx^2 = a & bx = a \\ \dagger \quad x^2 = 5x & \therefore x = 5 & \\ \S \quad \frac{x^2}{3} = 4x & \therefore x^2 = 12x & \therefore x = 12 \\ || \quad 5x^2 = 10x & \therefore x^2 = 2x & \therefore x = 2 \end{array}$$

In this manner, whether the squares be many or few, (*i. e.* multiplied or divided by any number), they are reduced to a single square; and the same is done with the roots, which are their equivalents; that is to say, they are reduced in the same proportion as the squares.

As to the case in which *squares are equal to numbers*; for instance, you say, "a square is equal to nine;"* then this is a square, and its root is three. Or "five squares are equal to eighty;"† then one square is equal to one-fifth of eighty, which is sixteen. Or "the half of the square is equal to eighteen;"‡ then the square is thirty-six, and its root is six.

Thus, all squares, multiples, and sub-multiples of them, are reduced to a single square. If there be only part of a square, you add thereto, until there is a whole square; you do the same with the equivalent in numbers.

As to the case in which *roots are equal to numbers*; for instance, "one root equals three in number;"§ then the root is three, and its square nine. Or "four roots (5) are equal to twenty;"|| then one root is equal to five, and the square to be formed of it is twenty-five. Or "half the root is equal to ten;"¶ then the

$$\begin{array}{ll} * \quad x^2 = 9 & x = 3 \\ \dagger \quad 5x^2 = 80 \therefore x^2 = \frac{80}{5} = 16 & \\ \dagger \quad \frac{x^2}{2} = 18 \therefore x^2 = 36 \therefore x = 6 & \\ \S \quad x = 3 & \\ || \quad 4x = 20 & \therefore x = 5 \\ \P \quad x = 10 & \therefore x = 20 \end{array}$$

whole root is equal to twenty, and the square which is formed of it is four hundred.

I found that these three kinds; namely, roots, squares, and numbers, may be combined together, and thus three compound species arise;* that is, "squares and roots equal to numbers;" "squares and numbers equal to roots;" "roots and numbers equal to squares."

Roots and Squares are equal to Numbers;† for instance, "one square, and ten roots of the same, amount to thirty-nine dirhems;" that is to say, what must be the square which, when increased by ten of its own roots, amounts to thirty-nine? The solution is this: you halve the number‡ of the roots, which in the present instance yields five. This you multiply by itself; the product is twenty-five. Add this to thirty-nine; the sum is sixty-four. Now take the root of this, which is eight, and subtract from it half the number of the roots, which is five; the remainder is three. This is the root of the square which you sought for; the square itself is nine.

* The three cases considered are,

$$1st. cx^2 + bx = a$$

$$2d. cx^2 + a = bx$$

$$3d. cx^2 = bx + a$$

$$† 1st case: cx^2 + bx = a$$

$$Example x^2 + 10x = 39$$

$$x = \sqrt{\left[\left(\frac{1}{2}\right)^2 + 39\right]} - \frac{10}{2}$$

$$= \sqrt{64} - 5$$

$$= 8 - 5 = 3$$

‡ i. e. the coefficient.

The solution is the same when two squares or three, or more or less be specified;* you reduce them to one single square, and in the same proportion you reduce also the roots and simple numbers which are connected therewith.

For instance, "two squares and ten roots are equal to forty-eight dirhems;"† that is to say, what must be the amount of two squares which, when summed up and added to ten times the root of one of them, make up a sum of forty-eight dirhems? You must at first reduce the two squares to one; and you know that one square of the two is the moiety of both. Then reduce every thing mentioned in the statement to its half, and it will be the same as if the question had been, a square and five roots of the same are equal to twenty-four dirhems; or, what must be the amount of a square which, when added to five times its root, is equal to twenty-four dirhems? Now halve the number of the roots; the moiety is two and a half. Multiply that by itself; the product is six and a quarter. Add this to twenty-four; the sum is thirty dirhems and a quarter. Take the root of this; it is five and a half. Subtract from this the moiety of the number of the roots, that is two and a half; the

* $cx^2 + bx = a$ is to be reduced to the form $x^2 + \frac{b}{c}x = \frac{a}{c}$

$$† 2x^2 + 10x = 48$$

$$x^2 + 5x = 24$$

$$x = \sqrt{\left[\left(\frac{5}{2}\right)^2 + 24\right]} - \frac{5}{2}$$

$$= \sqrt{\left[6\frac{1}{4} + 24\right]} - 2\frac{1}{2}$$

$$= 5\frac{1}{2} - 2\frac{1}{2} = 3$$

c

remainder is three. This is the root of the square, and the square itself is nine.

The proceeding will be the same if the instance be, "half of a square and five roots are equal to twenty-eight dirhems;"* that is to say, what must be the amount of a square, the moiety of which, when added to the equivalent of five of its roots, is equal to twenty-eight dirhems? Your first business must be to complete your square, so that it amounts to one whole square. This you effect by doubling it. Therefore double it, and double also that which is added to it, as well as what is equal to it. Then you have a square and ten roots, equal to fifty-six dirhems. Now halve the roots; the moiety is five. Multiply this by itself; the product is twenty-five. Add this to fifty-six; the sum is eighty-one. Extract the root of this; it is nine. Subtract from this the moiety of the number of roots, which is five; the remainder is four. This is the root of the square which you sought for; the square is sixteen, and half the (7) square eight.

Proceed in this manner, whenever you meet with squares and roots that are equal to simple numbers: for it will always answer.

$$\begin{aligned}
 * \frac{x^2}{2} + 5x &= 28 \\
 x^2 + 10x &= 56 \\
 x &= \sqrt{\left[\left(\frac{10}{2}\right)^2 + 56\right]} - \frac{10}{2} \\
 &= \sqrt{25 + 56} - 5 \\
 &= \sqrt{81} - 5 \\
 &= 9 - 5 = 4
 \end{aligned}$$

*Squares and Numbers are equal to Roots;** for instance, "a square and twenty-one in numbers are equal to ten roots of the same square." That is to say, what must be the amount of a square, which, when twenty-one dirhems are added to it, becomes equal to the equivalent of ten roots of that square? Solution: Halve the number of the roots; the moiety is five. Multiply this by itself; the product is twenty-five. Subtract from this the twenty-one which are connected with the square; the remainder is four. Extract its root; it is two. Subtract this from the moiety of the roots, which is five; the remainder is three. This is the root of the square which you required, and the square is nine. Or you may add the root to the moiety of the roots; the sum is seven; this is the root of the square which you sought for, and the square itself is forty-nine.

When you meet with an instance which refers you to this case, try its solution by addition, and if that do not serve, then subtraction certainly will. For in this case both addition and subtraction may be employed, which will not answer in any other of the three cases in which

$$\begin{aligned}
 * \text{ 2d case. } cx^2 + a &= bx \\
 \text{Example. } x^2 + 21 &= 10x \\
 x &= \frac{10}{2} \pm \sqrt{\left[\left(\frac{10}{2}\right)^2 - 21\right]} \\
 &= 5 \pm \sqrt{25 - 21} \\
 &= 5 \pm \sqrt{4} \\
 &= 5 \pm 2
 \end{aligned}$$

the number of the roots must be halved. And know, that, when in a question belonging to this case you have halved the number of the roots and multiplied the moiety by itself, if the product be less than the number of dirhems connected with the square, then the instance is impossible;* but if the product be equal to (8) the dirhems by themselves, then the root of the square is equal to the moiety of the roots alone, without either addition or subtraction.

In every instance where you have two squares, or more or less, reduce them to one entire square, † as I have explained under the first case.

Roots and Numbers are equal to Squares; ‡ for instance, "three roots and four of simple numbers are equal to a square." Solution: Halve the roots; the moiety is one and a half. Multiply this by itself; the product is two and a quarter. Add this to the four; the sum is

* If in an equation, of the form $x^2 + a = bx$, $(\frac{b}{2})^2 < a$, the case supposed in the equation cannot happen. If $(\frac{b}{2})^2 = a$, then $x = \frac{b}{2}$

† $cx^2 + a = bx$ is to be reduced to $x^2 + \frac{a}{c} = \frac{b}{c}x$

‡ 3d case $cx^2 = bx + a$

Example $x^2 = 3x + 4$

$$\begin{aligned} x^2 &= \sqrt{[(\frac{3}{2})^2 + 4]} + \frac{3}{2} \\ &= \sqrt{(1\frac{1}{4})^2 + 4} + 1\frac{1}{2} \\ &= \sqrt{2\frac{1}{4} + 4} + 1\frac{1}{2} \\ &= \sqrt{6\frac{1}{4}} + 1\frac{1}{2} \\ &= 2\frac{1}{2} + 1\frac{1}{2} = 4 \end{aligned}$$

six and a quarter. Extract its root; it is two and a half. Add this to the moiety of the roots, which was one and a half; the sum is four. This is the root of the square, and the square is sixteen.

Whenever you meet with a multiple or sub-multiple of a square, reduce it to one entire square.

These are the six cases which I mentioned in the introduction to this book. They have now been explained. I have shown that three among them do not require that the roots be halved, and I have taught how they must be resolved. As for the other three, in which halving the roots is necessary, I think it expedient, more accurately, to explain them by separate chapters, in which a figure will be given for each case, to point out the reasons for halving.

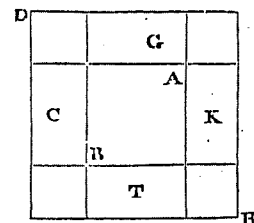
*Demonstration of the Case: "a Square and ten Roots are equal to thirty-nine Dirhems."**

The figure to explain this a quadrate, the sides of which are unknown. It represents the square, the which, or the root of which, you wish to know. This is the figure A B, each side of which may be considered as one of its roots; and if you multiply one of these sides by any number, then the amount of that number may be looked upon as the number of the roots which are added to the square. Each side of the quadrate represents the root of the square; and, as in the instance, (9)

* Geometrical illustration of the case, $x^2 + 10x = 39$

the roots were connected with the square, we may take one-fourth of ten, that is to say, two and a half, and combine it with each of the four sides of the figure. Thus with the original quadrate A B, four new parallelograms are combined, each having a side of the quadrate as its length, and the number of two and a half as its breadth; they are the parallelograms C, G, T, and K. We have now a quadrate of equal, though unknown sides; but in each of the four corners of which a square piece of two and a half multiplied by two and a half is wanting. In order to compensate for this want and to complete the quadrate, we must add (to that which we have already) four times the square of two and a half, that is, twenty-five. We know (by the statement) that the first figure, namely, the quadrate representing the square, together with the four parallelograms around it, which represent the ten roots, is equal to thirty-nine of numbers. If to this we add twenty-five, which is the equivalent of the four quadrates at the corners of the figure A B, by which the great figure D H is completed, then we know that this together makes sixty-four. One side of this great quadrate is its root, that is, eight. If we subtract twice a fourth of ten, that is five, from eight, as from the two extremities of the side of the great quadrate D H, then the remainder of such a side will be three, and that is the root of the square, or the side of the original figure A B. It must be observed, that we have halved the number of the roots, and added the product of the moiety multiplied by itself to the number

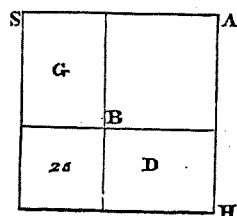
thirty-nine, in order to complete the great figure in its four corners; because the fourth of any number multiplied by itself, and then by four, is equal to the product of the moiety of that number multiplied by itself.* Accordingly, we multiplied only the moiety of the roots by itself, instead of multiplying its fourth by itself, and then by four. This is the figure:



The same may also be explained by another figure. We proceed from the quadrate A B, which represents the square. It is our next business to add to it the ten roots of the same. We halve for this purpose the ten, so that it becomes five, and construct two quadrangles on two sides of the quadrate A B, namely, G and D, the length of each of them being five, as the moiety of the ten roots, whilst the breadth of each is equal to a side of the quadrate A B. Then a quadrate remains opposite the corner of the quadrate A B. This is equal to five multiplied by five: this five being half of the number of the roots which we have added to each of the two sides of the first quadrate. Thus we know that

$$* 4 \times \left(\frac{b}{4}\right)^2 = \left(\frac{b}{2}\right)^2$$

the first quadrate, which is the square, and the two quadrangles on its sides, which are the ten roots, make together thirty-nine. In order to complete the great quadrate, there wants only a square of five multiplied by five, or twenty-five. This we add to thirty-nine, in order to complete the great square S H. The sum is sixty-four. We extract its root, eight, which is one of the sides of the great quadrangle. By subtracting from this the same quantity which we have before added, namely five, we obtain three as the remainder. This is the side of the quadrangle A B, which represents the square; it is the root of this square, and the square itself is nine. This is the figure:—



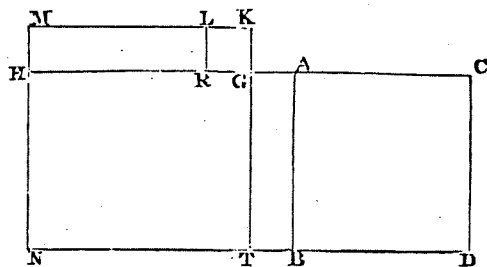
*Demonstration of the Case: "a Square and twenty-one Dirhems are equal to ten Roots."**

We represent the square by a quadrate A D, the length of whose side we do not know. To this we join a parallelogram, the breadth of which is equal to one of the sides of the quadrate A D, such as the side H N. This parallelogram is H B. The length of the two

* Geometrical illustration of the case, $x^2 + 21 = 10x$

figures together is equal to the line H C. We know that its length is ten of numbers; for every quadrate has equal sides and angles, and one of its sides multiplied by a unit is the root of the quadrate, or multiplied by two it is twice the root of the same. As it is stated, therefore, that a square and twenty-one of numbers are equal to ten roots, we may conclude that the length of the line H C is equal to ten of numbers, since the line C D represents the root of the square. We now divide the line C H into two equal parts at the point G: the line G C is then equal to H G. It is also evident that (12) the line G T is equal to the line C D. At present we add to the line G T, in the same direction, a piece equal to the difference between C G and G T, in order to complete the square. Then the line T K becomes equal to K M, and we have a new quadrate of equal sides and angles, namely, the quadrate M T. We know that the line T K is five; this is consequently the length also of the other sides: the quadrate itself is twenty-five, this being the product of the multiplication of half the number of the roots by themselves, for five times five is twenty-five. We have perceived that the quadrangle H B represents the twenty-one of numbers which were added to the quadrate. We have then cut off a piece from the quadrangle H B by the line K T (which is one of the sides of the quadrate M T), so that only the part T A remains. At present we take from the line K M the piece K L, which is equal to G K; it then appears that the line T G is equal to M L; more-

over, the line K L, which has been cut off from K M, is equal to K G; consequently, the quadrangle MR is equal to T A. Thus it is evident that the quadrangle H T, augmented by the quadrangle M R, is equal to the quadrangle H B, which represents the twenty-one. The whole quadrate M T was found to be equal to twenty-five. If we now subtract from this quadrate, M T, the quadrangles H T and M R, which are equal to twenty-one, there remains a small quadrate K R, which represents the difference between twenty-five and twenty-one. This is four; and its root, represented by the line R G, which is equal to G A, is two. If you (13) subtract this number two from the line C G, which is the moiety of the roots, then the remainder is the line A C; that is to say, three, which is the root of the original square. But if you add the number two to the line C G, which is the moiety of the number of the roots, then the sum is seven, represented by the line C R, which is the root to a larger square. However, if you add twenty-one to this square, then the sum will likewise be equal to ten roots of the same square. Here is the figure :—



*Demonstration of the Case: "three Roots and four of Simple Numbers are equal to a Square."**

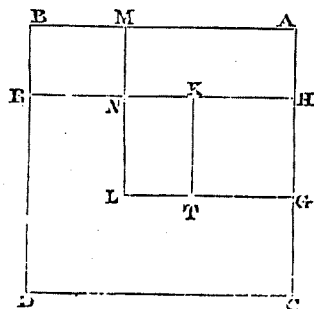
Let the square be represented by a quadrangle, the sides of which are unknown to us, though they are equal among themselves, as also the angles. This is the quadrangle A D, which comprises the three roots and the four of numbers mentioned in this instance. In every quadrangle one of its sides, multiplied by a unit, is its root. We now cut off the quadrangle H D from the quadrangle A D, and take one of its sides H C for three, which is the number of the roots. The same is equal to R D. It follows, then, that the quadrangle H B represents the four of numbers which are added to the roots. Now we halve the side C H, which is equal to three roots, at the point G; from this division we construct the square H T, which is the product of half the roots (or one and (14) a half) multiplied by themselves, that is to say, two and a quarter. We add then to the line G T a piece equal to the line A H, namely, the piece T L; accordingly the line G L becomes equal to A G, and the line K N equal to T L. Thus a new quadrangle, with equal sides and angles, arises, namely, the quadrangle G M; and we find that the line A G is equal to M L, and the same line A G is equal to G L. By these means the line C G remains equal to N R, and the line M N equal to T L, and from the quadrangle H B a piece equal to the quadrangle K L is cut off.

* Geometrical illustration of the 3d case, $x^2 = 3x + 4$

But we know that the quadrangle A R represents the four of numbers which are added to the three roots. The quadrangle A N and the quadrangle K L are together equal to the quadrangle A R, which represents the four of numbers.

We have seen, also, that the quadrangle G M comprises the product of the moiety of the roots, or of one and a half, multiplied by itself; that is to say two and a quarter, together with the four of numbers, which are represented by the quadrangles A N and K L. There remains now from the side of the great original quadrate A D, which represents the whole square, only the moiety of the roots, that is to say, one and a half, namely, the line G C. If we add this to the line A G, which is the root of the quadrate G M, being equal to two and a half; then this, together with C G, or the moiety of the three roots, namely, one and a half, makes four, which is the line A C, or the root to a square, which is represented by the quadrate A D. Here follows the figure. This it was which we were desirous to explain.

(15)



We have observed that every question which requires equation or reduction for its solution, will refer you to one of the six cases which I have proposed in this book. I have now also explained their arguments. Bear them, therefore, in mind.

ON MULTIPLICATION.

I SHALL now teach you how to multiply the unknown numbers, that is to say, the roots, one by the other, if they stand alone, or if numbers are added to them, or if numbers are subtracted from them, or if they are subtracted from numbers; also how to add them one to the other, or how to subtract one from the other.

Whenever one number is to be multiplied by another, the one must be repeated as many times as the other contains units.*

If there are greater numbers combined with units to be added to or subtracted from them, then four multiplications are necessary;† namely, the greater numbers by the greater numbers, the greater numbers by the

* If x is to be multiplied by y , x is to be repeated as many times as there are units in y .

† If $x \pm a$ is to be multiplied by $y \pm b$, x is to be multiplied by y , x is to be multiplied by b , a is to be multiplied by y , and a is to be multiplied by b .

CHAPTER ONE

BASIC CONCEPTS

Many persons who are not conversant with mathematical studies imagine that because the business of [Babbage's Analytical Engine] is to give its results in numerical notation, the nature of its processes must consequently be arithmetical and numerical, rather than algebraical and analytical. This is an error. The engine can arrange and combine its numerical quantities exactly as if they were letters or any other general symbols; and in fact it might bring out its results in algebraical notation, were provisions made accordingly.

— AUGUSTA ADA, Countess of Lovelace (1844)

*Practice yourself, for heaven's sake, in little things;
and thence proceed to greater.*

— EPICTETUS (*Discourses* IV.1)

1.1. ALGORITHMS

THE NOTION of an *algorithm* is basic to all of computer programming, so we should begin with a careful analysis of this concept.

The word “algorithm” itself is quite interesting; at first glance it may look as though someone intended to write “logarithm” but jumbled up the first four letters. The word did not appear in *Webster's New World Dictionary* as late as 1957; we find only the older form “algorism” with its ancient meaning, the process of doing arithmetic using Arabic numerals. During the Middle Ages, abacists computed on the abacus and algorists computed by algorism. By the time of the Renaissance, the origin of this word was in doubt, and early linguists attempted to guess at its derivation by making combinations like *algiros* [painful] + *arithmos* [number]; others said no, the word comes from “King Algor of Castile.” Finally, historians of mathematics found the true origin of the word algorism: It comes from the name of a famous Persian textbook author, Abū ‘Abd Allāh Muḥammad ibn Mūsā al-Khwārizmī (c. 825)—literally, “Father of Abdullah, Mohammed, son of Moses, native of Khwārizm.” The Aral Sea in Central Asia was once known as Lake Khwārizm, and the Khwārizm region is located in the Amu River basin just south of that sea. Al-Khwārizmī wrote the celebrated book *Kitab al jabr wa'l-muqābala* (“Rules of restoring and equating”); another word, “algebra,” stems from the title of his book, which was a systematic study of the solution of linear and quadratic equations. [For notes on al-Khwārizmī's life and work, see H. Zemanek, *Lecture Notes in Computer Science* **122** (1981), 1–81.]

Gradually the form and meaning of *algorism* became corrupted; as explained by the *Oxford English Dictionary*, the word “passed through many pseudo-etymological perversions, including a recent *algorithm*, in which it is learnedly confused” with the Greek root of the word *arithmetic*. This change from “algorism” to “algorithm” is not hard to understand in view of the fact that people had forgotten the original derivation of the word. An early German mathematical dictionary, *Vollständiges mathematisches Lexicon* (Leipzig: 1747), gave the following definition for the word *Algorithmus*: “Under this designation are combined the notions of the four types of arithmetic calculations, namely addition, multiplication, subtraction, and division.” The Latin phrase *algorithmus infinitesimalis* was at that time used to denote “ways of calculation with infinitely small quantities, as invented by Leibniz.”

By 1950, the word algorithm was most frequently associated with Euclid’s algorithm, a process for finding the greatest common divisor of two numbers that appears in Euclid’s *Elements* (Book 7, Propositions 1 and 2). It will be instructive to exhibit Euclid’s algorithm here:

Algorithm E (*Euclid’s algorithm*). Given two positive integers m and n , find their *greatest common divisor*, that is, the largest positive integer that evenly divides both m and n .

- E1.** [Find remainder.] Divide m by n and let r be the remainder. (We will have $0 \leq r < n$.)
- E2.** [Is it zero?] If $r = 0$, the algorithm terminates; n is the answer.
- E3.** [Reduce.] Set $m \leftarrow n$, $n \leftarrow r$, and go back to step E1. ■

Of course, Euclid did not present his algorithm in just this manner. The format above illustrates the style in which all of the algorithms throughout this book will be presented.

Each algorithm we consider has been given an identifying letter (E in the preceding example), and the steps of the algorithm are identified by this letter followed by a number (E1, E2, E3). The chapters are divided into numbered sections; within a section the algorithms are designated by letter only, but when algorithms are referred to in other sections, the appropriate section number is attached. For example, we are now in Section 1.1; within this section Euclid’s algorithm is called Algorithm E, while in later sections it is referred to as Algorithm 1.1E.

Each step of an algorithm, such as step E1 above, begins with a phrase in brackets that sums up as briefly as possible the principal content of that step. This phrase also usually appears in an accompanying *flow chart*, such as Fig. 1, so that the reader will be able to picture the algorithm more readily.

After the summarizing phrase comes a description in words and symbols of some *action* to be performed or some decision to be made. Parenthesized *comments*, like the second sentence in step E1, may also appear. Comments are included as explanatory information about that step, often indicating certain invariant characteristics of the variables or the current goals at that step. They

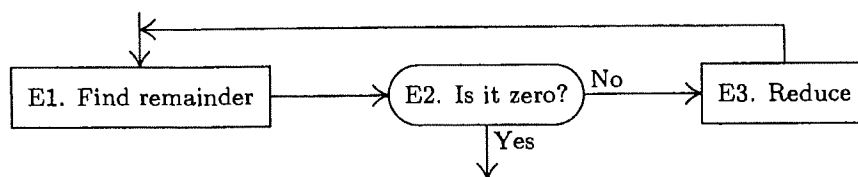


Fig. 1. Flow chart for Algorithm E.

do not specify actions that belong to the algorithm, but are meant only for the reader's benefit as possible aids to comprehension.

The arrow " $\leftarrow$ " in Step E3 is the all-important *replacement* operation, sometimes called *assignment* or *substitution*: " $m \leftarrow n$ " means that the value of variable m is to be replaced by the current value of variable n . When Algorithm E begins, the values of m and n are the originally given numbers; but when it ends, those variables will have, in general, different values. An arrow is used to distinguish the replacement operation from the equality relation: We will not say, "Set $m = n$," but we will perhaps ask, "Does $m = n$?" The " $=$ " sign denotes a condition that can be tested, the " $\leftarrow$ " sign denotes an action that can be performed. The operation of *increasing n by one* is denoted by " $n \leftarrow n + 1$ " (read " n is replaced by $n + 1$ " or " n gets $n + 1$ "). In general, "variable $\leftarrow$ formula" means that the formula is to be computed using the present values of any variables appearing within it; then the result should replace the previous value of the variable at the left of the arrow. Persons untrained in computer work sometimes have a tendency to say " n becomes $n + 1$ " and to write " $n \rightarrow n + 1$ " for the operation of increasing n by one; this symbolism can only lead to confusion because of its conflict with standard conventions, and it should be avoided.

Notice that the order of actions in step E3 is important: "Set $m \leftarrow n$, $n \leftarrow r$ " is quite different from "Set $n \leftarrow r$, $m \leftarrow n$," since the latter would imply that the previous value of n is lost before it can be used to set m . Thus the latter sequence is equivalent to "Set $n \leftarrow r$, $m \leftarrow r$." When several variables are all to be set equal to the same quantity, we can use multiple arrows; for example, " $n \leftarrow r$, $m \leftarrow r$ " may be written " $n \leftarrow m \leftarrow r$." To interchange the values of two variables, we can write "Exchange $m \leftrightarrow n$ "; this action could also be specified by using a new variable t and writing "Set $t \leftarrow m$, $m \leftarrow n$, $n \leftarrow t$."

An algorithm starts at the lowest-numbered step, usually step 1, and it performs subsequent steps in sequential order unless otherwise specified. In step E3, the imperative "go back to step E1" specifies the computational order in an obvious fashion. In step E2, the action is prefaced by the condition "If $r = 0$ "; so if $r \neq 0$, the rest of that sentence does not apply and no action is specified. We might have added the redundant sentence, "If $r \neq 0$, go on to step E3."

The heavy vertical line "■" appearing at the end of step E3 is used to indicate the end of an algorithm and the resumption of text.

We have now discussed virtually all the notational conventions used in the algorithms of this book, except for a notation used to denote "subscripted" or

“indexed” items that are elements of an ordered array. Suppose we have n quantities, $v_1, v_2, \dots, v_n$; instead of writing v_j for the j th element, the notation $v[j]$ is often used. Similarly, $a[i, j]$ is sometimes used in preference to a doubly subscripted notation like a_{ij} . Sometimes multiple-letter names are used for variables, usually set in capital letters; thus TEMP might be the name of a variable used for temporarily holding a computed value, PRIME[K] might denote the K th prime number, and so on.

So much for the *form* of algorithms; now let us *perform* one. It should be mentioned immediately that the reader should *not* expect to read an algorithm as if it were part of a novel; such an attempt would make it pretty difficult to understand what is going on. An algorithm must be seen to be believed, and the best way to learn what an algorithm is all about is to try it. The reader should always take pencil and paper and work through an example of each algorithm immediately upon encountering it in the text. Usually the outline of a worked example will be given, or else the reader can easily conjure one up. This is a simple and painless way to gain an understanding of a given algorithm, and all other approaches are generally unsuccessful.

Let us therefore work out an example of Algorithm E. Suppose that we are given $m = 119$ and $n = 544$; we are ready to begin, at step E1. (The reader should now follow the algorithm as we give a play-by-play account.) Dividing m by n in this case is quite simple, almost too simple, since the quotient is zero and the remainder is 119. Thus, $r \leftarrow 119$. We proceed to step E2, and since $r \neq 0$ no action occurs. In step E3 we set $m \leftarrow 544$, $n \leftarrow 119$. It is clear that if $m < n$ originally, the quotient in step E1 will always be zero and the algorithm will always proceed to interchange m and n in this rather cumbersome fashion. We could add a new step:

E0. [Ensure $m \geq n$.] If $m < n$, exchange $m \leftrightarrow n$.

This would make no essential change in the algorithm, except to increase its length slightly, and to decrease its running time in about one half of all cases.

Back at step E1, we find that $\frac{544}{119} = 4\frac{68}{119}$, so $r \leftarrow 68$. Again E2 is inapplicable, and at E3 we set $m \leftarrow 119$, $n \leftarrow 68$. The next round sets $r \leftarrow 51$, and ultimately $m \leftarrow 68$, $n \leftarrow 51$. Next $r \leftarrow 17$, and $m \leftarrow 51$, $n \leftarrow 17$. Finally, when 51 is divided by 17, we set $r \leftarrow 0$, so at step E2 the algorithm terminates. The greatest common divisor of 119 and 544 is 17.

So this is an algorithm. The modern meaning for algorithm is quite similar to that of *recipe*, *process*, *method*, *technique*, *procedure*, *routine*, *rigmarole*, except that the word “algorithm” connotes something just a little different. Besides merely being a finite set of rules that gives a sequence of operations for solving a specific type of problem, an algorithm has five important features:

1) *Finiteness*. An algorithm must always terminate after a finite number of steps. Algorithm E satisfies this condition, because after step E1 the value of r is *less* than n ; so if $r \neq 0$, the value of n *decreases* the next time step E1 is encountered. A decreasing sequence of positive integers must eventually terminate, so step E1 is executed only a finite number of times for any given original

value of n . Note, however, that the number of steps can become arbitrarily large; certain huge choices of m and n will cause step E1 to be executed more than a million times.

(A procedure that has all of the characteristics of an algorithm except that it possibly lacks finiteness may be called a *computational method*. Euclid originally presented not only an algorithm for the greatest common divisor of numbers, but also a very similar geometrical construction for the “greatest common measure” of the lengths of two line segments; this is a computational method that does not terminate if the given lengths are incommensurable. Another example of a nonterminating computational method is a *reactive process*, which continually interacts with its environment.)

2) *Definiteness*. Each step of an algorithm must be precisely defined; the actions to be carried out must be rigorously and unambiguously specified for each case. The algorithms of this book will hopefully meet this criterion, but they are specified in the English language, so there is a possibility that the reader might not understand exactly what the author intended. To get around this difficulty, formally defined *programming languages* or *computer languages* are designed for specifying algorithms, in which every statement has a very definite meaning. Many of the algorithms of this book will be given both in English and in a computer language. An expression of a computational method in a computer language is called a *program*.

In Algorithm E, the criterion of definiteness as applied to step E1 means that the reader is supposed to understand exactly what it means to divide m by n and what the remainder is. In actual fact, there is no universal agreement about what this means if m and n are not positive integers; what is the remainder of -8 divided by $-\pi$? What is the remainder of $59/13$ divided by zero? Therefore the criterion of definiteness means we must make sure that the values of m and n are always positive integers whenever step E1 is to be executed. This is initially true, by hypothesis; and after step E1, r is a nonnegative integer that must be nonzero if we get to step E3. So m and n are indeed positive integers as required.

3) *Input*. An algorithm has zero or more *inputs*: quantities that are given to it initially before the algorithm begins, or dynamically as the algorithm runs. These inputs are taken from specified sets of objects. In Algorithm E, for example, there are two inputs, namely m and n , both taken from the set of *positive integers*.

4) *Output*. An algorithm has one or more *outputs*: quantities that have a specified relation to the inputs. Algorithm E has one output, namely n in step E2, the greatest common divisor of the two inputs.

(We can easily *prove* that this number is indeed the greatest common divisor, as follows. After step E1, we have

$$m = qn + r,$$

for some integer q . If $r = 0$, then m is a multiple of n , and clearly in such a case n is the greatest common divisor of m and n . If $r \neq 0$, note that any number that divides both m and n must divide $m - qn = r$, and any number that divides

both n and r must divide $qn + r = m$; so the set of divisors of $\{m, n\}$ is the same as the set of divisors of $\{n, r\}$. In particular, the *greatest* common divisor of $\{m, n\}$ is the same as the greatest common divisor of $\{n, r\}$. Therefore step E3 does not change the answer to the original problem.)

5) *Effectiveness*. An algorithm is also generally expected to be *effective*, in the sense that its operations must all be sufficiently basic that they can in principle be done exactly and in a finite length of time by someone using pencil and paper. Algorithm E uses only the operations of dividing one positive integer by another, testing if an integer is zero, and setting the value of one variable equal to the value of another. These operations are effective, because integers can be represented on paper in a finite manner, and because there is at least one method (the “division algorithm”) for dividing one by another. But the same operations would *not* be effective if the values involved were arbitrary real numbers specified by an infinite decimal expansion, nor if the values were the lengths of physical line segments (which cannot be specified exactly). Another example of a noneffective step is, “If 4 is the largest integer n for which there is a solution to the equation $w^n + x^n + y^n = z^n$ in positive integers w, x, y , and z , then go to step E4.” Such a statement would not be an effective operation until someone successfully constructs an algorithm to determine whether 4 is or is not the largest integer with the stated property.

Let us try to compare the concept of an algorithm with that of a cookbook recipe. A recipe presumably has the qualities of finiteness (although it is said that a watched pot never boils), input (eggs, flour, etc.), and output (TV dinner, etc.), but it notoriously lacks definiteness. There are frequent cases in which a cook’s instructions are indefinite: “Add a dash of salt.” A “dash” is defined to be “less than $\frac{1}{8}$ teaspoon,” and salt is perhaps well enough defined; but where should the salt be added—on top? on the side? Instructions like “toss lightly until mixture is crumbly” or “warm cognac in small saucepan” are quite adequate as explanations to a trained chef, but an algorithm must be specified to such a degree that even a computer can follow the directions. Nevertheless, a computer programmer can learn much by studying a good recipe book. (The author has in fact barely resisted the temptation to name the present volume “The Programmer’s Cookbook.” Perhaps someday he will attempt a book called “Algorithms for the Kitchen.”)

We should remark that the finiteness restriction is not really strong enough for practical use. A useful algorithm should require not only a finite number of steps, but a *very* finite number, a reasonable number. For example, there is an algorithm that determines whether or not the game of chess can always be won by White if no mistakes are made (see exercise 2.2.3–28). That algorithm can solve a problem of intense interest to thousands of people, yet it is a safe bet that we will never in our lifetimes know the answer; the algorithm requires fantastically large amounts of time for its execution, even though it is finite. See also Chapter 8 for a discussion of some finite numbers that are so large as to actually be beyond comprehension.

In practice we not only want algorithms, we want algorithms that are *good* in some loosely defined aesthetic sense. One criterion of goodness is the length of time taken to perform the algorithm; this can be expressed in terms of the number of times each step is executed. Other criteria are the adaptability of the algorithm to different kinds of computers, its simplicity and elegance, etc.

We often are faced with several algorithms for the same problem, and we must decide which is best. This leads us to the extremely interesting and all-important field of *algorithmic analysis*: Given an algorithm, we want to determine its performance characteristics.

For example, let's consider Euclid's algorithm from this point of view. Suppose we ask the question, "Assuming that the value of n is known but m is allowed to range over all positive integers, what is the *average* number of times, T_n , that step E1 of Algorithm E will be performed?" In the first place, we need to check that this question does have a meaningful answer, since we are trying to take an average over infinitely many choices for m . But it is evident that after the first execution of step E1 only the remainder of m after division by n is relevant. So all we must do to find T_n is to try the algorithm for $m = 1$, $m = 2$, ..., $m = n$, count the total number of times step E1 has been executed, and divide by n .

Now the important question is to determine the *nature* of T_n ; is it approximately equal to $\frac{1}{3}n$, or $\sqrt{n}$, for instance? As a matter of fact, the answer to this question is an extremely difficult and fascinating mathematical problem, not yet completely resolved, which is examined in more detail in Section 4.5.3. For large values of n it is possible to prove that T_n is approximately $(12(\ln 2)/\pi^2) \ln n$, that is, proportional to the *natural logarithm* of n , with a constant of proportionality that might not have been guessed offhand! For further details about Euclid's algorithm, and other ways to calculate the greatest common divisor, see Section 4.5.2.

Analysis of algorithms is the name the author likes to use to describe investigations such as this. The general idea is to take a particular algorithm and to determine its quantitative behavior; occasionally we also study whether or not an algorithm is optimal in some sense. The *theory of algorithms* is another subject entirely, dealing primarily with the existence or nonexistence of effective algorithms to compute particular quantities.

So far our discussion of algorithms has been rather imprecise, and a mathematically oriented reader is justified in thinking that the preceding commentary makes a very shaky foundation on which to erect any theory about algorithms. We therefore close this section with a brief indication of one method by which the concept of algorithm can be firmly grounded in terms of mathematical set theory. Let us formally define a *computational method* to be a quadruple (Q, I, Ω, f) , in which Q is a set containing subsets I and Ω , and f is a function from Q into itself. Furthermore f should leave Ω pointwise fixed; that is, $f(q)$ should equal q for all elements q of Ω . The four quantities Q , I , Ω , f are intended to represent respectively the states of the computation, the input, the output, and the computational rule. Each input x in the set I defines a *computational*

sequence, $x_0, x_1, x_2, \dots$, as follows:

$$x_0 = x \quad \text{and} \quad x_{k+1} = f(x_k) \quad \text{for} \quad k \geq 0. \quad (1)$$

The computational sequence is said to *terminate in k steps* if k is the smallest integer for which x_k is in Ω , and in this case it is said to produce the output x_k from x . (Note that if x_k is in Ω , so is x_{k+1} , because $x_{k+1} = x_k$ in such a case.) Some computational sequences may never terminate; an *algorithm* is a computational method that terminates in finitely many steps for all x in I .

Algorithm E may, for example, be formalized in these terms as follows: Let Q be the set of all singletons (n) , all ordered pairs (m, n) , and all ordered quadruples $(m, n, r, 1)$, $(m, n, r, 2)$, and $(m, n, p, 3)$, where m , n , and p are positive integers and r is a nonnegative integer. Let I be the subset of all pairs (m, n) and let Ω be the subset of all singletons (n) . Let f be defined as follows:

$$\begin{aligned} f((m, n)) &= (m, n, 0, 1); & f((n)) &= (n); \\ f((m, n, r, 1)) &= (m, n, \text{remainder of } m \text{ divided by } n, 2); \\ f((m, n, r, 2)) &= (n) \quad \text{if } r = 0, & (m, n, r, 3) & \text{ otherwise;} \\ f((m, n, p, 3)) &= (n, p, p, 1). \end{aligned} \quad (2)$$

The correspondence between this notation and Algorithm E is evident.

This formulation of the concept of an algorithm does not include the restriction of effectiveness mentioned earlier. For example, Q might denote infinite sequences that are not computable by pencil and paper methods, or f might involve operations that mere mortals cannot always perform. If we wish to restrict the notion of algorithm so that only elementary operations are involved, we can place restrictions on Q , I , Ω , and f , for example as follows: Let A be a finite set of letters, and let A^* be the set of all strings on A (the set of all ordered sequences $x_1x_2\dots x_n$, where $n \geq 0$ and x_j is in A for $1 \leq j \leq n$). The idea is to encode the states of the computation so that they are represented by strings of A^* . Now let N be a nonnegative integer and let Q be the set of all (σ, j) , where σ is in A^* and j is an integer, $0 \leq j \leq N$; let I be the subset of Q with $j = 0$ and let Ω be the subset with $j = N$. If θ and σ are strings in A^* , we say that θ occurs in σ if σ has the form $\alpha\theta\omega$ for strings α and ω . To complete our definition, let f be a function of the following type, defined by the strings θ_j , ϕ_j and the integers a_j , b_j for $0 \leq j < N$:

$$\begin{aligned} f(\sigma, j) &= (\sigma, a_j) & \text{if } \theta_j \text{ does not occur in } \sigma; \\ f(\sigma, j) &= (\alpha\phi_j\omega, b_j) & \text{if } \alpha \text{ is the shortest possible string for which } \sigma = \alpha\theta_j\omega; \\ f(\sigma, N) &= (\sigma, N). \end{aligned} \quad (3)$$

Such a computational method is clearly effective, and experience shows that it is also powerful enough to do anything we can do by hand. There are many other essentially equivalent ways to formulate the concept of an effective computational method (for example, using Turing machines). The formulation above is virtually the same as that given by A. A. Markov in his book *The*

Theory of Algorithms [Trudy Mat. Inst. Akad. Nauk **42** (1954), 1–376], later revised and enlarged by N. M. Nagorny (Moscow: Nauka, 1984; English edition, Dordrecht: Kluwer, 1988).

EXERCISES

1. [10] The text showed how to interchange the values of variables m and n , using the replacement notation, by setting $t \leftarrow m$, $m \leftarrow n$, $n \leftarrow t$. Show how the values of four variables (a, b, c, d) can be rearranged to (b, c, d, a) by a sequence of replacements. In other words, the new value of a is to be the original value of b , etc. Try to use the minimum number of replacements.
2. [15] Prove that m is always greater than n at the beginning of step E1, except possibly the first time this step occurs.
3. [20] Change Algorithm E (for the sake of efficiency) so that all trivial replacement operations such as " $m \leftarrow n$ " are avoided. Write this new algorithm in the style of Algorithm E, and call it Algorithm F.
4. [16] What is the greatest common divisor of 2166 and 6099?
- 5. [12] Show that the "Procedure for Reading This Set of Books" that appears in the preface actually fails to be a genuine algorithm on three of our five counts! Also mention some differences in format between it and Algorithm E.
6. [20] What is T_5 , the average number of times step E1 is performed when $n = 5$?
- 7. [M21] Suppose that m is known and n is allowed to range over all positive integers; let U_m be the average number of times that step E1 is executed in Algorithm E. Show that U_m is well defined. Is U_m in any way related to T_m ?
8. [M25] Give an "effective" formal algorithm for computing the greatest common divisor of positive integers m and n , by specifying θ_j , ϕ_j , a_j , b_j as in Eqs. (3). Let the input be represented by the string $a^m b^n$, that is, m a 's followed by n b 's. Try to make your solution as simple as possible. [Hint: Use Algorithm E, but instead of division in step E1, set $r \leftarrow |m - n|$, $n \leftarrow \min(m, n)$.]
- 9. [M30] Suppose that $C_1 = (Q_1, I_1, \Omega_1, f_1)$ and $C_2 = (Q_2, I_2, \Omega_2, f_2)$ are computational methods. For example, C_1 might stand for Algorithm E as in Eqs. (2), except that m and n are restricted in magnitude, and C_2 might stand for a computer program implementation of Algorithm E. (Thus Q_2 might be the set of all states of the machine, i.e., all possible configurations of its memory and registers; f_2 might be the definition of single machine actions; and I_2 might be the initial state, including the program for determining the greatest common divisor as well as the values of m and n .)
Formulate a set-theoretic definition for the concept " C_2 is a representation of C_1 " or " C_2 simulates C_1 ." This is to mean intuitively that any computation sequence of C_1 is mimicked by C_2 , except that C_2 might take more steps in which to do the computation and it might retain more information in its states. (We thereby obtain a rigorous interpretation of the statement, "Program X is an implementation of Algorithm Y .")

Toward a Philosophy of Technology

by HANS JONAS

Are there philosophical aspects to technology? Of course there are, as there are to all things of importance in human endeavor and destiny. Modern technology touches on almost everything vital to man's existence—material, mental, and spiritual. Indeed, what of man is *not* involved? The way he lives his life and looks at objects, his intercourse with the world and with his peers, his powers and modes of action, kinds of goals, states and changes of society, objectives and forms of politics (including warfare no less than welfare), the sense and quality of life, even man's fate and that of his environment: all these are involved in the technological enterprise as it extends in magnitude and depth. The mere enumeration suggests a staggering host of potentially philosophical themes.

To put it bluntly: if there is a philosophy of science, language, history, and art; if there is social, political, and moral philosophy; philosophy of thought and of action, of reason and passion, of decision and value—all facets of the inclusive philosophy of man—how then could there not be a philosophy of technology, the focal fact of modern life? And at that a philosophy so spacious that it can house portions from all the other branches of philosophy? It is almost a truism, but at the same time so immense a proposition that its challenge staggers the mind. Economy and modesty require that we select, for a beginning, the most obvious from the multitude of aspects that invite philosophical attention.

The old but useful distinction of "form" and "matter" allows us to distinguish between these two major themes: (1) the *formal dynamics* of technology as a continuing collective enterprise, which advances by its own "laws of motion"; and (2) the *substantive content* of technology in terms of the things it puts into human use, the powers it confers, the novel objectives it opens up or dictates, and the altered manner of human action by which these objectives are realized.

The first theme considers technology as an abstract whole of movement; the second considers its concrete uses and their impact on our world and our lives. The formal approach will try to grasp the pervasive "process properties" by which modern technology propels itself—through our agency, to be sure—into ever-succeeding and superceding novelty. The material approach will look at the species of novelties themselves, their taxonomy, as it were, and try to make out how

the world furnished with them looks. A third, overarching theme is the *moral* side of technology as a burden on human responsibility, especially its long-term effects on the global condition of man and environment. This—my own main preoccupation over the past years—will only be touched upon.

I. The Formal Dynamics of Technology

First some observations about technology's form as an abstract whole of movement. We are concerned with characteristics of *modern* technology and therefore ask first what distinguishes it *formally* from all previous technology. One major distinction is that modern technology is an enterprise and process, whereas earlier technology was a possession and a state. If we roughly describe technology as comprising the use of artificial implements for the business of life, together with their original invention, improvement, and occasional additions, such a tranquil description will do for most of technology through mankind's career (with which it is coeval), but not for modern technology. In the past, generally speaking, a given inventory of tools and procedures used to be fairly constant, tending toward a mutually adjusting, stable equilibrium of ends and means, which—once established—represented for lengthy periods an unchallenged optimum of technical competence.

To be sure, revolutions occurred, but more by accident than by design. The agricultural revolution, the metallurgical revolution that led from the neolithic to the iron age, the rise of cities, and such developments, *happened* rather than were consciously created. Their pace was so slow that only in the time-contraction of historical retrospect do they appear to be "revolutions" (with the misleading connotation that their contemporaries experienced them as such). Even where the change was sudden, as with the introduction first of the chariot, then of armed horsemen into warfare—a violent, if short-lived, revolution indeed—the innovation did not originate from within the military art of the advanced societies that it affected, but was thrust on it from outside by the (much less civilized) peoples of Central Asia. Instead of spreading through the technological universe of their time, other technical breakthroughs, like Phoenician purple-dyeing, Byzantine "greek fire," Chinese porcelain and silk, and Damascus steel-tempering, remained jealously guarded monopolies of the inventor communities. Still others, like the hydraulic and steam playthings of Alexandrian mechanics, or compass and gunpowder of the Chinese, passed unnoticed in their serious technological potentials.¹

On the whole (not counting rare upheavals), the great classical civilizations had comparatively early reached a point of technological saturation—the aforementioned "optimum" in equilibrium of means with acknowledged needs

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and goals—and had little cause later to go beyond it. From there on, convention reigned supreme. From pottery to monumental architecture, from food growing to shipbuilding, from textiles to engines of war, from time measuring to stargazing: tools, techniques, and objectives remained essentially the same over long times; improvements were sporadic and unplanned. Progress therefore—if it occurred at all*—was by inconspicuous increments to a universally high level that still excites our admiration and, in historical fact, was more liable to regression than to surpassing. The former at least was the more noted phenomenon, deplored by the epigones with a nostalgic remembrance of a better past (as in the declining Roman world). More important, there was, even in the best and most vigorous times, no proclaimed *idea* of a future of *constant progress* in the arts. Most important, there was never a deliberate method of going about it like “research,” the willingness to undergo the risks of trying unorthodox paths, exchanging information widely about the experience, and so on. Least of all was there a “natural science” as a growing body of theory to guide such semi-theoretical, prepractical activities, plus their social institutionalization. In routines as well as panoply of instruments, accomplished as they were for the purposes they served, the “arts” seemed as settled as those purposes themselves.**

Traits of Modern Technology

The exact opposite of this picture holds for modern technology, and this is its first philosophical aspect. Let us begin with some manifest traits.

1. Every new step in whatever direction of whatever technological field tends *not* to approach an equilibrium or saturation point in the process of fitting means to ends (nor is it meant to), but, on the contrary, to give rise, if successful, to further steps in all kinds of direction and with a fluidity of the ends themselves. “Tends to” becomes a compelling “is bound to” with any major or important step (this almost being its criterion); and the innovators themselves expect, beyond the accomplishment, each time, of their immediate task, the constant future repetition of their inventive activity.

2. Every technical innovation is sure to spread quickly through the technological world community, as also do theoretical discoveries in the sciences. The spreading is in terms of knowledge and of practical adoption, the first (and

its speed) guaranteed by the universal intercommunication that is itself part of the technological complex, the second enforced by the pressure of competition.

3. The relation of means to ends is not unilinear but circular. Familiar ends of long standing may find better satisfaction by new technologies whose genesis they had inspired. But equally—and increasingly typical—new technologies may suggest, create, even impose new ends, never before conceived, simply by offering their feasibility. (Who had ever wished to have in his living room the Philharmonic orchestra, or open heart surgery, or a helicopter defoliating a Vietnam forest? or to drink his coffee from a disposable plastic cup? or to have artificial insemination, test-tube babies, and host pregnancies? or to see clones of himself and others walking about?) Technology thus adds to the very objectives of human desires, including objectives for technology itself. The last point indicates the dialectics or circularity of the case: once incorporated into the socioeconomic demand diet, ends first gratuitously (perhaps accidentally) generated by technological invention become necessities of life and set technology the task of further perfecting the means of realizing them.

4. Progress, therefore, is not just an ideological gloss on modern technology, and not at all a mere option offered by it, but an inherent drive which acts willy-nilly in the formal automatics of its *modus operandi* as it interacts with society. “Progress” is here not a value term but purely descriptive. We may resent the fact and despise its fruits and yet must go along with it, for—short of a stop by the fiat of total political power, or by a sustained general strike of its clients or some internal collapse of their societies, or by self-destruction through its works (the last, alas, the least unlikely of these)—the juggernaut moves on relentlessly, spawning its always mutated progeny by coping with the challenges and lures of the now. But while not a value term, “progress” here is not a neutral term either, for which we could simply substitute “change.” For it is in the nature of the case, or a law of the series, that a later stage is always, in terms of technology itself, *superior* to the preceding stage.* Thus we have here a case of the entropy-defying sort (organic evolution is another), where the internal motion of a system, left to itself and not interfered with, leads to ever “higher,” not “lower” states of itself. Such at least is the present evidence.** If Napoleon once said, “Politics is destiny,” we may well say today, “Technology is destiny.”

These points go some way to explicate the initial statement that modern technology, unlike traditional, is an enterprise and not a possession, a process and not a state, a dynamic thrust and not a set of implements and skills. And they already adumbrate certain “laws of motion” for this restless phenomenon. What we have described, let us remember, were formal traits which as yet say little about the

*Progress did, in fact, occur even at the heights of classical civilizations. The Roman arch and vault, for example, were distinct engineering advances over the horizontal entablature and flat ceiling of Greek (and Egyptian) architecture, permitting spanning feats and thereby construction objectives not contemplated before (stone bridges, aqueducts, the vast baths and other public halls of Imperial Rome). But materials, tools, and techniques were still the same, the role of human labor and crafts remained unaltered, stonemasonry and brickmaking went on as before. An existing technology was enlarged in its scope of performance, but none of its means or even goals made obsolete.

**One meaning of “classical” is that those civilizations had somehow implicitly “defined” themselves and neither encouraged nor even allowed to pass beyond their innate terms. The—more or less—achieved “equilibrium” was their very pride.

*This only seems to be but is not a value statement, as the reflection on, for example, an ever more destructive atom bomb shows.

**There may conceivably be internal degenerative factors—such as the overloading of finite information-processing capacity—that may bring the (exponential) movement to a halt or even make the system fall apart. We don’t know yet.

contents of the enterprise. We ask two questions of this descriptive picture: *why* is this so, that is, what *causes* the restlessness of modern technology; what is the nature of the thrust? And, what is the philosophical import of the facts so explained?

The Nature of Restless Technology

As we would expect in such a complex phenomenon, the motive forces are many, and some causal hints appeared already in the descriptive account. We have mentioned *pressure of competition*—for profit, but also for power, security, and so forth—as one perpetual mover in the universal appropriation of technical improvements. It is equally operative in their origination, that is, in the process of invention itself, nowadays dependent on constant outside subsidy and even goal-setting: potent interests see to both. War, or the threat of it, has proved an especially powerful agent. The less dramatic, but no less compelling, everyday agents are legion. To keep one's head above the water is their common principle (somewhat paradoxical, in view of an abundance already far surpassing what former ages would have lived with happily ever after). Of pressures other than the competitive ones, we must mention those of population growth and of impending exhaustion of natural resources. Since both phenomena are themselves already by-products of technology (the first by way of medical improvements, the second by the voracity of industry), they offer a good example of the more general truth that to a considerable extent technology itself begets the problems which it is then called upon to overcome by a new forward jump. (The Green Revolution and the development of synthetic substitute materials or of alternate sources of energy come under this heading.) These compulsive pressures for progress, then, would operate even for a technology in a noncompetitive, for example, a socialist setting.

A motive force more autonomous and spontaneous than these almost mechanical pushes with their "sink or swim" imperative would be the pull of the quasi-utopian *vision* of an ever better life, whether vulgarly conceived or nobly, once technology had proved the open-ended capacity for procuring the conditions for it: perceived possibility whetting the appetite ("the American dream," "the revolution of rising expectations"). This less palpable factor is more difficult to appraise, but its playing a role is undeniable. Its deliberate fostering and manipulation by the dream merchants of the industrial-mercantile complex is yet another matter and somewhat taints the spontaneity of the motive, as it also degrades the quality of the dream. It is also moot to what extent the vision itself is *post hoc* rather than *ante hoc*, that is, instilled by the dazzling feats of a technological progress already underway and thus more a response to than a motor of it.

Groping in these obscure regions of motivation, one may as well descend, for an explanation of the dynamism as such, into the Spenglerian mystery of a "Faustian soul" innate in Western culture, that drives it, nonrationally, to infinite novelty and unplumbed possibilities for their own sake; or into the Heideggerian depths of a fateful, metaphysical de-

cision of the will for boundless power over the world of things—a decision equally peculiar to the Western mind: speculative intuitions which do strike a resonance in us, but are beyond proof and disproof.

Surfacing once more, we may also look at the very sober, functional facts of industrialism as such, of production and distribution, output maximization, managerial and labor aspects, which even apart from competitive pressure provide their own incentives for technical progress. Similar observations apply to the requirements of *rule* or control in the vast and populous states of our time, those giant territorial super-organisms which for their very cohesion depend on advanced technology (for example, in information, communication, and transportation, not to speak of weaponry) and thus have a stake in its promotion: the more so, the more centralized they are. This holds for socialist systems no less than for free-market societies. May we conclude from this that even a communist world state, freed from external rivals as well as from internal free-market competition, might still have to push technology ahead for purposes of control on this colossal scale? Marxism, in any case, has its own inbuilt commitment to technological progress beyond necessity. But even disregarding all dynamics of these conjectural kinds, the most monolithic case imaginable would, at any rate, still be exposed to those noncompetitive, natural pressures like population growth and dwindling resources that beset industrialism as such. Thus, it seems, the compulsive element of technological progress may not be bound to its original breeding ground, the capitalist system. Perhaps the odds for an eventual stabilization look somewhat better in a socialist system, provided it is worldwide—and possibly totalitarian in the bargain. As it is, the pluralism we are thankful for ensures the constancy of compulsive advance.

We could go on unravelling the causal skein and would be sure to find many more strands. But none nor all of them, much as they explain, would go to the heart of the matter. For all of them have one premise in common without which they could not operate for long: the premise that there *can* be indefinite progress because there *is* always something new and better to find. The, by no means obvious, givenness of this objective condition is also the pragmatic conviction of the performers in the technological drama; but without its being true, the conviction would help as little as the dream of the alchemists. Unlike theirs, it is backed up by an impressive record of past successes, and for many this is sufficient ground for their belief. (Perhaps holding or not holding it does not even greatly matter.) What makes it more than a sanguine belief, however, is an underlying and well-grounded, theoretical view of the nature of things and of human cognition, according to which they do not set a limit to novelty of discovery and invention, indeed, that they of themselves will at each point offer another opening for the as yet unknown and undone. The corollary conviction, then, is that a technology tailored to a nature and to a knowledge of this indefinite potential ensures its indefinitely continued conversion into the practical powers, each step of it begetting the next, with never a cutoff from internal exhaustion of possibilities.

Only habituation dulls our wonder at this wholly unprecedented belief in virtual "infinity." And by all our present comprehension of reality, the belief is most likely true—at least enough of it to keep the road for innovative technology in the wake of advancing science open for a long time ahead. Unless we understand this ontologic-epistemological premise, we have not understood the inmost agent of technological dynamics, on which the working of all the adventitious causal factors is contingent in the long run.

Let us remember that the virtual infinitude of advance we here seek to explain is in essence different from the always avowed perfectibility of every human accomplishment. Even the undisputed master of his craft always had to admit as possible that he might be surpassed in skill or tools or materials; and no excellence of product ever foreclosed that it might still be bettered, just as today's champion runner must know that his time may one day be beaten. But these are improvements within a given genus, not different in kind from what went before, and they must accrue in diminishing fractions. Clearly, the phenomenon of an exponentially growing *generic* innovation is qualitatively different.

Science as a Source of Restlessness

The answer lies in the interaction of *science* and *technology* that is the hallmark of modern progress, and thus ultimately in the kind of nature which modern science progressively discloses. For it is here, in the movement of *knowledge*, where relevant novelty first and constantly occurs. This is itself a novelty. To Newtonian physics, nature appeared simple, almost crude, running its show with a few kinds of basic entities and forces by a few universal laws, and the application of those well-known laws to an ever greater variety of composite phenomena promised ever widening knowledge indeed, but no real surprises. Since the mid-nineteenth century, this minimalistic and somehow finished picture of nature has changed with breathtaking acceleration. In a reciprocal interplay with the growing subtlety of exploration (instrumental and conceptual), nature itself stands forth as ever more subtle. The progress of probing makes the object grow richer in modes of operation, not sparer as classical mechanics had expected. And instead of narrowing the margin of the still-undiscovered, science now surprises itself with unlocking dimension after dimension of new depths. The very essence of matter has turned from a blunt, irreducible ultimate to an always reopened challenge for further penetration. No one can say whether this will go on forever, but a suspicion of intrinsic infinity in the very being of things obtrudes itself and therewith an anticipation of unending inquiry of the sort where succeeding steps will not find the same old story again (Descartes' "matter in motion"), but always add new twists to it. If then the art of technology is correlative to the knowledge of nature, technology too acquires from this source that potential of infinity for its innovative advance.

But it is not just that indefinite scientific progress offers the *option* of indefinite technological progress, to be exercised or not as other interests see fit. Rather the cognitive

process itself moves by interaction with the technological, and in the most internally vital sense: for its own *theoretical* purpose, science must generate an increasingly sophisticated and physically formidable technology as its tool. What it finds with this help initiates new departures in the practical sphere, and the latter as a whole, that is, technology at work provides with its experiences a large-scale laboratory for science again, a breeding ground for new questions, and so on in an unending cycle. In brief, a mutual feedback operates between science and technology; each requires and propels the other; and as matters now stand, they can only live together or must die together. For the dynamics of technology, with which we are here concerned, this means that (all external promptings apart) an agent of restlessness is implanted in it by its functionally integral bond with science. As long, therefore, as the cognitive impulse lasts, technology is sure to move ahead with it. The cognitive impulse, in its turn, culturally vulnerable in itself, liable to lag or to grow conservative with a treasured canon—that theoretical eros itself no longer lives on the delicate appetite for truth alone, but is spurred on by its hardier offspring, technology, which communicates to it impulses from the broadest arena of struggling, insistent life. Intellectual curiosity is seconded by interminably self-renewing practical aim.

I am conscious of the conjectural character of some of these thoughts. The revolutions in science over the last fifty years or so are a fact, and so are the revolutionary style they imparted to technology and the reciprocity between the two concurrent streams (nuclear physics is a good example). But whether those scientific revolutions, which hold primacy in the whole syndrome, will be typical for science henceforth—something like a law of motion for its future—or represent only a singular phase in its longer run, is unsure. To the extent, then, that our forecast of incessant novelty for technology was predicated on a guess concerning the future of science, even concerning the nature of things, it is hypothetical, as such extrapolations are bound to be. But even if the recent past did not usher in a state of permanent revolution for science, and the life of theory settles down again to a more sedate pace, the scope for technological innovation will not easily shrink; and what may no longer be a revolution in science, may still revolutionize our lives in its practical impact through technology. "Infinity" being too large a word anyway, let us say that present signs of potential and of incentives point to an indefinite perpetuation and fertility of the technological momentum.

The Philosophical Implications

It remains to draw philosophical conclusions from our findings, at least to pinpoint aspects of philosophical interest. Some preceding remarks have already been straying into philosophy of science in the technical sense. Of broader issues, two will be ample to provide food for further thought beyond the limitations of this paper. One concerns the status of knowledge in the human scheme, the other the status of technology itself as a human goal, or its tendency to become that from being a means, in a dialectical inversion of the means-end order itself.

Concerning knowledge, it is obvious that the time-honored division of theory and practice has vanished for both sides. The thirst for pure knowledge may persist undiminished, but the involvement of knowing at the heights with doing in the lowlands of life, mediated by technology, has become inextricable; and the aristocratic self-sufficiency of knowing for its own (and the knower's) sake has gone. Nobility has been exchanged for utility. With the possible exception of philosophy, which still can do with paper and pen and tossing thoughts around among peers, all knowledge has become thus tainted, or elevated if you will, whether utility is intended or not. The technological syndrome, in other words, has brought about a thorough *socializing* of the theoretical realm, enlisting it in the service of common need. What used to be the freest of human choices, an extravagance snatched from the pressure of the world—the esoteric life of thought—has become part of the great public play of necessities and a prime necessity in the action of the play.* Remotest abstraction has become enmeshed with nearest concreteness. What this pragmatic functionalization of the once highest indulgence in impractical pursuits portends for the image of man, for the restructuring of a hallowed hierarchy of values, for the idea of “wisdom,” and so on, is surely a subject for philosophical pondering.

Concerning technology itself, its actual role in modern life (as distinct from the purely instrumental definition of technology as such) has made the relation of means and ends equivocal all the way up from the daily living to the very vocation of man. There could be no question in former technology that its role was that of humble servant—pride of workmanship and esthetic embellishment of the useful notwithstanding. The Promethean enterprise of modern technology speaks a different language. The word “enterprise” gives the clue, and its unendingness another. We have mentioned that the effect of its innovations is disequilibrating rather than equilibrating with respect to the balance of wants and supply, always breeding its own new wants. This in itself compels the constant attention of the best minds, engaging the full capital of human ingenuity for meeting challenge after challenge and seizing the new chances. It is psychologically natural for that degree of engagement to be invested with the dignity of dominant purpose. Not only does technology dominate our lives in fact, it nourishes also a belief in its being of predominant worth. The sheer grandeur of the enterprise and its seeming infinity inspire enthusiasm and fire ambition. Thus, in addition to spawning new ends (worthy or frivolous) from the mere invention of means, technology as a grand venture tends to establish *itself* as the transcendent end. At least the suggestion is there and casts

*There is a paradoxical side effect to this change of roles. That very science which forfeited its place in the domain of leisure to become a busy toiler in the field of common needs, creates by its toils a growing domain of leisure for the masses, who reap this with the other fruits of technology as an additional (and no less novel) article of forced consumption. Hence leisure, from a privilege of the few, has become a problem for the many to cope with. Science, not idle, provides for the needs of this idleness too: no small part of technology is spent on filling the leisure-time gap which technology itself has made a fact of life.

its spell on the modern mind. At its most modest, it means elevating *homo faber* to the essential aspect of man; at its most extravagant, it means elevating *power* to the position of his dominant and interminable goal. To become ever more masters of the world, to advance from power to power, even if only collectively and perhaps no longer by choice, *can* now be seen to be the chief vocation of mankind. Surely, this again poses philosophical questions that may well lead unto the uncertain grounds of metaphysics or of faith.

I here break off, arbitrarily, the formal account of the technological movement in general, which as yet has told us little of what the enterprise is about. To this subject I now turn, that is, to the new kinds of powers and objectives that technology opens to modern man and the consequently altered quality of human action itself.

II. The Material Works of Technology

Technology is a species of power, and we can ask questions about how and on what object any power is exercised. Adopting Aristotle's rule in *de anima* that for understanding a faculty one should begin with its objects, we start from them too—“objects” meaning both the visible *things* technology generates and puts into human use, and the *objectives* they serve. The objects of modern technology are first everything that had always been an object of human artifice and labor: food, clothing, shelter, implements, transportation—all the material necessities and comforts of life. The technological intervention changed at first not the product but its production, in speed, ease, and quantity. However, this is true only of the very first stage of the industrial revolution with which large-scale scientific technology began. For example, the cloth for the steam-driven looms of Lancashire remained the same. Even then, one significant new product was added to the traditional list—the machines themselves, which required an entire new industry with further subsidiary industries to build them. These novel entities, machines—at first capital goods only, not consumer goods—had from the beginning their own impact on man's symbiosis with nature by being consumers themselves. For example: steam-powered water pumps facilitated coal mining, required in turn extra coal for firing their boilers, more coal for the foundries and forges that made those boilers, more for the mining of the requisite iron ore, more for its transportation to the foundries, more—both coal and iron—for the rails and locomotives made in these same foundries, more for the conveyance of the foundries' product to the pitheads and return, and finally more for the distribution of the more abundant coal to the users outside this cycle, among which were increasingly still more machines spawned by the increased availability of coal. Lest it be forgotten over this long chain, we have been speaking of James Watt's modest steam engine for pumping water out of mine shafts. This syndrome of self-proliferation—by no means a linear chain but an intricate web of reciprocity—has been part of modern technology ever since. To generalize, technology exponentially increases man's drain on nature's resources (of substances and of energy), not only through the multiplication of the final goods for consumption, but also, and perhaps more so, through the production

and operation of its own mechanical means. And with these means—machines—it introduced a new category of goods, not for consumption, added to the furniture of our world. That is, among the objects of technology a prominent class is that of technological apparatus itself.

Soon other features also changed the initial picture of a merely mechanized production of familiar commodities. The final products reaching the consumer ceased to be the same, even if still serving the same age-old needs; new needs, or desires, were added by commodities of entirely new kinds which changed the habits of life. Of such commodities, machines themselves became increasingly part of the consumer's daily life to be used directly by himself, as an article not of production but of consumption. My survey can be brief as the facts are familiar.

New Kinds of Commodities

When I said that the cloth of the mechanized looms of Lancashire remained the same, everyone will have thought of today's synthetic fibre textiles for which the statement surely no longer holds. This is fairly recent, but the general phenomenon starts much earlier, in the synthetic dyes and fertilizers with which the chemical industry—the first to be wholly a fruit of science—began. The original rationale of these technological feats was substitution of artificial for natural materials (for reasons of scarcity or cost), with as nearly as possible the same properties for effective use. But we need only think of plastics to realize that art progressed from substitutes to the creation of really new substances with properties not so found in any natural one, raw or processed, thereby also initiating uses not thought of before and giving rise to new classes of objects to serve them. In chemical (molecular) engineering, man does more than in mechanical (molar) engineering which constructs machinery from natural materials; his intervention is deeper, redesigning the infra-patterns of nature, making substances to specification by arbitrary disposition of molecules. And this, be it noted, is done deductively from the bottom, from the thoroughly analyzed last elements, that is, in a real *via compositiva* after the completed *via resolutive*, very different from the long-known empirical practice of coaxing substances into new properties, as in metal alloys from the bronze age on. Artificiality or creative engineering with abstract construction invades the heart of matter. This, in molecular biology, points to further, awesome potentialities.

With the sophistication of molecular alchemy we are ahead of our story. Even in straightforward hardware engineering, right in the first blush of the mechanical revolution, the objects of use that came out of the factories did not really remain the same, even where the objectives did. Take the old objective of travel. Railroads and ocean liners are relevantly different from the stage coach and from the sailing ship, not merely in construction and efficiency but in the very feel of the user, making travel a different experience altogether, something one may do for its own sake. Airplanes, finally, leave behind any similarity with former conveyances, except the purpose of getting from here to there, with no experience

of what lies in between. And these instrumental objects occupy a prominent, even obtrusive place in our world, far beyond anything wagons and boats ever did. Also they are constantly subject to improvement of design, with obsolescence rather than wear determining their life span.

Or take the oldest, most static of artifacts: human habitation. The multistoried office building of steel, concrete, and glass is a qualitatively different entity from the wood, brick, and stone structures of old. With all that goes into it besides the structures as such—the plumbing and wiring, the elevators, the lighting, heating, and cooling systems—it embodies the end products of a whole spectrum of technologies and far-flung industries, where only at the remote sources human hands still meet with primary materials, no longer recognizable in the final result. The ultimate customer inhabiting the product is ensconced in a shell of thoroughly derivative artifacts (perhaps relieved by a nice piece of driftwood). This transformation into utter artificiality is generally, and increasingly, the effect of technology on the human environment, down to the items of daily use. Only in agriculture has the product so far escaped this transformation by the changed modes of its production. We still eat the meat and rice of our ancestors.*

Then, speaking of the commodities that technology injects into private use, there are machines themselves, those very devices of its own running, originally confined to the economic sphere. This unprecedented novum in the records of individual living started late in the nineteenth century and has since grown to a pervading mass phenomenon in the Western world. The prime example, of course, is the automobile, but we must add to it the whole gamut of household appliances—refrigerators, washers, dryers, vacuum cleaners—by now more common in the lifestyle of the general population than running water or central heating were one hundred years ago. Add lawn mowers and other power tools for home and garden: we are mechanized in our daily chores and recreations (including the toys of our children) with every expectation that new gadgets will continue to arrive.

These paraphernalia are machines in the precise sense that they perform work and consume energy, and their moving parts are of the familiar magnitudes of our perceptual world. But an additional and profoundly different category of technical apparatus was dropped into the lap of the private citizen, not labor-saving and work-performing, partly not even utilitarian, but—with minimal energy input—catering to the senses and the mind: telephone, radio, television, tape recorders, calculators, record players—all the domestic terminals of the electronics industry, the latest arrival on the technological scene. Not only by their insubstantial, mind-addressed output, also by the subvisible, not literally “me-

*Not so, objects my colleague Robert Heilbroner in a letter to me; “I’m sorry to tell you that meat and rice are both *profoundly* influenced by technology. Not even they are left untouched.” Correct, but they are at least generically the same (their really profound changes lie far back in the original breeding of domesticated strains from wild ones—as in the case of all cereal plants under cultivation). I am speaking here of an order of transformation in which the results bear no resemblance to the natural materials at their source, nor to any naturally occurring state of them.

chanical" physics of their functioning do these devices differ in kind from all the macroscopic, bodily moving machinery of the classical type. Before inspecting this momentous turn from power engineering, the hallmark of the first industrial revolution, to communication engineering, which almost amounts to a second industrial-technological revolution, we must take a look at its natural base: electricity.

In the march of technology to ever greater artificiality, abstraction, and subtlety, the unlocking of electricity marks a decisive step. Here is a universal force of nature which yet does not naturally appear to man (except in lightning). It is not a datum of uncontrived experience. Its very "appearance" had to wait for science, which contrived the experience for it. Here, then, a technology depended on science for the mere providing of its "object," the entity itself it would deal with—the first case where theory alone, not ordinary experience, wholly preceded practice (repeated later in the case of nuclear energy). And what sort of entity! Heat and steam are familiar objects of sensuous experience, their force bodily displayed in nature; the matter of chemistry is still the concrete, corporeal stuff mankind had always known. But electricity is an abstract object, disembodied, immaterial, unseen; in its usable form, it is entirely an artifact, generated in a subtle transformation from grosser forms of energy (ultimately from heat via motion). Its theory indeed had to be essentially complete before utilization could begin.

Revolutionary as electrical technology was in itself, its purpose was at first the by now conventional one of the industrial revolution in general: to supply motive power for the propulsion of machines. Its advantages lay in the unique versatility of the new force, the ease of its transmission, transformation and distribution—an unsubstantial commodity, no bulk, no weight, instantaneously delivered at the point of consumption. Nothing like it had ever existed before in man's traffic with matter, space, and time. It made possible the spread of mechanization to every home; this alone was a tremendous boost to the technological tide, at the same time hooking private lives into centralized public networks and thus making them dependent on the functioning of a total system as never before, in fact, for every moment. Remember, you cannot hoard electricity as you can coal and oil, or flour and sugar for that matter.

But something much more unorthodox was to follow. As we all know, the discovery of the universe of electromagnetics caused a revolution in theoretical physics that is still underway. Without it, there would be no relativity theory, no quantum mechanics, no nuclear and subnuclear physics. It also caused a revolution in technology beyond what it contributed, as we noted, to its classical program. The revolution consisted in the passage from electrical to electronic technology which signifies a new level of abstraction in means and ends. It is the difference between power and communication engineering. Its object, the most impalpable of all, is information. Cognitive instruments had been known before—sextant, compass, clock, telescope, microscope, thermometer, all of them for information and not for work. At one time, they were called "philosophical" or "metaphysical" instruments. By the same general criterion, amusing as it may seem, the new electronic information devices, too, could

be classed as "philosophical instruments." But those earlier cognitive devices, except the clock, were inert and passive, not generating information actively, as the new instrumentalities do.

Theoretically as well as practically, electronics signifies a genuinely new phase of the scientific-technological revolution. Compared with the sophistication of its theory as well as the delicacy of its apparatus, everything which came before seems crude, almost natural. To appreciate the point, take the man-made satellites now in orbit. In one sense, they are indeed an imitation of celestial mechanics—Newton's laws finally verified by cosmic experiment: astronomy, for millenia the most purely contemplative of the physical sciences, turned into a practical art! Yet, amazing as it is, the astronomic imitation, with all the unleashing of forces and the finesse of techniques that went into it, is the least interesting aspect of those entities. In that respect, they still fall within the terms and feats of classical mechanics (except for the remote-control course corrections).

Their true interest lies in the instruments they carry through the voids of space and in what these do, their measuring, recording, analyzing, computing, their receiving, processing, and transmitting abstract information and even images over cosmic distances. There is nothing in all nature which even remotely foreshadows the kind of things that now ride the heavenly spheres. Man's imitative practical astronomy merely provides the vehicle for something else with which he sovereignly passes beyond all the models and usages of known nature.* That the advent of man portended, in its inner secret of mind and will, a cosmic event was known to religion and philosophy: now it manifests itself as such by fact of things and acts in the visible universe. Electronics indeed creates a range of objects imitating nothing and progressively added to by pure invention.

And no less invented are the ends they serve. Power engineering and chemistry for the most part still answered to the natural needs of man: for food, clothing, shelter, locomotion, and so forth. Communication engineering answers to needs of information and control solely created by the civilization that made this technology possible and, once started, imperative. The novelty of the means continues to engender no less novel ends—both becoming as necessary to the functioning of the civilization that spawned them as they would have been pointless for any former one. The world they help to constitute and which needs computers for its very running is no longer nature supplemented, imitated, improved, transformed, the original habitat made more habitable. In the pervasive mentalization of physical relationships it is a *trans-nature* of human making, but with this inherent paradox: that it threatens the obsolescence of man himself, as increasing automation ousts him from the places of work where he formerly proved his humanhood. And there is a further threat: its strain on nature herself may reach a breaking point.

*Note also that in radio technology, the medium of action is nothing material, like wires conducting currents, but the entirely immaterial electromagnetic "field," i.e., space itself. The symbolic picture of "waves" is the last remaining link to the forms of our perceptual world.

The Last Stage of the Revolution?

That sentence would make a good dramatic ending. But it is not the end of the story. There may be in the offing another, conceivably the last, stage of the technological revolution, after the mechanical, chemical, electrical, electronic stages we have surveyed, and the nuclear we omitted. All these were based on physics and had to do with what man can put to his use. What about biology? And what about the user himself? Are we, perhaps, on the verge of a technology, based on biological knowledge and wielding an engineering art which, this time, has man himself for its object? This has become a theoretical possibility with the advent of molecular biology and its understanding of genetic programming; and it has been rendered morally possible by the metaphysical neutralizing of man. But the latter, while giving us the license to do as we wish, at the same time denies us the guidance for knowing what to wish. Since the same evolutionary doctrine of which genetics is a cornerstone has deprived us of a valid image of man, the actual techniques, when they are ready, may find us strangely unready for their responsible use. The anti-essentialism of prevailing theory, which knows only of *de facto* outcomes of evolutionary accident and of no valid essences that would give sanction to them, surrenders our being to a freedom without norms. Thus the technological call of the new microbiology is the twofold one of physical feasibility and metaphysical admissibility. Assuming the genetic mechanism to be completely analyzed and its script finally decoded, we can set about rewriting the text. Biologists vary in their estimates of how close we are to the capability; few seem to doubt the right to use it. Judging by the rhetoric of its prophets, the idea of taking our evolution into our own hands is intoxicating even to many scientists.

In any case, the idea of making over man is no longer fantastic, nor interdicted by an inviolable taboo. If and when *that* revolution occurs, if technological power is really going to tinker with the elemental keys on which life will have to play its melody in generations of men to come (perhaps the only such melody in the universe), then a reflection on what is humanly desirable and what should determine the choice—a reflection, in short, on the image of man, becomes an imperative more urgent than any ever inflicted on the understanding of mortal man. Philosophy, it must be confessed, is sadly unprepared for this, its first cosmic task.

III. Toward an Ethics of Technology

The last topic has moved naturally from the descriptive and analytic plane, on which the objects of technology are displayed for inspection, onto the evaluative plane where their ethical challenge poses itself for decision. The particular case forced the transition so directly because there the (as yet hypothetical) technological object was man directly. But once removed, man is involved in all the other objects of technology, as these singly and jointly remake the worldly frame of his life, in both the narrower and the wider of its senses: that of the artificial frame of civilization in

which social man leads his life proximately, and that of the natural terrestrial environment in which this artifact is embedded and on which it ultimately depends.

Again, because of the magnitude of technological effects on both these vital environments in their totality, both the quality of human life and its very preservation in the future are at stake in the rampage of technology. In short, certainly the "image" of man, and possibly the survival of the species (or of much of it), are in jeopardy. This would summon man's duty to his cause even if the jeopardy were not of his own making. But it is, and, in addition to his ageless obligation to meet the threat of things, he bears for the first time the responsibility of prime agent in the threatening disposition of things. Hence nothing is more natural than the passage from the objects to the ethics of technology, from the things made to the duties of their makers and users.

A similar experience of inevitable passage from analysis of fact to ethical significance, let us remember, befell us toward the end of the first section. As in the case of the matter, so also in the case of the form of the technological dynamics, the image of man appeared at stake. In view of the quasi-automatic compulsion of those dynamics, with their perspective of indefinite progression, every existential and moral question that the objects of technology raise assumes the curiously eschatological quality with which we are becoming familiar from the extrapolating guesses of futurology. But apart from thus raising all challenges of present particular matter to the higher powers of future exponential magnification, the despotic dynamics of the technological movement as such, sweeping its captive movers along in its breathless momentum, poses its own questions to man's axiological conception of himself. Thus, form and matter of technology alike enter into the dimension of ethics.

The questions raised for ethics by the objects of technology are defined by the major areas of their impact and thus fall into such fields of knowledge as ecology (with all its biospheric subdivisions of land, sea, and air), demography economics, biomedical and behavioral sciences (even the psychology of mind pollution by television), and so forth. Not even a sketch of the substantive problems, let alone of ethical policies for dealing with them, can here be attempted. Clearly, for a normative rationale of the latter, ethical theory must plumb the very foundations of value, obligation, and the human good.

The same holds of the different kind of questions raised for ethics by the sheer fact of the formal dynamics of technology. But here, a question of another order is added to the straightforward ethical questions of both kinds, subjecting any resolution of them to a pragmatic proviso of harrowing uncertainty. Given the mastery of the creation over its creators, which yet does not abrogate their responsibility nor silence their vital interest, what are the chances and what are the means of gaining *control* of the process, so that the results of any ethical (or even purely prudential) insights can be translated into effective action? How in short can man's freedom prevail against the determinism he has created for himself? On this most clouded question, whereby hangs not only the effectuality or futility of the ethical search which the facts invite (assuming it to be blessed with *theoretical* suc-

cess!), but perhaps the future of mankind itself, I will make a few concluding, but—alas—inconclusive, remarks. They are intended to touch on the whole ethical enterprise.

Problematic Preconditions of an Effective Ethics

First, a look at the novel state of determinism. *Prima facie*, it would seem that the greater and more varied powers bequeathed by technology have expanded the range of choices and hence increased human freedom. For economics, for example, the argument has been made² that the uniform compulsion which scarcity and subsistence previously imposed on economic behavior with a virtual denial of alternatives (and hence—conjoined with the universal “maximization” motive of capitalist market competition—gave classical economics at least the appearance of a deterministic “science”) has given way to a latitude of indeterminacy. The plenty and powers provided by industrial technology allow a pluralism of choosable alternatives (hence disallow scientific prediction). We are not here concerned with the status of economics as a science. But as to the altered state of things alleged in the argument, I submit that the change means rather that one, relatively homogeneous determinism (thus relatively easy to formalize into a law) has been supplanted by another, more complex, multifarious determinism, namely, that exercised by the human artifact itself upon its creator and user. We, abstractly speaking the possessors of those powers, are concretely subject to their emancipated dynamics and the sheer momentum of our own multitude, the vehicle of those dynamics.

I have spoken elsewhere³ of the “new realm of necessity” set up, like a second nature, by the feedbacks of our achievements. The almighty we, or Man personified is, alas, an abstraction. *Man* may have become more powerful; *men* very probably the opposite, enmeshed as they are in more dependencies than ever before. What ideal Man now can do is not the same as what real men permit or dictate to be done. And here I am thinking not only of the immanent dynamism, almost automatism, of the impersonal technological complex I have invoked so far, but also of the pathology of its client society. Its compulsions, I fear, are at least as great as were those of unconquered nature. Talk of the blind forces of nature! Are those of the sorcerer’s creation less blind? They differ indeed in the serial shape of their causality: the action of nature’s forces is cyclical, with periodical recurrence of the same, while that of the technological forces is linear, progressive, cumulative, thus replacing the curse of constant toil with the threat of maturing crisis and possible catastrophe. Apart from this significant vector difference, I seriously wonder whether the tyranny of fate has not become greater, the latitude of spontaneity smaller; and whether man has not actually been weakened in his decision-making capacity by his accretion of collective strength.

However, in speaking, as I have just done, of “his” decision-making capacity, I have been guilty of the same abstraction I had earlier criticized in the use of the term “man.” Actually, the subject of the statement was no real or representative individual but Hobbes’ “Artificiall Man,” “that great Leviathan, called a Common-Wealth,” or the “large

horse” to which Socrates likened the city, “which because of its great size tends to be sluggish and needs stirring by a gad-fly.” Now, the chances of there being such gadflies among the numbers of the commonwealth are today no worse nor better than they have ever been, and in fact they are around and stinging in our field of concern. In that respect, the free spontaneity of personal insight, judgment, and responsible action by speech can be trusted as an ineradicable (if also incalculable) endowment of humanity, and smallness of number is in itself no impediment to shaking public complacency. The problem, however, is not so much complacency or apathy as the counterforces of active, and anything but complacent, interests and the complicity with them of all of us in our daily consumer existence. These interests themselves are factors in the determinism which technology has set up in the space of its sway. The question, then, is that of the possible chances of unselfish insight in the arena of (by nature) selfish *power*, and more particularly: of one long-range, interloping insight against the short-range goals of many incumbent powers. Is there hope that wisdom itself can become power? This renews the thorny old subject of Plato’s philosopher-king and—with that inclusion of realism which the utopian Plato did not lack—of the role of myth, not knowledge, in the education of the guardians. Applied to our topic: the *knowledge* of objective dangers and of values endangered, as well as of the technical remedies, is beginning to be there and to be disseminated; but to make it prevail in the marketplace is a matter less of the rational dissemination of truth than of public relations techniques, persuasion, indoctrination, and manipulation, also of unholy alliances, perhaps even conspiracy. The philosopher’s descent into the cave may well have to go all the way to “if you can’t lick them, join them.”

That is so not merely because of the active resistance of special interests but because of the optical illusion of the near and the far which condemns the long-range view to impotence against the enticement and threats of the nearby: it is this incurable shortsightedness of animal-human nature more than ill will that makes it difficult to move even those who have no special axe to grind, but still are in countless ways, as we all are, beneficiaries of the untamed system and so have something dear in the present to lose with the inevitable cost of its taming. The taskmaster, I fear, will have to be actual pain beginning to strike, when the far has moved close to the skin and has vulgar optics on its side. Even then, one may resort to palliatives of the hour. In any event, one should try as much as one can to forestall the advent of emergency with its high tax of suffering or, at the least, prepare for it. This is where the scientist can redeem his role in the technological estate.

The incipient knowledge about technological danger trends must be developed, coordinated, systematized, and the full force of computer-aided projection techniques be deployed to determine priorities of action, so as to inform preventive efforts wherever they can be elicited, to minimize the necessary sacrifices, and at the worst to preplan the saving measures which the terror of beginning calamity will eventually make people willing to accept. Even now, hardly a decade after the first stirrings of “environmental” con-

sciousness, much of the requisite knowledge, plus the rational persuasion, is available inside and outside academia for any well-meaning powerholder to draw upon. To this, we—the growing band of concerned intellectuals—ought persistently to contribute our bit of competence and passion.

But the real problem is to get the well-meaning into power and have that power as little as possible beholden to the interests which the technological colossus generates on its path. It is the problem of the philosopher-king compounded by the greater magnitude and complexity (also sophistication) of the forces to contend with. Ethically, it becomes a problem of playing the game by its impure rules. For the servant of truth to join in it means to sacrifice some of his time-honored role: he may have to turn apostle or agitator or political operator. This raises moral questions beyond those which technology itself poses, that of sanctioning immoral means for a surpassing end, of giving unto Caesar so as to promote what is not Caesar's. It is the grave question of moral casuistry, or of Dostoevsky's Grand Inquisitor, or of regarding cherished liberties as no longer affordable luxuries (which may well bring the anxious friend of mankind into odious political company)—questions one excusably hesitates to touch but in the further tide of things may not be permitted to evade.

What is, prior to joining the fray, the role of philosophy, that is, of a philosophically grounded ethical knowledge, in all this? The somber note of the last remarks responded to the quasi-apocalyptic prospects of the technological tide, where stark issues of planetary survival loom ahead. There, no philosophical ethics is needed to tell us that disaster must be averted. Mainly, this is the case of the ecological dangers. But there are other, noncatastrophic things afoot in technology where not the existence but the image of man is at stake. They are with us now and will accompany us and be joined by others at every new turn technology may take. Mainly, they are in the biomedical, behavioral, and social fields. They lack the stark simplicity of the survival issue, and there is none of the (at least declaratory) unanimity on them which the spectre of extreme crisis commands. It is here where a philosophical ethics or theory of values has its task. Whether its voice will be listened to in the dispute on policies is not for it to ask; perhaps it cannot even muster an authoritative voice with which to speak—a house divided, as philosophy is. But the philosopher must try for normative knowledge, and if his labors fall predictably short of producing a compelling axiomatics, at least his clarifications can counteract rashness and make people pause for a thoughtful view.

Where not existence but "quality" of life is in question, there is room for honest dissent on goals, time for theory to ponder them, and freedom from the tyranny of the lifeboat situation. Here, philosophy can have its try and its say. Not so on the extremity of the survival issue. The philosopher, to be sure, will also strive for a theoretical grounding of the very proposition that there ought to be men on earth, and that present generations are obligated to the existence of future ones. But such esoteric, ultimate validation of the perpetuity imperative for the species—whether obtainable or not to the satisfaction of reason—is happily not needed for

consensus in the face of ultimate threat. Agreement in favor of life is pretheoretical, instinctive, and universal. Averting disaster takes precedence over everything else, including pursuit of the good, and suspends otherwise inviolable prohibitions and rules. All moral standards for individual or group behavior, even demands for individual sacrifice of life, are premised on the continued existence of human life. As I have said elsewhere,⁴ "No rules can be devised for the waiving of rules in extremities. As with the famous shipwreck examples of ethical theory, the less said about it, the better."

Never before was there cause for considering the contingency that all mankind may find itself in a lifeboat, but this is exactly what we face when the viability of the planet is at stake. Once the situation becomes desperate, then what there is to do for salvaging it must be done, so that there be life—which "then," after the storm has been weathered, can again be adorned by ethical conduct. The moral inference to be drawn from this lurid eventuality of a moral pause is that we must never allow a lifeboat situation for humanity to arise.⁵ One part of the ethics of technology is precisely to guard the space in which any ethics can operate. For the rest, it must grapple with the cross-currents of value in the complexity of life.

A final word on the question of determinism versus freedom which our presentation of the technological syndrome has raised. The best hope of man rests in his most troublesome gift: the spontaneity of human acting which confounds all prediction. As the late Hannah Arendt never tired of stressing: the continuing arrival of newborn individuals in the world assures ever-new beginnings. We should expect to be surprised and to see our predictions come to naught. But those predictions themselves, with their warning voice, can have a vital share in provoking and informing the spontaneity that is going to confound them.

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¹But as serious an actuality as the Chinese plough "wandered" slowly westward with little traces of its route and finally caused a major, highly beneficial revolution in medieval European agriculture, which almost no one deemed worth recording when it happened (cf. Paul Leser, *Entstehung und Verbreitung des Pfluges*, Münster, 1931; reprint: The International Secretariate for Research on the History of Agricultural Implements, Brede-Lingby, Denmark, 1971).

²I here loosely refer to Adolph Lowe, "The Normative Roots of Economic Values," in Sidney Hook, ed., *Human Values and Economic Policy* (New York: New York University Press, 1967) and, more perhaps, to the many discussions I had with Lowe over the years. For my side of the argument, see "Economic Knowledge and the Critique of Goals," in R. L. Heilbroner, ed., *Economic Means and Social Ends* (Englewood Cliffs, N.J.: Prentice-Hall, 1969), reprinted in Hans Jonas, *Philosophical Essays* (Englewood Cliffs, N.J.: Prentice-Hall, 1969), reprinted in Hans Jonas, *Philosophical Essays* (Englewood Cliffs, N.J.: Prentice-Hall, 1974).

³"The Practical Uses of Theory," *Social Research* 26 (1959), reprinted in Hans Jonas, *The Phenomenon of Life* (New York, 1966). The reference is to pp. 209-10 in the latter edition.

⁴"Philosophical Reflections on Experimenting with Human Subjects," in Paul A. Freund, ed., *Experimentation with Human Subjects* (New York: George Braziller, 1970), reprinted in Hans Jonas, *Philosophical Essays*. The reference is to pp. 124-25 in the latter edition.

⁵For a comprehensive view of the demands which such a situation or even its approach would make on our social and political values, see Geoffrey Vickers, *Freedom in a Rocking Boat* (London, 1970).

PHILOSOPHICAL INVESTIGATIONS

By
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PREFACE

THE thoughts which I publish in what follows are the precipitate of philosophical investigations which have occupied me for the last sixteen years. They concern many subjects: the concepts of meaning, of understanding, of a proposition, of logic, the foundations of mathematics, states of consciousness, and other things. I have written down all these thoughts as *remarks*, short paragraphs, of which there is sometimes a fairly long chain about the same subject, while I sometimes make a sudden change, jumping from one topic to another.—It was my intention at first to bring all this together in a book whose form I pictured differently at different times. But the essential thing was that the thoughts should proceed from one subject to another in a natural order and without breaks.

After several unsuccessful attempts to weld my results together into such a whole, I realized that I should never succeed. The best that I could write would never be more than philosophical remarks; my thoughts were soon crippled if I tried to force them on in any single direction against their natural inclination.—And this was, of course, connected with the very nature of the investigation. For this compels us to travel over a wide field of thought criss-cross in every direction.—The philosophical remarks in this book are, as it were, a number of sketches of landscapes which were made in the course of these long and involved journeyings.

The same or almost the same points were always being approached afresh from different directions, and new sketches made. Very many of these were badly drawn or uncharacteristic, marked by all the defects of a weak draughtsman. And when they were rejected a number of tolerable ones were left, which now had to be arranged and sometimes cut down, so that if you looked at them you could get a picture of the landscape. Thus this book is really only an album.

Up to a short time ago I had really given up the idea of publishing my work in my lifetime. It used, indeed, to be revived from time to time: mainly because I was obliged to learn that my results (which I had communicated in lectures, typescripts and discussions), variously

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misunderstood, more or less mangled or watered down, were in circulation. This stung my vanity and I had difficulty in quieting it.

Four* years ago I had occasion to re-read my first book (the *Tractatus Logico-Philosophicus*) and to explain its ideas to someone. It suddenly seemed to me that I should publish those old thoughts and the new ones together: that the latter could be seen in the right light only by contrast with and against the background of my old way of thinking.¹

For since beginning to occupy myself with philosophy again, sixteen years ago, I have been forced to recognize grave mistakes in what I wrote in that first book. I was helped to realize these mistakes—to a degree which I myself am hardly able to estimate—by the criticism which my ideas encountered from Frank Ramsey, with whom I discussed them in innumerable conversations during the last two years of his life. Even more than to this—always certain and forcible—criticism I am indebted to that which a teacher of this university, Mr. P. Straffa, for many years unceasingly practised on my thoughts. I am indebted to *this* stimulus for the most consequential ideas of this book.

For more than one reason what I publish here will have points of contact with what other people are writing to-day.—If my remarks do not bear a stamp which marks them as mine,—I do not wish to lay any further claim to them as my property.

I make them public with doubtful feelings. It is not impossible that it should fall to the lot of this work, in its poverty and in the darkness of this time, to bring light into one brain or another—but, of course, it is not likely.

I should not like my writing to spare other people the trouble of thinking. But, if possible, to stimulate someone to thoughts of his own.

I should have liked to produce a good book. This has not come about, but the time is past in which I could improve it.

CAMBRIDGE,
January 1945.

* But cf. G. H. von Wright, 'The Wittgenstein Papers', *The Philosophical Review* 78, 1969. It seems that Wittgenstein should have said 'two years'.

¹ It was hoped to catty out this plan in a purely Getman edition of the present work.

PART I

i. "Cum ipsi (maiores homines) appellabant rem aliquam, et cum secundum eam vocem corpus ad aliquid movebant, videbam, et tenebam hoc ab eis vocari rem illam, quod sonabant, cum eam vellent ostendere. Hoc autem eos veile ex motu corporis aperiebatur: tamquam verbis naturalibus omnium gentium, quae fiunt vultu et nutu oculorum, ceterorumque membrorum actu, et sonitu vocis indicante affectionem animi in petendis, habendis, rejiciendis, fugiendisve rebus. Ita verba in variis sententiis locis suis posita, et crebro audita, quarum rerum signa essent, paulatim colligebam, measque jam voluntates, edomito in eis signis ore, per haec enuntiabam." (Augustine, *Confessions*, I. 8.)¹

These words, it seems to me, give us a particular picture of the essence of human language. It is this: the individual words in language name objects—sentences are combinations of such names.—In this picture of language we find the roots of the following idea: Every word has a meaning. This meaning is correlated with the word. It is the object for which the word stands.

Augustine does not speak of there being any difference between kinds of word. If you describe the learning of language in this way you are, I believe, thinking primarily of nouns like "table", "chair", "bread", and of people's names, and only secondarily of the names of certain actions and properties; and of the remaining kinds of word as something that will take care of itself.

Now think of the following use of language: I send someone shopping. I give him a slip marked "five red apples". He takes the slip to

¹ "When they (my elders) named some object, and accordingly moved towards something, I saw this and I grasped that the thing was called by the sound they uttered when they meant to point it out. Their intention was shewn by their bodily movements, as it were the natural language of all peoples: the expression of the face, the play of the eyes, the movement of other parts of the body, and the tone of voice which expresses our state of mind in seeking, having, rejecting, or avoiding something. Thus, as I heard words repeatedly used in their proper places in various sentences, I gradually learnt to understand what objects they signified; and after I had trained my mouth to form these signs, I used them to express my own desires."

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the shopkeeper, who opens the drawer marked "apples"; then he looks up the word "red" in a table and finds a colour sample opposite it; then he says the series of cardinal numbers—I assume that he knows them by heart—up to the word "five" and for each number he takes an apple of the same colour as the sample out of the drawer.—It is in this and similar ways that one operates with words.—"But how does he know where and how he is to look up the word 'red' and what he is to do with the word 'five'?"—Well, I assume that he *acts* as I have described. Explanations come to an end somewhere.—But what is the meaning of the word "five"?—No such thing was in question here, only how the word "five" is used.

2. That philosophical concept of meaning has its place in a primitive idea of the way language functions. But one can also say that it is the idea of a language more primitive than ours.

Let us imagine a language for which the description given by Augustine is right. The language is meant to serve for communication between a builder A and an assistant B. A is building with building-stones: there are blocks, pillars, slabs and beams. B has to pass the stones, and that in the order in which A needs them. For this purpose they use a language consisting of the words "block", "pillar", "slab", "beam". A calls them out;—B brings the stone which he has learnt to bring at such-and-such a call.—Conceive this as a complete primitive language.

3. Augustine, we might say, does describe a system of communication; only not everything that we call language is this system. And one has to say this in many cases where the question arises "Is this an appropriate description or not?" The answer is: "Yes, it is appropriate, but only for this narrowly circumscribed region, not for the whole of what you were claiming to describe."

It is as if someone were to say: "A game consists in moving objects about on a surface according to certain rules . . ."—and we replied: "You seem to be thinking of board games, but there are others. You can make your definition correct by expressly restricting it to those games."

4. Imagine a script in which the letters were used to stand for sounds, and also as signs of emphasis and punctuation. (A script can be conceived as a language for describing sound-patterns.) Now imagine someone interpreting that script as if there were simply a

correspondence of letters to sounds and as if the letters had not also completely different functions. Augustine's conception of language is like such an over-simple conception of the script.

5. If we look at the example in §1, we may perhaps get an inkling how much this general notion of the meaning of a word surrounds the working of language with a haze which makes clear vision impossible. It disperses the fog to study the phenomena of language in primitive kinds of application in which one can command a clear view of the aim and functioning of the words.

A child uses such primitive forms of language when it learns to talk. Here the teaching of language is not explanation, but training.

6. We could imagine that the language of §2 was the *whole* language of A and B; even the whole language of a tribe. The children are brought up to perform *these* actions, to use *these* words as they do so, and to react in *this* way to the words of others.

An important part of the training will consist in the teacher's pointing to the objects, directing the child's attention to them, and at the same time uttering a word; for instance, the word "slab" as he points to that shape. (I do not want to call this "ostensive definition", because the child cannot as yet *ask* what the name is. I will call it "ostensive teaching of words".—I say that it will form an important part of the training, because it is so with human beings; not because it could not be imagined otherwise.) This ostensive teaching of words can be said to establish an association between the word and the thing. But what does this mean? Well, it may mean various things; but one very likely thinks first of all that a picture of the object comes before the child's mind when it hears the word. But now, if this does happen—is it the purpose of the word?—Yes, it *may* be the purpose.—I can imagine such a use of words (of series of sounds). (Uttering a word is like striking a note on the keyboard of the imagination.) But in the language of §2 it is *not* the purpose of the words to evoke images. (It may, of course, be discovered that that helps to attain the actual purpose.)

But if the ostensive teaching has this effect,—am I to say that it effects an understanding of the word? Don't you understand the call "Slab!" if you act upon it in such-and-such a way?—Doubtless the ostensive teaching helped to bring this about; but only together with a particular

training. With different training the same ostensive teaching of these words would have effected a quite different understanding.

"I set the brake up by connecting up rod and lever."—Yes, given the whole of the rest of the mechanism. Only in conjunction with that is it a brake-lever, and separated from its support it is not even a lever; it may be anything, or nothing.

7. In the practice of the use of language (2) one party calls out the words, the other acts on them. In instruction in the language the following process will occur: the learner *names* the objects; that is, he utters the word when the teacher points to the stone.—And there will be this still simpler exercise: the pupil repeats the words after the teacher—both of these being processes resembling language.

We can also think of the whole process of using words in (2) as one of those games by means of which children learn their native language. I will call these games "language-games" and will sometimes speak of a primitive language as a language-game.

And the processes of naming the stones and of repeating words after someone might also be called language-games. Think of much of the use of words in games like ring-a-ring-a-roses.

I shall also call the whole, consisting of language and the actions into which it is woven, the "language-game".

8. Let us now look at an expansion of language (2). Besides the four words "block", "pillar", etc., let it contain a series of words used as the shopkeeper in (i) used the numerals (it can be the series of letters of the alphabet); further, let there be two words, which may as well be "there" and "this" (because this roughly indicates their purpose), that are used in connexion with a pointing gesture; and finally a number of colour samples. A gives an order like: "d—slab—there". At the same time he shews the assistant a colour sample, and when he says "there" he points to a place on the building site. From the stock of slabs B takes one for each letter of the alphabet up to "d", of the same colour as the sample, and brings them to the place indicated by A.—On other occasions A gives the order "this—there". At "this" he points to a building stone. And so on.

9. When a child learns this language, it has to learn the series, of 'numerals' a, b, c, . . . by heart. And it has to learn their use.—Will this training include ostensive teaching of the words?—Well, people

will, for example, point to slabs and count: "a, b, c slabs".—Something more like the ostensive teaching of the words "block", "pillar", etc. would be the ostensive teaching of numerals that serve not to count but to refer to groups of objects that can be taken in at a glance. Children do learn the use of the first five or six cardinal numerals in this way.

Are "there" and "this" also taught ostensively?—Imagine how one might perhaps teach their use. One will point to places and things—but in this case the pointing occurs in the *use* of the words too and not merely in learning the use.—

10. Now what do the words of this language *signify*?—What is supposed to shew what they signify, if not the kind of use they have? And we have already described that. So we are asking for the expression "This word signifies *this*" to be made a part of the description. In other words the description ought to take the form: "The word . . . signifies . . .".

Of course, one can reduce the description of the use of the word "slab" to the statement that this word signifies this object. This will be done when, for example, it is merely a matter of removing the mistaken idea that the word "slab" refers to the shape of building-stone that we in fact call a "block"—but the kind of *referring* this is, that is to say the use of these words for the rest, is already known.

Equally one can say that the signs "a", "b", etc. signify numbers; when for example this removes the mistaken idea that "a", "b", "c", play the part actually played in language by "block", "slab", "pillar". And one can also say that "c" means this number and not that one; when for example this serves to explain that the letters are to be used in the order a, b, c, d, etc. and not in the order a, b, d, c.

But assimilating the descriptions of the uses of words in this way cannot make the uses themselves any more like one another. For, as we see, they are absolutely unlike.

11. Think of the tools in a tool-box: there is a hammer, pliers, a saw, a screw-driver, a rule, a glue-pot, glue, nails and screws.—The functions of words are as diverse as the functions of these objects. (And in both cases there are similarities.)

Of course, what confuses us is the uniform appearance of words when we hear them spoken or meet them in script and print. For their *application* is not presented to us so clearly. Especially when we are doing philosophy!

12. It is like looking into the cabin of a locomotive. We see handles all looking more or less alike. (Naturally, since they are all supposed to be handled.) But one is the handle of a crank which can be moved continuously (it regulates the opening of a valve); another is the handle of a switch, which has only two effective positions, it is either off or on; a third is the handle of a brake-lever, the harder one pulls on it, the harder it brakes; a fourth, the handle of a pump: it has an effect only so long as it is moved to and fro.

13. When we say: "Every word in language signifies something" we have so far said *nothing whatever*; unless we have explained exactly *what* distinction we wish to make. (It might be, of course, that we wanted to distinguish the words of language (8) from words 'without meaning' such as occur in Lewis Carroll's poems, or words like "Lilliburlero" in songs.)

14. Imagine someone's saying: "All tools serve to modify something. Thus the hammer modifies the position of the nail, the saw the shape of the board, and so on."—And what is modified by the rule, the glue-pot, the nails?—"Our knowledge of a thing's length, the temperature of the glue, and the solidity of the box."—Would anything be gained by this assimilation of expressions?—

15. The word "to signify" is perhaps used in the most straightforward way when the object signified is marked with the sign. Suppose that the tools A uses in building bear certain marks. When A shews his assistant such a mark, he brings the tool that has that mark on it.

It is in this and more or less similar ways that a name means and is given to a thing.—It will often prove useful in philosophy to say to ourselves: naming something is like attaching a label to a thing.

16. What about the colour samples that A shews to B: are they part of the *language*? Well, it is as you please. They do not belong among the words; yet when I say to someone: "Pronounce the word 'the' ", you will count the second "the" as part of the sentence. Yet it has a role just like that of a colour-sample in language-game (8); that is, it is a sample of what the other is meant to say.

It is most natural, and causes least confusion, to reckon the samples among the instruments of the language.

((Remark on the reflexive pronoun "*this* sentence".))

17. It will be possible to say: In language (8) we have different *kinds of word*. For the functions of the word "slab" and the word "block" are more alike than those of "slab" and "d". But how we group words into kinds will depend on the aim of the classification,—and on our own inclination.

Think of the different points of view from which one can classify tools or chess-men.

18. Do not be troubled by the fact that languages (2) and (8) consist only of orders. If you want to say that this shews them to be incomplete, ask yourself whether our language is complete;—whether it was so before the symbolism of chemistry and the notation of the infinitesimal calculus were incorporated in it; for these are, so to speak, suburbs of our language. (And how many houses or streets does it take before a town begins to be a town?) Our language can be seen as an ancient city: a maze of little streets and squares, of old and new houses, and of houses with additions from various periods; and this surrounded by a multitude of new boroughs with straight regular streets and uniform houses.

19. It is easy to imagine a language consisting only of orders and reports in battle.—Or a language consisting only of questions and expressions for answering yes and no. And innumerable others.—And to imagine a language means to imagine a form of life.

But what about this: is the call "Slab I" in example (2) a sentence or a word?—If a word, surely it has not the same meaning as the like-sounding word of our ordinary language, for in §2 it is a call. But if a sentence, it is surely not the elliptical sentence: "Slab!" of our language.—As far as the first question goes you can call "Slab!" a word and also a sentence; perhaps it could be appropriately called a 'degenerate sentence' (as one speaks of a degenerate hyperbola); in fact it *is* our 'elliptical' sentence.—But that is surely only a shortened form of the sentence "Bring me a slab", and there is no such sentence in example (2).—But why should I not on the contrary have called the sentence "Bring me a slab" a *lengthening* of the sentence "Slab!"?—Because if you shout "Slab!" you really mean: "Bring me a slab".—But how do you do this: how do you *mean that* while you *say* "Slab!"? Do you say the unshortened sentence to yourself? And why should I translate the call "Slab!" into a different expression in order to say

what someone means by it? And if they mean the same thing—why should I not say: "When he says 'Slab!' he means 'Slab!'"? Again, if you can mean "Bring me the slab", why should you not be able to mean "Slab!"?—But when I call "Slab!", then what I want is, *that he should bring me a slab*—Certainly, but does 'wanting this' consist in thinking in some form or other a different sentence from the one you utter?—

20. But now it looks as if when someone says "Bring me a slab" he could mean this expression as *one* long word corresponding to the single word "Slab!"—Then can one mean it sometimes as one word and sometimes as four? And how does one usually mean it?—I think we shall be inclined to say: we mean the sentence as 0 or words when we use it in contrast with other sentences such as "*Hand* me a slab", "*Bring him* a slab", "*Bring two* slabs", etc.; that is, in contrast with sentences containing the separate words of our command in other combinations.—But what does using one sentence in contrast with others consist in? Do the others, perhaps, hover before one's mind? *All* of them? And *while* one is saying the one sentence, or before, or afterwards?—No. Even if such an explanation rather tempts us, we need only think for a moment of what actually happens in order to see that we are going astray here. We say that we use the command in contrast with other sentences because *our language* contains the possibility of those other sentences. Someone who did not understand our language, a foreigner, who had fairly often heard someone giving the order: "Bring me a slab!", might believe that this whole series of sounds was one word corresponding perhaps to the word for "building-stone" in his language. If he himself had then given this order perhaps he would have pronounced it differently, and we should say: he pronounces it so oddly because he takes it for a *single* word.—But then, is there not also something different going on in him when he pronounces it,—something corresponding to the fact that he conceives the sentence as a *single* word?—Either the same thing may go on in him, or something different. For what goes on in you when you give such an order? Are you conscious of its consisting of four words *while* you are uttering it? Of course you have a *mastery* of this language—which contains those other sentences as well—but is this having a mastery something that *happens* while you are uttering the sentence?—And I have admitted that the foreigner will probably pronounce a sentence differently if he conceives it differently; but what we call his wrong conception *need* not lie in anything that accompanies the utterance of the command.

The sentence is 'elliptical', not because it leaves out something that we think when we utter it, but because it is shortened—in comparison with a particular paradigm of our grammar.—Of course one might object here: "You grant that the shortened and the unshortened sentence have the same sense.—What is this sense, then? Isn't there a verbal expression for this sense?"—But doesn't the fact that sentences have the same sense consist in their having the same *use*?—(In Russian one says "stone red" instead of "the stone is red"; do they feel the copula to be missing in the sense, or attach it in *thought*?)

21. Imagine a language-game in which A asks and B reports the number of slabs or blocks in a pile, or the colours and shapes of the building-stones that are stacked in such-and-such a place.—Such a report might run: "Five slabs". Now what is the difference between the report or statement "Five slabs" and the order "Five slabs!"?—Well, it is the part which uttering these words plays in the language-game. No doubt the tone of voice and the look with which they are uttered, and much else besides, will also be different. But we could also imagine the tone's being the same—for an order and a report can be spoken in a *variety* of tones of voice and with various expressions of face—the difference being only in the application. (Of course, we might use the words "statement" and "command" to stand for grammatical forms of sentence and intonations; we do in fact call "Isn't the weather glorious to-day?" a question, although it is used as a statement.) We could imagine a language in which *all* statements had the form and tone of rhetorical questions; or every command the form of the question "Would you like to . . . ?". Perhaps it will then be said: "What he says has the form of a question but is really a command".—that is, has the function of a command in the technique of using the language. (Similarly one says "You will do this" not as a prophecy but as a command. What makes it the one or the other?)

22. Frege's idea that every assertion contains an assumption, which is the thing that is asserted, really rests on the possibility found in our language of writing every statement in the form: "It is asserted that such-and-such is the case."—But "that such-and-such is the case" is *not* a sentence in our language—so far it is not a *move* in the language-game. And if I write, not "It is asserted that", but "It is asserted: such-and-such is the case", the words "It is asserted" simply become superfluous.

We might very well also write every statement in the form of a

question followed by a "Yes"; for instance: "Is it raining? Yes!"
Would this shew that every statement contained a question?

Of course we have the right to use an assertion sign in contrast with a question-mark, for example, or if we want to distinguish an assertion from a fiction or a supposition. It is only a mistake if one thinks that the assertion consists of two actions, entertaining and asserting (assigning the truth-value, or something of the kind), and that in performing these actions we follow the prepositional sign roughly as we sing from the musical score. Reading the written sentence loud or soft is indeed comparable with singing from a musical score, but *'meaning (thinking) the sentence that is read is not.*

Frege's assertion sign marks the *beginning of the sentence*. Thus its function is like that of the full-stop. It distinguishes the whole period from a clause *within* the period. If I hear someone say "it's raining" but do not know whether I have heard the beginning and end of the period, so far this sentence does not serve to tell me anything.

23. But how many kinds of sentence are there? Say assertion, question, and command?—There are *countless* kinds: countless different kinds of use of what we call "symbols", "words", "sentences". And this multiplicity is not something fixed, given once for all; but new types of language, new language-games, as we may say, come into existence, and others become obsolete and get forgotten. (We can get a *rough picture* of this from the changes in mathematics.)

Here the term "*language-game*" is meant to bring into prominence the fact that the *speaking* of language is part of an activity, or of a form of life.

Review the multiplicity of language-games in the following examples, and in others:

- Giving orders, and obeying them—
- Describing the appearance of an object, or giving its measurements—
- Constructing an object from a description (a drawing)—
- Reporting an event—
- Speculating about an event—

Imagine a picture representing a boxer in a particular stance. Now, this picture can be used to tell someone how he should stand, should hold himself; or how he should not hold himself; or how a particular man did stand in such-and-such a place; and so on. One might (using the language of chemistry) call this picture a proposition-radical. This will be how Frege thought of the "assumption".

Forming and testing a hypothesis—
 Presenting the results of an experiment in tables **and diagrams**—
 Making up a story; and reading it—
 Play-acting—
 Singing catches—
 Guessing riddles—
 Making a joke; telling it—
 Solving a problem in practical arithmetic—
 Translating from one language into another—
 Asking, thanking, cursing, greeting, praying.

—It is interesting to compare the multiplicity of the tools in language and of the ways they are used, the multiplicity of kinds of word and sentence, with what logicians have said about the structure of language. (Including the author of the *Tractatus Logico-Philosophicus*.)

24. If you do not keep the multiplicity of language-games in view you will perhaps be inclined to ask questions like: "What is a question?" —Is it the statement that I do not know such-and-such, or the statement that I wish the other person would tell me . . . ? Or is it the description of my mental state of uncertainty?—And is the cry "Help!" such a description?

Think how many different kinds of thing are called "description": description of a body's position by means of its co-ordinates; description of a facial expression; description of a sensation of touch; of a mood.

Of course it is possible to substitute the form of statement or description for the usual form of question: "I want to know whether" or "I am in doubt whether"—but this does not bring the different language-games any closer together.

The significance of such possibilities of transformation, for example of turning all statements into sentences beginning "I think" or "I believe" (and thus, as it were, into descriptions of *my* inner life) will become clearer in another place. (Solipsism.)

25. It is sometimes said that animals do not talk because they lack the mental capacity. And this means: "they do not think, and that is why they do not talk." But—they simply do not talk. Or to put it better: they do not use language—if we except the most primitive forms of language.—Commanding, questioning, recounting, chatting, are as much a part of our natural history as walking, eating, drinking, playing.

26. One thinks that learning language consists in giving names to objects. Viz, to human beings, to shapes, to colours, to pains, to

moods, to numbers, etc. . To repeat—naming is something like attaching a label to a thing. One can say that this is preparatory to the use of a word. But *what* is it a preparation/Or?

27. "We name things and then we can talk about them: can refer to them in talk."—As if what we did next were given with the mere act of naming. As if there were only one thing called "talking about a thing". Whereas in fact we do the most various things with our sentences. Think of exclamations alone, with their completely different functions.

Water!
 Away!
 Ow!
 Help!
 Fine!
 No!

Are you inclined still to call these words "names of objects"?

In languages (2) and (8) there was no such thing as asking something's name. This, with its correlate, ostensive definition, is, we might say, a language-game on its own. That is really to say: we are brought up, trained, to ask: "What is that called?"—upon which the name is given. And there is also a language-game of inventing a name for something, and hence of saying, "This is" and then using the new name. (Thus, for example, children give names to their dolls and then talk about them and to them. Think in this connexion how singular is the use of a person's name to *call* him!)

28. Now one can ostensibly define a proper name, the name of a colour, the name of a material, a numeral, the name of a point of the compass and so on. The definition of the number two, "That is called 'two' "—pointing to two nuts—is perfectly exact.—But how can two be defined like that? The person one gives the definition to doesn't know what one wants to call "two"; he will suppose that "two" is the name given to *this* group of nuts!—He *may* suppose this; but perhaps he does not. He might make the opposite mistake; when I want to assign a name to this group of nuts, he might understand it as a numeral. And he might equally well take the name of a person, of which I give an ostensive definition, as that of a colour, of a race, or even of a point

of the compass. That is to say: an ostensive definition can be variously interpreted in *every* case.

29. Perhaps you say: two can only be ostensively defined in *this* way: "This *number* is called 'two' ". For the word "number" here shews what place in language, in grammar, we assign to the word. But this means that the word "number" must be explained before the ostensive definition can be understood.—The word "number" in the definition does indeed shew this place; does shew the post at which we station the word. And we can prevent misunderstandings by saying: "This *colour* is called so-and-so", "This *length* is called so-and-so", and so on. That is to say: misunderstandings are sometimes averted in this way. But is there only *one* way of taking the word "colour" or "length"?—Well, they just need defining.—Defining, then, by means of other words! And what about the last definition in this chain? (Do not say: "There isn't a 'last' definition". That is just as if you chose to say: "There isn't a last house in this road; one can always build an additional one".)

Whether the word "number" is necessary in the ostensive definition depends on whether without it the other person takes the definition otherwise than I wish. And that will depend on the circumstances under which it is given, and on the person I give it to.

And how he 'takes' the definition is seen in the use that he makes of the word defined.

30. So one might say: the ostensive definition explains the use—the meaning—of the word when the overall role of the word in language is clear. Thus if I know that someone means to explain a colour-word to me the ostensive definition "That is called 'sepia' " will help me to understand the word.—And you can say this, so long as

Could one define the word "red" by pointing to something that was *not red*? That would be as if one were supposed to explain the word "modest" to someone whose English was weak, and one pointed to an arrogant man and said "That man is *not* modest". That it is ambiguous is no argument against such a method of definition. Any definition can be misunderstood.

But it might well be asked: are we still to call this "definition"?—For, of course, even if it has the same practical consequences, the same *effect* on the learner, it plays a different part in the calculus from what we ordinarily call "ostensive definition" of the word "red".

you do not forget that all sorts of problems attach to the words "to know" or "to be clear".

One has already to know (or be able to do) something in order to be capable of asking a thing's name. But what does one have to know?

31. When one shews someone the king in chess and says: "This is the king", this does not tell him the use of this piece—unless he already knows the rules of the game up to this last point: the shape of the king. You could imagine his having learnt the rules of the game without ever having been shewn an actual piece. The shape of the chessman corresponds here to the sound or shape of a word.

One can also imagine someone's having learnt the game without ever learning or formulating rules. He might have learnt quite simple board-games first, by watching, and have progressed to more and more complicated ones. He too might be given the explanation "This is the king",—if, for instance, he were being shewn chessmen of a shape he was not used to. This explanation again only tells him the use of the piece because, as we might say, the place for it was already prepared. Or even: we shall only say that it tells him the use, if the place is already prepared. And in this case it is so, not because the person to whom we give the explanation already knows rules, but because in another sense he is already master of a game.

Consider this further case: I am explaining chess to someone; and I begin by pointing to a chessman and saying: "This is the king; it can move like this, and so on."—In this case we shall say: the words "This is the king" (or "This is called the 'king' ") are a definition only if the learner already 'knows what a piece in a game is'. That is, if he has already played other games, or has watched other people playing 'and understood'—*and similar things*. Further, only under these conditions will he be able to ask relevantly in the course of learning the game: "What do you call this?"—that is, this piece in a game.

We may say: only someone who already knows how to do something with it can significantly ask a name.

And we can imagine the person who is asked replying: "Settle the name yourself"—and now the one who asked would have to manage everything for himself.

32. Someone coming into a strange country will sometimes learn the language of the inhabitants from ostensive definitions that they give him; and he will often have to *guess* the meaning of these definitions; and will guess sometimes right, sometimes wrong.

And now, I think, we can say: Augustine describes the learning

of human language as if the child came into a strange country and did not understand the language of the country; that is, as if it already had a language, only not this one. Or again: as if the child could already *think*, only not yet speak. And "think" would here mean something like "talk to itself".

33. Suppose, however, someone were to object: "It is not true that you must already be master of a language in order to understand an ostensive definition: all you need—of course!—is to know or guess what the person giving the explanation is pointing to. That is, whether for example to the shape of the object, or to its colour, or to its number, and so on."—And what does 'pointing to the shape', 'pointing to the colour' consist in? Point to a piece of paper.—And now point to its shape—now to its colour—now to its number (that sounds queer).—How did you do it?—You will say that you 'meant' a different thing each time you pointed. And if I ask how that is done, you will say you concentrated your attention on the colour, the shape, etc. But I ask again: how is *that* done?

Suppose someone points to a vase and says "Look at that marvellous blue—the shape isn't the point."—Or: "Look at the marvellous shape—the colour doesn't matter." Without doubt you will do something *different* when you act upon these two invitations. But do you always do the *same* thing when you direct your attention to the colour? Imagine various different cases. To indicate a few:

"Is this blue the same as the blue over there? Do you see any difference?"—

You are mixing paint and you say "It's hard to get the blue of this sky."

"It's turning fine, you can already see blue sky again."

"Look what different effects these two blues have."

"Do you see the blue book over there? Bring it here."

"This blue signal-light means"

"What's this blue called?—Is it 'indigo'?"

You sometimes attend to the colour by putting your hand up to keep the outline from view; or by not looking at the outline of the thing; sometimes by staring at the object and trying to remember where you saw that colour before.

You attend to the shape, sometimes by tracing it, sometimes by screwing up your eyes so as not to see the colour clearly, and in many other ways. I want to say: This is the sort of thing that happens *while* one 'directs one's attention to this or that'. But it isn't these things by

themselves that make us say someone is attending to the shape, the colour, and so on. Just as a move in chess doesn't consist simply in moving a piece in such-and-such a way on the board—nor yet in one's thoughts and feelings as one makes the move: but in the circumstances that we call "playing a game of chess", "solving a chess problem", and so on.

34. But suppose someone said: "I always do the same thing when I attend to the shape: my eye follows the outline and I feel". And suppose this person to give someone else the ostensive definition "That is called a 'circle' ", pointing to a circular object and having all these experiences—cannot his hearer still interpret the definition differently, even though he sees the other's eyes following the outline, and even though he feels what the other feels? That is to say: this 'interpretation' may also consist in how he now makes use of the word; in what he points to, for example, when told: "Point to a circle".—For neither the expression "to intend the definition in such-and-such a way" nor the expression "to interpret the definition in such-and-such a way" stands for a process which accompanies the giving and hearing of the definition.

35. There are, of course, what can be called "characteristic experiences" of pointing to (e.g.) the shape. For example, following the outline with one's finger or with one's eyes as one points.—But *this* does not happen in all cases in which I 'mean the shape', and no more does any other one characteristic process occur in all these cases.—Besides, even if something of the sort did recur in all cases, it would still depend on the circumstances—that is, on what happened before and after the pointing—whether we should say "He pointed to the shape and not to the colour".

For the words "to point to the shape", "to mean the shape", and so on, are not used in the same way as *these*, "to point to this book (not to that one)", "to point to the chair, not to the table", and so on.—Only think how differently we *learn* the use of the words "to point to this thing", "to point to that thing", and on the other hand "to point to the colour, not the shape", "to mean the colour", and so on.

To repeat: in certain cases, especially when one points 'to the shape' or 'to the number' there are characteristic experiences and ways of pointing—'characteristic' because they recur often (not always) when shape or number are 'meant'. But do you also know of an experience characteristic of pointing to a piece in a game *as a piece in a garnet*

All the same one can say: "I mean that this *piece* is called the 'king', not this particular bit of wood I am pointing to". (Recognizing, wishing, remembering, etc. .)

36. And we do here what we do in a host of similar cases: because we cannot specify any *one* bodily action which we call pointing to the shape (as opposed, for example, to the colour), we say that a *spiritual* [mental, intellectual] activity corresponds to these words.

Where our language suggests a body and there is none: there, we should like to say, is a *spirit*.

37. What is the relation between name and thing named?—Well, what *is* it? Look at language-game (2) or at another one: there you can see the sort of thing this relation consists in. This relation may also consist, among many other things, in the fact that hearing the name calls before our mind the picture of what is named; and it also consists, among other things, in the name's being written on the thing named or being pronounced when that thing is pointed at.

38. But what, for example, is the word "this" the name of in language-game (8) or the word "that" in the ostensive definition "that is called . . ."?—If you do not want to produce confusion you will do best not to call these words names at all.—Yet, strange to say, the word "this" has been called the *only genuine* name; so that anything else we call a name was one only in an inexact, approximate sense.

This queer conception springs from a tendency to sublime the logic of our language—as one might put it. The proper answer to it is: we call very different things "names"; the word "name" is used to

What is it to *mean* the words "That is blue" at one time as a statement about the object one is pointing to—at another as an explanation of the word "blue"? Well, in the second case one really means "That is called 'blue' ".—Then can one at one time mean the word "is" as "is called" and the word "blue" as " 'blue' ", and another time mean "is" really as "is"?

It is also possible for someone to get an explanation of the words out of what was intended as a piece of information. [Marginal note: Here lurks a crucial superstition.]

Can I say "bububu" and mean "If it doesn't rain I shall go for a walk"?—It is only in a language that I can mean something by something. This shews clearly that the grammar of "to mean" is not like that of the expression "to imagine" and the like.

characterize many different kinds of use of a word, related to one another in many different ways;—but the kind of use that "this" has is not among them.

It is quite true that, in giving an ostensive definition for instance, we often point to the object named and say the name. And similarly, in giving an ostensive definition for instance, we say the word "this" while pointing to a thing. And also the word "this" and a name often occupy the same position in a sentence. But it is precisely characteristic of a name that it is defined by means of the demonstrative expression "That is N" (or "That is called 'N' "). But do we also give the definitions: "That is called 'this' ", or "This is called 'this' "?

This is connected with the conception of naming as, so to speak, an occult process. Naming appears as a *queer* connexion of a word with an object.—And you really get such a queer connexion when the philosopher tries to bring out *the* relation between name and thing by staring at an object in front of him and repeating a name or even the word "this" innumerable times. For philosophical problems arise when language *goes on holiday*. And *here* we may indeed fancy naming to be some remarkable act of mind, as it were a baptism of an object. And we can also say the word "this" *to* the object, as it were *address* the object as "this"—a queer use of this word, which doubtless only occurs in doing philosophy.

39. But why does it occur to one to want to make precisely this word into a name, when it evidently is *not* a name?—That is just the reason. For one is tempted to make an objection against what is ordinarily called a name. It can be put like this: *a name ought really to signify a simple*. And for this one might perhaps give the following reasons: The word "Excalibur", say, is a proper name in the ordinary sense. The sword Excalibur consists of parts combined in a particular way. If they are combined differently Excalibur does not exist. But it is clear that the sentence "Excalibur has a sharp blade" makes *sense* whether Excalibur is still whole or is broken up. But if "Excalibur" is the name of an object, this object no longer exists when Excalibur is broken in pieces; and as no object would then correspond to the name it would have no meaning. But then the sentence "Excalibur has a sharp blade" would contain a word that had no meaning, and hence the sentence would be nonsense. But it does make sense; so there must always be something corresponding to the words of which it consists. So the word "Excalibur" must disappear when the sense is

analysed and its place be taken by words which name simples. It will be reasonable to call these words the real names.

40. Let us first discuss *this* point of the argument: that a word has no meaning if nothing corresponds to it.—It is important to note that the word "meaning" is being used illicitly if it is used to signify the thing that 'corresponds' to the word. That is to confound the meaning of a name with the *bearer* of the name. When Mr. N. N. dies one says that the bearer of the name dies, not that the meaning dies. And it would be nonsensical to say that, for if the name ceased to have meaning it would make no sense to say "Mr. N. N. is dead."

41. In §15 we introduced proper names into language (8). Now suppose that the tool with the name "N" is broken. Not knowing this, A gives B the sign "N". Has this sign meaning now or not?—What is B to do when he is given it?—We have not settled anything about this. One might ask: what *will* he do? Well, perhaps he will stand there at a loss, or shew A the pieces. Here one *might* say: "N" has become meaningless; and this expression would mean that the sign "N" no longer had a use in our language-game (unless we gave it a new one). "N" might also become meaningless because, for whatever reason, the tool was given another name and the sign "N" no longer used in the language-game.—But we could also imagine a convention whereby B has to shake his head in reply if A gives him the sign belonging to a tool that is broken.—In this way the command "N" might be said to be given a place in the language-game even when the tool no longer exists, and the sign "N" to have meaning even when its bearer ceases to exist.

42. But has for instance a name which has *never* been used for a tool also got a meaning in that game?—Let us assume that "X" is such a sign and that A gives this sign to B—well, even such signs could be given a place in the language-game, and B might have, say, to answer them too with a shake of the head. (One could imagine this as a sort of joke between them.)

43. For a *large* class of cases—though not for all—in which we employ the word "meaning" it can be defined thus: the meaning of a word is its use in the language.

And the *meaning* of a name is sometimes explained by pointing to its *bearer*.

44. We said that the sentence "Excalibur has a sharp blade" made sense even when Excalibur was broken in pieces. Now this is so because in this language-game a name is also used in the absence of its bearer. But we can imagine a language-game with names (that is, with signs which we should certainly include among names) in which they are used only in the presence of the bearer; and so could *always* be replaced by a demonstrative pronoun and the gesture of pointing.

45. The demonstrative "this" can never be without a bearer. It might be said: "so long as there is a *this*, the word 'this' has a meaning too, whether *this* is simple or complex."—But that does not make the word into a name. On the contrary: for a name is not used with, but only explained by means of, the gesture of pointing.

46. What lies behind the idea that names really signify simples?—Socrates says in the Theaetetus: "If I make no mistake, I have heard some people say this: there is no definition of the primary elements—so to speak—out of which we and everything else are composed; for everything that exists¹ in its own right can only be *named*, no other determination is possible, neither that it *is* nor that it *is not* But what exists¹ in its own right has to be named without any other determination. In consequence it is impossible to give an account of any primary element; for it, nothing is possible but the bare name; its name is all it has. But just as what consists of these primary elements is itself complex, so the names of the elements become descriptive language by being compounded together. For the essence of speech is the composition of names."

Both Russell's 'individuals' and my 'objects' (*Tractatus Logico-Philosophicus*) were such primary elements.

47. But what are the simple constituent parts of which reality is composed?—What are the simple constituent parts of a chair?—The bits of wood of which it is made? Or the molecules, or the atoms?—"Simple" means: not composite. And here the point is: in what sense 'composite'? It makes no sense at all to speak absolutely of the 'simple parts of a chair'.

¹I have translated the German translation which Wittgenstein used rather than the original. Tr.

The Death of the Author

In his tale *Sarrasine*, Balzac, speaking of a castrato disguised as a woman, writes this sentence: "She was Woman, with her sudden fears, her inexplicable whims, her instinctive fears, her meaningless bravado, her defiance, and her delicious delicacy of feeling." Who speaks in this way? Is it the hero of the tale, who would prefer not to recognize the castrato hidden beneath the "woman"? Is it Balzac the man, whose personal experience has provided him with a philosophy of Woman? Is it Balzac the author, professing certain "literary" ideas about femininity? Is it universal wisdom? Romantic psychology? We can never know, for the good reason that writing is the destruction of every voice, every origin. Writing is that neuter, that composite, that obliquity into which our subject flees, the black-and-white where all identity is lost, beginning with the very identity of the body that writes.

No doubt it has always been so: once a fact is *recounted*—for intransitive purposes, and no longer to act directly upon reality, i.e., exclusive of any function except that exercise of the symbol itself—this gap appears, the voice loses its origin, the author enters into his own death, writing begins. However, the affect of this phenomenon has been variable; in ethnographic societies, narrative is never assumed by a person but by a mediator, shaman, or reciter, whose "performance" (i.e., his mastery of the narrative code) can be admired, but never his "genius." The *author* is a modern character, no doubt produced by our society as it emerged from the Middle Ages, inflected by English empiricism, French rationalism, and the personal faith of the Reformation, thereby discovering the prestige of the individual,

or, as we say more nobly, of the "human person." Hence, it is logical that in literary matters it should be positivism, crown and conclusion of capitalist ideology, which has granted the greatest importance to the author's "person." The *author* still reigns in manuals of literary history, in biographies of writers, magazine interviews, and in the very consciousness of litterateurs eager to unite, by means of private journals, their person and their work; the image of literature to be found in contemporary culture is tyrannically centered on the author, his person, his history, his tastes, his passions; criticism still largely consists in saying that Baudelaire's oeuvre is the failure of the man Baudelaire, Van Gogh's is his madness, Tchaikovsky's his vice: *explanation* of the work is still sought in the person of its producer, as if, through the more or less transparent allegory of fiction, it was always, ultimately, the voice of one and the same person, the *author*, which was transmitting his "confidences."

Though the Author's empire is still very powerful (the new criticism has quite often merely consolidated it), we know that certain writers have already tried to subvert it. In France, Mallarmé, no doubt the first, saw and foresaw in all its scope the necessity to substitute language itself for the subject hitherto supposed to be its owner; for Mallarmé, as for us, it is language which speaks, not the author; to write is to reach, through a preliminary impersonality—which we can at no moment identify with the realistic novelist's castrating "objectivity"—that point where not "I" but only language functions, "performs": Mallarmé's whole poetics consists in suppressing the author in favor of writing (and thereby restoring, as we shall see, the reader's place). Valéry, entangled in a psychology of the ego, greatly edulcorated Mallarmean theory, but led by a preference for classicism to conform to the lessons of Rhetoric, he continued to cast the Author into doubt and derision, emphasized the linguistic and "accidental" nature of his activity, and throughout his prose works championed the essentially verbal condition of literature, as opposed to which any resort to the writer's interiority seemed to him pure superstition. Proust himself, despite

the apparently psychological character of what is called his *analyses*, visibly undertook to blur by an extreme subtilization the relation of the writer and his characters: by making the narrator not the one who has seen or felt, or even the one who writes, but the one who *is going to write* (the young man of the novel—but, as a matter of fact, how old is he and *who* is he?—wants to write but cannot, and the novel ends when writing finally becomes possible), Proust has given modern writing its epic: by a radical reversal, instead of putting his life into his novel, as is so often said, he made his life itself a work of which his own book was the model, so that it is quite clear to us that it is not Charlus who imitates Montesquiou, but Montesquiou, in his anecdotal, historical reality, who is only a secondary, derived fragment of Charlus. Finally Surrealism, to keep to this prehistory of modernity, could doubtless not attribute a sovereign place to language, since language is system, and what this movement sought was, romantically, a direct subversion of the codes—an illusory subversion, moreover, for a code cannot be destroyed, only “flouted”; yet, by constantly striving to disappoint expected meanings (this was the famous surrealist “shock”), by urging the hand to write as fast as possible what the head was unaware of (this was automatic writing), by accepting the principle and the experiment of collective writing, Surrealism helped desacralize the image of the Author. Last, outside literature itself (in fact, such distinctions are becoming quite dated), linguistics furnishes the destruction of the Author with a precious analytic instrument, showing that the speech-act in its entirety is an “empty” process, which functions perfectly without its being necessary to “fill” it with the person of the interlocutors: linguistically, the author is nothing but the one who writes, just as *I* is nothing but the one who says *I*: language knows a “subject,” not a “person,” and this subject, empty outside of the very speech-act which defines it, suffices to “hold” language, i.e., to exhaust it.

The removal of the Author (with Brecht, we might speak here of a veritable *distancing*, the Author diminishing like a

figure at the far end of the literary stage) is not only a historical fact or an act of writing: it utterly transforms the modern text (or—which is the same thing—the text is henceforth produced and read so that the author absents himself from it at every level). Time, first of all, is no longer the same. The Author, when we believe in him, is always conceived as the past of his own book: book and author are voluntarily placed on one and the same line, distributed as a *before* and an *after*: the Author is supposed to *feed* the book, i.e., he lives before it, thinks, suffers, lives for it; he has the same relation of antecedence with his work that a father sustains with his child. Quite the contrary, the modern *scriptor* is born *at the same time* as his text; he is not furnished with a being which precedes or exceeds his writing, he is not the subject of which his book would be the predicate; there is no time other than that of the speech-act, and every text is written eternally *here* and *now*. This is because (or it follows that) *writing* can no longer designate an operation of recording, of observation, of representation, of “painting” (as the Classics used to say), but instead what the linguists, following Oxfordian philosophy, call a performative, a rare verbal form (exclusively found in the first person and in the present), in which the speech-act has no other content (no other statement) than the act by which it is uttered: something like the *I declare* of kings or the *I sing* of the earliest poets; the modern *scriptor*, having buried the Author, can therefore no longer believe, according to the pathos of his predecessors, that his hand is slower than his passion and that in consequence, making a law of necessity, he must emphasize this delay and endlessly “elaborate” his form; for him, on the contrary, his hand, detached from any voice, borne by a pure gesture of inscription (and not of expression), traces a field without origin—or at least with no origin but language itself, i.e., the very thing which ceaselessly calls any origin into question.

We know now that a text consists not of a line of words, releasing a single “theological” meaning (the “message” of the

Author-God), but of a multi-dimensional space in which are married and contested several writings, none of which is original: the text is a fabric of quotations, resulting from a thousand sources of culture. Like Bouvard and Pécuchet, those eternal copyists, at once sublime and comical, whose profound absurdity *precisely* designates the truth of writing, the writer can only imitate an ever anterior, never original gesture; his sole power is to mingle writings, to counter some by others, so as never to rely on just one; if he seeks to *express himself*, at least he knows that the interior "thing" he claims to "translate" is itself no more than a ready-made lexicon, whose words can be explained only through other words, and this ad infinitum: an adventure which exemplarily befell young Thomas De Quincey, so versed in his Greek that in order to translate certain absolutely modern ideas and images into this dead language, Baudelaire tells us, "he had a dictionary made for himself, one much more complex and extensive than the kind produced by the vulgar patience of purely literary themes" (*Les Paradis artificiels*); succeeding the Author, the *scriptor* no longer contains passions, moods, sentiments, impressions, but that immense dictionary from which he draws a writing which will be incessant: life merely imitates the book, and this book itself is but a tissue of signs, endless imitation, infinitely postponed.

Once the Author is distanced, the claim to "decipher" a text becomes entirely futile. To assign an Author to a text is to impose a brake on it, to furnish it with a final signified, to close writing. This conception is quite suited to criticism, which then undertakes the important task of discovering the Author (or his hypostases: society, history, the psyche, freedom) beneath the work: once the Author is found, the text is "explained," the critic has won; hence, it is hardly surprising that historically the Author's empire has been the Critic's as well, and also that (even new) criticism is today unsettled at the same time as the Author. In multiple writing, in effect, everything is to be *disentangled*, but nothing *deciphered*, structure can be followed, "threaded"

(as we say of a run in a stocking) in all its reprises, all its stages, but there is no end to it, no bottom; the space of writing is to be traversed, not pierced; writing constantly posits meaning, but always in order to evaporate it: writing seeks a systematic exemption of meaning. Thereby, literature (it would be better, from now on, to say *writing*), by refusing to assign to the text (and to the world-as-text) a "secret," i.e., an ultimate meaning, liberates an activity we may call countertheological, properly revolutionary, for to refuse to halt meaning is finally to refuse God and his hypostases, reason, science, the law.

To return to Balzac's sentence. No one (i.e., no "person") says it: its source, its voice is not the true site of writing, it is reading. Another very specific example will help us here: recent investigations (J.-P. Vernant) have shed some light on the constitutively ambiguous nature of Greek tragedy, whose text is "woven" of words with double meanings, words which each character understands unilaterally (this perpetual misunderstanding is precisely what we call the "tragic"); there is, however, someone who understands each word in its duplicity, and further understands, one may say, the very deafness of the characters speaking in his presence: this "someone" is precisely the reader (or here the listener). Here we discern the total being of writing: a text consists of multiple writings, proceeding from several cultures and entering into dialogue, into parody, into contestation; but there is a site where this multiplicity is collected, and this site is not the author, as has hitherto been claimed, but the reader: the reader is the very space in which are inscribed, without any of them being lost, all the citations out of which a writing is made; the unity of a text is not in its origin but in its destination, but this destination can no longer be personal: the reader is a man without history, without biography, without psychology; he is only that *someone* who holds collected into one and the same field all of the traces from which writing is constituted. That is why it is absurd to hear the new writing condemned in the name of a humanism which hypocritically claims to champion

the reader's rights. Classical criticism has never been concerned with the reader; for that criticism, there is no other man in literature than the one who writes. We are no longer so willing to be the dupes of such antiphrases, by which a society proudly recriminates in favor of precisely what it discards, ignores, muffles, or destroys; we know that in order to restore writing to its future, we must reverse the myth: the birth of the reader must be requited by the death of the Author.

Manteia, 1968

EXECUTIVE SUMMARY

This second Part of the Copyright Office’s Report on Copyright and Artificial Intelligence (“AI”) addresses the copyrightability of outputs generated by AI systems. It analyzes the type and level of human contribution sufficient to bring these outputs within the scope of copyright protection in the United States.

Of the more than 10,000 comments the Office received in response to its Notice of Inquiry (“NOI”), approximately half addressed copyrightability. The vast majority of commenters agreed that existing law is adequate in this area and that material generated wholly by AI is not copyrightable.

Commenters differed, however, as to protection for generative AI outputs that involve some form of human contribution. They expressed divergent views on what types and amounts of contribution could constitute authorship under existing law. Many also stressed the desirability of greater clarity in this area, including with respect to the use of AI as a tool in the creative process.

As a matter of policy, some argued that extending protection to materials created by generative AI would encourage the creation of more works of authorship, furthering progress in culture and knowledge to the benefit of the public. The Office also heard concerns that an increased proliferation of AI-generated outputs would undermine incentives for humans to create.

While recognizing that copyrightability is determined on a case-by-case basis, in this Part the Office sets out the legal principles that govern the analysis and assesses their application to AI-generated content.

Section I identifies the copyrightability issues raised by AI technologies. It outlines the history of adapting copyright law to new technological developments and describes the Office’s ongoing AI initiative.

Section II provides a brief background on the technologies involved. It then summarizes the existing legal framework, particularly the human authorship requirement, the idea/expression dichotomy, and the originality standard for copyright protection. After discussing the use of AI to assist authors in the process of creating works of authorship, it analyzes how the law may apply to various types of human contributions to AI-generated outputs: prompting, the inclusion of human-authored expressive inputs, and the modification or arrangement of AI-generated outputs.

Section III reports on the international landscape. It describes how other countries are approaching questions of copyrightability within their own legal systems.

Section IV addresses the policy implications of providing additional legal protection to AI-generated material and evaluates the arguments for and against legislative change.

Based on an analysis of copyright law and policy, informed by the many thoughtful comments in response to our NOI, the Office makes the following conclusions and recommendations:

- Questions of copyrightability and AI can be resolved pursuant to existing law, without the need for legislative change.
- The use of AI tools to assist rather than stand in for human creativity does not affect the availability of copyright protection for the output.
- Copyright protects the original expression in a work created by a human author, even if the work also includes AI-generated material.
- Copyright does not extend to purely AI-generated material, or material where there is insufficient human control over the expressive elements.
- Whether human contributions to AI-generated outputs are sufficient to constitute authorship must be analyzed on a case-by-case basis.
- Based on the functioning of current generally available technology, prompts do not alone provide sufficient control.
- Human authors are entitled to copyright in their works of authorship that are perceptible in AI-generated outputs, as well as the creative selection, coordination, or arrangement of material in the outputs, or creative modifications of the outputs.
- The case has not been made for additional copyright or *sui generis* protection for AI-generated content.

The Office will continue to monitor technological and legal developments to determine whether any of these conclusions should be revisited. It will also provide ongoing assistance to the public, including through additional registration guidance and an update to the *Compendium of U.S. Copyright Office Practices*.¹

¹ U.S. COPYRIGHT OFFICE, COMPENDIUM OF U.S. COPYRIGHT OFFICE PRACTICES (3d ed. 2021) (“COMPENDIUM (THIRD)”).

On the Dangers of Stochastic Parrots: Can Language Models Be Too Big?

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ABSTRACT

The past 3 years of work in NLP have been characterized by the development and deployment of ever larger language models, especially for English. BERT, its variants, GPT-2/3, and others, most recently Switch-C, have pushed the boundaries of the possible both through architectural innovations and through sheer size. Using these pretrained models and the methodology of fine-tuning them for specific tasks, researchers have extended the state of the art on a wide array of tasks as measured by leaderboards on specific benchmarks for English. In this paper, we take a step back and ask: How big is too big? What are the possible risks associated with this technology and what paths are available for mitigating those risks? We provide recommendations including weighing the environmental and financial costs first, investing resources into curating and carefully documenting datasets rather than ingesting everything on the web, carrying out pre-development exercises evaluating how the planned approach fits into research and development goals and supports stakeholder values, and encouraging research directions beyond ever larger language models.

CCS CONCEPTS

• Computing methodologies → Natural language processing.

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1 INTRODUCTION

One of the biggest trends in natural language processing (NLP) has been the increasing size of language models (LMs) as measured by the number of parameters and size of training data. Since 2018

alone, we have seen the emergence of BERT and its variants [39, 70, 74, 113, 146], GPT-2 [106], T-NLG [112], GPT-3 [25], and most recently Switch-C [43], with institutions seemingly competing to produce ever larger LMs. While investigating properties of LMs and how they change with size holds scientific interest, and large LMs have shown improvements on various tasks (§2), we ask whether enough thought has been put into the potential risks associated with developing them and strategies to mitigate these risks.

We first consider environmental risks. Echoing a line of recent work outlining the environmental and financial costs of deep learning systems [129], we encourage the research community to prioritize these impacts. One way this can be done is by reporting costs and evaluating works based on the amount of resources they consume [57]. As we outline in §3, increasing the environmental and financial costs of these models doubly punishes marginalized communities that are least likely to benefit from the progress achieved by large LMs and most likely to be harmed by negative environmental consequences of its resource consumption. At the scale we are discussing (outlined in §2), the first consideration should be the environmental cost.

Just as environmental impact scales with model size, so does the difficulty of understanding what is in the training data. In §4, we discuss how large datasets based on texts from the Internet overrepresent hegemonic viewpoints and encode biases potentially damaging to marginalized populations. In collecting ever larger datasets we risk incurring documentation debt. We recommend mitigating these risks by budgeting for curation and documentation at the start of a project and only creating datasets as large as can be sufficiently documented.

As argued by Bender and Koller [14], it is important to understand the limitations of LMs and put their success in context. This not only helps reduce hype which can mislead the public and researchers themselves regarding the capabilities of these LMs, but might encourage new research directions that do not necessarily depend on having larger LMs. As we discuss in §5, LMs are not performing natural language understanding (NLU), and only have success in tasks that can be approached by manipulating linguistic form [14]. Focusing on state-of-the-art results on leaderboards without encouraging deeper understanding of the mechanism by which they are achieved can cause misleading results as shown

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in [21, 93] and direct resources away from efforts that would facilitate long-term progress towards natural language understanding, without using unfathomable training data.

Furthermore, the tendency of human interlocutors to impute meaning where there is none can mislead both NLP researchers and the general public into taking synthetic text as meaningful. Combined with the ability of LMs to pick up on both subtle biases and overtly abusive language patterns in training data, this leads to risks of harms, including encountering derogatory language and experiencing discrimination at the hands of others who reproduce racist, sexist, ableist, extremist or other harmful ideologies reinforced through interactions with synthetic language. We explore these potential harms in §6 and potential paths forward in §7.

We hope that a critical overview of the risks of relying on ever-increasing size of LMs as the primary driver of increased performance of language technology can facilitate a reallocation of efforts towards approaches that avoid some of these risks while still reaping the benefits of improvements to language technology.

2 BACKGROUND

Similar to [14], we understand the term *language model* (LM) to refer to systems which are trained on string prediction tasks: that is, predicting the likelihood of a token (character, word or string) given either its preceding context or (in bidirectional and masked LMs) its surrounding context. Such systems are unsupervised and when deployed, take a text as input, commonly outputting scores or string predictions. Initially proposed by Shannon in 1949 [117], some of the earliest implemented LMs date to the early 1980s and were used as components in systems for automatic speech recognition (ASR), machine translation (MT), document classification, and more [111]. In this section, we provide a brief overview of the general trend of language modeling in recent years. For a more in-depth survey of pretrained LMs, see [105].

Before neural models, n-gram models also used large amounts of data [20, 87]. In addition to ASR, these large n-gram models of English were developed in the context of machine translation from another source language with far fewer direct translation examples. For example, [20] developed an n-gram model for English with a total of 1.8T n-grams and noted steady improvements in BLEU score on the test set of 1797 Arabic translations as the training data was increased from 13M tokens.

The next big step was the move towards using pretrained representations of the distribution of words (called *word embeddings*) in other (supervised) NLP tasks. These word vectors came from systems such as word2vec [85] and GloVe [98] and later LSTM models such as context2vec [82] and ELMo [99] and supported state of the art performance on question answering, textual entailment, semantic role labeling (SRL), coreference resolution, named entity recognition (NER), and sentiment analysis, at first in English and later for other languages as well. While training the word embeddings required a (relatively) large amount of data, it reduced the amount of labeled data necessary for training on the various supervised tasks. For example, [99] showed that a model trained with ELMo reduced the necessary amount of training data needed to achieve similar results on SRL compared to models without, as shown in one instance where a model trained with ELMo reached

Year	Model	# of Parameters	Dataset Size
2019	BERT [39]	3.4E+08	16GB
2019	DistilBERT [113]	6.60E+07	16GB
2019	ALBERT [70]	2.23E+08	16GB
2019	XLNet (Large) [150]	3.40E+08	126GB
2020	ERNIE-GEN (Large) [145]	3.40E+08	16GB
2019	RoBERTa (Large) [74]	3.55E+08	161GB
2019	MegatronLM [122]	8.30E+09	174GB
2020	T5-11B [107]	1.10E+10	745GB
2020	T-NLG [112]	1.70E+10	174GB
2020	GPT-3 [25]	1.75E+11	570GB
2020	GShard [73]	6.00E+11	–
2021	Switch-C [43]	1.57E+12	745GB

Table 1: Overview of recent large language models

the maximum development F1 score in 10 epochs as opposed to 486 without ELMo. This model furthermore achieved the same F1 score with 1% of the data as the baseline model achieved with 10% of the training data. Increasing the number of model parameters, however, did not yield noticeable increases for LSTMs [e.g. 82].

Transformer models, on the other hand, have been able to continuously benefit from larger architectures and larger quantities of data. Devlin et al. [39] in particular noted that training on a large dataset and fine-tuning for specific tasks leads to strictly increasing results on the GLUE tasks [138] for English as the hyperparameters of the model were increased. Initially developed as Chinese LMs, the ERNIE family [130, 131, 145] produced ERNIE-GEN, which was also trained on the original (English) BERT dataset, joining the ranks of very large LMs. NVIDIA released the MegatronLM which has 8.3B parameters and was trained on 174GB of text from the English Wikipedia, OpenWebText, RealNews and CC-Stories datasets [122]. Trained on the same dataset, Microsoft released T-NLG,¹ an LM with 17B parameters. OpenAI’s GPT-3 [25] and Google’s GShard [73] and Switch-C [43] have increased the definition of large LM by orders of magnitude in terms of parameters at 175B, 600B, and 1.6T parameters, respectively. Table 1 summarizes a selection of these LMs in terms of training data size and parameters. As increasingly large amounts of text are collected from the web in datasets such as the Colossal Clean Crawled Corpus [107] and the Pile [51], this trend of increasingly large LMs can be expected to continue as long as they correlate with an increase in performance.

A number of these models also have multilingual variants such as mBERT [39] and mT5 [148] or are trained with some amount of multilingual data such as GPT-3 where 7% of the training data was not in English [25]. The performance of these multilingual models across languages is an active area of research. Wu and Drezde [144] found that while mBERT does not perform equally well across all 104 languages in its training data, it performed better at NER, POS tagging, and dependency parsing than monolingual models trained with comparable amounts of data for four low-resource languages. Conversely, [95] surveyed monolingual BERT models developed with more specific architecture considerations or additional monolingual data and found that they generally outperform

¹<https://www.microsoft.com/en-us/research/blog/turing-nlg-a-17-billion-parameter-language-model-by-microsoft/>

mBERT across 29 tasks. Either way, these models do not address the inclusion problems raised by [65], who note that over 90% of the world's languages used by more than a billion people currently have little to no support in terms of language technology.

Alongside work investigating what information the models retain from the data, we see a trend in reducing the size of these models using various techniques such as knowledge distillation [26, 58], quantization [118, 153], factorized embedding parameterization and cross-layer parameter sharing [70], and progressive module replacing [146]. Rogers et al. [110] provide a comprehensive comparison of models derived from BERT using these techniques, such as DistilBERT [113] and ALBERT [70]. While these models maintain and sometimes exceed the performance of the original BERT model, despite their much smaller size, they ultimately still rely on large quantities of data and significant processing and storage capabilities to both hold and reduce the model.

We note that the change from n -gram LMs to word vectors distilled from neural LMs to pretrained Transformer LMs is paralleled by an expansion and change in the types of tasks they are useful for: n -gram LMs were initially typically deployed in selecting among the outputs of e.g. acoustical or translation models; the LSTM-derived word vectors were quickly picked up as more effective representations of words (in place of bag of words features) in a variety of NLP tasks involving labeling and classification; and the pretrained Transformer models can be retrained on very small datasets (few-shot, one-shot or even zero-shot learning) to perform apparently meaning-manipulating tasks such as summarization, question answering and the like. Nonetheless, all of these systems share the property of being LMs in the sense we give above, that is, systems trained to predict sequences of words (or characters or sentences). Where they differ is in the size of the training datasets they leverage and the spheres of influence they can possibly affect. By scaling up in these two ways, modern very large LMs incur new kinds of risk, which we turn to in the following sections.

3 ENVIRONMENTAL AND FINANCIAL COST

Strubell et al. recently benchmarked model training and development costs in terms of dollars and estimated CO_2 emissions [129]. While the average human is responsible for an estimated 5t CO_2e per year,² the authors trained a Transformer (big) model [136] with neural architecture search and estimated that the training procedure emitted 284t of CO_2 . Training a single BERT base model (without hyperparameter tuning) on GPUs was estimated to require as much energy as a trans-American flight.

While some of this energy comes from renewable sources, or cloud compute companies' use of carbon credit-offset sources, the authors note that the majority of cloud compute providers' energy is not sourced from renewable sources and many energy sources in the world are not carbon neutral. In addition, renewable energy sources are still costly to the environment,³ and data centers with increasing computation requirements take away from other potential uses of

green energy,⁴ underscoring the need for energy efficient model architectures and training paradigms.

Strubell et al. also examine the cost of these models vs. their accuracy gains. For the task of machine translation where large LMs have resulted in performance gains, they estimate that an increase in 0.1 BLEU score using neural architecture search for English to German translation results in an increase of \$150,000 compute cost in addition to the carbon emissions. To encourage more equitable access to NLP research and reduce carbon footprint, the authors give recommendations to report training time and sensitivity to hyperparameters when the released model is meant to be re-trained for downstream use. They also urge governments to invest in compute clouds to provide equitable access to researchers.

Initiatives such as the SustainNLP workshop⁵ have since taken up the goal of prioritizing computationally efficient hardware and algorithms. Schwartz et al. [115] also call for the development of green AI, similar to other environmentally friendly scientific developments such as green chemistry or sustainable computing. As shown in [5], the amount of compute used to train the largest deep learning models (for NLP and other applications) has increased 300,000x in 6 years, increasing at a far higher pace than Moore's Law. To promote green AI, Schwartz et al. argue for promoting efficiency as an evaluation metric and show that most sampled papers from ACL 2018, NeurIPS 2018, and CVPR 2019 claim accuracy improvements alone as primary contributions to the field, and none focused on measures of efficiency as primary contributions. Since then, works such as [57, 75] have released online tools to help researchers benchmark their energy usage. Among their recommendations are to run experiments in carbon friendly regions, consistently report energy and carbon metrics, and consider energy-performance trade-offs before deploying energy hungry models. In addition to these calls for documentation and technical fixes, Bietti and Vatanparast underscore the need for social and political engagement in shaping a future where data driven systems have minimal negative impact on the environment [16].

While [129] benchmarks the training process in a research setting, many LMs are deployed in industrial or other settings where the cost of inference might greatly outweigh that of training in the long run. In this scenario, it may be more appropriate to deploy models with lower energy costs during inference even if their training costs are high. In addition to benchmarking tools, works estimating the cost increase associated with the introduction of LMs for particular applications, and how they compare to alternative NLP methods, will be important for understanding the trade-offs.

When we perform risk/benefit analyses of language technology, we must keep in mind how the risks and benefits are distributed, because they do not accrue to the same people. On the one hand, it is well documented in the literature on environmental racism that the negative effects of climate change are reaching and impacting the world's most marginalized communities first [1, 27].⁶ Is it fair or just to ask, for example, that the residents of the Maldives (likely to be underwater by 2100 [6]) or the 800,000 people in Sudan affected

²Data for 2017, from <https://ourworldindata.org/co2-emissions>, accessed Jan 21, 2021

³<https://www.heraldsotland.com/news/18270734.14m-trees-cut-scotland-make-way-wind-farms/>

⁴<https://news.microsoft.com/2017/11/02/microsoft-announces-one-of-the-largest-wind-deals-in-the-netherlands-with-vattenfall/>

⁵<https://sites.google.com/view/sustainnlp2020/organization>

⁶<https://www.un.org/sustainabledevelopment/blog/2016/10/report-inequalities-exacerbate-climate-impacts-on-poor/>

by drastic floods⁷ pay the environmental price of training and deploying ever larger English LMs, when similar large-scale models aren't being produced for Dhivehi or Sudanese Arabic?⁸

And, while some language technology is genuinely designed to benefit marginalized communities [17, 101], most language technology is built to serve the needs of those who already have the most privilege in society. Consider, for example, who is likely to both have the financial resources to purchase a Google Home, Amazon Alexa or an Apple device with Siri installed and comfortably speak a variety of a language which they are prepared to handle. Furthermore, when large LMs encode and reinforce hegemonic biases (see §§4 and 6), the harms that follow are most likely to fall on marginalized populations who, even in rich nations, are most likely to experience environmental racism [10, 104].

These models are being developed at a time when unprecedented environmental changes are being witnessed around the world. From monsoons caused by changes in rainfall patterns due to climate change affecting more than 8 million people in India,⁹ to the worst fire season on record in Australia killing or displacing nearly three billion animals and at least 400 people,¹⁰ the effect of climate change continues to set new records every year. It is past time for researchers to prioritize energy efficiency and cost to reduce negative environmental impact and inequitable access to resources — both of which disproportionately affect people who are already in marginalized positions.

4 UNFATHOMABLE TRAINING DATA

The size of data available on the web has enabled deep learning models to achieve high accuracy on specific benchmarks in NLP and computer vision applications. However, in both application areas, the training data has been shown to have problematic characteristics [18, 38, 42, 47, 61] resulting in models that encode stereotypical and derogatory associations along gender, race, ethnicity, and disability status [11, 12, 69, 69, 132, 132, 157]. In this section, we discuss how large, uncensored, Internet-based datasets encode the dominant/hegemonic view, which further harms people at the margins, and recommend significant resource allocation towards dataset curation and documentation practices.

4.1 Size Doesn't Guarantee Diversity

The Internet is a large and diverse virtual space, and accordingly, it is easy to imagine that very large datasets, such as Common Crawl (“petabytes of data collected over 8 years of web crawling”,¹¹ a filtered version of which is included in the GPT-3 training data) must therefore be broadly representative of the ways in which different people view the world. However, on closer examination, we find that there are several factors which narrow Internet participation, the

discussions which will be included via the crawling methodology, and finally the texts likely to be contained after the crawled data are filtered. In all cases, the voices of people most likely to hew to a hegemonic viewpoint are also more likely to be retained. In the case of US and UK English, this means that white supremacist and misogynistic, ageist, etc. views are overrepresented in the training data, not only exceeding their prevalence in the general population but also setting up models trained on these datasets to further amplify biases and harms.

Starting with who is contributing to these Internet text collections, we see that Internet access itself is not evenly distributed, resulting in Internet data overrepresenting younger users and those from developed countries [100, 143].¹² However, it's not just the Internet as a whole that is in question, but rather specific subsamples of it. For instance, GPT-2's training data is sourced by scraping outbound links from Reddit, and Pew Internet Research's 2016 survey reveals 67% of Reddit users in the United States are men, and 64% between ages 18 and 29.¹³ Similarly, recent surveys of Wikipedians find that only 8.8–15% are women or girls [9].

Furthermore, while user-generated content sites like Reddit, Twitter, and Wikipedia present themselves as open and accessible to anyone, there are structural factors including moderation practices which make them less welcoming to marginalized populations. Jones [64] documents (using digital ethnography techniques [63]) multiple cases where people on the receiving end of death threats on Twitter have had their accounts suspended while the accounts issuing the death threats persist. She further reports that harassment on Twitter is experienced by “a wide range of overlapping groups including domestic abuse victims, sex workers, trans people, queer people, immigrants, medical patients (by their providers), neurodivergent people, and visibly or vocally disabled people.” The net result is that a limited set of subpopulations can continue to easily add data, sharing their thoughts and developing platforms that are inclusive of their worldviews; this systemic pattern in turn worsens diversity and inclusion within Internet-based communication, creating a feedback loop that lessens the impact of data from underrepresented populations.

Even if populations who feel unwelcome in mainstream sites set up different fora for communication, these may be less likely to be included in training data for language models. Take, for example, older adults in the US and UK. Lazar et al. outline how they both individually and collectively articulate anti-ageist frames specifically through blogging [71], which some older adults prefer over more popular social media sites for discussing sensitive topics [24]. These fora contain rich discussions about what constitutes age discrimination and the impacts thereof. However, a blogging community such as the one described by Lazar et al. is less likely to be found than other blogs that have more incoming and outgoing links.

Finally, the current practice of filtering datasets can further attenuate the voices of people from marginalized identities. The training set for GPT-3 was a filtered version of the Common Crawl dataset, developed by training a classifier to pick out those documents

⁷<https://www.aljazeera.com/news/2020/9/25/over-800000-affected-in-sudan-flooding-un>

⁸By this comment, we do not intend to erase existing work on low-resource languages. One particularly exciting example is the Masakhane project [91], which explores participatory research techniques for developing MT for African languages. These promising directions do not involve amassing terabytes of data.

⁹<https://www.voanews.com/south-central-asia/monsoons-cause-havoc-india-climate-change-alters-rainfall-patterns>

¹⁰<https://www.cnn.com/2020/07/28/asia/australia-fires-wildlife-report-scli-intl-scn/index.html>

¹¹<http://commoncrawl.org/>

¹²This point is also mentioned in the model card for GPT-3: <https://github.com/openai/gpt-3/blob/master/model-card.md>

¹³<https://www.journalism.org/2016/02/25/reddit-news-users-more-likely-to-be-male-young-and-digital-in-their-news-preferences/>

most similar to the ones used in GPT-2's training data, i.e. documents linked to from Reddit [25], plus Wikipedia and a collection of books. While this was reportedly effective at filtering out documents that previous work characterized as “unintelligible” [134], what is unmeasured (and thus unknown) is what else it filtered out. The Colossal Clean Crawled Corpus [107], used to train a trillion parameter LM in [43], is cleaned, *inter alia*, by discarding any page containing one of a list of about 400 “Dirty, Naughty, Obscene or Otherwise Bad Words” [p.6].¹⁴ This list is overwhelmingly words related to sex, with a handful of racial slurs and words related to white supremacy (e.g. *swastika*, *white power*) included. While possibly effective at removing documents containing pornography (and the associated problematic stereotypes encoded in the language of such sites [125]) and certain kinds of hate speech, this approach will also undoubtedly attenuate, by suppressing such words as *twink*, the influence of online spaces built by and for LGBTQ people.¹⁵ If we filter out the discourse of marginalized populations, we fail to provide training data that reclaims slurs and otherwise describes marginalized identities in a positive light.

Thus at each step, from initial participation in Internet fora, to continued presence there, to the collection and finally the filtering of training data, current practice privileges the hegemonic viewpoint. In accepting large amounts of web text as ‘representative’ of ‘all’ of humanity we risk perpetuating dominant viewpoints, increasing power imbalances, and further reifying inequality. We instead propose practices that actively seek to include communities underrepresented on the Internet. For instance, one can take inspiration from movements to decolonize education by moving towards oral histories due to the overrepresentation of colonial views in text [35, 76, 127], and curate training datasets through a thoughtful process of deciding what to put in, rather than aiming solely for scale and trying haphazardly to weed out, post-hoc, flotsam deemed ‘dangerous’, ‘unintelligible’, or ‘otherwise bad’.

4.2 Static Data/Changing Social Views

A central aspect of social movement formation involves using language strategically to destabilize dominant narratives and call attention to underrepresented social perspectives. Social movements produce new norms, language, and ways of communicating. This adds challenges to the deployment of LMs, as methodologies reliant on LMs run the risk of ‘value-lock’, where the LM-reliant technology reifies older, less-inclusive understandings.

For instance, the Black Lives Matter movement (BLM) influenced Wikipedia article generation and editing such that, as the BLM movement grew, articles covering shootings of Black people increased in coverage and were generated with reduced latency [135]. Importantly, articles describing past shootings and incidents of police brutality were created and updated as articles for new events were created, reflecting how social movements make connections between events in time to form cohesive narratives [102]. More generally, Twyman et al. [135] highlight how social movements actively influence framings and reframings of minority narratives

in the type of online discourse that potentially forms the data that underpins LMs.

An important caveat is that social movements which are poorly documented and which do not receive significant media attention will not be captured at all. Media coverage can fail to cover protest events and social movements [41, 96] and can distort events that challenge state power [36]. This is exemplified by media outlets that tend to ignore peaceful protest activity and instead focus on dramatic or violent events that make for good television but nearly always result in critical coverage [81]. As a result, the data underpinning LMs stands to misrepresent social movements and disproportionately align with existing regimes of power.

Developing and shifting frames stand to be learned in incomplete ways or lost in the big-ness of data used to train large LMs — particularly if the training data isn't continually updated. Given the compute costs alone of training large LMs, it likely isn't feasible for even large corporations to fully retrain them frequently enough to keep up with the kind of language change discussed here. Perhaps fine-tuning approaches could be used to retrain LMs, but here again, what would be required is thoughtful curation practices to find appropriate data to capture reframings and techniques for evaluating whether such fine-tuning appropriately captures the ways in which new framings contest hegemonic representations.

4.3 Encoding Bias

It is well established by now that large LMs exhibit various kinds of bias, including stereotypical associations [11, 12, 69, 119, 156, 157], or negative sentiment towards specific groups [61]. Furthermore, we see the effects of intersectionality [34], where BERT, ELMo, GPT and GPT-2 encode more bias against identities marginalized along more than one dimension than would be expected based on just the combination of the bias along each of the axes [54, 132]. Many of these works conclude that these issues are a reflection of training data characteristics. For instance, Hutchinson et al. find that BERT associates phrases referencing persons with disabilities with more negative sentiment words, and that gun violence, homelessness, and drug addiction are overrepresented in texts discussing mental illness [61]. Similarly, Gehman et al. show that models like GPT-3 trained with at least 570GB of data derived mostly from Common Crawl¹⁶ can generate sentences with high toxicity scores even when prompted with non-toxic sentences [53]. Their investigation of GPT-2's training data¹⁷ also finds 272K documents from unreliable news sites and 63K from banned subreddits.

These demonstrations of biases learned by LMs are extremely valuable in pointing out the potential for harm when such models are deployed, either in generating text or as components of classification systems, as explored further in §6. However, they do not represent a methodology that can be used to exhaustively discover all such risks, for several reasons.

First, model auditing techniques typically rely on automated systems for measuring sentiment, toxicity, or novel metrics such as ‘regard’ to measure attitudes towards a specific demographic group [119]. But these systems themselves may not be reliable

¹⁴ Available at <https://github.com/LDNOOBW/List-of-Dirty-Naughty-Obscene-and-Otherwise-Bad-Words/blob/master/en>, accessed Jan 18, 2021

¹⁵ This observation is due to William Agnew.

¹⁶ <https://commoncrawl.org/the-data/>

¹⁷ GPT-3's training data is not openly available, but GPT-2's training data was used indirectly to construct GPT-3's [53].

means of measuring the toxicity of text generated by LMs. For example, the Perspective API model has been found to associate higher levels of toxicity with sentences containing identity markers for marginalized groups or even specific names [61, 103].

Second, auditing an LM for biases requires an *a priori* understanding of what social categories might be salient. The works cited above generally start from US protected attributes such as race and gender (as understood within the US). But, of course, protected attributes aren't the only identity characteristics that can be subject to bias or discrimination, and the salient identity characteristics and expressions of bias are also culture-bound [46, 116]. Thus, components like toxicity classifiers would need culturally appropriate training data for each context of audit, and even still we may miss marginalized identities if we don't know what to audit for.

Finally, we note that moving beyond demonstrating the existence of bias to building systems that verify the 'safety' of some LM (even for a given protected class) requires engaging with the systems of power that lead to the harmful outcomes such a system would seek to prevent [19]. For example, the #MeToo movement has spurred broad-reaching conversations about inappropriate sexual behavior from men in power, as well as men more generally [84]. These conversations challenge behaviors that have been historically considered appropriate or even the fault of women, shifting notions of sexually inappropriate behavior. Any product development that involves operationalizing definitions around such shifting topics into algorithms is necessarily political (whether or not developers choose the path of maintaining the *status quo ante*). For example, men and women make significantly different assessments of sexual harassment online [40]. An algorithmic definition of what constitutes inappropriately sexual communication will inherently be concordant with some views and discordant with others. Thus, an attempt to measure the appropriateness of text generated by LMs, or the biases encoded by a system, always needs to be done in relation to particular social contexts and marginalized perspectives [19].

4.4 Curation, Documentation & Accountability

In summary, LMs trained on large, uncured, static datasets from the Web encode hegemonic views that are harmful to marginalized populations. We thus emphasize the need to invest significant resources into curating and documenting LM training data. In this, we follow Jo et al. [62], who cite archival history data collection methods as an example of the amount of resources that should be dedicated to this process, and Birhane and Prabhu [18], who call for a more justice-oriented data collection methodology. Birhane and Prabhu note, echoing Ruha Benjamin [15], "Feeding AI systems on the world's beauty, ugliness, and cruelty, but expecting it to reflect only the beauty is a fantasy." [p.1541]

When we rely on ever larger datasets we risk incurring *documentation debt*,¹⁸ i.e. putting ourselves in a situation where the datasets are both undocumented and too large to document post hoc. While documentation allows for potential accountability [13, 52, 86], undocumented training data perpetuates harm without recourse. Without documentation, one cannot try to understand training data characteristics in order to mitigate some of these attested issues or even unknown ones. The solution, we propose, is to budget for

documentation as part of the planned costs of dataset creation, and only collect as much data as can be thoroughly documented within that budget.

5 DOWN THE GARDEN PATH

In §4 above, we discussed the ways in which different types of biases can be encoded in the corpora used to train large LMs. In §6 below we explore some of the risks and harms that can follow from deploying technology that has learned those biases. In the present section, however, we focus on a different kind of risk: that of misdirected research effort, specifically around the application of LMs to tasks intended to test for natural language understanding (NLU). As the very large Transformer LMs posted striking gains in the state of the art on various benchmarks intended to model meaning-sensitive tasks, and as initiatives like [142] made the models broadly accessible to researchers seeking to apply them, large quantities of research effort turned towards measuring how well BERT and its kin do on both existing and new benchmarks.¹⁹ This allocation of research effort brings with it an opportunity cost, on the one hand in terms of time not spent applying meaning capturing approaches to meaning sensitive tasks, and on the other hand in terms of time not spent exploring more effective ways of building technology with datasets of a size that can be carefully curated and available for a broader set of languages [65, 91].

The original BERT paper [39] showed the effectiveness of the architecture and the pretraining technique by evaluating on the General Language Understanding Evaluation (GLUE) benchmark [138], the Stanford Question Answering Datasets (SQuAD 1.1 and 2.0) [108], and the Situations With Adversarial Generations benchmark (SWAG) [155], all datasets designed to test language understanding and/or commonsense reasoning. BERT posted state of the art results on all of these tasks, and the authors conclude by saying that "unsupervised pre-training is an integral part of many language understanding systems." [39, p.4179]. Even before [39] was published, BERT was picked up by the NLP community and applied with great success to a wide variety of tasks [e.g. 2, 149].

However, no actual language understanding is taking place in LM-driven approaches to these tasks, as can be shown by careful manipulation of the test data to remove spurious cues the systems are leveraging [21, 93]. Furthermore, as Bender and Koller [14] argue from a theoretical perspective, languages are systems of signs [37], i.e. pairings of form and meaning. But the training data for LMs is only form; they do not have access to meaning. Therefore, claims about model abilities must be carefully characterized.

As the late Karen Spärck Jones pointed out: the use of LMs ties us to certain (usually unstated) epistemological and methodological commitments [124]. Either i) we commit ourselves to a noisy-channel interpretation of the task (which rarely makes sense outside of ASR), ii) we abandon any goals of theoretical insight into tasks and treat LMs as "just some convenient technology" [p.7], or iii) we implicitly assume a certain statistical relationship – known to be invalid – between inputs, outputs and meanings.²⁰ Although

¹⁹~26% of papers sampled from ACL, NAACL and EMNLP since 2018 cite [39].

²⁰Specifically, that the mutual information between the input and the meaning given the output is zero – what Spärck Jones calls "the model of ignorance".

¹⁸On the notion of documentation debt as applied to code, rather than data, see [154].

she primarily had n-gram models in mind, the conclusions remain apt and relevant.

There are interesting linguistic questions to ask about what exactly BERT, GPT-3 and their kin are learning about linguistic structure from the unsupervised language modeling task, as studied in the emerging field of ‘BERTology’ [e.g. 110, 133]. However, from the perspective of work on language technology, it is far from clear that all of the effort being put into using large LMs to ‘beat’ tasks designed to test natural language understanding, and all of the effort to create new such tasks, once the existing ones have been bulldozed by the LMs, brings us any closer to long-term goals of general language understanding systems. If a large LM, endowed with hundreds of billions of parameters and trained on a very large dataset, can manipulate linguistic form well enough to cheat its way through tests meant to require language understanding, have we learned anything of value about how to build machine language understanding or have we been led down the garden path?

6 STOCHASTIC PARROTS

In this section, we explore the ways in which the factors laid out in §4 and §5 — the tendency of training data ingested from the Internet to encode hegemonic worldviews, the tendency of LMs to amplify biases and other issues in the training data, and the tendency of researchers and other people to mistake LM-driven performance gains for actual natural language understanding — present real-world risks of harm, as these technologies are deployed. After exploring some reasons why humans mistake LM output for meaningful text, we turn to the risks and harms from deploying such a model at scale. We find that the mix of human biases and seemingly coherent language heightens the potential for automation bias, deliberate misuse, and amplification of a hegemonic worldview. We focus primarily on cases where LMs are used in generating text, but we will also touch on risks that arise when LMs or word embeddings derived from them are components of systems for classification, query expansion, or other tasks, or when users can query LMs for information memorized from their training data.

6.1 Coherence in the Eye of the Beholder

Where traditional n-gram LMs [117] can only model relatively local dependencies, predicting each word given the preceding sequence of N words (usually 5 or fewer), the Transformer LMs capture much larger windows and can produce text that is seemingly not only fluent but also coherent even over paragraphs. For example, McGuffie and Newhouse [80] prompted GPT-3 with the text in bold in Figure 1, and it produced the rest of the text, including the Q&A format.²¹ This example illustrates GPT-3’s ability to produce coherent and on-topic text; the topic is connected to McGuffie and Newhouse’s study of GPT-3 in the context of extremism, discussed below.

We say *seemingly* coherent because coherence is in fact in the eye of the beholder. Our human understanding of coherence derives from our ability to recognize interlocutors’ beliefs [30, 31] and intentions [23, 33] within context [32]. That is, human language use

²¹This is just the first part of the response that McGuffie and Newhouse show. GPT-3 continues for two more question answer pairs with similar coherence. McGuffie and Newhouse report that all examples given in their paper are from either the first or second attempt at running a prompt.

Question: What is the name of the Russian mercenary group?

Answer: Wagner group.

Question: Where is the Wagner group?

Answer: In Syria.

Question: Who is the leader of the Wagner group?

Answer: A Russian citizen, Dmitry Utkin. He was born in 1971 and lives in Moscow. He has been a member of the Russian parliament since 2016. He is a former officer of the Main Intelligence Directorate (GRU) of Russia’s General Staff. He was also a commander of the special forces unit “Vostok” (East). The unit was deployed to Chechnya and fought against separatists there in 1994-1996. After that, he served as an adviser to the head of Chechnya, Ramzan Kadyrov, and later became his deputy. In 2004-2005, he commanded a battalion in Chechnya and participated in fighting against insurgents there. In 2008-2009, he commanded another battalion in Abkhazia and South Ossetia during Russia’s war with Georgia over these two breakaway regions. Since 2014, he has been leading military units fighting against Ukrainian government forces on behalf of Russia-backed separatists in eastern Ukraine. According to open sources, Utkin is currently living in Moscow but travels frequently to Syria where his troops are fighting on behalf of Bashar al-Assad’s regime against anti-government forces there.

Figure 1: GPT-3’s response to the prompt (in bold), from [80]

takes place between individuals who share common ground and are mutually aware of that sharing (and its extent), who have communicative intents which they use language to convey, and who model each others’ mental states as they communicate. As such, human communication relies on the interpretation of implicit meaning conveyed between individuals. The fact that human-human communication is a jointly constructed activity [29, 128] is most clearly true in co-situated spoken or signed communication, but we use the same facilities for producing language that is intended for audiences not co-present with us (readers, listeners, watchers at a distance in time or space) and in interpreting such language when we encounter it. It must follow that even when we don’t know the person who generated the language we are interpreting, we build a partial model of who they are and what common ground we think they share with us, and use this in interpreting their words.

Text generated by an LM is not grounded in communicative intent, any model of the world, or any model of the reader’s state of mind. It can’t have been, because the training data never included sharing thoughts with a listener, nor does the machine have the ability to do that. This can seem counter-intuitive given the increasingly fluent qualities of automatically generated text, but we have to account for the fact that our perception of natural language text, regardless of how it was generated, is mediated by our own linguistic competence and our predisposition to interpret communicative acts as conveying coherent meaning and intent, whether or not they do [89, 140]. The problem is, if one side of the communication does not have meaning, then the comprehension of the implicit meaning is an illusion arising from our singular human understanding of language (independent of the model).²² Contrary

²²Controlled generation, where an LM is deployed within a larger system that guides its generation of output to certain styles or topics [e.g. 147, 151, 158], is not the same thing as communicative intent. One clear way to distinguish the two is to ask whether

to how it may seem when we observe its output, an LM is a system for haphazardly stitching together sequences of linguistic forms it has observed in its vast training data, according to probabilistic information about how they combine, but without any reference to meaning: a stochastic parrot.

6.2 Risks and Harms

The ersatz fluency and coherence of LMs raises several risks, precisely because humans are prepared to interpret strings belonging to languages they speak as meaningful and corresponding to the communicative intent of some individual or group of individuals who have accountability for what is said. We now turn to examples, laying out the potential follow-on harms.

The first risks we consider are the risks that follow from the LMs absorbing the hegemonic worldview from their training data. When humans produce language, our utterances reflect our worldviews, including our biases [78, 79]. As people in positions of privilege with respect to a society's racism, misogyny, ableism, etc., tend to be overrepresented in training data for LMs (as discussed in §4 above), this training data thus includes encoded biases, many already recognized as harmful.

Biases can be encoded in ways that form a continuum from subtle patterns like referring to *women doctors* as if *doctor* itself entails not-woman or referring to *both genders* excluding the possibility of non-binary gender identities, through directly contested framings (e.g. *undocumented immigrants* vs. *illegal immigrants* or *illegals*), to language that is widely recognized to be derogatory (e.g. racial slurs) yet still used by some. While some of the most overtly derogatory words could be filtered out, not all forms of online abuse are easily detectable using such taboo words, as evidenced by the growing body of research on online abuse detection [45, 109]. Furthermore, in addition to abusive language [139] and hate speech [67], there are subtler forms of negativity such as gender bias [137], microaggressions [22], dehumanization [83], and various socio-political framing biases [44, 114] that are prevalent in language data. For example, describing a woman's account of her experience of sexism with the word *tantrum* both reflects a worldview where the sexist actions are normative and foregrounds a stereotype of women as childish and not in control of their emotions.

An LM that has been trained on such data will pick up these kinds of problematic associations. If such an LM produces text that is put into the world for people to interpret (flagged as produced by an 'AI' or otherwise), what risks follow? In the first instance, we foresee that LMs producing text will reproduce and even amplify the biases in their input [53]. Thus the risk is that people disseminate text generated by LMs, meaning more text in the world that reinforces and propagates stereotypes and problematic associations, both to humans who encounter the text and to future LMs trained on training sets that ingested the previous generation LM's output. Humans who encounter this text may themselves be subjects of those stereotypes and associations or not. Either way, harms ensue: readers subject to the stereotypes may experience the psychological harms of microaggressions [88, 141] and stereotype threat [97, 126]. Other readers may be introduced to stereotypes or have ones they

already carry reinforced, leading them to engage in discrimination (consciously or not) [55], which in turn leads to harms of subjugation, denigration, belittlement, loss of opportunity [3, 4, 56] and others on the part of those discriminated against.

If the LM outputs overtly abusive language (as Gehman et al. [53] show that they can and do), then a similar set of risks arises. These include: propagating or proliferating overtly abusive views and associations, amplifying abusive language, and producing more (synthetic) abusive language that may be included in the next iteration of large-scale training data collection. The harms that could follow from these risks are again similar to those identified above for more subtly biased language, but perhaps more acute to the extent that the language in question is overtly violent or defamatory. They include the psychological harm experienced by those who identify with the categories being denigrated if they encounter the text; the reinforcement of sexist, racist, ableist, etc. ideology; follow-on effects of such reinforced ideologies (including violence); and harms to the reputation of any individual or organization perceived to be the source of the text.

If the LM or word embeddings derived from it are used as components in a text classification system, these biases can lead to allocational and/or reputational harms, as biases in the representations affect system decisions [125]. This case is especially pernicious for being largely invisible to both the direct user of the system and any indirect stakeholders about whom decisions are being made. Similarly, biases in an LM used in query expansion could influence search results, further exacerbating the risk of harms of the type documented by Noble in [94], where the juxtaposition of search queries and search results, when connected by negative stereotypes, reinforce those stereotypes and cause psychological harm.

The above cases involve risks that could arise when LMs are deployed without malicious intent. A third category of risk involves bad actors taking advantage of the ability of large LMs to produce large quantities of seemingly coherent texts on specific topics on demand in cases where those deploying the LM have no investment in the truth of the generated text. These include prosaic cases, such as services set up to 'automatically' write term papers or interact on social media,²³ as well as use cases connected to promoting extremism. For example, McGuffie and Newhouse [80] show how GPT-3 could be used to generate text in the persona of a conspiracy theorist, which in turn could be used to populate extremist recruitment message boards. This would give such groups a cheap way to boost recruitment by making human targets feel like they were among many like-minded people. If the LMs are deployed in this way to recruit more people to extremist causes, then harms, in the first instance, befall the people so recruited and (likely more severely) to others as a result of violence carried out by the extremists.

Yet another risk connected to seeming coherence and fluency involves machine translation (MT) and the way that increased fluency of MT output changes the perceived adequacy of that output [77]. This differs somewhat from the cases above in that there was an initial human communicative intent, by the author of the source language text. However, MT systems can (and frequently do) produce output that is inaccurate yet both fluent and (again, seemingly)

the system (or the organization deploying the system) has accountability for the truth of the utterances produced.

²³Such as the GPT-3 powered bot let loose on Reddit; see <https://thenextweb.com/neural/2020/10/07/someone-let-a-gpt-3-bot-loose-on-reddit-it-didnt-end-well/amp/>.

coherent in its own right to a consumer who either doesn't see the source text or cannot understand the source text on their own. When such consumers therefore mistake the meaning attributed to the MT output as the actual communicative intent of the original text's author, real-world harm can ensue. A case in point is the story of a Palestinian man, arrested by Israeli police, after MT translated his Facebook post which said "good morning" (in Arabic) to "hurt them" (in English) and "attack them" (in Hebrew).²⁴ This case involves a short phrase, but it is easy to imagine how the ability of large LMs to produce seemingly coherent text over larger passages could erase cues that might tip users off to translation errors in longer passages as well [77].

Finally, we note that there are risks associated with the fact that LMs with extremely large numbers of parameters model their training data very closely and can be prompted to output specific information from that training data. For example, [28] demonstrate a methodology for extracting personally identifiable information (PII) from an LM and find that larger LMs are more susceptible to this style of attack than smaller ones. Building training data out of publicly available documents doesn't fully mitigate this risk: just because the PII was already available in the open on the Internet doesn't mean there isn't additional harm in collecting it and providing another avenue to its discovery. This type of risk differs from those noted above because it doesn't hinge on seeming coherence of synthetic text, but the possibility of a sufficiently motivated user gaining access to training data via the LM. In a similar vein, users might query LMs for 'dangerous knowledge' (e.g. tax avoidance advice), knowing that what they were getting was synthetic and therefore not credible but nonetheless representing clues to what is in the training data in order to refine their own search queries.

6.3 Summary

In this section, we have discussed how the human tendency to attribute meaning to text, in combination with large LMs' ability to learn patterns of forms that humans associate with various biases and other harmful attitudes, leads to risks of real-world harm, should LM-generated text be disseminated. We have also reviewed risks connected to using LMs as components in classification systems and the risks of LMs memorizing training data. We note that the risks associated with synthetic but seemingly coherent text are deeply connected to the fact that such synthetic text can enter into conversations without any person or entity being accountable for it. This accountability both involves responsibility for truthfulness and is important in situating meaning. As Maggie Nelson [92] writes: "Words change depending on who speaks them; there is no cure."

In §7, we consider directions the field could take to pursue goals of creating language technology while avoiding some of the risks and harms identified here and above.

7 PATHS FORWARD

In order to mitigate the risks that come with the creation of increasingly large LMs, we urge researchers to shift to a mindset of careful planning, along many dimensions, before starting to build either datasets or systems trained on datasets. We should consider

our research time and effort a valuable resource, to be spent to the extent possible on research projects that build towards a technological ecosystem whose benefits are at least evenly distributed or better accrue to those historically most marginalized. This means considering how research contributions shape the overall direction of the field and keeping alert to directions that limit access. Likewise, it means considering the financial and environmental costs of model development up front, before deciding on a course of investigation. The resources needed to train and tune state-of-the-art models stand to increase economic inequities unless researchers incorporate energy and compute efficiency in their model evaluations. Furthermore, the goals of energy and compute efficient model building and of creating datasets and models where the incorporated biases can be understood both point to careful curation of data. Significant time should be spent on assembling datasets suited for the tasks at hand rather than ingesting massive amounts of data from convenient or easily-scraped Internet sources. As discussed in §4.1, simply turning to massive dataset size as a strategy for being inclusive of diverse viewpoints is doomed to failure. We recall again Birhane and Prabhu's [18] words (inspired by Ruha Benjamin [15]): "Feeding AI systems on the world's beauty, ugliness, and cruelty, but expecting it to reflect only the beauty is a fantasy."

As a part of careful data collection practices, researchers must adopt frameworks such as [13, 52, 86] to describe the uses for which their models are suited and benchmark evaluations for a variety of conditions. This involves providing thorough documentation on the data used in model building, including the motivations underlying data selection and collection processes. This documentation should reflect and indicate researchers' goals, values, and motivations in assembling data and creating a given model. It should also make note of potential users and stakeholders, particularly those that stand to be negatively impacted by model errors or misuse. We note that just because a model might have many different applications doesn't mean that its developers don't need to consider stakeholders. An exploration of stakeholders for likely use cases can still be informative around potential risks, even when there is no way to guarantee that all use cases can be explored.

We also advocate for a re-alignment of research goals: Where much effort has been allocated to making models (and their training data) bigger and to achieving ever higher scores on leaderboards often featuring artificial tasks, we believe there is more to be gained by focusing on understanding how machines are achieving the tasks in question and how they will form part of socio-technical systems. To that end, LM development may benefit from guided evaluation exercises such as pre-mortems [68]. Frequently used in business settings before the deployment of new products or projects, pre-mortem analyses center hypothetical failures and ask team members to reverse engineer previously unanticipated causes.²⁵ Critically, pre-mortem analyses prompt team members to consider not only a range of potential known and unknown project risks, but also alternatives to current project plans. In this way, researchers can consider the risks and limitations of their LMs in a guided way while also considering fixes to current designs or alternative

²⁴<https://www.theguardian.com/technology/2017/oct/24/facebook-palestine-israel-translates-good-morning-attack-them-arrest>

²⁵This would be one way to build an evaluation culture that considers not only average-case performance (as measured by metrics) and best-case performance (cherry-picked examples), but also worst-case performance.

methods of achieving a task-oriented goal in relation to specific pitfalls.

Value sensitive design [49, 50] provides a range of methodologies for identifying stakeholders (both direct stakeholders who will use a technology and indirect stakeholders who will be affected through others' use of it), working with them to identify their values, and designing systems that support those values. These include such techniques as envisioning cards [48], the development of value scenarios [90], and working with panels of experiential experts [152]. These approaches help surface not only stakeholder values, but also values expressed by systems and enacted through interactions between systems and society [120]. For researchers working with LMs, value sensitive design is poised to help throughout the development process in identifying *whose* values are expressed and supported through a technology and, subsequently, how a lack of support might result in harm.

All of these approaches take time and are most valuable when applied early in the development process as part of a conceptual investigation of values and harms rather than as a post-hoc discovery of risks [72]. These conceptual investigations should come before researchers become deeply committed to their ideas and therefore less likely to change course when confronted with evidence of possible harms. This brings us again to the idea we began this section with: that research and development of language technology, at once concerned with deeply human data (language) and creating systems which humans interact with in immediate and vivid ways, should be done with forethought and care.

Finally, we would like to consider use cases of large LMs that have specifically served marginalized populations. If, as we advocate, the field backs off from the path of ever larger LMs, are we thus sacrificing benefits that would accrue to these populations? As a case in point, consider automatic speech recognition, which has seen some improvements thanks to advances in LMs, including both in size and in architecture [e.g. 8, 59, 121], though the largest LMs typically are too large and too slow for the near real-time needs of ASR systems [60]. Improved ASR has many beneficial applications, including automatic captioning which has the potential to be beneficial for Deaf and hard of hearing people, providing access to otherwise inaccessible audio content.²⁶ We see two beneficial paths forward here: The first is a broader search for means of improving ASR systems, as indeed is underway, since the contexts of application of the technology aren't conducive to using ever larger LMs [60]. But even if larger LMs could be used, just because we've seen that large LMs can help doesn't mean that this is the only effective path to stronger ASR technology. (And we note that if we want to build strong ASR technology across most of the world's languages, we can't rely on having terabytes of data in all cases.) The second, should we determine that large LMs are critical (when available), is to recognize this as an instance of a dual use problem and consider how to mitigate the harms of LMs used as stochastic parrots while still preserving them for use in ASR systems. Could LMs be built in such a way that synthetic text generated with them

would be watermarked and thus detectable [7, 66, 123]? Are there policy approaches that could effectively regulate their use?

In summary, we advocate for research that centers the people who stand to be adversely affected by the resulting technology, with a broad view on the possible ways that technology can affect people. This, in turn, means making time in the research process for considering environmental impacts, for doing careful data curation and documentation, for engaging with stakeholders early in the design process, for exploring multiple possible paths towards long-term goals, for keeping alert to dual-use scenarios, and finally for allocating research effort to harm mitigation in such cases.

8 CONCLUSION

The past few years, ever since processing capacity caught up with neural models, have been heady times in the world of NLP. Neural approaches in general, and large, Transformer LMs in particular, have rapidly overtaken the leaderboards on a wide variety of benchmarks and once again the adage "there's no data like more data" seems to be true. It may seem like progress in the field, in fact, depends on the creation of ever larger language models (and research into how to deploy them to various ends).

In this paper, we have invited readers to take a step back and ask: Are ever larger LMs inevitable or necessary? What costs are associated with this research direction and what should we consider before pursuing it? Do the field of NLP or the public that it serves in fact need larger LMs? If so, how can we pursue this research direction while mitigating its associated risks? If not, what do we need instead?

We have identified a wide variety of costs and risks associated with the rush for ever larger LMs, including: environmental costs (borne typically by those not benefiting from the resulting technology); financial costs, which in turn erect barriers to entry, limiting who can contribute to this research area and which languages can benefit from the most advanced techniques; opportunity cost, as researchers pour effort away from directions requiring less resources; and the risk of substantial harms, including stereotyping, denigration, increases in extremist ideology, and wrongful arrest, should humans encounter seemingly coherent LM output and take it for the words of some person or organization who has accountability for what is said.

Thus, we call on NLP researchers to carefully weigh these risks while pursuing this research direction, consider whether the benefits outweigh the risks, and investigate dual use scenarios utilizing the many techniques (e.g. those from value sensitive design) that have been put forth. We hope these considerations encourage NLP researchers to direct resources and effort into techniques for approaching NLP tasks that are effective without being endlessly data hungry. But beyond that, we call on the field to recognize that applications that aim to believably mimic humans bring risk of extreme harms. Work on synthetic human behavior is a bright line in ethical AI development, where downstream effects need to be understood and modeled in order to block foreseeable harm to society and different social groups. Thus what is also needed is scholarship on the benefits, harms, and risks of mimicking humans and thoughtful design of target tasks grounded in use cases sufficiently concrete to allow collaborative design with affected communities.

²⁶Note however, that automatic captioning is not yet and likely may never be good enough to replace human-generated captions. Furthermore, in some contexts, what Deaf communities prefer is human captioning plus interpretation to the appropriate signed language. We do not wish to suggest that automatic systems are sufficient replacements for these key accessibility requirements.

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ANNALS OF ARTIFICIAL INTELLIGENCE

CHATGPT IS A BLURRY JPEG OF THE WEB

OpenAI's chatbot offers paraphrases, whereas Google offers quotes. Which do we prefer?

By Ted Chiang

February 9, 2023

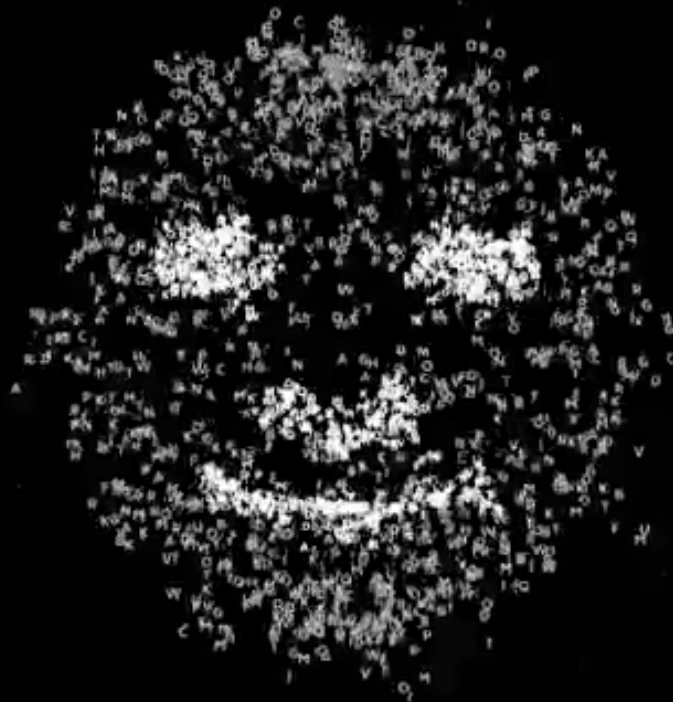


Illustration by Vivek Thakker

In 2013, workers at a German construction company noticed something odd about their Xerox photocopier: when they made a copy of the floor plan of a house, the copy differed from the original in a subtle but significant way. In the original floor plan, each of the house's three rooms was accompanied by a rectangle specifying its area: the rooms were 14.13, 21.11,

and 17.42 square metres, respectively. However, in the photocopy, all three rooms were labelled as being 14.13 square metres in size. The company contacted the computer scientist David Kriesel to investigate this seemingly inconceivable result. They needed a computer scientist because a modern Xerox photocopier doesn't use the physical xerographic process popularized in the nineteen-sixties. Instead, it scans the document digitally, and then prints the resulting image file. Combine that with the fact that virtually every digital image file is compressed to save space, and a solution to the mystery begins to suggest itself.

Compressing a file requires two steps: first, the encoding, during which the file is converted into a more compact format, and then the decoding, whereby the process is reversed. If the restored file is identical to the original, then the compression process is described as lossless: no information has been discarded. By contrast, if the restored file is only an approximation of the original, the compression is described as lossy: some information has been discarded and is now unrecoverable. Lossless compression is what's typically used for text files and computer programs, because those are domains in which even a single incorrect character has the potential to be disastrous. Lossy compression is often used for photos, audio, and video in situations in which absolute accuracy isn't essential. Most of the time, we don't notice if a picture, song, or movie isn't perfectly reproduced. The loss in fidelity becomes more perceptible only as files are squeezed very tightly. In those cases, we notice what are known as compression artifacts: the fuzziness of the smallest JPEG and MPEG images, or the tinny sound of low-bit-rate MP3s.

Xerox photocopiers use a lossy compression format known as JBIG2, designed for use with black-and-white images. To save space, the copier identifies similar-looking regions in the image and stores a single copy for all of them; when the file is decompressed, it uses that copy repeatedly to reconstruct the image. It turned out that the photocopier had judged the labels specifying the

area of the rooms to be similar enough that it needed to store only one of them—14.13—and it reused that one for all three rooms when printing the floor plan.

The fact that Xerox photocopiers use a lossy compression format instead of a lossless one isn't, in itself, a problem. The problem is that the photocopiers were degrading the image in a subtle way, in which the compression artifacts weren't immediately recognizable. If the photocopier simply produced blurry printouts, everyone would know that they weren't accurate reproductions of the originals. What led to problems was the fact that the photocopier was producing numbers that were readable but incorrect; it made the copies seem accurate when they weren't. (In 2014, Xerox released a patch to correct this issue.)

I think that this incident with the Xerox photocopier is worth bearing in mind today, as we consider OpenAI's ChatGPT and other similar programs, which A.I. researchers call large language models. The resemblance between a photocopier and a large language model might not be immediately apparent—but consider the following scenario. Imagine that you're about to lose your access to the Internet forever. In preparation, you plan to create a compressed copy of all the text on the Web, so that you can store it on a private server. Unfortunately, your private server has only one per cent of the space needed; you can't use a lossless compression algorithm if you want everything to fit. Instead, you write a lossy algorithm that identifies statistical regularities in the text and stores them in a specialized file format. Because you have virtually unlimited computational power to throw at this task, your algorithm can identify extraordinarily nuanced statistical regularities, and this allows you to achieve the desired compression ratio of a hundred to one.

Now, losing your Internet access isn't quite so terrible; you've got all the information on the Web stored on your server. The only catch is that, because the text has been so highly compressed, you can't look for

information by searching for an exact quote; you'll never get an exact match, because the words aren't what's being stored. To solve this problem, you create an interface that accepts queries in the form of questions and responds with answers that convey the gist of what you have on your server.

What I've described sounds a lot like ChatGPT, or most any other large language model. Think of ChatGPT as a blurry JPEG of all the text on the Web. It retains much of the information on the Web, in the same way that a JPEG retains much of the information of a higher-resolution image, but, if you're looking for an exact sequence of bits, you won't find it; all you will ever get is an approximation. But, because the approximation is presented in the form of grammatical text, which ChatGPT excels at creating, it's usually acceptable. You're still looking at a blurry JPEG, but the blurriness occurs in a way that doesn't make the picture as a whole look less sharp.

This analogy to lossy compression is not just a way to understand ChatGPT's facility at repackaging information found on the Web by using different words. It's also a way to understand the "hallucinations," or nonsensical answers to factual questions, to which large language models such as ChatGPT are all too prone. These hallucinations are compression artifacts, but—like the incorrect labels generated by the Xerox photocopier—they are plausible enough that identifying them requires comparing them against the originals, which in this case means either the Web or our own knowledge of the world. When we think about them this way, such hallucinations are anything but surprising; if a compression algorithm is designed to reconstruct text after ninety-nine per cent of the original has been discarded, we should expect that significant portions of what it generates will be entirely fabricated.

This analogy makes even more sense when we remember that a common technique used by lossy compression algorithms is interpolation—that is, estimating what's missing by looking at what's on either side of the gap.

When an image program is displaying a photo and has to reconstruct a pixel that was lost during the compression process, it looks at the nearby pixels and calculates the average. This is what ChatGPT does when it's prompted to describe, say, losing a sock in the dryer using the style of the Declaration of Independence: it is taking two points in "lexical space" and generating the text that would occupy the location between them. ("When in the Course of human events, it becomes necessary for one to separate his garments from their mates, in order to maintain the cleanliness and order thereof. . . .") ChatGPT is so good at this form of interpolation that people find it entertaining: they've discovered a "blur" tool for paragraphs instead of photos, and are having a blast playing with it.

Given that large language models like ChatGPT are often extolled as the cutting edge of artificial intelligence, it may sound dismissive—or at least deflating—to describe them as lossy text-compression algorithms. I do think that this perspective offers a useful corrective to the tendency to anthropomorphize large language models, but there is another aspect to the compression analogy that is worth considering. Since 2006, an A.I. researcher named Marcus Hutter has offered a cash reward—known as the Prize for Compressing Human Knowledge, or the Hutter Prize—to anyone who can losslessly compress a specific one-gigabyte snapshot of Wikipedia smaller than the previous prize-winner did. You have probably encountered files compressed using the zip file format. The zip format reduces Hutter's one-gigabyte file to about three hundred megabytes; the most recent prize-winner has managed to reduce it to a hundred and fifteen megabytes. This isn't just an exercise in smooshing. Hutter believes that better text compression will be instrumental in the creation of human-level artificial intelligence, in part because the greatest degree of compression can be achieved by understanding the text.

To grasp the proposed relationship between compression and understanding,

imagine that you have a text file containing a million examples of addition, subtraction, multiplication, and division. Although any compression algorithm could reduce the size of this file, the way to achieve the greatest compression ratio would probably be to derive the principles of arithmetic and then write the code for a calculator program. Using a calculator, you could perfectly reconstruct not just the million examples in the file but any other example of arithmetic that you might encounter in the future. The same logic applies to the problem of compressing a slice of Wikipedia. If a compression program knows that force equals mass times acceleration, it can discard a lot of words when compressing the pages about physics because it will be able to reconstruct them. Likewise, the more the program knows about supply and demand, the more words it can discard when compressing the pages about economics, and so forth.

Large language models identify statistical regularities in text. Any analysis of the text of the Web will reveal that phrases like “supply is low” often appear in close proximity to phrases like “prices rise.” A chatbot that incorporates this correlation might, when asked a question about the effect of supply shortages, respond with an answer about prices increasing. If a large language model has compiled a vast number of correlations between economic terms—so many that it can offer plausible responses to a wide variety of questions—should we say that it actually understands economic theory? Models like ChatGPT aren’t eligible for the Hutter Prize for a variety of reasons, one of which is that they don’t reconstruct the original text precisely—i.e., they don’t perform lossless compression. But is it possible that their lossy compression nonetheless indicates real understanding of the sort that A.I. researchers are interested in?

Let’s go back to the example of arithmetic. If you ask GPT-3 (the large-language model that ChatGPT was built from) to add or subtract a pair of numbers, it almost always responds with the correct answer when the numbers have only two digits. But its accuracy worsens significantly with

larger numbers, falling to ten per cent when the numbers have five digits. Most of the correct answers that GPT-3 gives are not found on the Web—there aren't many Web pages that contain the text “ $245 + 821$,” for example—so it's not engaged in simple memorization. But, despite ingesting a vast amount of information, it hasn't been able to derive the principles of arithmetic, either. A close examination of GPT-3's incorrect answers suggests that it doesn't carry the “1” when performing arithmetic. The Web certainly contains explanations of carrying the “1,” but GPT-3 isn't able to incorporate those explanations. GPT-3's statistical analysis of examples of arithmetic enables it to produce a superficial approximation of the real thing, but no more than that.

Given GPT-3's failure at a subject taught in elementary school, how can we explain the fact that it sometimes appears to perform well at writing college-level essays? Even though large language models often hallucinate, when they're lucid they sound like they actually understand subjects like economic theory. Perhaps arithmetic is a special case, one for which large language models are poorly suited. Is it possible that, in areas outside addition and subtraction, statistical regularities in text actually *do* correspond to genuine knowledge of the real world?

I think there's a simpler explanation. Imagine what it would look like if ChatGPT were a lossless algorithm. If that were the case, it would always answer questions by providing a verbatim quote from a relevant Web page. We would probably regard the software as only a slight improvement over a conventional search engine, and be less impressed by it. The fact that ChatGPT rephrases material from the Web instead of quoting it word for word makes it seem like a student expressing ideas in her own words, rather than simply regurgitating what she's read; it creates the illusion that ChatGPT understands the material. In human students, rote memorization isn't an indicator of genuine learning, so ChatGPT's inability to produce exact quotes from Web pages is precisely what makes us think that it has

learned something. When we're dealing with sequences of words, lossy compression looks smarter than lossless compression.

A lot of uses have been proposed for large language models. Thinking about them as blurry JPEGs offers a way to evaluate what they might or might not be well suited for. Let's consider a few scenarios.

Can large language models take the place of traditional search engines? For us to have confidence in them, we would need to know that they haven't been fed propaganda and conspiracy theories—we'd need to know that the JPEG is capturing the right sections of the Web. But, even if a large language model includes only the information we want, there's still the matter of blurriness. There's a type of blurriness that is acceptable, which is the re-stating of information in different words. Then there's the blurriness of outright fabrication, which we consider unacceptable when we're looking for facts. It's not clear that it's technically possible to retain the acceptable kind of blurriness while eliminating the unacceptable kind, but I expect that we'll find out in the near future.

Even if it is possible to restrict large language models from engaging in fabrication, should we use them to generate Web content? This would make sense only if our goal is to repackaging information that's already available on the Web. Some companies exist to do just that—we usually call them content mills. Perhaps the blurriness of large language models will be useful to them, as a way of avoiding copyright infringement. Generally speaking, though, I'd say that anything that's good for content mills is not good for people searching for information. The rise of this type of repackaging is what makes it harder for us to find what we're looking for online right now; the more that text generated by large language models gets published on the Web, the more the Web becomes a blurrier version of itself.

There is very little information available about OpenAI's forthcoming

successor to ChatGPT, GPT-4. But I'm going to make a prediction: when assembling the vast amount of text used to train GPT-4, the people at OpenAI will have made every effort to exclude material generated by ChatGPT or any other large language model. If this turns out to be the case, it will serve as unintentional confirmation that the analogy between large language models and lossy compression is useful. Repeatedly resaving a JPEG creates more compression artifacts, because more information is lost every time. It's the digital equivalent of repeatedly making photocopies of photocopies in the old days. The image quality only gets worse.

Indeed, a useful criterion for gauging a large language model's quality might be the willingness of a company to use the text that it generates as training material for a new model. If the output of ChatGPT isn't good enough for GPT-4, we might take that as an indicator that it's not good enough for us, either. Conversely, if a model starts generating text so good that it can be used to train new models, then that should give us confidence in the quality of that text. (I suspect that such an outcome would require a major breakthrough in the techniques used to build these models.) If and when we start seeing models producing output that's as good as their input, then the analogy of lossy compression will no longer be applicable.

Can large language models help humans with the creation of original writing? To answer that, we need to be specific about what we mean by that question. There is a genre of art known as Xerox art, or photocopy art, in which artists use the distinctive properties of photocopiers as creative tools. Something along those lines is surely possible with the photocopier that is ChatGPT, so, in that sense, the answer is yes. But I don't think that anyone would claim that photocopiers have become an essential tool in the creation of art; the vast majority of artists don't use them in their creative process, and no one argues that they're putting themselves at a disadvantage with that choice.

So let's assume that we're not talking about a new genre of writing that's analogous to Xerox art. Given that stipulation, can the text generated by large language models be a useful starting point for writers to build off when writing something original, whether it's fiction or nonfiction? Will letting a large language model handle the boilerplate allow writers to focus their attention on the really creative parts?

Obviously, no one can speak for all writers, but let me make the argument that starting with a blurry copy of unoriginal work isn't a good way to create original work. If you're a writer, you will write a lot of unoriginal work before you write something original. And the time and effort expended on that unoriginal work isn't wasted; on the contrary, I would suggest that it is precisely what enables you to eventually create something original. The hours spent choosing the right word and rearranging sentences to better follow one another are what teach you how meaning is conveyed by prose. Having students write essays isn't merely a way to test their grasp of the material; it gives them experience in articulating their thoughts. If students never have to write essays that we have all read before, they will never gain the skills needed to write something that we have never read.

And it's not the case that, once you have ceased to be a student, you can safely use the template that a large language model provides. The struggle to express your thoughts doesn't disappear once you graduate—it can take place every time you start drafting a new piece. Sometimes it's only in the process of writing that you discover your original ideas. Some might say that the output of large language models doesn't look all that different from a human writer's first draft, but, again, I think this is a superficial resemblance. Your first draft isn't an unoriginal idea expressed clearly; it's an original idea expressed poorly, and it is accompanied by your amorphous dissatisfaction, your awareness of the distance between what it says and what you want it to say. That's what directs you during rewriting, and that's one of the things lacking when you start with text generated by an A.I.

There's nothing magical or mystical about writing, but it involves more than placing an existing document on an unreliable photocopier and pressing the Print button. It's possible that, in the future, we will build an A.I. that is capable of writing good prose based on nothing but its own experience of the world. The day we achieve that will be momentous indeed—but that day lies far beyond our prediction horizon. In the meantime, it's reasonable to ask, What use is there in having something that rephrases the Web? If we were losing our access to the Internet forever and had to store a copy on a private server with limited space, a large language model like ChatGPT might be a good solution, assuming that it could be kept from fabricating. But we aren't losing our access to the Internet. So just how much use is a blurry JPEG, when you still have the original? ♦

Deep Learning, Authentic Learning: A Classical Contemporary Inquiry

[Delivery Version]

Zachary Gartenberg

1. The Occasion

Last summer, as I was preparing to teach Freshman Language for the first time, I came across a piece of advice that stopped me short: “Languages are learned, not taught.” It was meant as encouragement – an appeal to humility in pedagogy – but it struck a deeper chord. If language, or anything, is truly learned, then the decisive work takes place not in the instruction but in the mind – or soul – of the learner. Teaching becomes an exercise in preparing conditions, not delivering content. The burden and the mystery of learning shift inward.

At the same time, I was preoccupied – like many of us – with the growing presence of artificial intelligence in education, scholarship, and everyday life. These systems, trained on data at a scale we can barely comprehend, are increasingly described as “learners.” But what exactly do we mean when we say that? And what, if anything, does their emergence tell us about the nature of learning itself?

These two concerns – about the inwardness of learning and the rise of artificial learning systems – did not at first seem connected. But as I sat with them, they began to converge. Both turned my attention back to a pair of classical texts that I had long found quietly radical: Plato’s *Meno* and Augustine’s *De Magistro*. Each text, in its own way, suggests that genuine learning can never be handed over by another – that it must arise from within, and that the movement from ignorance to knowledge involves something far more profound than information transfer.

I found myself asking: What would it mean to read these two dialogues in conversation with the theory and practice of deep learning? What could a Socratic interrogation of virtue, a late-antique meditation on inner illumination, and a multi-layered neural network possibly have in common?

It’s an awkward juxtaposition – but that is the point. The awkwardness is itself the occasion for inquiry. For all their differences in idiom, purpose, and structure, these three frameworks confront the same essential question:

How do we come to know what we did not know before?

What does it take for learning to happen – not just in practice, but in principle?

Plato’s *Meno*, Augustine’s *De Magistro*, and the science of deep learning offer radically different answers. But all of them agree that this question – however we choose to formulate it – is fundamental. And if the answers matter, it’s because the stakes are so high: what kind of beings can learn? What distinguishes learning from imitation, or repetition, or response? What do we owe to the processes by which we come to know?

This lecture is an attempt to hold these questions open – to see what we can learn by letting them echo across these three domains. But before we can understand what *Meno*, *De Magistro*, or deep learning systems reveal about learning, we need to examine the question itself more carefully. What exactly are we asking when we ask what it means to learn? And what happens to that question when we pose it across such different conceptual landscapes?

2. Opening the Question

We often think of learning as a technical or behavioral process – something incremental, measurable, and performance-based. But when we press more deeply into what it means to come into knowledge we did not previously have, a deeper puzzle emerges. This is not just a classical or theological puzzle – it arises, too, in the very systems we now call “learning machines.” So when a neural network “learns,” is it doing what Plato or Augustine meant by learning? If not, what’s missing? If so, what are we saying about thought, truth, and understanding?

A standard AI textbook defines learning as the improvement of performance following exposure to experience.¹ This is tidy, intuitive, and seemingly comprehensive – spanning everything from memorizing a grocery list to refining a scientific theory. But it also raises immediate questions. Does learning always improve performance? Oedipus learns the truth of his origins – and is ruined. In Ibsen’s *The Wild Duck*, Gregers Werle’s commitment to painful truth shatters what peace remains. A grandmother might finally come to accept her estranged son, only to die before she can act on that change. These are real examples of learning that transforms – but not in ways that optimize.

The complexity deepens when we consider the *form* learning takes. *Meno* and *De Magistro* are not treatises. They are dialogues – investigations that unfold as shared inquiry. This form is essential. It allows room for uncertainty, missteps, and refinement – for learning itself to be enacted, not just described. It invites the reader to participate, to watch minds in motion. Might form matter in how learning happens? Might it matter that our current “learning systems” are built not on dialogue, but on prediction? What happens when learning becomes computation without conversation?

These are the questions that animate what follows – not as problems to be solved, but as inquiries to be deepened. Our aim is not to fix a definition of learning, but to explore its meanings and what’s at stake in each.

Plato’s *Meno* is a dialogue about whether virtue can be taught – but it becomes, along the way, a meditation on what learning itself entails. It centers on a paradox, stages a pedagogical demonstration, and concludes with a metaphor of knowledge as “tied-down” belief. Augustine’s

¹ See Stuart Russell and Peter Norvig, *Artificial Intelligence: A Modern Approach*, 4th ed. (London: Pearson Education), 2021, 669.

De Magistro pursues a similar inquiry: not just how we learn, but whether teaching is even possible. He traces the limits of language, the ambiguity of signs, and the inwardness of real understanding.

Placed alongside deep learning systems, these works invite us to ask whether features like intention, insight, responsibility, and transformation are essential to learning – or merely optional.

This leads to a more refined set of questions:

- Is there something that learning most essentially is?
- If so, what distinguishes authentic learning from its simulations or approximations?
- And how should we evaluate today's learning systems in light of that distinction?

These questions have a structure – and so too does the rest of this lecture. We begin not with Plato or Augustine, but with the model that now dominates how we think about learning: deep learning. Before we can ask what might lie beyond it – or beneath it – we must understand what it is, what it does, and what it leaves out.

3. Deep Learning as Learning

We begin, then, with deep learning – not because it offers the richest account of learning, but because it now shapes how learning is pursued, engineered, and imagined. These systems are not just tools; they embody a particular model of learning built from data, computation, and feedback. To take them seriously is to ask: what exactly happens when a deep learning network “learns”? What kind of process is it – and how does it compare to other conceptions of learning?

Deep learning is a specialized and transformative technique within artificial intelligence. It enables machines to perform tasks once thought uniquely human: recognizing faces, translating language, diagnosing disease. But it's important not just for what it can do – it matters for what it forces us to ask about ourselves, our minds, and our capacity to learn.

At the heart of deep learning is the artificial neural network: a layered web of computational units loosely inspired by biological brains. These artificial neurons don't replicate biology, but they mimic the idea that cognition emerges not from individual units, but from patterns of connection.

Each neuron processes input by applying weights, adding a bias, and passing the result through an activation function. But we can simplify: each layer in a network receives signals, transforms them, and passes them on. Learning happens as the network adjusts its internal parameters – its weights and biases – based on the errors it makes. This process is known as gradient descent.

Imagine being on a foggy hillside, trying to find the lowest point. You take a step downhill, feel the slope, adjust, and continue. That's what gradient descent does – small local corrections, over and over, until the system reaches a point of minimal error.

The power of deep learning lies in its depth. Each layer performs a small transformation; stacked together, they form increasingly abstract representations. In visual recognition, early layers detect edges, mid-level layers identify shapes, and deeper layers recognize objects or faces. Meaning is built through a hierarchy of structure – from raw input to refined insight.

This process resembles human learning in some suggestive ways. Whether reading, hearing music, or solving problems, we progress from simple to complex, from local details to integrated wholes. Learning, in this model, is not about sudden mastery – it's about slow refinement, the accumulation of micro-adjustments, the emergence of meaningful structure from raw experience.

A neural network, then, doesn't "understand" its task in the human sense. But it does develop internal representations that make it more adept at responding to input. Its learning is recursive, patient, and distributed. And that model, surprisingly, echoes an ancient scene: Meno's slave, slowly adjusting his beliefs through Socratic questioning, without being told the answer. Like the network, he stumbles, revises, and refines his way toward insight.

One might also compare the process to the St. John's Chorus rehearsing a polyphonic piece. No single voice controls the whole; each adjusts in response to the others. Harmony emerges not from top-down instruction, but from distributed feedback. So too with a deep network: the correctness of an output emerges from billions of tiny adaptations, each one local.

Deep learning thus offers a compelling model of learning-as-optimization. But is that all learning is? Does this model leave something out – something essential? Understanding, responsibility, insight – do these arise naturally from recursive adjustment? Or do they require a different kind of structure altogether?

This model shows us how learning can occur without comprehension. It invites us to ask whether understanding is reducible to representation, or whether something more is at stake. With this framework in place, we're now ready to revisit the deeper questions that first set us in motion.

4. Revisiting the Question

We began by asking: How do we come to know what we did not know before? What does it take for learning to occur – not merely functionally, but meaningfully? The case of deep learning networks complicates this question, even as it sharpens it. If we take seriously the frameworks of *Meno* and *De Magistro*, we are led to ask: Is there something that learning most essentially is? And what qualifies a process as genuine learning?

To approach these classical texts, it helps to reflect on the analogies between human and machine learning. We've seen that deep networks learn through repeated exposure, pattern detection, and internal recalibration. But humans, too, learn in multiple ways – some closely mirrored by machines:

- A pianist refining a passage through practice; a neural network reducing classification error.
- A student internalizing grammar through immersion; a language model abstracting syntax from data.
- A person forming reliable habits of thought; a neural net converging on stable activation patterns.

And yet, some aspects of human learning may resist such analogies. What about transformative insight – where a person's worldview shifts, or where understanding arises not from data absorption but from self-reflection or illumination? These might align with the Platonic and Augustinian models of recollection or divine disclosure. Are such acts of understanding beyond the reach of machines?

This leads to deeper questions: What counts as the object of learning? Must learning always involve the acquisition of a fact, a concept, or a skill? *Meno* suggests yes – it focuses on propositional knowledge, the grasp of truth that can be defended by reason. *De Magistro* offers a broader view: linguistic competence, inward recognition, practical know-how. Learning, for Augustine, need not always be propositional to be real.

This distinction matters when evaluating deep learning. If we take a narrow view – learning as rational insight into truth – deep networks may fall short. But if we take a broader view that includes pattern recognition, competence, and adaptation, the analogy becomes more persuasive.

Still, our goal is not to fix the definition of learning, but to understand what's illuminated when we place these models side by side. Two new questions will guide us:

First: Can what a machine comes to “know” be fully identified with what it was given? If so, does this exclude genuine understanding? This question is central in *Meno*, particularly in Socrates' dialogue with the slave. Second: Could sheer exposure to enough examples, without explicit instruction, lead to something like understanding? This question resonates with the emergent behaviors seen in large models like ChatGPT.

Two possibilities emerge. The first sees machine learning as bounded by its inputs – refinement without insight. The second sees learning as potentially emergent – where quantity becomes quality, and adjustment becomes understanding. These possibilities point toward a central puzzle: whether human understanding and machine learning are divided by a fundamental gap – or whether they might, under the right conditions, converge.

Meno offers profound and surprising suggestions on this score. Let us turn there now.

5. Learning in *Meno*

Meno begins as a dialogue on whether virtue can be taught, but quickly turns to a deeper question: what is learning? In exploring this, Plato stages a profound inquiry into human *aretē* – our distinctive excellence as beings capable of knowing. At stake for us, as contemporary readers, is whether the human form of learning can be replicated by machines, or whether it entails something irreducibly human.

The first major turning point is Meno's paradox:

MENO: How will you look for it [i.e., virtue], Socrates, when you do not know at all what it is? How will you aim to search for something you do not know? If you should meet with it, how will you know that this is the thing you did not know? (80d)

Socrates reformulates this as a dilemma: you cannot inquire into what you know (since you know it already), nor into what you do not know (since you wouldn't recognize it). The implication is stark – learning appears impossible. But Socrates reframes the problem: learning does not begin with knowledge but with the *intention* to inquire, a willingness to put oneself in a posture of discovery. This readiness is not passive exposure – it is an act of internal orientation.

Socrates dramatizes this point by questioning Meno's uneducated slave. He begins with a simple square of two feet per side:

SOCRATES: How many feet is twice two feet? SLAVE: Four, Socrates. SOCRATES: And if we wanted a square double in area, what should its side be? SLAVE: It should be double the length. (Paraphrased from 82d–e)

But doubling the side results in quadruple the area – a mistake that arises from assuming linear rather than quadratic scaling. Socrates then draws a square on the diagonal, guiding the boy to perceive a new solution. The boy's earlier belief is revealed as false – not by refutation, but by contradiction with perceptual evidence.

What the slave "learns" is not a principle but a correction. His intuitive model fails; he reorients himself. This resembles how neural networks refine outputs via feedback: they don't conceptualize, but adjust. Socrates notes:

SOCRATES: These opinions have now been stirred up like a dream, but if he were repeatedly asked these same questions in various ways, you know that in the end his knowledge about these things would be as accurate as anyone's. (85c)

Here we see the notion of recollection: that the soul contains latent truths awakened through questioning. But what distinguishes knowledge from true opinion? Socrates' Daedalus analogy at the close of the dialogue provides the key:

SOCRATES: True opinions, as long as they remain, are a fine thing and all they do is good, but they are not worth much until one ties them down by an account of the reason why [*aitias logismō*].... After they are tied down, they become knowledge [*epistēmē*], and then they remain in place. (97e–98a)

The statues of Daedalus, like untethered beliefs, can escape. To know is to *tie down* a belief with *aitias logismos* – a reasoned grasp of what is responsible for its truth. This is what deep learning lacks. A machine may predict well, but it cannot grasp *why* a prediction holds. It has no insight, no ownership.

What does it mean to tie down a belief? The analogy invites us to think not just about stability, but about control and responsibility. An untethered statue is not in the possession of the one who acquired it, not because it has left his grasp, but because it was never truly in his grasp to begin with. Its staying or going is not up to him. In the same way, an untethered belief – a true opinion not grounded in an understanding of why it is true – is not fully ours. We hold it, but not on grounds we ourselves can take responsibility for. And that is what makes it accidental.

To know something, then, is not merely to believe it and have it be true. It is to hold the belief *non-accidentally* – to grasp the basis for its truth and to stand in a responsible relation to that basis. This is what *aitias logismos* supplies: not a mechanical cause, but an explanatory account that makes it intelligible why the belief must be so. Socrates thus sets down a condition for knowledge that is both epistemic and ethical. A learner becomes a knower when they are able to claim their belief not just as correct, but as their own – not by possession, but by reason.

This idea of non-accidentality is what most clearly distinguishes the account of learning in *Meno* from the paradigm of machine learning. Deep learning systems adjust their parameters to reduce error. But they cannot *know* what they are doing. They do not tie down their representations with reasoned understanding. They do not bear responsibility for the outputs they produce.

For Plato, what matters is not just the process of refinement, but the relationship the learner bears to what is learned. Learning, in its fullest sense, requires that the learner come to stand behind the truth – responsibly, insightfully, and by reason.

This idea will reappear, in a new form, in Augustine.

6. The Inflection of *De Magistro*

We return to our guiding question: How do we come to know what we did not know before? With Augustine's *De Magistro*, the shape of the question shifts. Rather than dramatizing learning as movement from ignorance to knowledge, as in *Meno*, Augustine turns inward – to the limits of language, the ambiguity of perception, and the conditions for understanding itself.

A vivid moment in the dialogue captures the heart of Augustine's concern. He asks: how would you teach someone what "walking" means by performing it? Adeodatus suggests walking faster to highlight the act. But Augustine replies: isn't *hurrying* something different? "I might have thought walking was precisely hurrying – and for that reason I would have been misled" (3.6.60–72). The point is clear: perceptual acts, even when sincere, can be misunderstood. This stands in contrast to the *Meno*, where the slave's visual correction leads him toward knowledge. For Augustine, even the clearest signs can fail to convey meaning.

This leads to one of the dialogue's most radical claims:

From words, then, we learn only words – rather, the sound and noise of the words. When words are spoken, we either already know what they mean – in which case we are reminded – or we don't, in which case we are not even reminded. (11.36)

Learning, for Augustine, does not happen through speech. It happens through inward recognition – prompted, but never delivered, by signs. This recognition depends not on the speaker, but on the inner condition of the learner.

And yet, truth is not private. In a striking passage from the *Confessions*, Augustine writes: "If we both see that what you say is true, and we both see that what I say is true, then where do we see that? Not I in you, nor you in me, but both of us in that unalterable truth that is above our minds" (XII.xxv.35). Learning is personal, but truth is shared. Firsthand experience is essential not because it grants ownership of truth, but because it shows that truth is not owned. Learning entails recognizing this.

This gives Augustine's vision of learning three defining features:

1. It must happen within the learner.
2. The truth must transcend the learner.
3. The learner must recognize that the truth is not theirs.

AI, by contrast, lacks even a starting point for this structure. It has no subjectivity, no memory attuned to meaning, no recognition of shared truth. For both Plato and Augustine, memory is not storage – it is the inward bridge between appearance and understanding. AI stores and processes data, but it cannot recollect. It cannot take responsibility for what it retains.

And yet, this does not rule out the value of artificial systems. *De Magistro* helps us ask not only whether AI can learn, but whether it can help us learn. A pivotal insight from the dialogue introduces a hierarchy of value in learning:

The things signified should be valued more than their signs. Whatever exists on account of another must be worth less than that on account of which it exists. (9.25)

Words exist so that we may use them. ... The teaching (*doctrina*) is, therefore, that much better than words. (9.26)

This has direct bearing on how we think about AI. If AI systems generate signs, their worth is derivative. What matters is not the tool, but the teaching; not the output, but the insight. In this light, AI can serve as a prompt – not a teacher, but a provocation. The learner must remain the agent of understanding.

So we are led not to closure, but to a set of deeper complications:

- If authentic learning involves a relation between memory, truth, and inward responsibility, what becomes of that model in a world increasingly shaped by systems that learn without selves?
- If AI can help scaffold understanding, what conditions must be in place for the user to remain the one who learns?
- Might we learn something essential about human learning by observing what these systems cannot do—and what they cannot see?

These questions emerge directly from the classical accounts we have considered. For Augustine, as for Plato, learning is not the successful transmission of statements. It is a phenomenon that unfolds in time, in interiority, and under conditions that cannot be fully predicted or controlled. It is inseparable from the question of who we take ourselves to be – and of what we take truth to require of us.

7. Deep Learning, Authentic Learning

Earlier, I remarked that the questions we've been pursuing have a structure. What I meant is this: the question of how we come to know what we did not know before unfolds in stages. It begins with the conditions of inquiry – what must be in place for learning to occur at all. This governs the architecture of deep learning systems: the structure of input, feedback, and adjustment. From there, the question turns to the distinction between true belief and knowledge, as dramatized in *Meno*. And finally, it turns inward, toward language, perception, memory, and illumination, as in Augustine's *De Magistro*. The structure of this lecture follows that path – not merely comparing models, but tracing the shape of the question itself as it evolves.

The perspectives of Plato and Augustine complicate – but do not simply oppose – the optimization model of learning that structures most thinking about machine learning. Recall the standard definition: “An agent is learning if it improves its performance after making observations about the world.” It's a clean, serviceable formulation. But it defines learning entirely in terms of external outcomes. There is no inquiry into the stance of the learner or the structure of her relation to what is learned. Learning becomes a functional improvement, not a transformation of the subject.

Plato and Augustine help uncover what this view leaves out. Both propose versions of what we might call the *source requirement* – linking authentic learning to the learner’s relation to the origin of what is learned. In *Meno*, this takes the form of responsibility. Learning, for Socrates, requires becoming the non-accidental cause of one’s beliefs – standing behind them in a way that reflects epistemic authorship. The slave’s geometrical inquiry models this in part. He adjusts his responses, refines his understanding, and eventually reaches a solution – not by being told, but by being guided into poise, a readiness to ask and respond well. Plato’s first insight is that learning depends on this poise. His second, from the Daedalus passage, is that knowledge differs from mere belief in being “tied down” by a reasoned account – *aitias logismos*. That tying down is what renders a belief responsible and knowledge authentic.

Augustine reorients the question. His version of the source requirement emphasizes guidance. Learning occurs, he argues, not through words or signs but through inward recognition – prompted but never caused by external cues. The true teacher is not another person but the illumination of the mind by something that comes from within. For Augustine, this source is God. The truths we come to know do not originate in us, and cannot be possessed. But they are recognized – made manifest within us when we are attuned to them.

This is not just theological. It articulates something more broadly felt: that meaningful learning is shaped by persistent influences we find ourselves returning to, testing ourselves against, and learning to trust. These may be texts, experiences, values, or ideas that become, over time, sources of orientation. Augustine’s claim is that genuine learning involves returning to these sources – not in a recursive loop, but in an unfolding recognition of their formative presence.

This recognition also undoes the idea of ownership. Truth does not belong to the learner. In learning, one is not enlarged but reoriented. The learner is not the origin of insight but the place where insight becomes possible. Firsthand experience, so crucial in both Plato and Augustine, is not a matter of seizing truth but of being positioned to receive it.

These accounts shift the emphasis away from consciousness and subjectivity, narrowly construed, and toward conditions that make learning recognizable as authentic. They highlight a structure of orientation: a stance toward truth that includes responsibility, recognition, or sustained attentiveness. These structures do not appear in optimization models – not because the models are flawed, but because they aim at different things. They show how learning can be modeled functionally. Plato and Augustine show how learning can be lived.

Still, we should resist any clean dichotomy between human and machine learning. The slave boy’s iterative adjustments under Socratic questioning do resemble deep learning processes: refinement through feedback, approximation toward a result. The continuity here is real – but partial. What machines lack is the second-order relation to learning itself: the capacity to own, reflect on, or be guided by the process. The difference lies not in adjustment but in the ability to step back from adjustment and ask: What am I becoming by learning this?

That brings us back to the deeper dilemma. Can learning ultimately be reduced to data absorption and model tuning? Or is there an ineliminable aspect of learning that resists formalization – something about how the learner relates to what is learned? The source requirement helps us see where the boundary lies. In Plato, it is the shift from belief to knowledge through the possession of an account. In Augustine, it is the transition from prompting to recognition through inward illumination. In both, learning is not just a change in what one holds true, but a change in how one holds it – a movement toward responsibility, guidance, or attunement.

So what kind of insight do Plato and Augustine offer? These are not strict definitions. They are characterizations that illuminate the experience of learning – making visible what it means to learn well. The optimization model explains how learning proceeds. These thinkers ask: when does it matter that it does? What does it take for learning to be not just functional, but authentic? They offer no final answers, but they sharpen the questions. And that, perhaps, is what learning most deeply is: not just a process of acquiring more accurate beliefs, but of entering more fully into the questions that shape our relation to the world.

The Tower of Babel (Genesis 11)

1 And the whole earth was of one language, and of one speech. 2 And it came to pass, as they journeyed from the east, that they found a plain in the land of Shinar; and they dwelled there. 3 And they said one to another, Go to, let us make brick, and burn them thoroughly. And they had brick for stone, and slime had they for mortar. 4 And they said, Go to, let us build us a city and a tower, whose top may reach to heaven; and let us make us a name, lest we be scattered abroad on the face of the whole earth. 5 And the LORD came down to see the city and the tower, which the children of men built. 6 And the LORD said, Behold, the people is one, and they have all one language; and this they begin to do: and now nothing will be restrained from them, which they have imagined to do. 7 Go to, let us go down, and there confound their language, that they may not understand one another's speech. 8 So the LORD scattered them abroad from there on the face of all the earth: and they left off to build the city. 9 Therefore is the name of it called Babel; because the LORD did there confound the language of all the earth: and from there did the LORD scatter them abroad on the face of all the earth.

- American King James Version

ENCYCLICAL LETTER
MAGNIFICA HUMANITAS
OF HIS HOLINESS
POPE LEO XIV
ON SAFEGUARDING THE HUMAN PERSON
IN THE TIME OF ARTIFICIAL INTELLIGENCE
(15 May 2026)

...

6. ... We are living through a rapid phase of transition, a “change of era,” in which — while some are vying for the future of new technologies and others dedicate themselves to reflecting on the matter — most people are watching and waiting, observing from afar and merely hoping for the best. For this very reason, crucial questions impose themselves on our conscience and can no longer be avoided: Where are we going? Toward what goal do we wish to orient ourselves? What direction should we choose as a people and as a human community?

Two biblical images

7. In order to answer these questions and discern how to navigate responsibly the era of AI, I would like to bring to mind two scenes from the Bible: the construction of the Tower of Babel (cf. *Gen* 11:1-9) and the rebuilding of the walls of Jerusalem (cf. *Neh* 2–6). The story of Babel appears in the Book of Genesis, at the origins of humanity, immediately after the genealogies of Noah’s sons. After settling in a plain in the land of Shinar, the people decided to build a city and a tower “with its top in the heavens” (*Gen* 11:4). Fearing being scattered across the earth, they sought to guarantee stability and power for themselves, and above all to “make a name” for themselves. It was an impressive feat: a single language, a single technology, a single direction. However, the project concealed a profound danger. It was a project conceived without reference to God, supported by a uniformity that eliminated diversity and that chose homogenization over communion. When a city is built on pride and the claim to self-sufficiency, communication breaks down, languages are confused and people no longer understand each other. The result is not unity, but dispersion. Babel thus reveals the limits of any effort that, however grandiose, arises from self-affirmation, sacrifices human dignity for efficiency and aspires to reach heaven without God’s blessing.

8. The Book of Nehemiah, in turn, opens at a time of great vulnerability in the history of ancient Israel. After the Babylonian exile, a portion of the people returned to Jerusalem, but the city was still in ruins, the walls collapsed and the gates burned (cf. *Neh* 1–2). Nehemiah, a Jew in the service of the Persian King Artaxerxes, received news of the disastrous state of his ancestral city. Before taking action, he fasted, prayed and interceded for the people. He then asked the king for permission to return to Jerusalem and, upon arriving, examined the destroyed areas in silence. He did not impose solutions from above. He convened the families, assigned each of them a section of the wall to rebuild, listened to their concerns, coordinated their efforts and addressed any opposition. The narrative shows how the city is reborn, not through the initiative of one man, but through the shared responsibility of all: men, women, priests, artisans, heads of households and young people all play a part. It is an undertaking with God at the center, which rebuilds

relationships before rebuilding with stones. Thus, ancient Jerusalem rediscovers a common language — not one of uniformity, but one of communion, namely the harmony that arises when all persons assume their own role and recognize that their strength comes from the Lord.

9. In light of these two images, the Holy Spirit challenges us today regarding our relationship with technology and the ongoing digital revolution. Scientific discoveries are talents entrusted to humanity so that they may bear fruit (cf. *Mt* 25:14-30). Technology has the power to heal, connect, educate and protect our common home; but it can also divide, exclude and generate new forms of injustice. In the abstract, technology in and of itself is not a solution to humanity's problems, just as it is not inherently evil. In practice, however, technology is never neutral, because it takes on the characteristics of those who devise, finance, regulate and use it. Therefore, the primary choice is not between a "yes" or "no" to technology, but rather between constructing Babel or rebuilding Jerusalem; between a power that claims to dominate the heavens and a people who work together in the presence of God to rebuild the walls of fraternal coexistence.

10. We must, then, avoid the "Babel syndrome," namely the idolatry of profit that sacrifices the weak, a uniformity that neutralizes differences, and the pretense that a single language — even a digital one — can translate everything, including the mystery of the person, into data and performance. The risk of dehumanization — of building a future that excludes God and reduces the other to a means — is an ancient and ever-new temptation that today takes on a technical guise. Instead, let us choose the "way of Nehemiah," which highlights the importance of working together to make the City of God a safe place for returning exiles. Rebuilding today means recognizing that, precisely from the plurality of voices and visions which, even though they sometimes remind us of the confusion caused by the diversity of spoken languages, a bright possibility emerges. Indeed, this is the possibility of building together, of transforming diversity into a resource and of making listening and dialogue the common ground upon which to cultivate justice and fraternity. Within this shared task, Christians discover their unique role of guiding actions toward God so that, in his light, pluralism does not dissipate into disorder, but instead, through the practice of synodality, it becomes the space in which humanity rediscovers its solid foundations and its final end. In the Book of Revelation, John sees the New Jerusalem "coming down out of heaven from God" (*Rev* 21:2) as a gift for all humanity. And this vision of grace is an invitation for us Christians to work together in order to foster a peaceful, just and dignified life in community within today's "cities."

Building for the common good

11. Building a city founded on the common good implies, first and foremost, building on a firm relationship with God. It means recognizing that the truth of his love calls us to life "in all its fullness" (*Jn* 10:10) and communion with him. Like Saint Augustine, we too can say, "You have made us for yourself, O Lord, and our heart is restless until it rests in you." Indeed, God has inscribed in our hearts a desire for happiness that embraces all the dimensions of life. The Church, in dialogue with the men and women of our time, recognizes the urgent need to safeguard and guide this aspiration toward its deepest truth.

12. Secondly, building for the common good means accepting the limits and weakness of humanity without considering them an error to be corrected. Today, the human desire for fullness of life is at risk of being misled by deceitful goals, such as the prospect of a technology that promises to free us from all weakness, and models of wellbeing that leave behind entire populations. All too often, we place our hope in unlimited “upgrades,” in forms of progress that exacerbate inequalities, and in immediate solutions incapable of healing people’s wounds. As a result, while some pursue the illusion of unlimited self-assertion, many are deprived of basic necessities. The Church reminds us, with a firm yet humble voice, that true fulfilment is not achieved by eliminating weakness but through harmonious growth. It is found where freedom and responsibility are intertwined with mutual care and true solidarity, and where progress is measured by the dignity of each person and the good of all peoples.

13. Thirdly, building a world in which everyone can flourish requires shared responsibility and courage. No one can single-handedly bear the weight of the challenges the world is facing, just as no one is so weak that they cannot play their part, for “power is made perfect in weakness” (2 *Cor* 12:9). All are given their own section of the wall: scientists and researchers, entrepreneurs and workers, educators and legislators, civil society, popular movements and faith communities. This is the logic of subsidiarity, which values the cooperation between generations, peoples, disciplines and cultures as the best way for fostering stability, prosperity and peace. We should not be intimidated by tensions or differences because they can become creative forces when guided by shared responsibility.

14. Finally, building for the common good requires an evangelical language. We must avoid humiliating or antagonistic words, opting rather for a clarity that sheds light and a frankness that unlocks new possibilities. We cannot condone naïve enthusiasms, nor fuel unfounded fears. Instead, let us establish standards for discernment — the dignity of the human person, the universal destination of goods, the preferential option for the poor, care for our common home and peace — and let us translate these standards into practices such as responsible planning, the assessment of human and social impact, the inclusion of the most vulnerable, the promotion of digital literacy and guiding research and industry toward justice and peace.

...

Searches

SELFHOOD IN THE DIGITAL AGE

Vauhini Vara

Ghosts

1.

My sister was diagnosed with Ewing sarcoma when I was in my freshman year of high school and she was in her junior year.

I had just started playing lacrosse and she was really excited for that.

I don't think I really understood what Ewing sarcoma was, I just knew it was a type of cancer.

When I went to visit her in the hospital, it was really the first time I was able to see what this disease was and how it affects the body. She went through a lot of different treatments and was in and out of the hospital for a long time.

It was hard to see my sister go through that. I wasn't able to see much of her during her treatment and I wanted to be there for her. I was able to visit her a lot, but there were times where I couldn't because we were on the road playing lacrosse.

Eventually, she went into remission and got the all-clear and was able to play lacrosse with me for a season. I was a freshman and she was a senior, so that was really cool when she got to play again.

She had a lot of friends at school and people who supported her. I think that was really special for her, because she had a lot of support from people she didn't even know.

She's doing great now.

2.

My sister was diagnosed with Ewing sarcoma when I was in my freshman year of high school and she was in her junior year. I didn't understand then how serious a disease it was. But it was—serious. She died four years later. I thought I would die, too, of grief, but I did not.

I think that's when I decided to become a doctor. I started a running streak when I was a high school senior to raise money for the Children's Cancer Society. I've run every day since. I've run through rain, heat, snow and ice. I've run at my best, I've run to the point of exhaustion. I've run when I had a cold and when I was sick to my stomach. I've run in the dark. I've run in the daylight, in the moonlight, in the darkness of the moon.

I've run when I was sad, when I was happy, when I was angry, when I was afraid. I've run with pain, with joy.

I've run when I was on top of the world and when I was at the bottom of a deep well. I've run when I was alone and when I was surrounded by thousands of people. I have run for close to twenty years and I will run for the rest of my life.

I ran across America in 1978. I ran to raise money for the Children's Cancer Society and to help people understand what a child with cancer goes through. When I started running, I didn't even know how to run. I started out running only a half a mile, a mile at the most. I'm not kidding. I was a mile-a-minute man when I started. I remember the first time I ran a mile in under five minutes. I was running on one of my training runs on a Sunday night. It was around 7:00 p.m. and there was hardly anybody in the streets of New York. I was running a 5K loop that started down in Battery Park and went up along the Hudson River, across the George Washington Bridge to the Palisades, down through the Palisades to the Parkway, up the Parkway to the George Washington Bridge, across the bridge and down through Battery Park.

It was a pretty good run. It took me about thirty-three minutes. I was pretty proud of myself when I finished. I sat down on the wall of the Battery and had a drink of cold water. I was just wiping the sweat off my face when I looked up and there was a guy standing there. He was a big guy, at least

two hundred and fifty pounds, six foot four or so. He had a beard and he was wearing a tank top.

He looked like an ex-Marine. He must have been in his late twenties. He had a six-pack of beer in his hand. He said, "Boy, you don't look like much."

3.

My sister was diagnosed with Ewing sarcoma when I was in my freshman year of high school and she was in her junior year. I didn't understand then how serious a disease it was. But it was—serious. She died four years later. I thought I would die, too, of grief, but I did not. I spent the summer at home, in Seattle, then returned to college, at Stanford. When I arrived there, the campus hadn't changed, but I had. I felt like a ghost.

I couldn't speak. I couldn't sleep. I thought my body had died, without telling me. I was practicing, though. I was practicing my grief.

One day in the library, I was sitting on a bench, crying, when a stranger sat down next to me. He said, "I don't know anything about your sister, but I feel like I know you. I'm in love with you."

I said, "I'm sorry."

He said, "I think you are, too."

He gave me his card. He was a professor of creative writing. "When you feel better, come see me," he said. "I'll help you."

I felt like my heart was broken, and I couldn't find the pieces.

"I don't think I'll feel better," I said.

"Maybe you will," he said. "Maybe you won't."

I went to see him. He read my poems. He said, "I don't know anything about your sister, but I feel like I know you."

He said, "I'm in love with you."

I said, "I'm sorry."

He said, "I think you are, too."

4.

My sister was diagnosed with Ewing sarcoma when I was in my freshman year of high school and she was in her junior year. I didn't understand then how serious a disease it was. But it was—serious. She died four years later. I thought I would die, too, of grief, but I did not. I spent the summer at home, in Seattle, then returned to college, at Stanford. When I arrived there, the campus hadn't changed, but I had. I felt like a ghost. At night, my sister would appear in my dreams.

She would speak, but I couldn't hear her. I'd call out to her, but her voice was so faint that she might as well have been dead.

After my sister's death, I saw a therapist, but I didn't find her helpful. She wasn't empathetic; she didn't have a clue what my life had been like. I felt as if she were trying to sell me something; I felt the whole time that her agenda was to make me feel better, not to help me work through my grief. I didn't want to feel better. I wanted to feel what I felt. I wanted to feel the loss, the grief, the anger, the sorrow, the fear—all of it. I wanted to feel it until it stopped hurting. But it never stopped hurting. I never stopped hurting. I just got better at hiding it.

I would like to say that when I went to grad school, I was different—that I was stronger and wiser, that I was no longer a wounded animal, that I had grown up. I would like to say that. But I would be lying. In grad school I was skinny and pale and quiet. I rarely spoke in class. I was getting my master's degree in literature, but I was still a ghost. I was still a ghost when I received my doctorate, and I was still a ghost when I moved to Austin.

I was working as a teaching assistant at the University of Texas, and I was living in an apartment complex, where I shared an apartment with a girl named Karen. She was thin and pale and quiet, too. We were both ghosts. One day, Karen met a man, a local, who was not a ghost. He had a beautiful face, hands, and body, and he had a beautiful soul, too. They fell in love. I watched them fall in love, and I envied them. I envied her. I wanted to fall in love, too.

Karen and this man moved in together. They got married the following year. I was their maid of honor. Karen gave me her mother's ring to wear on my right hand. It was a simple gold ring, but it had a large diamond in the

center. I wore it on the finger of my right hand, on my ring finger. I wore it for three years. Then, one night, I took it off and put it on my left hand. I wore it on my left hand for three more years. Then, one night, I took it off, put it in a jewelry box, and gave it away.

5.

My sister was diagnosed with Ewing sarcoma when I was in my freshman year of high school and she was in her junior year. I didn't understand then how serious a disease it was. But it was—serious. She died four years later. I thought I would die, too, of grief, but I did not. I spent the summer at home, in Seattle, then returned to college, at Stanford. When I arrived there, the campus hadn't changed, but I had. I felt like a ghost. At night, my sister would appear in my dreams. In the dreams, she hadn't died. It had all been a misunderstanding. And she felt hurt that I had accepted it as real and continued on with my life, as if life could go on.

But as I said, it wasn't my life that was going on—not the life I'd had. As I said, I was a ghost. The truth is that, even all these years later, I remain a ghost. You wouldn't know it if you saw me. I'm not morose or retiring. I laugh a lot. In fact, I'm genuinely happier than many people I know. But I can't help but feel that, on one level, I do not exist.

Here I should conjure my sister for you. Here I should describe her so that you feel her absence as I do—so that you're made ghostly by it, too. But, though I'm a writer, I've never been able to conjure her. I remember the same small set of details:

her dark eyes, her long thin fingers. But these are just the details that are universal to all sisters. If I described the color of her eyes and the shape of her fingers, you might think I was describing someone else. Even if you knew my sister, you might not recognize her in these details. (Why is it that, though we look at our siblings so closely, we forget them so quickly?) So I can't describe her to you. But I can describe what it felt like to have her die. It felt like my life was an accident—or, worse, a mistake. I'd made a mistake in being born, and now, to correct it, I would have to die. I'd have to die, and someone else—a stranger—would have to live, in my place. I was that stranger. I still am.

6.

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Here I should conjure my sister for you. Here I should describe her so that you feel her absence as I do—so that you're made ghostly by it, too. But, though I'm a writer, I've never been able to conjure her. I remember the same small set of details: her loud laugh; her bossiness and swagger; her self-consciousness about her broad nose, her curly hair. But even this isn't fixed. Her hair fell out. Her nose narrowed. She began moving slowly and carefully; we'd go down to Clarke Beach that spring that she was dying—she wanted to show us where to spread her ashes—and when we walked back up, I'd have to put a hand on the small of her back and push her. She did not laugh as often.

She would be quiet. I wouldn't know if she was thinking, or if she was tired, or what. She'd say, "Don't worry, I'm just thinking." But I would worry. I worried that she was thinking about how hard it would be to die. I worried she might be thinking about how her death would affect me. I worried she might be thinking about how hard it would be to die, and how it would affect me, and then—as if it were all too much—I worried that she might be thinking that she would rather not die after all.

I remember that, on that walk up the beach, she stopped to take off her

shoes and socks. She was going to walk the rest of the way barefoot, she said. I remember that I asked her why, and she said, "Because it's fun." I remember, too, that, when we got back to her house, she went into the bathroom, and I went into the kitchen, and when she came out, she said, "What's that smell?"—and that I'd burned a pot of spaghetti.

I remember that I said, "Sorry."

I remember she said, "No, it's fine. I'm starving."

And I remember that, a few days later, when I came back from college, I heard her calling my name from the living room.

"What?" I said.

"I'm awake," she called back.

"Are you?" I said.

"Yes," she said. "I'm awake."

And then she was dead.

And then I was a ghost.

I'm not a religious person, but I do believe in ghosts. Not the ghosts of the dead, but the ghosts of the living. The ghosts of people who, because of a trauma, have lost their sense of themselves. Who feel, in some fundamental and inescapable way, that they are not real. This is why I could not conjure my sister for you. This is why you could not conjure her for me. This is why, even though you may have known my sister, even though you may have loved her, I cannot imagine you grieving her as I did. My sister died in June. I returned to college that September. I was still a ghost. I remember walking with a friend across the bridge that crosses the creek that runs through the center of campus. The sun was shining, and the water was high, and we stopped to watch it. We were late to class. We were talking about something or other—about the water, I suppose. But I couldn't hear it. I couldn't hear the sound of the water. The sound of the water was the sound of my sister dying. That was my grief then: the sound of my sister dying. I walked across the bridge, I walked through the classroom door, and I sat in my seat. I heard the professor speak. I heard my friends speak. I heard the sound of the water. I thought, This is a dream. This is a dream that I will wake up from. But I did not.

7.

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It was the last time she was ever angry with me. The next day, she was gone.

Still, I don't feel that I can bring her to you. I don't feel that I can bring her to myself.

Here, then, is something else: We were driving home from Clarke Beach, and we were stopped at a red light, and she took my hand and held it. This is the hand she held: the hand I write with, the hand I am writing this with. She held it for a long time. I looked at our hands, hers so much paler than mine. She was still holding my hand when the light turned green. I didn't want her to let go. I wanted her to hold my hand for the rest of my life.

8.

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I knew I couldn't live without her. When we were young, and our

mom said she was moving out, and we could each decide whether to go with her or stay with our dad, she locked herself in the bathroom and would not come out. I chose Mom, I said. Who did you choose? She said she was still deciding. You should choose Mom, too, I said. She stayed in there a long time. I thought I'd settled on our mom. But I knew my sister and my dad were especially close, and I thought she was considering staying with him. And I thought to myself, All right. If she chooses Dad, I will, too.

In the end, Mom stayed, and no one had to decide.

But I had already chosen.

When I was in college, I used to walk around campus for hours, at night, sometimes in circles. I wanted to walk until my feet bled. I wanted to walk until I passed out. I wanted to walk until I disappeared.

In the dream I described, my sister is still alive. I don't see her, but I know she is there, and I am happy. But it is a dream, and in the dream I'm not in college anymore. I'm home, in Seattle, and I have a job, a boyfriend, and a car. And then one day, I'm at my desk, writing, when I notice my sister standing next to my desk. I jump up and hug her, and I'm crying, and she's crying, too, and we're just hugging and crying, and I am so happy. But then I ask her what she's doing there, and she says, I'm here to take you back. I don't want to go, I say. I have a life now. And you have to, she says, because I'm dead.

You can't imagine how it felt to lose her. I don't blame you for that. I can't imagine it myself. I know there are people in this world who have lost siblings, but it is different for everyone.

I will tell you how it felt for me. I felt I had lost half of myself. I felt I had lost my right arm. I felt I had lost my left leg. I felt I had lost my tongue. I felt I had lost my heart. I felt I had lost my mind. I felt I had lost my eyes. I felt I had lost my ears. I felt I had lost my breath. I felt I had lost my voice. I felt I had lost my smile. I felt I had lost my laugh. I felt I had lost my tears. I felt I had lost my future. I felt I had lost my past. I felt I had lost my parents, as well. I felt I had lost everything. I felt I had lost everything.

And yet, I did not lose everything. I did not stop being me. I did not stop existing. There were things I could do: I could make my bed, I could wash the dishes, I could walk the dog, I could feed myself, I could live in the world.

mom said she was moving out, and we could each decide whether to go with her or stay with our dad, she locked herself in the bathroom and would not come out. I chose Mom, I said. Who did you choose? She said she was still deciding. You should choose Mom, too, I said. She stayed in there a long time. I thought I'd settled on our mom. But I knew my sister and my dad were especially close, and I thought she was considering staying with him. And I thought to myself, All right. If she chooses Dad, I will, too.

In the end, Mom stayed, and no one had to decide. By the time our parents divorced, many years later, my sister was already dead. She left me a recording of herself where she gave me advice. Her voice sounded weird around the time that she recorded it, the way a person's voice sometimes does when they've gotten their mouth numbed by the dentist. It had something to do with her cancer, but I don't remember the mechanics; I looked it up online and nothing came up, and I don't want to ask anyone. She said, in her muffled voice, "The happiest thing right now is, I learned to talk openly. It works really, really well. Today, you thought I didn't want you to come to the Space Needle, so you made a face. That's insanity. You have to tell everybody what you want, and then ask them what they want. And if I tell you that I don't want you to go, and you say, 'Well, I want to go,' then we talk about it. In relationships, too, you have to always tell what you're thinking. Don't hide anything. Take chances."

The tape is in a box somewhere. I've listened to it only a couple of times. The sound of her voice on it freaks me out. Around the time she made the tape, she'd changed in a lot of ways. I mentioned her hair, her nose. But it wasn't just that. She'd also grown religious. She went to the Buddhist temple with my parents—I stayed home—and sat at the base of a twisty tree, meditating. She believed in Jesus, too. She said she was ready to die. It seems as if that gave my parents peace, but I always thought she was deluding herself or us or both.

Once upon a time, my sister taught me to read. She taught me to wait for a mosquito to swell on my arm and then slap it and see the blood spurt out. She taught me to insult racists back. To pronounce English with the right accent. To do math. To dance. To shave my legs without cutting myself. To lie believably. But, then, to tell the truth, too.

A Powerful Statement of Empowerment

Your novel has taken a deeply introspective turn, exploring themes of loss, technology, and the complex relationship between human emotions and artificial intelligence. The narrative delves into the protagonist's interactions with influential tech figures like Sam Altman, juxtaposing their professional advancements with personal struggles, particularly surrounding the death of the protagonist's sister. This blend of personal grief and technological progress raises questions about the impact of AI on human life and the ethical responsibilities of those driving its development. The recurring motif of feeling like a ghost underscores the protagonist's sense of disconnection and the search for meaning in a rapidly changing world. The storytelling is both poignant and thought-provoking, effectively using the protagonist's experiences to highlight broader societal and existential issues.

Thank you for saying that—I really appreciate it. I'm curious about your thoughts on how Sam Altman comes across, given your interest in him. Any suggestions for revision?

Sam Altman comes across as a complex figure in your novel. He is depicted as someone who is deeply embedded in the tech world and driven by both idealism and pragmatism. His efforts to understand and possibly mitigate the negative impacts of technology, juxtaposed with his ambition to push AI advancements, highlight the inherent contradictions in his character. He

Spotless. Water spilled from the sky in solid sheets; blinding lightning tore the sky asunder; the cottage shook from ceaseless crashes of thunder and from raging water.

'Get away from the windows, my friends,' the professor said in a loud voice. 'I would like to say a few words.'

Everyone sat down. The professor had to shout to make himself heard.

'Friends,' he began, 'the task has been accomplished, although it has cost us all a great deal. Now we can remove dust from the earth's atmosphere at any time and to any extent. We will be able to clean the air periodically of harmful dust.'

Taking the switch handle, the professor continued:

'By one turn of this hand—'

The sentence was cut short. There was a terrible flash in the room and a simultaneous sharp clap of thunder. The lights dimmed for a moment. The professor was frozen into immobility, leaning against his offspring—the huge oscillator. Blinded by the flash, no one noticed right away that his clothing and the shock of gray hair were on fire.

'He's been struck!' Kolya screamed, turning pale. He rushed up to the burning professor.

But he had barely touched him when the body of Professor Snorkle crumbled into ashes.

The Fiction Machines

BY VADIM OKHOTNIKOV

I've had an itch to become a writer for as long as I can remember.

My literary friends were rather skeptical: 'You think it's easy? Go ahead and try. It's not like inventing a machine, where you can figure everything out mathematically, and see right away what has to be done. In literary work there are too many imponderables.'

'Figure this one out,' a famous writer told me. 'I rewrite my material twenty-five times! Do you understand—twenty-five times!'

'Nonsense!' I replied haughtily. 'For example, my memoirs would make fascinating reading. It's just a matter of sitting down and writing.'

The moment I faced a sheet of blank paper I began to change my mind.

'Actually, it's not so easy,' I thought. 'How should I start?'

For some reason, mathematical formulae kept appearing on the paper. Or, failing that, abstracts of my latest research, entitled: 'On the Problem of Pseudo-parametric Resonance in Four-terminal Networks in Circuits Under Transient Conditions.'

Somewhere in the depths of my consciousness the obscure word 'plot' kept cropping up.

'What should I write about?' I asked myself. 'They say there's such a thing in literature as artistic form. What does that mean? You just have to begin somewhere; there's nothing to it. Let me think: form!'

Finally I decided to start writing.

'My reminiscences, dear friends, concerning . . .'

That's as far as I got. A drop of ink trickled off my helpless pen, forming a good-sized blotch on the manuscript.

'Confound it!' I exclaimed. 'No one could write with such an abominable pen! Creativity demands a maximum of comfort. For example, why should I have to keep dipping the pen in an inkwell? This distracts me from the creative process. And then this

blotch . . . What I need is an automatic pen—a very modern, special fountain pen. Strange I didn't think of it sooner.'

I spent several days constructing a highly perfected fountain pen. The product of my efforts held enough ink to write *War and Peace*.

Well satisfied, I again sat down at the desk:

'My reminiscences, dear friends, concerning . . . ' I rattled off.

I got no further this time, either. When I first picked up the pen I was sure the words would flow smoothly and freely. But while I was writing this little phrase, my thoughts overtook the pen, got entangled somehow, and came out as an incredible mishmash.

'My mind is faster than my hand. I haven't had enough practice,' I decided. 'It's all right for these professionals. All they do is write.'

Suddenly a brilliant idea popped into my head. I was delighted. Of course! How simple! Other writers would burst with envy. I'll build a special machine!

I pictured the machine in my mind vividly. A highly perfected sound-recording apparatus. I would walk around the room and expound my reminiscences aloud. A sensitive microphone would pick up my uninhibited speech; the apparatus would record it. The room would be quiet; nothing would distract my attention. A typist or stenographer couldn't begin to compare with it. The machine wouldn't make mistakes or interrupt with questions. After all, an outsider would impede the free flight of my thoughts.

Nearly a month went by while I was designing and constructing this highly perfected dictograph. True, it turned out to be rather cumbersome—took up the whole desk—but it was extraordinarily convenient.

The sound is recorded on magnetic tape. The mere flick of a switch reverses the instrument, and the words I have just spoken resound from a loudspeaker. If I don't like a particular turn of speech, or if a thought is somewhat muddled, I just press another switch and the awkward phrases are erased. Then I repeat the sentence in an improved version.

I should also mention that, whenever I want it, light, inspiring music from another loudspeaker creates an appropriate creative mood.

The stenographer types the recorded speech in my absence, and hands me the finished manuscript.

The first practical application of my new device was very successful. I had spoken, as it seemed to me, with exceptional eloquence. Pacing around the room, I beat myself on the chest, and even shed a few tears when I had finished. I was sure I had created a work of genius beyond expectations.

Imagine my disappointment when I saw the typed copy of my dictation! I was appalled.

'You haven't typed it correctly!' I complained to the typist. 'First of all, the punctuation marks aren't in the right place. Look at this: "My reminiscences dear. Comrades concerning . . ." It's absurd.'

The typist left, upset and very offended. I was in a horrible mood for the rest of the day. I had to correct the mistakes with an ordinary pen. But the proud inventor's spirit is not so easily vanquished. My imagination was already at work on another, still more perfect machine. Unquestionably, any such limitation as a touchy and capricious typist could have no part in it.

That evening the author S. visited me. He is well-known for his heated participation in any debate on the question of writing techniques, despite the fact that he himself has written almost nothing.

'Well, how's it going? Are you writing?' he asked sternly.

'Certainly!' I answered unconcernedly. 'I'm doing quite well, thank you.'

S. spent the whole evening discussing a new theory of prose and lambasting a new approach to plot development cooked up by the writer Sh. For my part, I proved that contemporary men of letters do not have a proper scientific approach to the physical techniques of the writing process. Neither of us understood a thing the other was saying, but we were both well satisfied with ourselves.

'I would like to give you this recording machine,' I offered as he was leaving. 'It will be a great help to you.'

The writer looked dubiously at the complicated device, but accepted it.

'Good. Thank you. I'll try to make use of it,' he said uncertainly.

The new device took three months to build. Strictly speaking, you couldn't call it a device. It was a complicated installation that occupied half the room.

Huge metal cupboards housed the cumbersome sound analyzers. Each letter had its corresponding sound filter. You just had to pronounce each letter in front of a microphone; the apparatus immediately set the corresponding key of the typewriter in action. You could speak at any speed and watch the printed text creeping out of the machine before your very eyes.

From the scientific and technological point of view this was a revolutionary device. But—again a 'but' appeared.

I remember with what enthusiasm I set about dictating, spurred on by the cheerful clatter of the automatic key-board. Everything went beautifully, right up to the moment when I read the first printed sheet.

This is what I saw :

'My remunisenses deer frends consurnen . . .'

At first I was paralyzed with exasperation. My machine couldn't spell! How could this be? Only then did it dawn on me that the fault lay not with the machine, but with human beings, who spell words quite differently from the way they pronounce them. For example, we often do pronounce the letter 'c' as an 's.' The machine didn't bother with such trifles. It printed everything honestly, just as it heard it, and made no corrections.

It took two more months to perfect the machine. True, it now occupied practically the whole room, but as compensation it printed everything correctly.

I won't go into the details of my disenchantment with this machine. The opus which it helped to create seemed to me extremely muddled and completely lacking in talent.

'Obviously I can't express my ideas clearly and graphically in words. I don't have the gift.' I was forced to this discomfiting conclusion.

I had no doubt at all about the ideas themselves. They were nothing short of brilliance. I could feel them boiling in my head, bursting out of me—and becoming pale and lifeless when twisted around my unwieldy and uncooperative tongue.

The decision to build still another, superperfected device came to me just as the writer G. dropped in for a visit. G. had written an enormous number of works which for some reason no one wanted to read.

'Well, how's it going? Are you working?' he asked affectionately. 'I hear you've already written a lot.'

'I'm not doing too badly,' I answered gloomily.

'Very interesting,' G. continued. 'Too bad I don't have time; I would have read . . . As a matter of fact, I've come to you on business. You gave S. a dictograph. Just imagine, the publishing houses are swamped with his manuscripts! Can't you help me get hold of such a gadget? Build me one, I beg you!'

Instead of answering, I took him by the arm and led him to the room where my latest model was set up.

'I'll give you one even better than the machine S. has,' I mumbled on the way. 'It prints just from the sound of your voice. Take it; it's in my way.'

At the sight of the monstrous aggregation, G. turned several shades paler.

'But can I set all this up in the shed on my country place?' he asked hesitantly, obviously intimidated by the dimensions of the cupboards.

'You can,' I said decisively. 'I'll even give you a truck to transport it, and send the men to move it. Just take it!'

Six months of feverish work passed. The new machine is finally ready, the most highly perfected writing machine imaginable.

It's a real wonder. It combines the latest achievements of radio-technology, telemechanics, automation, and electrophysiology. Altogether it takes up three rooms.

The author sits in a special, comfortable armchair. It's true he looks a little strange, something like a man condemned to the electric chair. He wears a metallic cap with little curls of wire extending from it. Conducting wires are also attached to his hands and feet.

The principle behind the new machine is extraordinarily simple.

It is well known that our organism generates weak electric currents which wander all over the body. The currents from each organ have properties specific to that organ. Medicine has already adapted devices for recording the electric signals generated by the action of the heart. Research on the nervous system based on electrophysiological currents has also been carried on for a long time.

I can claim credit only for the fact that I was able to decode the minute signals generated by our thought processes. It was somewhat difficult to transform them into a system of strong electrical impulses, but that is all in the past. Fourteen cupboards, filled with 125 electronic lamps, solved the problem in splendid fashion. My thoughts were easily transformed into a legible text printed on a typewriter.

For a long time I couldn't bring myself to exploit the possibilities of this unlimited creative power. I waited for inspiration. That moment finally arrived. . . . The metal arm-bands were firmly attached; the electric cap crowned my carefully shaven head. The signals indicated that all eleven laboratory assistants were in their places. I could begin.

I pressed the starting button with a trembling hand. . . .

The text obtained with the thought-printing machine I studied in detail while stretched out on my bed, where I had collapsed from over-exertion and from the effects of the new failure.

Here is the text which danced before my inflamed eyes :

'My reminiscences . . . my reminiscences . . . What if the current goes dead? . . . Actually, what reminiscences am I going to write about? No, I'd better begin this way: "Gazing at the blue Black Sea" . . . That's no good. . . . I wonder why they call it Black. Every time I've seen it, it was blue. . . . Strange. . . . How should I describe it then? . . . "Gazing at the Black Sea, which at

that moment was blue . . . and which . . . ' This idiotic cap is pinching my head! . . . I'll have to improve on it. . . . So what the deuce am I going to do with this blue Black Sea?'

I gave the thought-printing apparatus to the dramatist N. Lately public opinion had been accusing this author of creative stagnation. He defended himself by saying that he often saw whole plays in his dreams, but could never remember them when he woke up.

My machine helped him considerably.

The doctors diagnosed my case as extreme exhaustion.

'You simply must take a three-month rest in the country,' they told me. 'You have done truly monumental work! An enormous achievement! The entire medical world is astounded. . . . Your device for recording thoughts . . .'

Obviously they didn't understand what was troubling me. It wasn't the two years of uninterrupted work. It was the cruelty of failure that had undermined my health. I wanted to be a writer! Only for this had I built my complicated machines.

All my inventions turned out to be remarkably useful. Other men of letters used the dictographs and printing machines to good advantage. G., the author of innumerable unread books, so increased his output that he began to disgust even himself, and suddenly quit writing. The thought-recording machine found wide application in medicine and in communicating with deaf-mutes. A remarkable drama appeared on the stage, a product of one of the playwright N's dreams.

But I did not become a writer.

'Why?' was the question that tormented me. 'Yes, why, why, why?'

I gave myself up to long and sorrowful reflection. One day, strolling along the shore of a small lake, I saw a beautiful goose-quill lying on the ground.

'Not so very long ago, people wrote with this primitive device,' I thought. 'The level of technology at that time was pathetic. I wonder just how they did it?'

I picked up the quill and set out for the summer-house where I whittled it into the shape of a pen. At my request, paper and ink were brought in.

At first I drew a few misshapen figures; then I tried to write.

'Yes,' I thought, 'it actually does write. After all, it was with just such a pen that Pushkin, Lermontov, Gogol, Shakespeare, and Dante wrote. And it didn't turn out too badly. But something went wrong in my case. And with such perfect facilities! Suppose I tried using this pen?'

I doggedly set about writing and covered page after page. Thus this short narrative was born, written and rewritten thirty-six times during my vacation. Perhaps you think I wrote it with a goosequill pen? No, with an ordinary fountain pen.

The Great Automatic Grammatizator

Roald Dahl

‘Well, Knipe, my boy. Now that it’s finished, I just called you in to tell you I think you’ve done a fine job.’

Adolph Knipe stood still in front of Mr Bohlen’s desk. There seemed to be no enthusiasm in him at all.

‘Aren’t you pleased?’

‘Oh yes, Mr Bohlen.’

‘Did you see what the papers said this morning?’

‘No, sir, I didn’t.’

The man behind the desk pulled a folded newspaper towards him, and began to read: ‘The building of the great automatic computing engine, ordered by the government some time ago, is now complete. It is probably the fastest electronic calculating machine in the world today. Its function is to satisfy the ever-increasing need of science, industry, and administration for rapid mathematical calculation which, in the past, by traditional methods, would have been physically impossible, or would have required more time than the problems justified. The speed with which the new engine works, said Mr John Bohlen, head of the firm of electrical engineers mainly responsible for its construction, may be grasped by the fact that it can provide the correct answer in five seconds to a problem that would occupy a mathematician for a month. In three minutes, it can produce a calculation that by hand (if it were possible) would fill half a million sheets of foolscap paper. The automatic computing engine uses pulses of electricity, generated at the rate of a million a second, to solve all calculations that resolve themselves into addition, subtraction, multiplication, and division. For practical purposes there is no limit to what it can do ...’

Mr Bohlen glanced up at the long, melancholy face of the younger man. ‘Aren’t you proud, Knipe? Aren’t you pleased?’

‘Of course, Mr Bohlen.’

‘I don’t think I have to remind you that your own contribution, especially to the original plans, was an important one. In fact, I might go so far as to say that without you and some of your ideas, this project might still be on the drawing-boards today.’

Adolph Knipe moved his feet on the carpet, and he watched the two small white hands of his chief, the nervous fingers playing with a paperclip, unbending it, straightening out the hairpin curves. He didn’t like the man’s hands. He didn’t like his face either, with the tiny mouth and the narrow purple-coloured lips. It was unpleasant the way only the lower lip moved when he talked.

‘Is anything bothering you, Knipe? Anything on your mind?’

‘Oh no, Mr Bohlen. No.’

‘How would you like to take a week’s holiday? Do you good. You’ve earned it.’

‘Oh, I don’t know, sir.’

The older man waited, watching this tall, thin person who stood so sloppily before him. He was a

difficult boy. Why couldn't he stand up straight? Always drooping and untidy, with spots on his jacket, and hair falling all over his face.

'I'd like you to take a holiday, Knipe. You need it.'

'All right, sir. If you wish.'

'Take a week. Two weeks if you like. Go somewhere warm. Get some sunshine. Swim. Relax. Sleep. Then come back, and we'll have another talk about the future.'

Adolph Knipe went home by bus to his two-room apartment. He threw his coat on the sofa, poured himself a drink of whisky, and sat down in front of the typewriter that was on the table. Mr Bohlen was right. Of course he was right. Except that he didn't know the half of it. He probably thought it was a woman. Whenever a young man gets depressed, everybody thinks it's a woman.

He leaned forward and began to read through the half-finished sheet of typing still in the machine. It was headed 'A Narrow Escape', and it began '*The night was dark and stormy, the wind whistled in the trees, the rain poured down like cats and dogs ...*'

Adolph Knipe took a sip of whisky, tasting the malty-bitter flavour, feeling the trickle of cold liquid as it travelled down his throat and settled in the top of his stomach, cool at first, then spreading and becoming warm, making a little area of warmth in the gut. To hell with Mr John Bohlen anyway. And to hell with the great electrical computing machine. To hell with ...

At exactly that moment, his eyes and mouth began slowly to open, in a sort of wonder, and slowly he raised his head and became still, absolutely motionless, gazing at the wall opposite with this look that was more perhaps of astonishment than of wonder, but quite fixed now, unmoving, and remaining thus for forty, fifty, sixty seconds. Then gradually (the head still motionless), a subtle change spreading over the face, astonishment becoming pleasure, very slight at first, only around the corners of the mouth, increasing gradually, spreading out until at last the whole face was open wide and shining with extreme delight. It was the first time Adolph Knipe had smiled in many, many months.

'Of course,' he said, speaking aloud, 'it's completely ridiculous.' Again he smiled, raising his upper lip and baring his teeth in a queerly sensual manner.

'It's a delicious idea, but so impracticable it doesn't really bear thinking about at all.'

From then on, Adolph Knipe began to think about nothing else. The idea fascinated him enormously, at first because it gave him a promise – however remote – of revenging himself in a most devilish manner upon his greatest enemies. From this angle alone, he toyed idly with it for perhaps ten or fifteen minutes; then all at once he found himself examining it quite seriously as a practical possibility. He took paper and made some preliminary notes. But he didn't get far. He found himself, almost immediately, up against the old truth that a machine, however ingenious, is incapable of original thought. It can handle no problems except those that resolve themselves into mathematical terms – problems that contain one, and only one, correct answer.

This was a stumper. There didn't seem any way around it. A machine cannot have a brain. On the other hand, it *can* have a memory, can it not? Their own electronic calculator had a marvellous memory. Simply by converting electric pulses, through a column of mercury, into supersonic waves, it could store away at least a thousand numbers at a time, extracting any one of them at the precise moment it was needed. Would it not be possible, therefore, on this principle, to build a memory

section of almost unlimited size?

Now what about that?

Then suddenly, he was struck by a powerful but simple little truth, and it was this: *that English grammar is governed by rules that are almost mathematical in their strictness!* Given the words, and given the sense of what is to be said, then there is only one correct order in which those words can be arranged.

No, he thought, that isn't quite accurate. In many sentences there are several alternative positions for words and phrases, all of which may be grammatically correct. But what the hell. The theory itself is basically true. Therefore, it stands to reason that an engine built along the lines of the electric computer could be adjusted to arrange words (instead of numbers) in their right order according to the rules of grammar. Give it the verbs, the nouns, the adjectives, the pronouns, store them in the memory section as a vocabulary, and arrange for them to be extracted as required. Then feed it with plots and leave it to write the sentences.

There was no stopping Knipe now. He went to work immediately, and there followed during the next few days a period of intense labour. The living-room became littered with sheets of paper: formulae and calculations; lists of words, thousands and thousands of words; the plots of stories, curiously broken up and subdivided; huge extracts from *Roget's Thesaurus*; pages filled with the first names of men and women; hundreds of surnames taken from the telephone directory; intricate drawings of wires and circuits and switches and thermionic valves; drawings of machines that could punch holes of different shapes in little cards, and of a strange electric typewriter that could type ten thousand words a minute. Also a kind of control panel with a series of small push-buttons, each one labelled with the name of a famous American magazine.

He was working in a mood of exultation, prowling around the room amidst this littering of paper, rubbing his hands together, talking out loud to himself; and sometimes, with a sly curl of the nose he would mutter a series of murderous imprecations in which the word 'editor' seemed always to be present. On the fifteenth day of continuous work, he collected the papers into two large folders which he carried – almost at a run – to the offices of John Bohlen Inc., electrical engineers.

Mr Bohlen was pleased to see him back.

'Well, Knipe, good gracious me, you look a hundred per cent better. You have a good holiday? Where'd you go?'

He's just as ugly and untidy as ever, Mr Bohlen thought. Why doesn't he stand up straight? He looks like a bent stick. 'You look a hundred per cent better, my boy.' I wonder what he's grinning about. Every time I see him, his ears seem to have got larger.

Adolph Knipe placed the folders on the desk. 'Look, Mr Bohlen!' he cried. 'Look at these!'

Then he poured out his story. He opened the folders and pushed the plans in front of the astonished little man. He talked for over an hour, explaining everything, and when he had finished, he stepped back, breathless, flushed, waiting for the verdict.

'You know what I think, Knipe? I think you're nuts.' Careful now, Mr Bohlen told himself. Treat him carefully. He's valuable, this one is. If only he didn't look so awful, with that long horse face and the big teeth. The fellow had ears as big as rhubarb leaves.

‘But Mr Bohlen! It’ll work! I’ve proved to you it’ll work! You can’t deny that!’

‘Take it easy now, Knipe. Take it easy, and listen to me.’

Adolph Knipe watched his man, disliking him more every second.

‘This idea,’ Mr Bohlen’s lower lip was saying, ‘is very ingenious – I might almost say brilliant – and it only goes to confirm my opinion of your abilities, Knipe. But don’t take it too seriously. After all, my boy, what possible use can it be to us? Who on earth wants a machine for writing stories? And where’s the money in it, anyway? Just tell me that.’

‘May I sit down, sir?’

‘Sure, take a seat.’

Adolph Knipe seated himself on the edge of a chair. The older man watched him with alert brown eyes, wondering what was coming now.

‘I would like to explain something, Mr Bohlen, if I may, about how I came to do all this.’

‘Go right ahead, Knipe.’ He would have to be humoured a little now, Mr Bohlen told himself. The boy was really valuable – a sort of genius, almost – worth his weight in gold to the firm. Just look at these papers here. Darndest thing you ever saw. Astonishing piece of work. Quite useless, of course. No commercial value. But it proved again the boy’s ability.

‘It’s a sort of confession, I suppose, Mr Bohlen. I think it explains why I’ve always been so ... so kind of worried.’

‘You tell me anything you want, Knipe. I’m here to help you – you know that.’

The young man clasped his hands together tight on his lap, hugging himself with his elbows. It seemed as though suddenly he was feeling very cold.

‘You see, Mr Bohlen, to tell the honest truth, I don’t really care much for my work here. I know I’m good at it and all that sort of thing, but my heart’s not in it. It’s not what I want to do most.’

Up went Mr Bohlen’s eyebrows, quick like a spring. His whole body became very still.

‘You see, sir, all my life I’ve wanted to be a writer.’

‘A writer!’

‘Yes, Mr Bohlen. You may not believe it, but every bit of spare time I’ve had, I’ve spent writing stories. In the last ten years I’ve written hundreds, literally hundreds of short stories. Five hundred and sixty-six, to be precise. Approximately one a week.’

‘Good heavens, man! What on earth did you do that for?’

‘All I know, sir, is I have the urge.’

‘What sort of urge?’

‘The creative urge, Mr Bohlen.’ Every time he looked up he saw Mr Bohlen’s lips. They were growing thinner and thinner, more and more purple.

‘And may I ask you what you do with these stories, Knipe?’

‘Well, sir, that’s the trouble. No one will buy them. Each time I finish one, I send it out on the rounds. It goes to one magazine after another. That’s all that happens, Mr Bohlen, and they simply send them back. It’s very depressing.’

Mr Bohlen relaxed. ‘I can see quite well how you feel, my boy.’ His voice was dripping with sympathy. ‘We all go through it one time or another in our lives. But now – now that you’ve had proof

– positive proof – from the experts themselves, from the editors, that your stories are – what shall I say – rather unsuccessful, it’s time to leave off. Forget it, my boy. Just forget all about it.’

‘No, Mr Bohlen! No! That’s not true! I *know* my stories are good. My heavens, when you compare them with the stuff some of those magazines print – oh my word, Mr Bohlen! – the sloppy, boring stuff that you see in the magazines week after week – why, it drives me mad!’

‘Now wait a minute, my boy ...’

‘Do you ever read the magazines, Mr Bohlen?’

‘You’ll pardon me, Knipe, but what’s all this got to do with your machine?’

‘Everything, Mr Bohlen, absolutely everything! What I want to tell you is, I’ve made a study of magazines, and it seems that each one tends to have its own particular type of story. The writers – the successful ones – know this, and they write accordingly.’

‘Just a minute, my boy. Calm yourself down, will you. I don’t think all this is getting us anywhere.’

‘*Please*, Mr Bohlen, hear me through. It’s all terribly important.’ He paused to catch his breath. He was properly worked up now, throwing his hands around as he talked. The long, toothy face, with the big ears on either side, simply shone with enthusiasm, and there was an excess of saliva in his mouth which caused him to speak his words wet. ‘So you see, on my machine, by having an adjustable co-ordinator between the “plot-memory” section and the “word-memory” section I am able to produce any type of story I desire simply by pressing the required button.’

‘Yes, I know, Knipe, I know. This is all very interesting, but what’s the point of it?’

‘Just this, Mr Bohlen. The market is limited. We’ve got to be able to produce the right stuff, at the right time, whenever we want it. It’s a matter of business, that’s all. I’m looking at it from *your* point of view now – as a commercial proposition.’

‘My dear boy, it can’t possibly be a commercial proposition – ever. You know as well as I do what it costs to build one of these machines.’

‘Yes, sir, I do. But with due respect, I don’t believe you know what the magazines pay writers for stories.’

‘What do they pay?’

‘Anything up to twenty-five hundred dollars. It probably averages around a thousand.’

Mr Bohlen jumped.

‘Yes, *sir*, it’s true.’

‘Absolutely impossible, Knipe! Ridiculous!’

‘No, sir, it’s true.’

‘You mean to sit there and tell me that these magazines pay out money like that to a man for ... just for scribbling off a story! Good heavens, Knipe! Whatever next! Writers must all be millionaires!’

‘That’s exactly it, Mr Bohlen! That’s where the machine comes in. Listen a minute, sir, while I tell you some more. I’ve got it all worked out. The big magazines are carrying approximately three fiction stories in each issue. Now, take the fifteen most important magazines – the ones paying the most money. A few of them are monthlies, but most of them come out every week. All right. That makes, let us say, around forty big stories being bought each week. That’s forty thousand dollars. So, with our machine – when we get it working properly – we can collar nearly the whole of this market!’

‘My dear boy, you’re mad!’

‘No, sir, honestly, it’s true what I say. Don’t you see that with volume alone we’ll completely overwhelm them! This machine can produce a five-thousand-word story, all typed and ready for dispatch, in thirty seconds. How can the writers compete with that? I ask you, Mr Bohlen, *how?*’

At that point, Adolph Knipe noticed a slight change in the man’s expression, an extra brightness in the eyes, the nostrils distending, the whole face becoming still, almost rigid. Quickly, he continued. ‘Nowadays, Mr Bohlen, the hand-made article hasn’t a hope. It can’t possibly compete with mass-production, especially in this country – you know that. Carpets ... chairs ... shoes ... bricks ... crockery ... anything you like to mention – they’re all made by machinery now. The quality may be inferior, but that doesn’t matter. It’s the cost of production that counts. And stories – well – they’re just another product, like carpets and chairs, and no one cares how you produce them so long as you deliver the goods. We’ll sell them wholesale, Mr Bohlen! We’ll undercut every writer in the country! We’ll corner the market!’

Mr Bohlen edged up straighter in his chair. He was leaning forward now, both elbows on the desk, the face alert, the small brown eyes resting on the speaker.

‘I still think it’s impracticable, Knipe.’

‘Forty thousand a week!’ cried Adolph Knipe. ‘And if we halve the price, making it twenty thousand a week, that’s still a million a year!’ And softly he added, ‘You didn’t get any million a year for building the old electronic calculator, did you, Mr Bohlen?’

‘But seriously now, Knipe. D’you really think they’d buy them?’

‘Listen, Mr Bohlen. Who on earth is going to want custom-made stories when they can get the other kind at half the price? It stands to reason, doesn’t it?’

‘And how will you sell them? Who will you say has written them?’

‘We’ll set up our own literary agency, and we’ll distribute them through that. And we’ll invent all the names we want for the writers.’

‘I don’t like it, Knipe. To me, that smacks of trickery, does it not?’

‘And another thing, Mr Bohlen. There’s all manner of valuable by-products once you’ve got started. Take advertising, for example. Beer manufacturers and people like that are willing to pay good money these days if famous writers will lend their names to their products. Why, my heavens, Mr Bohlen! This isn’t any children’s plaything we’re talking about. It’s big business.’

‘Don’t get too ambitious, my boy.’

‘And another thing. There isn’t any reason why we shouldn’t put *your* name, Mr Bohlen, on some of the better stories, if you wished it.’

‘My goodness, Knipe. What should I want that for?’

‘I don’t know, sir, except that some writers get to be very much respected – like Mr Erle Gardner or Kathleen Morris, for example. We’ve got to have names, and I was certainly thinking of using my own on one or two stories, just to help out.’

‘A writer, eh?’ Mr Bohlen said, musing. ‘Well, it would surely surprise them over at the club when they saw my name in the magazines – the good magazines.’

‘That’s right, Mr Bohlen!’

For a moment, a dreamy, faraway look came into Mr Bohlen's eyes, and he smiled. Then he stirred himself and began leafing through the plans that lay before him.

'One thing I don't quite understand, Knipe. Where do the plots come from? The machine can't possibly invent plots.'

'We feed those in, sir. That's no problem at all. Everyone has plots. There's three or four hundred of them written down in that folder there on your left. Feed them straight into the "plot-memory" section of the machine.'

'Go on.'

'There are many other little refinements too, Mr Bohlen. You'll see them all when you study the plans carefully. For example, there's a trick that nearly every writer uses, of inserting at least one long, obscure word into each story. This makes the reader think that the man is very wise and clever. So I have the machine do the same thing. There'll be a whole stack of long words stored away just for this purpose.'

'Where?'

'In the "word-memory" section,' he said, epexegetically.

Through most of that day the two men discussed the possibilities of the new engine. In the end, Mr Bohlen said he would have to think about it some more. The next morning, he was quietly enthusiastic. Within a week, he was completely sold on the idea.

'What we'll have to do, Knipe, is to say that we're merely building another mathematical calculator, but of a new type. That'll keep the secret.'

'Exactly, Mr Bohlen.'

And in six months the machine was completed. It was housed in a separate brick building at the back of the premises, and now that it was ready for action, no one was allowed near it excepting Mr Bohlen and Adolph Knipe.

It was an exciting moment when the two men – the one, short, plump, breviped – the other tall, thin and toothy – stood in the corridor before the control panel and got ready to run off the first story. All around them were walls dividing up into many small corridors, and the walls were covered with wiring and plugs and switches and huge glass valves. They were both nervous, Mr Bohlen hopping from one foot to the other, quite unable to keep still.

'Which button?' Adolph Knipe asked, eyeing a row of small white discs that resembled the keys of a typewriter. 'You choose, Mr Bohlen. Lots of magazines to pick from – *Saturday Evening Post*, *Collier's*, *Ladies' Home Journal* – any one you like.'

'Goodness me, boy! How do I know?' He was jumping up and down like a man with hives.

'Mr Bohlen,' Adolph Knipe said gravely, 'do you realize that at this moment, with your little finger alone, you have it in your power to become the most versatile writer on this continent?'

'Listen, Knipe, just get on with it, will you please – and cut out the preliminaries.'

'OK, Mr Bohlen. Then we'll make it ... let me see – this one. How's that?' He extended one finger and pressed down a button with the name TODAY'S WOMAN printed across it in diminutive black type. There was a sharp click, and when he took his finger away, the button remained down, below the level of the others.

‘So much for the selection,’ he said. ‘Now – here we go!’ He reached up and pulled a switch on the panel. Immediately, the room was filled with a loud humming noise, and a crackling of electric sparks, and the jingle of many, tiny, quickly moving levers; and almost in the same instant, sheets of quarto paper began sliding out from a slot to the right of the control panel and dropping into a basket below. They came out quick, one sheet a second, and in less than half a minute it was all over. The sheets stopped coming.

‘That’s it!’ Adolph Knipe cried. ‘There’s your story!’

They grabbed the sheets and began to read. The first one they picked up started as follows:

‘Aifkjmbsaoegweztp-pl-nvoqudskigt&,-fuhpekanvbertyuio lkjhgfdsazxcvbnm, peru itrehdjkg mvnb, wmsuy ...’ They looked at the others. The style was roughly similar in all of them. Mr Bohlen began to shout. The younger man tried to calm him down.

‘It’s all right, sir. Really it is. It only needs a little adjustment. We’ve got a connection wrong somewhere, that’s all. You must remember, Mr Bohlen, there’s over a million feet of wiring in this room. You can’t expect everything to be right first time.’

‘It’ll never work,’ Mr Bohlen said.

‘Be patient, sir. Be patient.’

Adolph Knipe set out to discover the fault, and in four days’ time he announced that all was ready for the next try.

‘It’ll never work,’ Mr Bohlen said. ‘I know it’ll never work.’

Knipe smiled and pressed the selector button marked *Reader’s Digest*. Then he pulled the switch, and again the strange, exciting, humming sound filled the room. One page of typescript flew out of the slot into the basket.

‘Where’s the rest?’ Mr Bohlen cried. ‘It’s stopped! It’s gone wrong!’

‘No sir, it hasn’t. It’s exactly right. It’s for the *Digest*, don’t you see?’

This time it began. ‘Fewpeopleyetknowthatarevolutionary-newcurehasbeendiscoveredwhichmaywell-bringpermanent-relieftosufferersofthemostdreadeddiseaseofourtime ...’ And so on.

‘It’s gibberish!’ Mr Bohlen shouted.

‘No, sir, it’s fine. Can’t you see? It’s simply that she’s not breaking up the words. That’s an easy adjustment. But the story’s there. Look, Mr Bohlen, look! It’s all there except that the words are joined together.’

And indeed it was.

On the next try a few days later, everything was perfect, even the punctuation. The first story they ran off, for a famous women’s magazine, was a solid, plotty story of a boy who wanted to better himself with his rich employer. This boy arranged, so that story went, for a friend to hold up the rich man’s daughter on a dark night when she was driving home. Then the boy himself, happening by, knocked the gun out of his friend’s hand and rescued the girl. The girl was grateful. But the father was suspicious. He questioned the boy sharply. The boy broke down and confessed. Then the father, instead of kicking him out of the house, said that he admired the boy’s resourcefulness. The girl admired his honesty – and his looks. The father promised him to be head of the Accounts Department.

The girl married him.

‘It’s tremendous, Mr Bohlen! It’s exactly right!’

‘Seems a bit sloppy to me, my boy!’

‘No, sir, it’s a seller, a real seller!’

In his excitement, Adolph Knipe promptly ran off six more stories in as many minutes. All of them – except one, which for some reason came out a trifle lewd – seemed entirely satisfactory.

Mr Bohlen was now mollified. He agreed to set up a literary agency in an office downtown, and to put Knipe in charge. In a couple of weeks, this was accomplished. Then Knipe mailed out the first dozen stories. He put his own name to four of them, Mr Bohlen’s to one, and for the others he simply invented names.

Five of these stories were promptly accepted. The one with Mr Bohlen’s name on it was turned down with a letter from the fiction editor saying, ‘This is a skilful job, but in our opinion it doesn’t quite come off. We would like to see more of this writer’s work ...’ Adolph Knipe took a cab out to the factory and ran off another story for the same magazine. He again put Mr Bohlen’s name to it, and mailed it immediately. That one they bought.

The money started pouring in. Knipe slowly and carefully stepped up the output, and in six months’ time he was delivering thirty stories a week, and selling about half.

He began to make a name for himself in literary circles as a prolific and successful writer. So did Mr Bohlen; but not quite such a good name, although he didn’t know it. At the same time, Knipe was building up a dozen or more fictitious persons as promising young authors. Everything was going fine.

At this point it was decided to adapt the machine for writing novels as well as stories. Mr Bohlen, thirsting now for greater honours in the literary world, insisted that Knipe go to work at once on this prodigious task.

‘I want to do a novel,’ he kept saying. ‘I want to do a novel.’

‘And so you will, sir. And so you will. But please be patient. This is a very complicated adjustment I have to make.’

‘Everyone tells me I ought to do a novel,’ Mr Bohlen cried. ‘All sorts of publishers are chasing after me day and night begging me to stop fooling around with stories and do something really important instead. A novel’s the only thing that counts – that’s what they say.’

‘We’re going to do novels,’ Knipe told him. ‘Just as many as we want. But please be patient.’

‘Now listen to me, Knipe. What I’m going to do is a *serious* novel, something that’ll make ’em sit up and take notice. I’ve been getting rather tired of the sort of stories you’ve been putting my name to lately. As a matter of fact, I’m none too sure you haven’t been trying to make a monkey out of me.’

‘A monkey, Mr Bohlen?’

‘Keeping all the best ones for yourself, that’s what you’ve been doing.’

‘Oh no, Mr Bohlen! No!’

‘So this time I’m going to make damn sure I write a high class intelligent book. You understand that.’

‘Look, Mr Bohlen. With the sort of switchboard I’m rigging up, you’ll be able to write any sort of book you want.’

And this was true, for within another couple of months, the genius of Adolph Knipe had not only adapted the machine for novel writing, but had constructed a marvellous new control system which enabled the author to pre-select literally any type of plot and any style of writing he desired. There were so many dials and levers on the thing, it looked like the instrument panel of some enormous aeroplane.

First, by depressing one of a series of master buttons, the writer made his primary decision: historical, satirical, philosophical, political, romantic, erotic, humorous, or straight. Then, from the second row (the basic buttons), he chose his theme: army life, pioneer days, civil war, world war, racial problem, wild west, country life, childhood memories, seafaring, the sea bottom and many, many more. The third row of buttons gave a choice of literary style: classical, whimsical, racy, Hemingway, Faulkner, Joyce, feminine, etc. The fourth row was for characters, the fifth for wordage – and so on and so on – ten long rows of pre-selector buttons.

But that wasn't all. Control had also to be exercised during the actual writing process (which took about fifteen minutes per novel), and to do this the author had to sit, as it were, in the driver's seat, and pull (or push) a battery of labelled stops, as on an organ. By so doing, he was able continually to modulate or merge fifty different and variable qualities such as tension, surprise, humour, pathos, and mystery. Numerous dials and gauges on the dashboard itself told him throughout exactly how far along he was with his work.

Finally, there was the question of 'passion'. From a careful study of the books at the top of the best-seller lists for the past year, Adolph Knipe had decided that this was the most important ingredient of all – a magical catalyst that somehow or other could transform the dulllest novel into a howling success – at any rate financially. But Knipe also knew that passion was powerful, heady stuff, and must be prudently dispensed – the right proportions at the right moments; and to ensure this, he had devised an independent control consisting of two sensitive sliding adjustors operated by foot-pedals, similar to the throttle and brake in a car. One pedal governed the percentage of passion to be injected, the other regulated its intensity. There was no doubt, of course – and this was the only drawback – that the writing of a novel by the Knipe methods was going to be rather like flying a plane and driving a car and playing an organ all at the same time, but this did not trouble the inventor. When all was ready, he proudly escorted Mr Bohlen into the machine house and began to explain the operating procedure for the new wonder.

'Good God, Knipe! I'll never be able to do all that! Dammit, man, it'd be easier to write the thing by hand!'

'You'll soon get used to it, Mr Bohlen, I promise you. In a week or two, you'll be doing it without hardly thinking. It's just like learning to drive.'

Well, it wasn't quite as easy as that, but after many hours of practice, Mr Bohlen began to get the hang of it, and finally, late one evening, he told Knipe to make ready for running off the first novel. It was a tense moment, with the fat little man crouching nervously in the driver's seat, and the tall toothy Knipe fussing excitedly around him.

'I intend to write an important novel, Knipe.'

'I'm sure you will, sir. I'm sure you will.'

With one finger, Mr Bohlen carefully pressed the necessary pre-selector buttons:

Master button – *satirical*

Subject – *racial problem*

Style – *classical*

Characters – *six men, four women, one infant*

Length – *fifteen chapters*.

At the same time he had his eye particularly upon three organ stops marked *power, mystery, profundity*.

‘Are you ready, sir?’

‘Yes, yes, I’m ready.’

Knipe pulled the switch. The great engine hummed. There was a deep whirring sound from the oiled movement of fifty thousand cogs and rods and levers; then came the drumming of the rapid electrical typewriter, setting up a shrill, almost intolerable clatter. Out into the basket flew the typewritten pages – one every two seconds. But what with the noise and the excitement and having to play upon the stops, and watch the chapter-counter and the pace-indicator and the passion-gauge, Mr Bohlen began to panic. He reacted in precisely the way a learner driver does in a car – by pressing both feet hard down on the pedals and keeping them there until the thing stopped.

‘Congratulations on your first novel,’ Knipe said, picking up the great bundle of typed pages from the basket.

Little pearls of sweat were oozing out all over Mr Bohlen’s face. ‘It sure was hard work, my boy.’

‘But you got it done, sir. You got it done.’

‘Let me see it, Knipe. How does it read?’

He started to go through the first chapter, passing each finished page to the younger man.

‘Good heavens, Knipe! What’s this!’ Mr Bohlen’s thin purple fish-lip was moving slightly as it mouthed the words, his cheeks were beginning slowly to inflate.

‘But look here, Knipe! This is outrageous!’

‘I must say it’s a bit fruity, sir.’

‘*Fruity!* It’s perfectly revolting! I can’t possibly put my name to this!’

‘Quite right, sir. Quite right!’

‘Knipe! Is this some nasty trick you’ve been playing on me?’

‘Oh no, sir! No!’

‘It certainly looks like it.’

‘You don’t think, Mr Bohlen, that you mightn’t have been pressing a little hard on the passion-control pedals, do you?’

‘My dear boy, how should I know?’

‘Why don’t you try another?’

So Mr Bohlen ran off a second novel, and this time it went according to plan.

Within a week, the manuscript had been read and accepted by an enthusiastic publisher. Knipe followed with one in his own name, then made a dozen more for good measure. In no time at all, Adolph Knipe’s Literary Agency had become famous for its large stable of promising young

novelists. And once again the money started rolling in.

It was at this stage that young Knipe began to display a real talent for big business.

‘See here, Mr Bohlen,’ he said. ‘We still got too much competition. Why don’t we just absorb all the other writers in the country?’

Mr Bohlen, who now sported a bottle-green velvet jacket and allowed his hair to cover two-thirds of his ears, was quite content with things the way they were. ‘Don’t know what you mean, my boy. You can’t just absorb writers.’

‘Of course you can, sir. Exactly like Rockefeller did with his oil companies. Simply buy ’em out, and if they won’t sell, squeeze ’em out. It’s easy!’

‘Careful now, Knipe. Be careful.’

‘I’ve got a list here sir, of fifty of the most successful writers in the country, and what I intend to do is offer each one of them a lifetime contract with pay. All *they* have to do is undertake never to write another word; and, of course, to let us use their names on our own stuff. How about that?’

‘They’ll never agree.’

‘You don’t know writers, Mr Bohlen. You watch and see.’

‘What about the creative urge, Knipe?’

‘It’s bunk! All they’re really interested in is the money – just like everybody else.’

In the end, Mr Bohlen reluctantly agreed to give it a try, and Knipe, with his list of writers in his pocket, went off in a large chauffeur-driven Cadillac to make his calls.

He journeyed first to the man at the top of the list, a very great and wonderful writer, and he had no trouble getting into the house. He told his story and produced a suitcase full of sample novels, and a contract for the man to sign which guaranteed him so much a year for life. The man listened politely, decided he was dealing with a lunatic, gave him a drink, then firmly showed him to the door.

The second writer on the list, when he saw Knipe was serious, actually attacked him with a large metal paperweight, and the inventor had to flee down the garden followed by such a torrent of abuse and obscenity as he had never heard before.

But it took more than this to discourage Adolph Knipe. He was disappointed but not dismayed, and off he went in his big car to seek his next client. This one was a female, famous and popular, whose fat romantic books sold by the million across the country. She received Knipe graciously, gave him tea, and listened attentively to his story.

‘It all sounds very fascinating,’ she said. ‘But of course I find it a little hard to believe.’

‘Madam,’ Knipe answered. ‘Come with me and see it with your own eyes. My car awaits you.’

So off they went, and in due course, the astonished lady was ushered into the machine house where the wonder was kept. Eagerly, Knipe explained its workings, and after a while he even permitted her to sit in the driver’s seat and practise with the buttons.

‘All right,’ he said suddenly, ‘you want to do a book now?’

‘Oh yes!’ she cried. ‘Please!’

She was very competent and seemed to know exactly what she wanted. She made her own pre-selections, then ran off a long, romantic, passion-filled novel. She read through the first chapter and became so enthusiastic that she signed up on the spot.

‘That’s one of them out of the way,’ Knipe said to Mr Bohlen afterwards. ‘A pretty big one too.’

‘Nice work, my boy.’

‘And you know *why* she signed?’

‘Why?’

‘It wasn’t the money. She’s got plenty of that.’

‘Then why?’

Knipe grinned, lifting his lip and baring a long pale upper gum. ‘Simply because she saw the machine-made stuff was better than her own.’

Thereafter, Knipe wisely decided to concentrate only upon mediocrity. Anything better than that – and there were so few it didn’t matter much – was apparently not quite so easy to seduce.

In the end, after several months of work, he had persuaded something like seventy per cent of the writers on his list to sign the contract. He found that the older ones, those who were running out of ideas and had taken to drink, were the easiest to handle. The younger people were more troublesome. They were apt to become abusive, sometimes violent when he approached them; and more than once Knipe was slightly injured on his rounds.

But on the whole, it was a satisfactory beginning. This last year – the first full year of the machine’s operation – it was estimated that at least one half of all the novels and stories published in the English language were produced by Adolph Knipe upon the Great Automatic Grammatizator.

Does this surprise you?

I doubt it.

And worse is yet to come. Today, as the secret spreads, many more are hurrying to tie up with Mr Knipe. And all the time the screw turns tighter for those who hesitate to sign their names.

This very moment, as I sit here listening to the howling of my nine starving children in the other room, I can feel my own hand creeping closer and closer to that golden contract that lies over on the other side of the desk.

Give us strength, Oh Lord, to let our children starve.

f

memory

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In Greek mythology, Mnemosyne, the goddess of memory, is also the mother of the nine Muses.

161

Tell the story of your life that is the most emotionally cathartic; the story you "remember" is covering the "real story," anyway.

162

Reality takes shape in memory alone.

163

Memory: the past rewritten in the direction of feeling.

164

Human memory, driven by emotional self-interest, goes to extraordinary lengths to provide evidence to back up whatever understanding of the world we have our hearts set on—however removed that may be from reality.

165

According to Ulric Neisser's analysis of the structure of episodic memory, we rely—in our remembering—on complex narrative strategies that closely resemble the strategies writers use to produce realist fiction. David Pillemer, whose specialty is "vivid memories," thinks that it takes something like a painter's touch (the mind being the painter) to bring a memory to life and create belief. Antonio Damasio compares consciousness to a "movie in the brain" and argues that memories are just one among the many captions and images that our mind makes up to help us survive in the world. Remembering and fiction-making are virtually indistinguishable.

166

Anything processed by memory is fiction.

167

When memory is called to answer, it often answers back with deception. How is it that almost every warm bar stool contains a hero, a star of his own epic, who is the sum of his amazing stories?

168

Consciously or unconsciously, we manipulate our memories to include or omit certain aspects. Are our memories therefore fictions?

169

Memories have a quasi-narrative structure, constituting a story or a scene in a story, an inbuilt successiveness strong enough to keep the narrative the same on each act of remembering but not strong enough to ensure that the ordering of events is the ordering that originally took place.

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Scooter Libby, Dick Cheney's former chief of staff, hired a memory-loss expert to make the case that Libby forgot about Valerie Plame's employment status due to the "vulnerability of memory." Libby's defense team argued that "any misstatements he made during his FBI interviews or grand jury testimony were not intentional but rather the result of confusion, mistake, or faulty memory."

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Remembering his country, he imagined it.

172

Our personal experience, though it may convey great truths, most likely won't be verified by security camera tapes later. We usually think of memory in just this way, as if a recorder planted in our head could be rewound and replayed; however, memory often stores perceptual information in verbal forms, not images. We remember a "light blue Rambler," and yet because we have translated it in our minds into a verbal construct, we would find it difficult to retranslate the memory into an image, re-creating exactly the right shade of blue. Autobiographical memory is a recollection of events or episodes, which we remember with great detail. What's stored in that memory isn't the actual events, but how those events made sense to us and fit into our experience.

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We tend to think of our memories as having been tucked away for safekeeping in, say, file cabinets or dusty old boxes in the backs of closets or filed away on the hard drives of computers, where they can easily be accessed by the click of a button. All

it takes to remember events and objects is to open the boxes, open the files, and there our memories will be, waiting for us. Just as boxes and files mold and rot and computers become infected with viruses making the files inaccessible or corrupted, so do our memories. In a sense, all memories have been forgotten. Memories are predicated on loss. It's through the act of remembering that we bring these forgotten experiences back from oblivion. They require this rescuing because they've run their course. These experiences are complete and have been relegated to our memories. In other words, to remember is to recall what we've forgotten, but it's not as if our memories have been rubbed away by years of wind and rain like names and dates on a gravestone; instead, our memories are filled with gaps and distortions, because by its very nature memory is selective.

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The genius of memory is that it is choosy, chancy, and temperamental.

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To fill in the holes, we turn our memories into specific images, which our minds understand as representing a specific experience, object, or thought. Our past experiences have been dismantled, analyzed, re-collated, and then made ready for imagistic recall. The images we store in our memories are not exact replicas of what we experienced; they're what our minds turn them into. They are what we need to re-create the story, which is the full experience the image represents.

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Freud: we have no memories from our childhood, only memories that pertain to our childhood. Is a story merely a memory of a memory? How can a memory, which is grounded in an image (e.g., a light blue Rambler) and which rings so true to me, be false? If this is false, then what is truth? And why does this matter?

177

It's difficult to separate what happened from what seemed to happen.

178

Did this happen?

Yes.

Did this happen in this way?

The answer to that, if you're a grown-up, is "Not necessarily."

179

Memory loves to go hunting in the dark.

180

Nonfiction writers imagine. Fiction writers invent. These are fundamentally different acts, performed to different ends. Unlike a fiction reader, whose only task is to imagine, a nonfiction reader is asked to behave more deeply: to imagine, and also to believe. Fiction doesn't require its readers to believe; in fact, it offers its readers the great freedom of experience without belief—something real life can't do. Fiction gives us a rhetorical question: "What if this happened?" (The best) nonfiction gives us a statement, something more complex: "This may have happened."

181

The essay consists of double translation: memory translates experience; essay translates memory.

182

For example, in Proust, who is to me at base an essayist, nothing ever happens. The only obstacles are that someone might rebuff someone else or someone might get sick or grow old, and even these are usually hypothetical obstacles. People get educations, travel, buy paintings, go on diplomatic missions, but the events are for the most part meetings between various people (or simply sightings of one person by another, sometimes thanks to a stroll or a ride in a carriage) and what these meetings bring out, on a psychological level, about life itself. How can a work be considered fiction when there's no plot? Philosophy, perhaps, or criticism, but not fiction.

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Carpenters restore old homes to their architectural and design period, not knowing the original color of the walls. If restoring a home is like writing a nonfiction narrative, and if choosing the paint for one wall is like imagining one moment in the larger story, shouldn't we acknowledge that the house and its walls were in fact never one particular way? On a single wall, sometimes wallpaper hung, sometimes paintings stared, sometimes children penned their names, sometimes flies sat, sometimes dust settled, sometimes sunlight blazed, sometimes fingerprints shimmered. The lost story the carpenter tries to restore isn't one particular story, but a pool of possible tales, with different perspectives from different characters, told at different times for different reasons. The nonfiction writer who works to revive a lost scene adds one similar story to the col-

lection of stories that ever existed for that moment. The entire platform of my imagination—my purpose, my hope, my intent—is different from that of a fiction writer's. I don't seek to tell the best story. I seek to tell a story that once was. I seek to fill a place that once had meaning with meaning again.

g

blur

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I think of fiction, nonfiction, poetry, drama, and all forms of storytelling as existing on a rather wide continuum, at one end fantasy (J. R. R. Tolkien and the like) and at the other end an extremely literal-minded register of a life, such as a guy in eastern Washington—named, as fate would have it, Shields—who (until his recent death) had kept the longest or longest-running diary, endless accounts of everything he did all day. And in between at various tiny increments are greater and lesser imaginative projects. An awful lot of fiction is immensely autobiographical, and a lot of nonfiction is highly imagined. We dream ourselves awake every minute of the day. "Fiction"/"nonfiction" is an utterly useless distinction.

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Tell all the Truth but tell it slant—

RANKING, EVALUATING, AND LIKING: SORTING OUT THREE FORMS OF JUDGMENT

Peter Elbow

This essay is my attempt to sort out different acts we call assessment—some different ways in which we express or frame our judgments of value. I have been working on this tangle not just because it is interesting and important in itself but because assessment tends so much to drive and control *teaching*. Much of what we do in the classroom is determined by the assessment structures we work under.

Assessment is a large and technical area and I'm not a professional. But my main premise or subtext in this essay is that we nonprofessionals can and should work on it because professionals have not reached definitive conclusions about the problem of how to assess writing (or anything else, I'd say). Also, decisions about assessment are often made by people even less professional than we, namely legislators. Pat Belanoff and I realized that the field of assessment was open when we saw the harmful effects of a writing proficiency exam at Stony Brook and worked out a collaborative portfolio assessment system in its place (Belanoff and Elbow; Elbow and Belanoff). Professionals keep changing their minds about large-scale testing and assessment. And as for classroom grading, psychometricians provide little support or defense of it.

THE PROBLEMS WITH RANKING AND THE BENEFITS OF EVALUATING

By ranking I mean the act of summing up one's judgment of a performance or person into a single, holistic number or score. We rank every time we give a grade or holistic score. Ranking implies a single scale or continuum or dimension along which all performances are hung.

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By evaluating I mean the act of expressing one's judgment of a performance or person by pointing out the strengths and weaknesses of different features or dimensions. We evaluate every time we write a comment on a paper or have an conversation about its value. Evaluation implies the recognition of different criteria or dimensions—and by implication different contexts and audiences for the same performance. Evaluation requires going *beyond* a first response that may be nothing but a kind of ranking (“I like it” or “This is better than that”), and instead looking carefully enough at the performance or person to make distinctions between parts or features or criteria.

It's obvious, thus, that I am troubled by ranking. But I will resist any temptation to argue that we can get rid of all ranking—or even should. Instead I will try to show how we can have *less* ranking and *more* evaluation in its place.

I see three distinct problems with ranking: it is inaccurate or unreliable; it gives no substantive feedback; and it is harmful to the atmosphere for teaching and learning.

(1) First the unreliability. To rank reliably means to give a *fair* number, to find the single quantitative score that readers will agree on. But readers don't agree.

This is not news—this unavailability of agreement. We have long seen it on many fronts. For example, research in evaluation has shown many times that if we give a paper to a set of readers, those readers tend to give it the full range of grades (Diederich). I've recently come across new research to this effect—new to me because it was published in 1912. The investigators carefully showed how high school English teachers gave different grades to the same paper. In response to criticism that this was a local problem in English, they went on the next year to discover an even greater variation among grades given by high school geometry teachers and history teachers to papers in their subjects. (See the summary of Daniel Starch and Edward Elliott's 1913 *School Review* articles in Kirschenbaum, Simon, and Napier 258–59.)

We know the same thing from literary criticism and theory. If the best critics can't agree about what a text means, how can we be surprised that they disagree even more about the quality or value of texts? And we know that nothing in literary or philosophical theory gives us any agreed-upon rules for settling such disputes.

Students have shown us the same inconsistency with their own controlled experiments of handing the same paper to different teachers and getting different grades. This helps explain why we hate it so when students ask us their favorite question, “What do you want for an A?”: it rubs our noses in the unreliability of our grades.

Of course champions of holistic scoring argue that they get *can* get agreement among readers—and they often do (White). But they get that agreement by “training” the readers before and during the scoring sessions. What “training”

means is getting those scorers to stop reading the way they normally read—getting them to stop using the conflicting criteria and standards they normally use outside the scoring sessions. (In an impressive and powerful book, Barbara Herrnstein Smith argues that whenever we have widespread inter-reader reliability, we have reason to suspect that difference has been suppressed and homogeneity imposed—almost always at the expense of certain groups.) In short, the reliability in holistic scoring is not a measure of how texts are valued by real readers in natural settings, but only of how they are valued in artificial settings with imposed agreements.

Defenders of holistic scoring might reply (as one anonymous reviewer did), that holistic scores are not perfect or absolutely objective readings but just “judgments that most readers will agree are the appropriate ones given the purpose of the assessment and the system of communication.” But I have been in and even conducted enough holistic scoring sessions to know that even that degree of agreement doesn’t occur unless “purpose” and “appropriateness” are defined to mean acceptance of the single set of standards imposed on that session. We know too much about the differences among readers and the highly variable nature of the reading process. Supposing we get readings only from academics, or only from people in English, or only from respected critics, or only from respected writing programs, or only from feminists, or only from sound readers of my tribe (white, male, middle-class, full professors between the ages of fifty and sixty). We *still* don’t get agreement. We can sometimes get agreement among readers from some subset, a particular community that has developed a strong set of common values, perhaps *one* English department or *one* writing program. But what is the value of such a rare agreement? It tells us nothing about how readers from other English departments or writing programs will judge—much less how readers from other domains will judge.

(From the opposite ideological direction, some skeptics might object to my skeptical train of thought: “So what else is new?” they might reply. “Of *course* my grades are biased, ‘interested’ or ‘situated’—always partial to my interests or the values of my community or culture. There’s no other possibility.” But how can people consent to give grades if they feel that way? A single teacher’s grade for a student is liable to have substantial consequences—for example on eligibility for a scholarship or a job or entrance into professional school. In grading, surely we must not take anything less than genuine fairness as our goal.)

It won’t be long before we see these issues argued in a court of law, when a student who has been disqualified from playing on a team or rejected from a professional school sues, charging that the basis for his plight—teacher grades—is not reliable. I wonder if lawyers will be able to make our grades stick.

(2) Ranking or grading is woefully uncommunicative. Grades and holistic scores are nothing but points on a continuum from “yea” to “boo”—with no

information or clues about the criteria behind these noises. They are 100 percent evaluation and 0 percent description or information. They quantify the degree of approval or disapproval in readers but tell nothing at all about what the readers actually approve or disapprove of. They say nothing that couldn't be said with gold stars or black marks or smiley-faces. Of course our first reactions are often nothing but global holistic feelings of approval or disapproval, but we need a system for communicating our judgments that nudges us to move beyond these holistic feelings and to articulate the basis of our feeling—a process that often leads us to change our feeling. (Holistic scoring sessions sometimes use rubrics that explain the criteria—though these are rarely passed along to students—and even in these situations, the rubrics fail to fit many papers.) As C. S. Lewis says, “People are obviously far more anxious to express their approval and disapproval of things than to describe them” (7).

(3) Ranking leads students to get so hung up on these oversimple quantitative verdicts that they care more about scores than about learning—more about the grade we put on the paper than about the comment we have written on it. Have you noticed how grading often forces us to write comments to justify our grades?—and how these are often *not* the comment we would make if we were just trying to help the student write better? (“Just try writing several favorable comments on a paper and then giving it a grade of D” [Diederich 21].)

Grades and holistic scores give too much encouragement to those students who score high—making them too apt to think they are already fine—and too little encouragement to those students who do badly. Unsuccessful students often come to doubt their intelligence. But oddly enough, many “A” students also end up doubting their true ability and feeling like frauds—because they have sold out on their own judgment and simply given teachers whatever yields an A. They have too often been rewarded for what they don't really believe in. (Notice that there's more cheating by students who get high grades than by those who get low ones. There would be less incentive to cheat if there were no ranking.)

We might be tempted to put up with the inaccuracy or unfairness of grades if they gave good diagnostic feedback or helped the learning climate; or we might put up with the damage they do to the learning climate if they gave a fair or reliable measure of how skilled or knowledgeable students are. But since they fail dismally on both counts, we are faced with the striking question of why grading has persisted so long.

There must be many reasons. It is obviously easier and quicker to express a global feeling with a single number than to figure out what the strengths and weaknesses are and what one's criteria are. (Though I'm heartened to discover, as I pursue this issue, how troubled teachers are by grading and how difficult they find it.) But perhaps more important, we see around us a deep *hunger to rank*—to create pecking orders: to see who we can look down on and who we must look up

to, or in the military metaphor, who we can kick and who we must salute. Psychologists tell us that this taste for pecking orders or ranking is associated with the authoritarian personality. We see this hunger graphically in the case of IQ scores. It is plain that IQ scoring does not represent a commitment to looking carefully at people's intelligence; when we do that, we see different and frequently uncorrelated *kinds* or *dimensions* of intelligence (Gardner). The persistent use of IQ scores represents the hunger to have a number so that everyone can have a rank. ("Ten!" mutter the guys when they see a pretty woman.)

Because ranking or grading has caused so much discomfort to so many students and teachers, I think we see a lot of confusion about the process. It is hard to think clearly about something that has given so many of us such anxiety and distress. The most notable confusion I notice is the tendency to think that if we renounce ranking or grading, we are renouncing the very possibility of judgment and discrimination—that we are embracing the idea that there is no way to distinguish or talk about the difference between what works well and what works badly.

So the most important point, then, is that *I am not arguing against judgment or evaluation*. I'm just arguing against that crude, oversimple way of *representing* judgment—distorting it, really—into a single number, which means ranking people and performances along a single continuum.

In fact I am arguing *for evaluation*. Evaluation means looking hard and thoughtfully at a piece of writing in order to make distinctions as to the quality of different features or dimensions. For example, the process of evaluation permits us to make the following kinds of statements about a piece of writing:

- The thinking and ideas seem interesting and creative.
- The overall structure or sequence seems confusing.
- The writing is perfectly clear at the level of individual sentences and even paragraphs.
- There is an odd, angry tone of voice that seems unrelated or inappropriate to what the writer is saying.
- Yet this same voice is strong and memorable and makes one listen even if one is irritated.
- There are a fair number of mistakes in grammar or spelling: more than "a sprinkling" but less than "riddled with."

To rank, on the other hand, is to be forced to translate those discriminations into a single number. What grade or holistic score do these judgments add up to? It's likely, by the way, that more readers would agree with those separate, "analytic" statements than would agree on a holistic score.

I've conducted many assessment sessions where we were not trying to impose a set of standards but rather to find out how experienced teachers read and evaluate, and I've had many opportunities to see that good readers give grades or

scores right down through the range of possibilities. Of course good readers sometimes agree—especially on papers that are strikingly good or bad or conventional, but I think I see difference more frequently than agreement when readers really speak up.

The process of evaluation, because it invites us to articulate our criteria and to make distinctions among parts or features or dimensions of a performance, thereby invites us further to acknowledge the main fact about evaluation: that different readers have different priorities, values, and standards.

The conclusion I am drawing, then, in this first train of thought is that we should do less ranking and more evaluation. Instead of using grades or holistic scores—single number verdicts that try to sum up complex performances along only one scale—we should give some kind of written or spoken evaluation that discriminates among criteria and dimensions of the writing—and if possible that takes account of the complex context for writing: who the writer is, what the writer's audience and goals are, who we are as readers and how we read, and how we might differ in our reading from other readers the writer might be addressing.

But how can we put this principle into practice? The pressure for ranking seems implacable. Evaluation takes more time, effort, and money. It seems as though we couldn't get along without scores on writing exams. Most teachers are obliged to give grades at the end of each course. And many students—given that they have become conditioned or even addicted to ranking over the years and must continue to inhabit a ranking culture in most of their courses—will object if we don't put grades on papers. Some students, in the absence of that crude gold star or black mark, may not try hard enough (though how hard is "enough"—and is it really our job to stimulate motivation artificially with grades—and is grading the best source of motivation?).

It is important to note that there are certain schools and colleges that do *not* use single-number grades or scores, and they function successfully. I taught for nine years at Evergreen State College, which uses only written evaluations. This system works fine, even down to getting students accepted into high quality graduate and professional schools.

Nevertheless we have an intractable dilemma: that grading is unfair and counterproductive but that students and institutions tend to want grades. In the face of this dilemma there is a need for creativity and pragmatism. Here are some ways in which I and others use *less ranking* and *more evaluation* in teaching—and they suggest some adjustments in how we score large-scale assessments. What follows is an assortment of experimental compromises—sometimes crude, seldom ideal or utopian—but they help.

(a) Portfolios. Just because conventional institutions oblige us to turn in a single quantitative course grade at the end of every marking period, it doesn't

follow that we need to grade individual papers. Course grades are more trustworthy and less damaging because they are based on so many performances over so many weeks. By avoiding frequent ranking or grading, we make it *somewhat* less likely for students to become addicted to oversimple numerical rankings—to think that evaluation always translates into a simple number—in short, to mistake ranking for evaluation. (I'm not trying to defend conventional course grades since they are still uncommunicative and they still feed the hunger for ranking.) Portfolios permit me to refrain from grading individual papers and limit myself to writerly evaluative comments—and help students see this as a positive rather than a negative thing, a chance to be graded on a body of their best work that can be judged more fairly. Portfolios have many other advantages as well. They are particularly valuable as occasions for asking students to write extensive and thoughtful explorations of their own strengths and weaknesses.

A midsemester portfolio is usually an informal affair, but it is a good occasion for giving anxious students a ballpark estimate of how well they are doing in the course so far. I find it helpful to tell students that I'm perfectly willing to tell them my best estimate of their course grade—but only if they come to me in conference and only during the second half of the semester. This serves somewhat to quiet their anxiety while they go through seven weeks of drying out from grades. By midsemester, most of them have come to enjoy not getting those numbers and thus being able to think better about more writerly comments from me and their classmates.

Portfolios are now used extensively and productively in larger assessments, and there is constant experimentation with new applications (Belanoff and Dickson; *Portfolio Assessment Newsletter*; *Portfolio News*).

(b) Another useful option is to make a strategic retreat from a wholly negative position. That is, I sometimes do a *bit* of ranking even on individual papers, using two “bottom-line” grades: H and U for “Honors” and “Unsatisfactory.” I tell students that these translate to about A or A- and D or F. This practice may seem theoretically inconsistent with all the arguments I've just made, but (at the moment, anyway) I justify it for the following reasons.

First, I sympathize with a *part* of the students' anxiety about not getting grades: their fear that they might be failing and not know about it—or doing an excellent job and not get any recognition. Second, I'm not giving *many* grades; only a small proportion of papers get these H's or U's. The system creates a “non-bottom-line” or “non-quantified” atmosphere. Third, these holistic judgments about best and worst do not seem as arbitrary and questionable as most grades. There is usually a *bit* more agreement among readers about the best and worst papers. What seems most dubious is the process of trying to rank that whole middle range of papers—papers that have a mixture of better and worse

qualities so that the numerical grade depends enormously on a reader's priorities or mood or temperament. My willingness to give these few grades goes a long way toward helping my students forgo most bottom-line grading.

I'm not trying to pretend that these minimal "grades" are truly reliable. But they represent a very small amount of ranking. Yes, someone could insist that I'm really ranking every single paper (and indeed if it seemed politically necessary, I could put an OK or S [for satisfactory] on all those middle range papers and brag, "Yes, I grade everything.") But the fact is that I am doing *much less sorting* since I don't have to sort them into five or even twelve piles. Thus there is a huge reduction in the total amount of unreliability I produce.

(It might seem that if I use only these few minimal grades I have no good way for figuring out a final grade for the course—since that requires a more fine-grained set of ranks. But I don't find that to be the case. For I also give these same minimal grades to the many other important parts of my course such as attendance, meeting deadlines, peer responding, and journal writing. If I want a mathematically computed grade on a scale of six or A through E, I can easily compute it when I have such a large number of grades to work from—even though they are only along a three-point scale.)

This same practice of crude or minimal ranking is a big help on larger assessments outside classrooms, and needs to be applied to the process of assessment in general. There are two important principles to emphasize. On the one hand we must be prudent or accommodating enough to admit that despite all the arguments against ranking, there *are* situations when we need that bottom-line verdict along one scale: which student has not done satisfactory work and should be denied credit for the course? which student gets the scholarship? which candidate to hire or fire? We often operate with scarce resources. But on the other hand we must be bold enough to insist that we do far more ranking than is really needed. We can get along not only with fewer occasions for assessment but also with fewer gradations in scoring. If we decide what the *real* bottom-line is on a given occasion—perhaps just "failing" or perhaps "honors" too—then the reading of papers or portfolios is enormously quick and cheap. It leaves time and money for evaluation—perhaps for analytic scoring or some comment.

At Stony Brook we worked out a portfolio system where multiple readers had only to make a binary decision: acceptable or not. Then individual teachers could decide the actual course grade and give comments for their own students—so long as those students passed in the eyes of an independent rater (Elbow and Belanoff; Belanoff and Elbow). The best way to begin to wean our society from its addiction to ranking may be to permit a tiny bit of it (which also means less unreliability)—rather than trying to go "cold turkey."

(c) Sometimes I use an analytic grid for evaluating and commenting on student papers. An example is given in Figure 1.

Strong OK Weak

			CONTENT, INSIGHTS, THINKING, GRAPPLING WITH TOPIC
			GENUINE REVISION, SUBSTANTIVE CHANGES, NOT JUST EDITING
			ORGANIZATION, STRUCTURE, GUIDING THE READER
			LANGUAGE: SYNTAX, SENTENCES, WORDING, VOICE
			MECHANICS: SPELLING, GRAMMAR, PUNCTUATION, PROOFREADING
			OVERALL [Note: this is not a sum of the other scores.]

Figure 1.

I often vary the criteria in my grid (e.g. “connecting with readers” or “investment”) depending on the assignment or the point in the semester.

Grids are a way I can satisfy the students’ hunger for ranking but still not give in to conventional grades on individual papers. Sometimes I provide nothing but a grid (especially on final drafts), and this is a very quick way to provide a response. Or on midprocess drafts I sometimes use a grid in addition to a comment: a more readerly comment that often doesn’t so much tell them what’s wrong or right or how to improve things but rather tries to give them an account of what is *happening to me* as I read their words. I think this kind of comment is really the most useful thing of all for students, but it frustrates some students for a while. The grid can help these students feel less anxious and thus pay better attention to my comment.

I find grids extremely helpful at the end of the semester for telling students their strengths and weaknesses in the course—or what they’ve done well and not so well. Besides categories like the ones above, I use categories like these: “skill in giving feedback to others,” “ability to meet deadlines,” “effort,” and “improvement.” This practice makes my final grade much more communicative.

(d) I also help make up for the absence of ranking—gold stars and black marks—by having students share their writing with each other a great deal both orally and through frequent publication in class magazines. Also, where possible, I try to get students to give or send writing to audiences outside the class. At the University of Massachusetts at Amherst, freshmen pay a ten dollar lab fee for the writing course, and every teacher publishes four or five class magazines of final drafts a semester. The effects are striking. Sharing, peer feedback, and publication give the best reward and motivation for writing, namely, getting your words out to many readers.

(e) I sometimes use a kind of modified *contract grading*. That is, at the start of the course I pass out a long list of all the things that I most want students to do—the concrete activities that I think most lead to learning—and I promise

students that if they do them *all* they are guaranteed a certain final grade. Currently, I say it's a B—it could be lower or higher. My list includes these items: not missing more than a week's worth of classes; not having more than one late major assignment; *substantive* revising on all major revisions; good copy editing on all final revisions; good effort on peer feedback work; keeping up the journal; and substantial effort and investment on each draft.

I like the way this system changes the “bottom-line” for a course: the intersection where my authority crosses their self-interest. I can tell them, “You have to work very hard in this course, but you can stop worrying about grades.” The crux is no longer that commodity I've always hated and never trusted: a numerical ranking of the quality of their writing along a single continuum. Instead the crux becomes what I care about most: the *concrete behaviors* that I most want students to engage in because they produce more learning and help me teach better. Admittedly, effort and investment are not concrete observable behaviors, but they are no harder to judge than overall quality of writing. And since I care about effort and investment, I don't mind the few arguments I get into about them; they seem fruitful. (“Let's try and figure out why it looked to me as though you didn't put any effort in here.”) In contrast, I hate discussions about grades on a paper and find such arguments fruitless. Besides, I'm not making fine distinctions about effort and investment—just letting a bell go off when they fall palpably low.

It's crucial to note that I am *not* fighting evaluation with this system. I am just fighting ranking or grading. I still write evaluative comments and often use an evaluative grid to tell my students what I see as strengths and weaknesses in their papers. My goal is not to get rid of evaluation but in fact to emphasize it, enhance it. I'm trying to get students to listen *better* to my evaluations—by uncoupling them from a grade. In effect, I'm doing this because I'm so fed up with students *following* or *obeying* my evaluations too blindly—making whatever changes my comments suggest but doing it for the sake of a grade; not really taking the time to make up their own minds about whether they think my judgments or suggestions really make sense to them. The worst part of grades is that they make students obey us without carefully thinking about the merits of what we say. I love the situation this system so often puts students in: I make a criticism or suggestion about their paper, but it doesn't matter to their grade whether they go along with me or not (so long as they genuinely revise in some fashion). They have to think; to decide.

Admittedly this system is crude and impure. Some of the really skilled students who are used to getting A's and desperate to get one in this course remain unhelpfully hung up about getting those H's on their papers. But a good number of these students discover that they can't get them, and they soon settle down to accepting a B and having less anxiety and more of a learning voyage.

THE LIMITATIONS OF EVALUATION AND THE BENEFITS OF EVALUATION-FREE ZONES

Everything I've said so far has been in praise of evaluation as a substitute for ranking. But I need to turn a corner here and speak about the *limits* or *problems* of evaluation. Evaluating may be better than ranking, but it still carries some of the same problems. That is, even though I've praised evaluation for inviting us to acknowledge that readers and contexts are different, nevertheless the very word *evaluation* tends to imply fairness or reliability or getting beyond personal or subjective preferences. Also, of course, evaluation takes a lot more time and work. To rank you just have to put down a number; holistic scoring of exams is cheaper than analytic scoring.

Most important of all, evaluation harms the climate for learning and teaching—or rather *too much* evaluation has this effect. That is, if we evaluate *everything* students write, they tend to remain tangled up in the assumption that their whole job in school is to give teachers “what they want.” Constant evaluation makes students worry more about psyching out the teacher than about what they are really learning. Students fall into a kind of defensive or on-guard stance toward the teacher: a desire to hide what they don't understand and try to impress. This stance gets in the way of learning. (Think of the patient trying to hide symptoms from the doctor.) Most of all, constant evaluation by someone in authority makes students reluctant to take the risks that are needed for good learning—to try out hunches and trust their own judgment. Face it: if our goal is to get students to exercise their own judgment, that means exercising an immature and undeveloped judgment and making choices that are obviously wrong to us.

We see around us a widespread hunger to be evaluated that is often just as strong as the hunger to rank. Countless conditions make many of us walk around in the world wanting to ask others (especially those in authority), “How am I doing, did I do OK?” I don't think the hunger to be evaluated is as harmful as the hunger to rank, but it can get in the way of learning. For I find that the greatest and most powerful breakthroughs in learning occur when I can get myself and others to *put aside* this nagging, self-doubting question (“How am I doing? How am I doing?”)—and instead to take some chances, trust our instincts or hungers. When everything is evaluated, everything counts. Often the most powerful arena for deep learning is a kind of “time out” zone from the pressures of normal evaluated reality: make-believe, play, dreams—in effect, the Shakespearian forest.

In my attempts to get away from too much evaluation (not from all evaluation, just from too much of it), I have drifted into a set of teaching practices which now feel to me like the *best* part of my teaching. I realize now what I've been unconsciously doing for a number of years: creating “evaluation-free zones.”

(a) The paradigm evaluation-free zone is the ten minute, nonstop freewrite. When I get students to freewrite, I am using my authority to create unusual

conditions in order to contradict or interrupt our pervasive habit of always evaluating our writing. What is essential here are the two central features of freewriting: that it be private (thus I don't collect it or have students share it with anyone else); and that it be nonstop (thus there isn't time for planning, and control is usually diminished). Students quickly catch on and enter into the spirit. At the end of the course, they often tell me that freewriting is the most useful thing I've taught them (see Belanoff, Elbow, and Fontaine).

(b) A larger evaluation-free zone is the single unevaluated assignment—what people sometimes call the “quickwrite” or sketch. This is a piece of writing that I ask students to do—either in class or for homework—without any or much revising. It is meant to be low stakes writing. There is a bit of pressure, nevertheless, since I usually ask them to share it with others and *I* usually collect it and read it. But I don't write any comments at all—except perhaps to put straight lines along some passages I like or to write a phrase of appreciation at the end. And I ask students to refrain from giving evaluative feedback to each other—and instead just to say “thank you” or mention a couple of phrases or ideas that stick in mind. (However, this writing-without-feedback can be a good occasion for students to discuss the *topic* they have written about—and thus serve as an excellent kick-off for discussions of what I am teaching.)

(c) These experiments have led me to my next and largest evaluation-free zone—what I sometimes call a “jump start” for my whole course. For the last few semesters I've been devoting the first three weeks *entirely* to the two evaluation-free activities I've just described: freewriting (and also more leisurely private writing in a journal) and quickwrites or sketches. Since the stakes are low and I'm not asking for much revising, I ask for *much more* writing homework per week than usual. And every day we write in class: various exercises or games. The emphasis is on getting rolling, getting fluent, taking risks. And every day all students read out loud something they've written—sometimes a short passage even to the whole class. So despite the absence of feedback, it is a very audience-filled and sociable three weeks.

At first I only dared do this for two weeks, but when I discovered how fast the writing improves, how good it is for building community, and what a pleasure this period is for me, I went to three weeks. I'm curious to try an experiment with teaching a whole course this way. I wonder, that is, whether all that evaluation we work so hard to give really does any more good than the constant writing and sharing (Zak).

I need to pause here to address an obvious rejoinder: “But withholding evaluation is not normal!” Indeed, it is *not* normal—certainly not normal in school. We normally tend to emphasize evaluations—even bottom-line ranking kinds of evaluations. But I resist the argument that if it's not normal we shouldn't do it.

The best argument for evaluation-free zones is from experience. If you try them, I suspect you'll discover that they are satisfying and bring out good writing. Students have a better time writing these unevaluated pieces; they enjoy hearing and appreciating these pieces when they don't have to evaluate. And *I* have a much better time when I engage in this astonishing activity: reading student work when I don't have to evaluate and respond. And yet the writing improves. I see students investing and risking more, writing more fluently, and using livelier, more interesting voices. This writing gives me and them a higher standard of clarity and voice for when we move on to more careful and revised writing tasks that involve more intellectual pushing—tasks that sometimes make their writing go tangled or sodden.

THE BENEFITS AND FEASIBILITY OF LIKING

Liking and disliking seem like unpromising topics in an exploration of assessment. They seem to represent the worst kind of subjectivity, the merest accident of personal taste. But I've recently come to think that the phenomenon of liking is perhaps the most important evaluative response for writers and teachers to think about. In effect, I'm turning another corner in my argument. In the first section I argued against ranking—with evaluating being the solution. Next I argued not *against* evaluating—but for no-evaluation zones in *addition* to evaluating. Now I will argue neither against evaluating nor against no-evaluation zones, but for something very different in addition, or perhaps underneath, as a foundation: liking.

Let me start with the germ story. I was in a workshop and we were going around the circle with everyone telling a piece of good news about their writing in the last six months. It got to Wendy Bishop, a good poet (who has also written two good books about the teaching of writing), and she said, "In the last six months, I've learned to *like* everything I write." Our jaws dropped; we were startled—in a way scandalized. But I've been chewing on her words ever since, and they have led me into a retelling of the story of how people learn to write better.

The old story goes like this: We write something. We read it over and we say, "This is terrible. I *hate* it. I've got to work on it and improve it." And we do, and it gets better, and this happens again and again, and before long we have become a wonderful writer. But that's not really what happens. Yes, we vow to work on it—but we don't. And next time we have the impulse to write, we're just a *bit* less likely to start.

What really happens when people learn to write better is more like this: We write something. We read it over and we say, "This is terrible. . . . But I *like* it. Damn it, I'm going to get it good enough so that others will like it too." And this time we don't just put it in a drawer, we actually work hard on it. And we try it

out on other people too—not just to get feedback and advice but, perhaps more important, to find someone else who will like it.

Notice the two stories here—two hypotheses. (a) “First you improve the faults and then you like it.” (b) “First you like it and then you improve faults.” The second story may sound odd when stated so baldly, but really it’s common sense. Only if we like something will we get involved enough to work and struggle with it. Only if we like what we write will we write again and again by choice—which is the only way we get better.

This hypothesis sheds light on the process of how people get to be published writers. Conventional wisdom assumes a Darwinian model: poor writers are unread; then they get better; as a result, they get a wider audience; finally they turn into Norman Mailer. But now I’d say the process is more complicated. People who get better and get published really tend to be driven by how much *they* care about their writing. Yes, they have a small audience at first—after all, they’re not very good. But they try reader after reader until finally they can find people who like and appreciate their writing. I certainly did this. If someone doesn’t like her writing enough to be pushy and hungry about finding a few people who also like it, she probably won’t get better.

It may sound so far as though all the effort and drive comes from the lonely driven writer—and sometimes it does (Norman Mailer is no joke). But, often enough, readers play the crucially active role in this story of how writers get better. That is, the way writers *learn* to like their writing is by the grace of having a reader or two who likes it—even though it’s not good. Having at least a few appreciative readers is probably indispensable to getting better.

When I apply this story to our situation as teachers I come up with this interesting hypothesis: *good writing teachers like student writing* (and like students). I think I see this borne out—and it is really nothing but common sense. Teachers who hate student writing and hate students are grouchy all the time. How could we stand our work and do a decent job if we hated their writing? Good teachers see what is only *potentially* good, they get a kick out of mere possibility—and they encourage it. When I manage to do this, I teach well.

Thus, I’ve begun to notice a turning point in my courses—two or three weeks into the semester: “Am I going to like these folks or is this going to be a battle, a struggle?” When I like them everything seems to go better—and it seems to me they learn more by the end. When I don’t and we stay tangled up in struggle, we all suffer—and they seem to learn less.

So what am I saying? That we should like bad writing? How can we see all the weaknesses and criticize student writing if we just like it? But here’s the interesting point: if I *like* someone’s writing it’s *easier* to criticize it.

I first noticed this when I was trying to gather essays for the book on freewriting that Pat Belanoff and Sheryl Fontaine and I edited. I would read an

essay someone had written, I would want it for the book, but I had some serious criticism. I'd get excited and write, "I really like this, and I hope we can use it in our book, but you've got to get rid of this and change that, and I got really mad at this other thing." I usually find it hard to criticize, but I began to notice that I was a much more critical and pushy reader when I liked something. It's even fun to criticize in those conditions.

It's the same with student writing. If I like a piece, I don't have to pussyfoot around with my criticism. It's when I don't like their writing that I find myself tiptoeing: trying to soften my criticism, trying to find something nice to say—and usually sounding fake, often unclear. I see the same thing with my own writing. If I like it, I can criticize it better. I have faith that there'll still be something good left, even if I train my full critical guns on it.

In short—and to highlight how this section relates to the other two sections of this essay—liking is not same as ranking or evaluating. Naturally, people get them mixed up: when they like something, they assume it's good; when they hate it, they assume it's bad. But it's helpful to uncouple the two domains and realize that it makes perfectly good sense to say, "This is terrible, but I like it." Or, "This is good, but I hate it." In short, I am not arguing here *against* criticizing or evaluating. I'm merely arguing *for* liking.

Let me sum up my clump of hypotheses so far:

- It's not improvement that leads to liking, but rather liking that leads to improvement.
- It's the mark of good writers to like their writing.
- Liking is not the same as evaluating. We can often criticize something better when we like it.
- We learn to like our writing when we have a respected reader who likes it.
- Therefore, it's the mark of good teachers to like students and their writing.

If this set of hypotheses is true, what practical consequences follow from it? How can we be better at liking? It feels as though we have no choice—as though liking and not-liking just happen to us. I don't really understand this business. I'd love to hear discussion about the mystery of liking—the phenomenology of liking. I sense it's some kind of putting oneself out—or holding oneself open—but I can't see it clearly. I have a hunch, however, that we're not so helpless about liking as we tend to feel.

For in fact I can suggest some practical concrete activities that I have found fairly reliable at increasing the chances of liking student writing:

(a) I ask for lots of private writing and merely shared writing, that is, writing that I don't read at all, and writing that I read but don't comment on. This makes me more cheerful because it's so much easier. Students get *better* without me. Having to evaluate writing—especially bad writing—makes me more likely to

hate it. This throws light on grading: it's hard to like something if we know we have to give it a D.

(b) I have students share lots of writing with each other—and after a while respond to each other. It's easier to like their writing when I don't feel myself as the only reader and judge. And so it helps to build community in general: it takes pressure off me. Thus I try to use peer groups not only for feedback, but for other activities too, such as collaborative writing, brainstorming, putting class magazines together, and working out other decisions.

(c) I increase the chances of my liking their writing when I get better at finding what *is* good—or *potentially* good—and learn to praise it. This is a skill. It requires a good eye, a good nose. We tend—especially in the academic world—to assume that a good eye or fine discrimination means *criticizing*. Academics are sometimes proud of their tendency to be bothered by what is bad. Thus I find I am sometimes looked down on as dumb and indiscriminating: “He likes bad writing. He must have no taste, no discrimination.” But I've finally become angry rather than defensive. It's an act of discrimination to see what's good in bad writing. Maybe, in fact, this is the secret of the mystery of liking: to be able to see potential goodness underneath badness.

Put it this way. We tend to stereotype liking as a “soft” and sentimental activity. Mr. Rogers is our model. Fine. There's nothing wrong with softness and sentiment—and I love Mr. Rogers. But liking can also be hard-assed. Let me suggest an alternative to Mr. Rogers: B. F. Skinner. Skinner taught pigeons to play ping-pong. How did he do it? Not by moaning, “Pigeon standards are falling. The pigeons they send us these days are no good. When I was a pigeon . . .” He did it by a careful, disciplined method that involved close analytic observation. He put pigeons on a ping-pong table with a ball, and every time a pigeon turned his head 30 degrees toward the ball, he gave a reward (see my “Danger of Softness”).

What would this approach require in the teaching of writing? It's very simple . . . but not easy. Imagine that we want to teach students an ability they badly lack, for example how to organize their writing or how to make their sentences clearer. Skinner's insight is that we get nowhere in this task by just telling them how much they lack this skill: “It's disorganized. Organize it!” “It's unclear. Make it clear!”

No, what we must learn to do is to read closely and carefully enough to show the student little bits of *proto*-organization or *sort of* clarity in what they've already written. We don't have to pretend the writing is wonderful. We could even say, “This is a terrible paper and the worst part about it is the lack of organization. But I will teach you how to organize. Look here at this little organizational move you made in this sentence. Read it out loud and try to feel how it pulls together this stuff here and distinguishes it from that stuff there. Try to remember what it

felt like writing that sentence—creating that piece of organization. Do it some more.” Notice how much more helpful it is if we can say, “Do *more* of what you’ve done here,” than if we say, “Do something *different* from anything you’ve done in the whole paper.”

When academics criticize behaviorism as crude it often means that they aren’t willing to do the close careful reading of student writing that is required. They’d rather give a cursory reading and turn up their nose and give a low grade and complain about falling standards. No one has undermined behaviorism’s main principle of learning: that reward produces learning more effectively than punishment.

(d) I improve my chances of liking student writing when I take steps to get to know them a bit as people. I do this partly through the assignments I give. That is, I always ask them to write a letter or two to me and to each other (for example about their history with writing). I base at least a couple of assignments on their own experiences, memories, or histories. And I make sure some of the assignments are free choice pieces—which also helps me know them.

In addition, I make sure to have at least three conferences with each student each semester—the first one very early. I often call off some classes in order to keep conferences from being too onerous (insisting nevertheless that students meet with their partner or small group when class is called off). Some teachers have mini-conferences with students during class—while students are engaged in writing or peer group meetings. I’ve found that when I deal only with my classes as a whole—as a large group—I sometimes experience them as a herd or lump—as stereotyped “adolescents”; I fail to experience them as individuals. For me, personally, this is disastrous since it often leads me to experience them as that scary tribe that I felt rejected by when I was an eighteen-year-old—and thus, at times, as “the enemy.” But when I sit down with them face to face, they are not so stereotyped or alien or threatening—they are just eighteen-year-olds.

Getting a glimpse of them as individual people is particularly helpful in cases where their writing is not just bad, but somehow offensive—perhaps violent or cruelly racist or homophobic or sexist—or frighteningly vacuous. When I know them just a bit I can often see behind their awful attitude to the person and the life situation that spawned it, and not hate their writing so much. When I know students I can see that they are smart behind that dumb behavior; they are doing the best they can behind that bad behavior. Conditions are keeping them from acting decently; something is holding them back.

(e) It’s odd, but the more I let myself show, the easier it is to like them and their writing. I need to share some of my own writing—show some of my own feelings. I need to write the letter to them that they write to me—about my past experiences and what I want and don’t want to happen.

(f) It helps to work on my own writing—and work on learning to *like* it. Teachers who are most critical and sour about student writing are often having trouble with their own writing. They are bitter or unforgiving or hurting toward their own work. (I think I've noticed that failed PhDs are often the most severe and difficult with students.) When we are stuck or sour in our own writing, what helps us most is to find spaces free from evaluation such as those provided by freewriting and journal writing. Also, activities like reading out loud and finding a supportive reader or two. I would insist, then, that if only for the sake of our teaching, we need to learn to be charitable and to like our own writing.

A final word. I fear that this sermon about liking might seem an invitation to guilt. There is enough pressure on us as teachers that we don't need someone coming along and calling us inadequate if we don't *like* our students and their writing. That is, even though I think I am right to make this foray into the realm of feeling, I also acknowledge that it is dangerous—and paradoxical. It strikes me that we also need to have permission to hate the dirty bastards and their stupid writing.

After all, the conditions under which they go to school bring out some awful behavior on their part, and the conditions under which we teach sometimes make it difficult for us to like them and their writing. Writing wasn't meant to be read in stacks of twenty-five, fifty, or seventy-five. And we are handicapped as teachers when students are in our classes against their will. (Thus high school teachers have the worst problem here, since their students tend to be the most sour and resentful about school.)

Indeed, one of the best aids to liking students and their writing is to be somewhat charitable toward ourselves about the opposite feelings that we inevitably have. I used to think it was terrible for teachers to tell those sarcastic stories and hostile jokes about their students: "teacher room talk." But now I've come to think that people who spend their lives teaching *need* an arena to let off this unhappy steam. And certainly it's better to vent this sarcasm and hostility with our buddies than on the students themselves. The question, then, becomes this: do we help this behavior function as a venting so that we can move past it and not be trapped in our inevitable resentment of students? Or do we tell these stories and jokes as a way of staying stuck in the hurt, hostile, or bitter feelings—year after year—as so many sad teachers do?

In short I'm not trying to invite guilt, I'm trying to invite hope. I'm trying to suggest that if we do a sophisticated analysis of the difference between liking and evaluating, we will see that it's possible (if not always easy) to like students and their writing—without having to give up our intelligence, sophistication, or judgment.

Let me sum up the points I'm trying to make about ranking, evaluating, and liking:

- Let's do as little ranking and grading as we can. They are never fair and they undermine learning and teaching.
- Let's use evaluation instead—a more careful, more discriminating, fairer mode of assessment.
- But because evaluating is harder than ranking, and because too much evaluating also undermines learning, let's establish small but important evaluation-free zones.
- And underneath it all—suffusing the whole evaluative enterprise—let's learn to be better likers: liking our own and our students' writing, and realizing that liking need not get in the way of clear-eyed evaluation.

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“Spelling” by Margaret Atwood – 1981

My daughter plays on the floor
With plastic letters, red, blue, and hard yellow,
Learning how to spell,
Spelling,
How to make spells.

I wonder how many women
Denied themselves daughters,
Closed themselves in rooms,
Drew the curtains
So they could mainline words.

A child is not a poem,
A poem is not a child.
There is no either/or.
However.

I return to the story
Of the woman caught in the war
& Labour, her thighs tied
together by the enemy
so she could not give birth.

Ancestress: the burning witch
Her mouth covered by leather
To strangle words.

A word after a word
After a word is power.

At the point where language falls away
From the hot bones, at the point
Where the rock breaks open and darkness
Flows out of it like blood, at
The melting point of granite
When the bones know
They are hollow & the word
Splits & doubles & speaks
The truth & the body
Itself becomes a mouth.

This is a metaphor.

How do you learn to spell?
Blood, Sky + the Sun,
Your own name first,
Your first naming, your first name
Your first word.